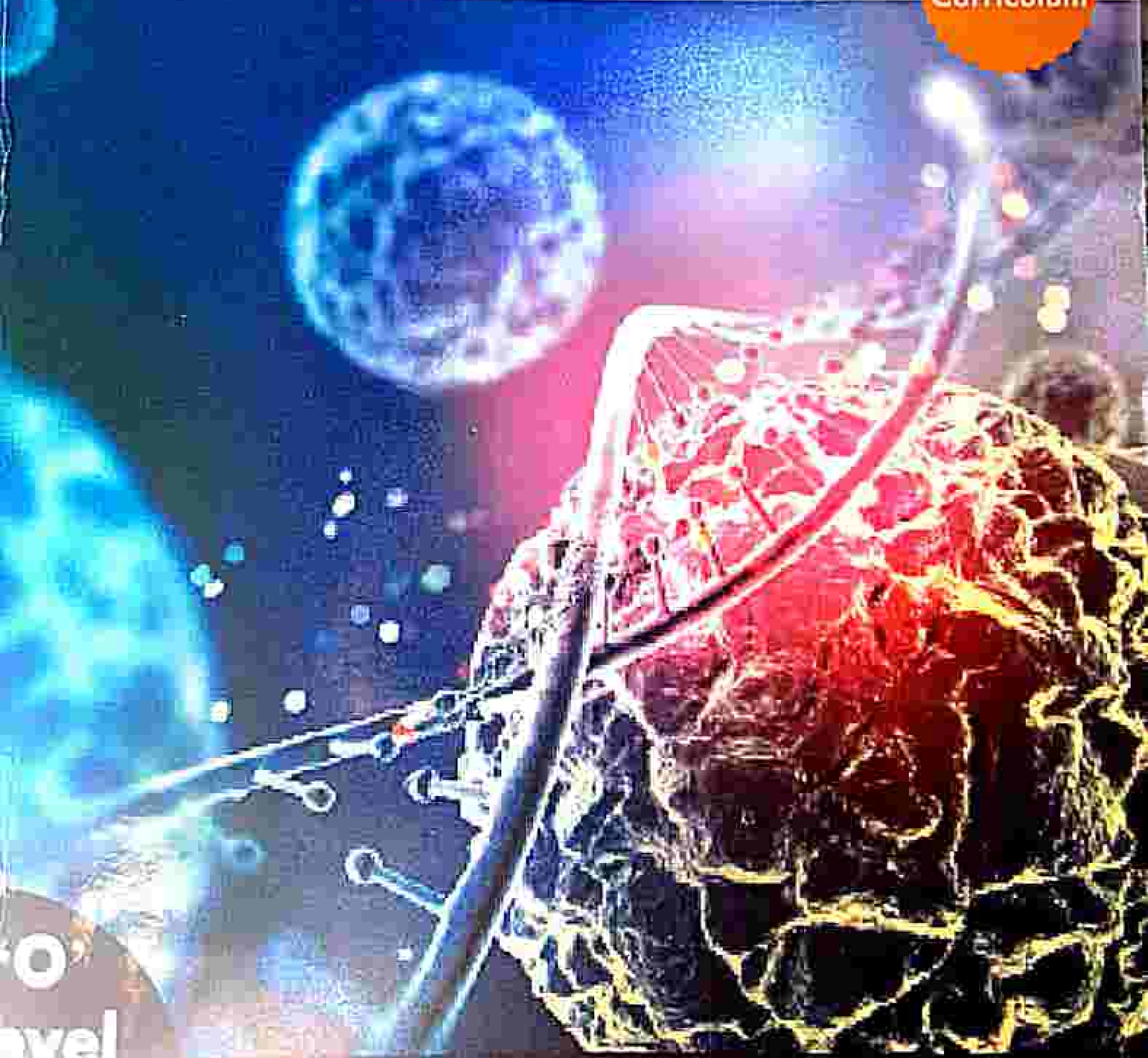
CPS

Secondary Combined Science

'O' Level Revision Book

New Curriculum



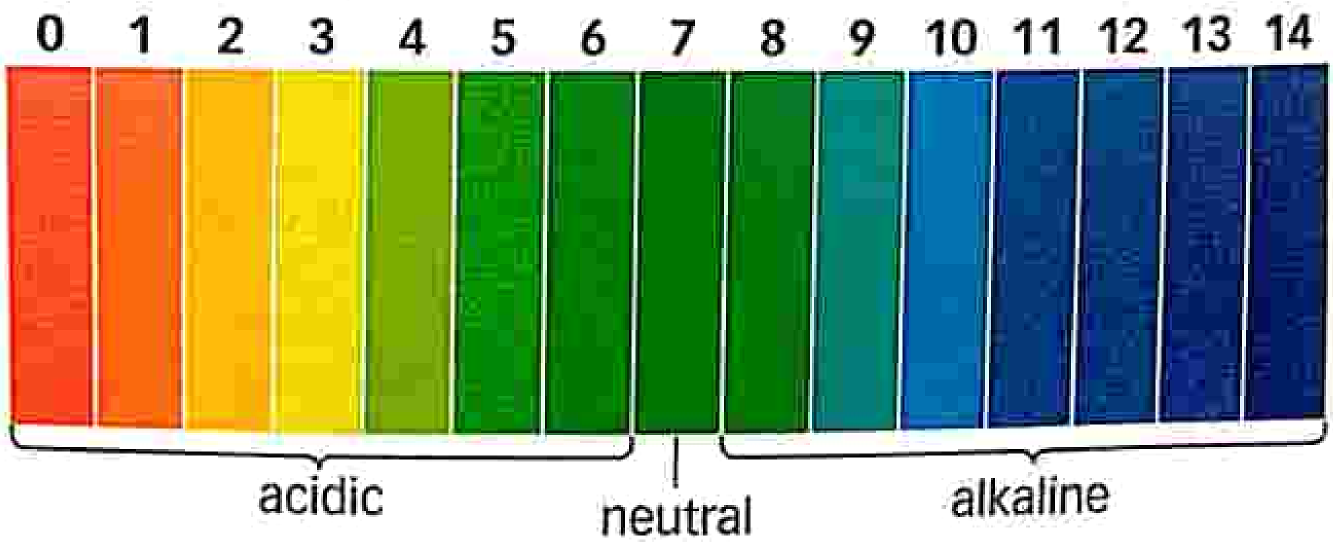
Petro F Tambama A

The Periodic Table of the Elements

↓Period / Group→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

pH scale



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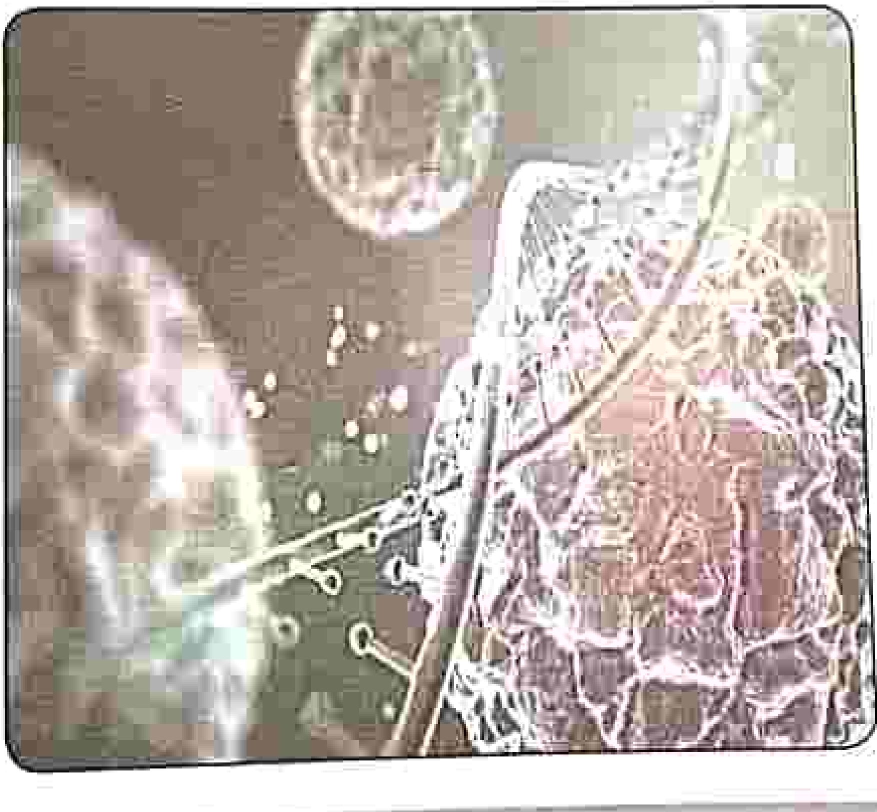

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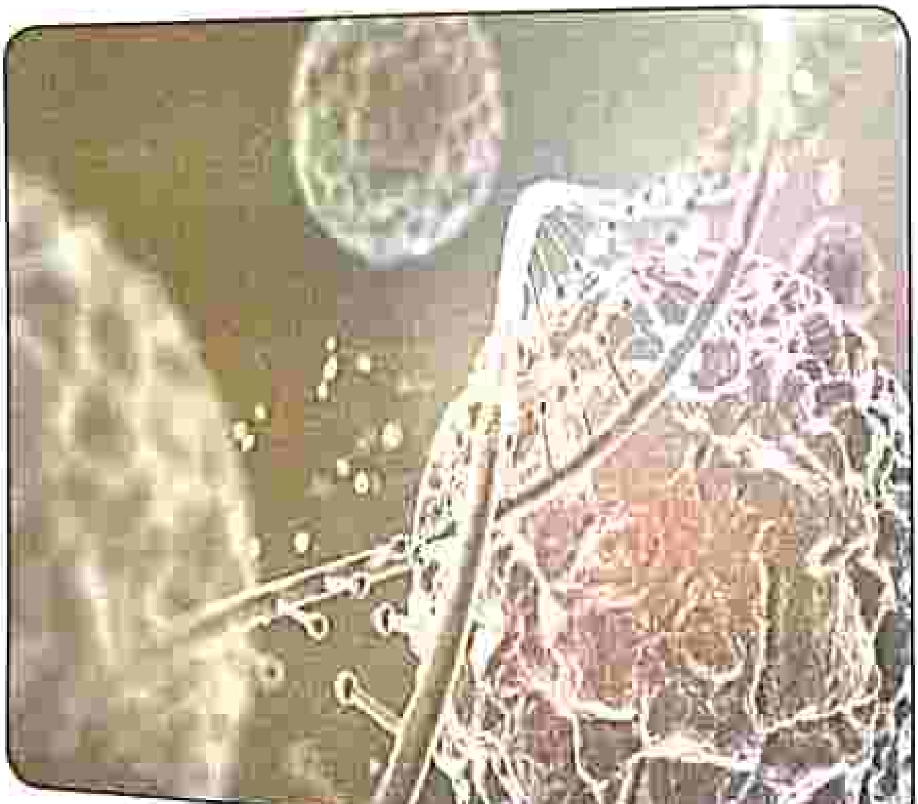


CHAPTER 1

Cell structure and function

Key concepts

- Identifying specialised cells
- Structure and function of specialised cells
- Using a microscope to observe cell structure

Summary notes

Specialised cells

- Specialised cells have distinct structures which enable them to adapt to their functions in an organism.

Red blood cells

The function of red blood cells is to transport oxygen around the body.

- Mature Red blood cells do not have a nucleus and this enables the cells to carry more oxygen inside them.
- The Bi-concave disc shape of the red blood cells increases the surface area for the absorption of oxygen because the centre of the cell is thinner than the edge.
- Haemoglobin which is a red pigment, readily combines with oxygen increasing the oxygen carrying capacity of the cell.

Fig 1.1 Red blood cells

Muscle cell

- The function of the Muscle cell is to bring about movement.
- Muscle cells are elastic, thus have the ability to change their shape from long and thin when they relax, to short and fat when they contract and this allows movement to take place.

When they relax

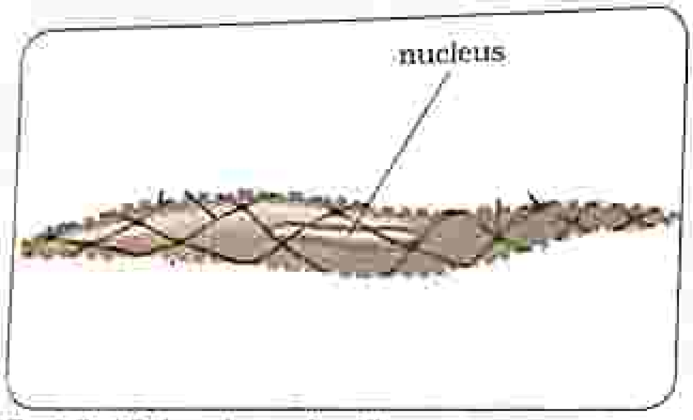


Fig 1.2 a) Relaxed muscle cells

When they contract

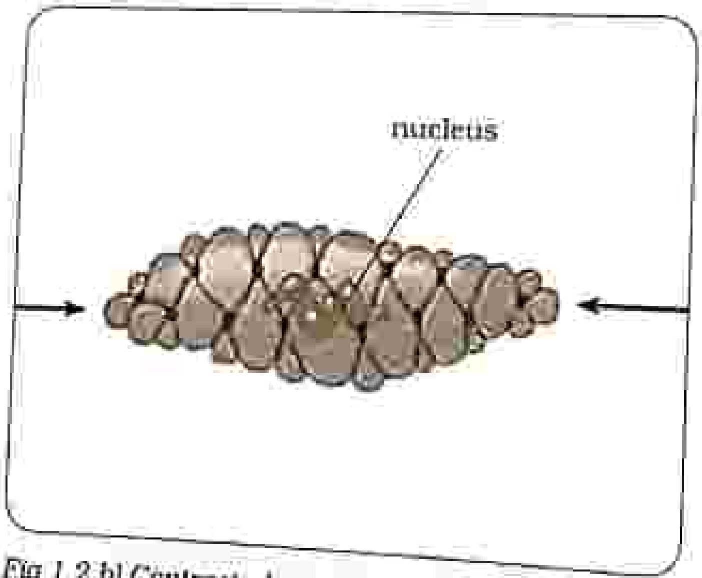


Fig 1.2 b) Contracted muscle cells

Palisade cell

- The function of the palisade cell is for enabling photosynthesis to take place.
- Palisade cells are packed with chloroplasts which contain the green pigment called chlorophyll which is responsible for trapping as much light as possible for photosynthesis to take place.
- Palisade cells are long and rectangular in shape so that many can fit into the palisade layer of the leaf hence increasing the rate of photosynthesis.
- Palisade cells are found in the palisade layer of the leaf.

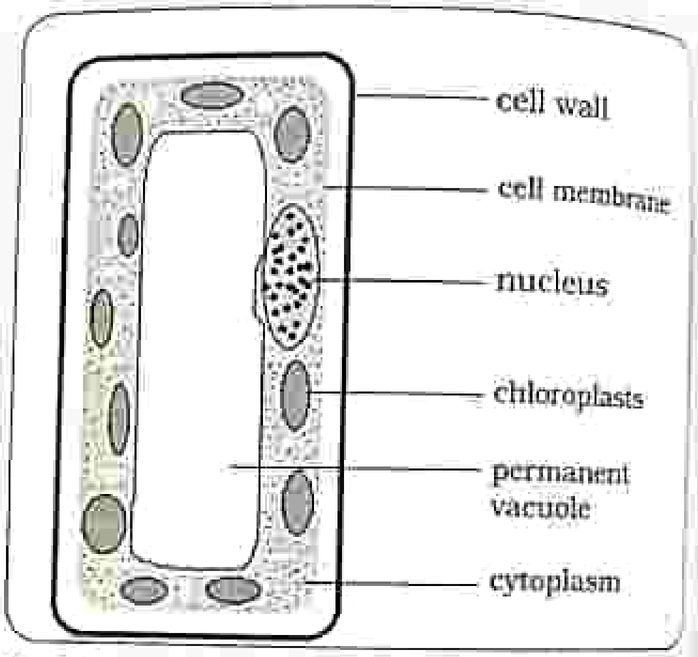


Fig 1.3 Palisade cell

Root hair cell

- The function of the root hair cell is to absorb water and dissolved ions from the soil into the root.
- The cell has a hair-like projection to help penetration between the soil particles and increase the surface area for the absorption of water and dissolved ions.
- As the root hair cell penetrates between soil particles, it can stick to the soil particles holding the soil firmly in place, thereby helping to anchor the plant in the ground.

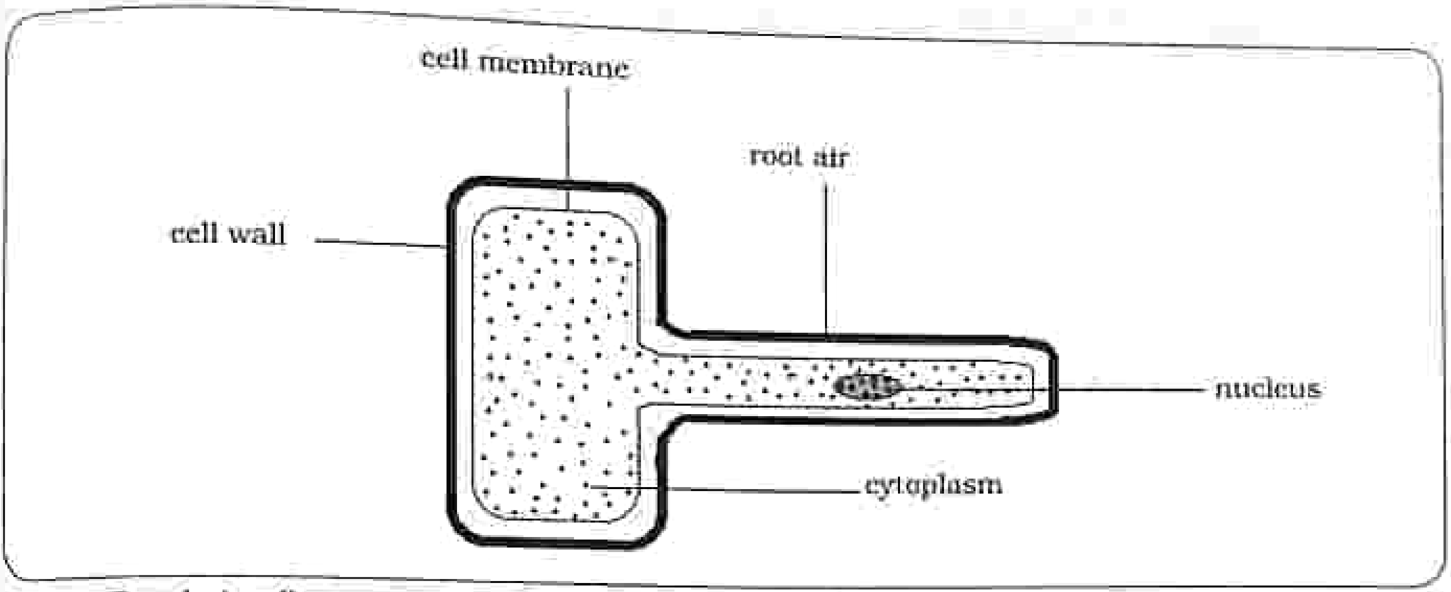


Fig 1.4: Root hair cell

Using a microscope to view cell structure

Practical activity 1

Aim : To observe cell structure of specialised cells.

Apparatus :
1. Light microscope
2. Prepared slides for specialised cells (red blood cell, muscle cell, palisade cell and root hair cell)

OR

3. Bioset

Method

1. Examine different cell types under a microscope or on a bioset.

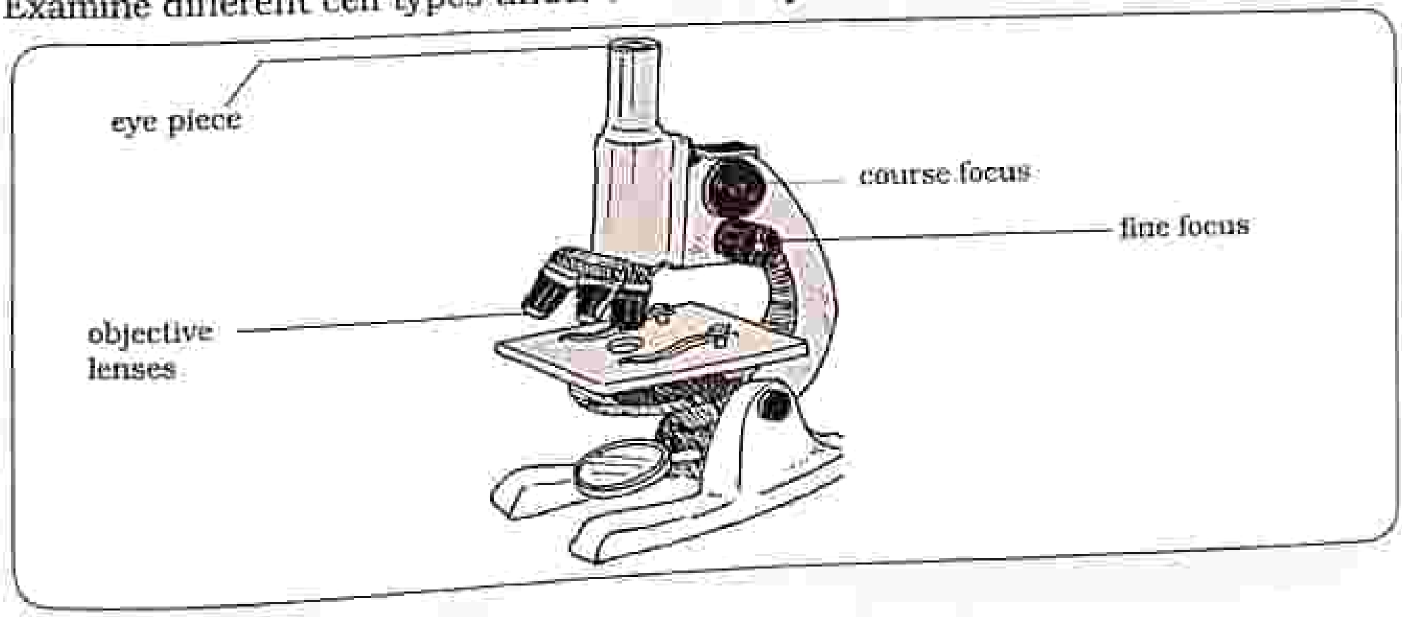


Fig 1.5: Microscope

Practical questions

1. Draw a diagram of each type of cell as you observe through the microscope or on a bioset and label the visible structures?
2. Identify the type of specialised cells as observed through the microscope or on a bioset?

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

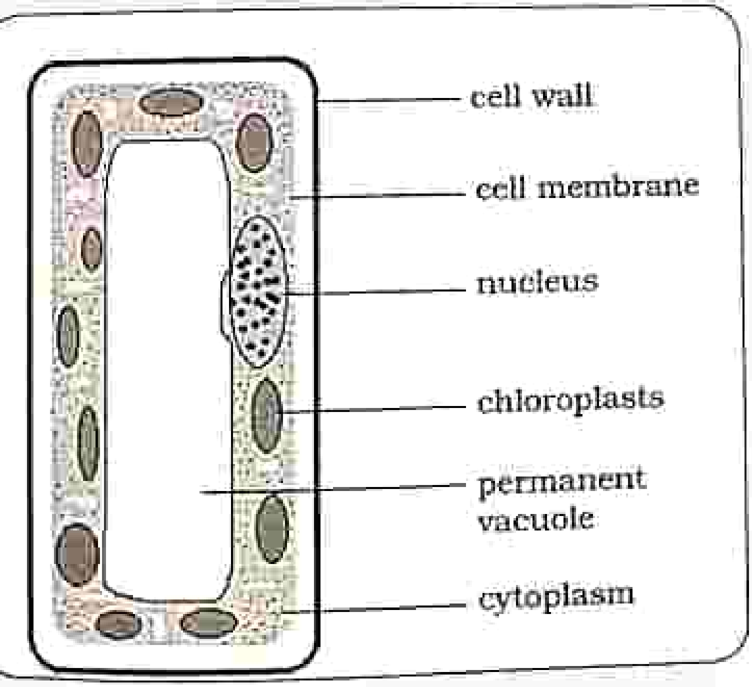
Multiple choice

1. What is the function of the nucleus in a cell?
   A. controls the cell's activities
   B. stores sugars
   C. controls what enters and leave the cell
   D. carries out photosynthesis
2. What is the function of the cell membrane in a cell?
   A. controls the cell's activities
   B. controls what enters and leaves the cell
   C. stores sugars and salts
   D. carries out photosynthesis
3. What is the function of the vacuole in the cell?
   A. it controls the movement of water in the cell
   B. controls the cell's activities
   C. surrounds the cell
   D. stores sugars and salts
4. Name three structures which are common in both plant and animal cells
   A. nucleus, cell wall, cytoplasm
   B. cell membrane, cytoplasm, permanent vacuole
   C. cytoplasm, cell membrane, nucleus
   D. cell wall, permanent vacuole, cell membrane

5. What is the function of the palisade cell?
   A. to absorb water and dissolved ions from the soil into the root
   B. bring about movement
   C. to transport oxygen around the body
   D. enabling photosynthesis to take place

6. Palisade cells are found in which layer of the leaf?
   A. upper epidermis
   B. lower epidermis
   C. palisade mesophyll
   D. spongy mesophyll

7. The diagram shows a palisade cell.



What is the function of the chloroplasts in the cell?
   A. carry oxygen
   B. contain chlorophyll which traps light for the process of photosynthesis
   C. contract and relax
   D. absorb water

8. Red blood cells have a bi-concave shape. How is this shape related to the function of the cell?
   A. has the ability to change shape
   B. brings about movement
   C. increases surface area for the absorption of oxygen
   D. absorbs a lot of light

9. Muscle cells have the ability to ______________ so that they allow movement to take place.
   A. absorb water and dissolved ions
   B. change their shape from long and thin to short and fat
   C. take in more light
   D. penetrate through the soil

10. The root hair cell has hair-like projection on each cell to______________
   A. help penetrate between soil particles and increase the surface area for the absorption of water and dissolved ions
   B. bring about movement
   C. trap light for photosynthesis
   D. become elastic

Structured questions

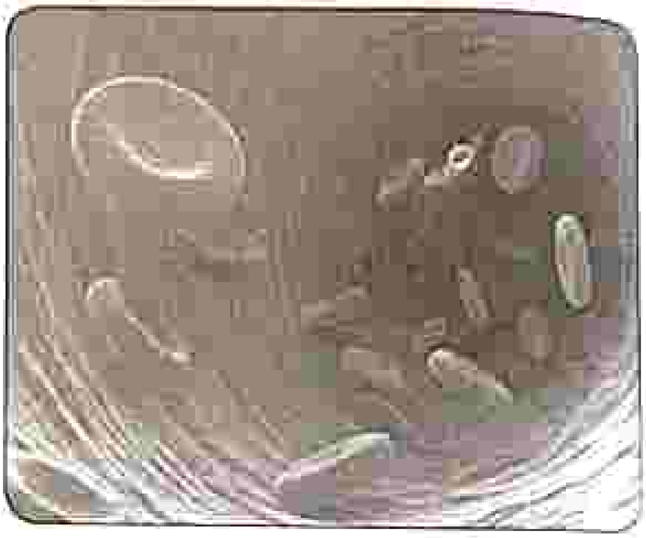
1. Identify three structures which are common to both plant and animal cells? [3]
2. Which structures are only found in plant cells? [2]
3. Specialised cells have distinct functions?
   a) State the functions of the following specialised cells by completing the table

Specialized cell	Function
Palisade cell	
Root hair cell	
Muscle cell	
Red blood cell	

[4]

   b) For each specialised cell given in question 3a) describe how the structure of the cell relates to its function [8]

4. The diagram shows some red blood cells.



   i) State a feature which is present in a muscle cell but is absent in a red blood cell [1]
   ii) Describe how the absence of the feature stated in 4 i) make the cell adapt to its function? [1]
5. Draw and label the specialised cells listed below
   i) Palisade cell [3]
   ii) Root hair cell [3]
   iii) Muscle cell [3]
   iv) Red blood cell [3]

CHAPTER 2

Ecosystems

Key concepts

- ecosystem
- components of an ecosystem
- Natural ecosystem
  - Energy flows
  - Food chains
  - Food webs
  - Pyramids of biomass
  - Carbon cycle
  - Nitrogen cycle
- Artificial ecosystem
  - Bio-diversity
  - Problems caused by limited bio-diversity
  - Advantages of bio-diversity

Summary notes

Ecosystem

- An ecosystem is a self-supporting system of interdependent organisms and their physical environment.

Components of an ecosystem

- An ecosystem contains:
  1. Physical component
  2. Biological component
- The physical and biological components of an ecosystem depend on each other meaning that they interact, therefore disturbing one part of the ecosystem may have a negative impact on the other part of the ecosystem.

Physical components

These are non-living parts of the ecosystem which include sunlight, soil and air.

- Sunlight – used to make food during photosynthesis. It is the ultimate source of energy in the ecosystem.
- Soil – it provides plants with nutrients and water so that they grow well. Soil is a habitat for some organisms like ants and worms.
- Air – it is essential in an ecosystem. Components of air include:

– Carbon-dioxide – needed by plants for photosynthesis

Note: carbon-dioxide is a by-product of respiration.

– Oxygen – living organisms need oxygen for respiration

Note: oxygen is a by-product of photosynthesis

– Nitrogen – for plants and animals to make proteins, nitrogen must be present. However, plants and animals cannot absorb nitrogen gas directly from the environment therefore the nitrogen is converted by bacteria and fungi.

– Water vapour – the amount of water vapor in the air varies. Living organisms need water to survive in the ecosystem. Plants also need water to make food for the ecosystem.

Biological components

- These are the living parts of an ecosystem and these include macro-organisms which are large organisms that are visible using our naked eyes for example plants and animals.
- They also include micro-organisms and these are very small and invisible to the naked eye for example bacteria and fungi.

Natural ecosystem

- It is an ecosystem with no human interference.
- Most natural ecosystems are in a state of equilibrium (balance) because of a variety of producers and consumers living together. Examples of natural ecosystems are rivers and deserts.
- The biological components of an ecosystem can be divided into:
  a) Producers – these are green plants which use light energy from the sun to make their own food by photosynthesis. Examples include grasses, trees and shrubs.
  b) Consumers – these depend on other organisms for food. They are not able to make their own food.
  Consumers are classified into three:
    i) Herbivores – these are animals that feed on plants and they are the primary consumers of an ecosystem. Examples include caterpillar, kudu and zebra
    ii) Carnivores – these are animals which feed on other animals. They feed on the primary consumers. These can be secondary or tertiary consumers of an ecosystem. Examples include lion, hyena and vulture.
    iii) Omnivores – these are animals which feed on both plants and animals. Examples include human beings and monkeys.
  c) Decomposers – they feed on dead plants and animals. They break down dead organic matter into simple nutrients in the soil which are then absorbed by plants when they make their own food. Examples include bacteria, fungi and earthworms.

Note:organisms in a natural ecosystem depend on one another for survival.

Energy flow

- The sun is the ultimate source of energy in the ecosystem. Energy from the sun is used by the producers (green plants) for photosynthesis.
- The energy is passed on from the producers to the consumers during feeding.
- Energy flows from dead plants and animals to decomposers during decomposition.
- At each trophic level, organisms use some of energy and some of the energy is lost to the environment for example in sweat, urine and faeces.

Food chain

- It shows feeding relationships between organisms in an ecosystem.

- They also show how energy flows from one organism to another. Energy is transferred from one link to another in a food chain.
- A food chain starts with a producer (green plant).
- A food chain is linked using arrows.

Example

1. Grass	→	Insects	→	Lizards	→	Birds
2. Leaf	→	Caterpillar	→	Bird	→	Hawk
3. Algae	→	fly larvae	→	Dwarf bream	→	Fish Eagle
Producer		**Primary consumer**		**Secondary consumer**		**Tertiary consumer**

Food web

- Food chains are linked to form food webs.
- A food web with many organisms is more stable.
- The source of food is always a producer (green plants)

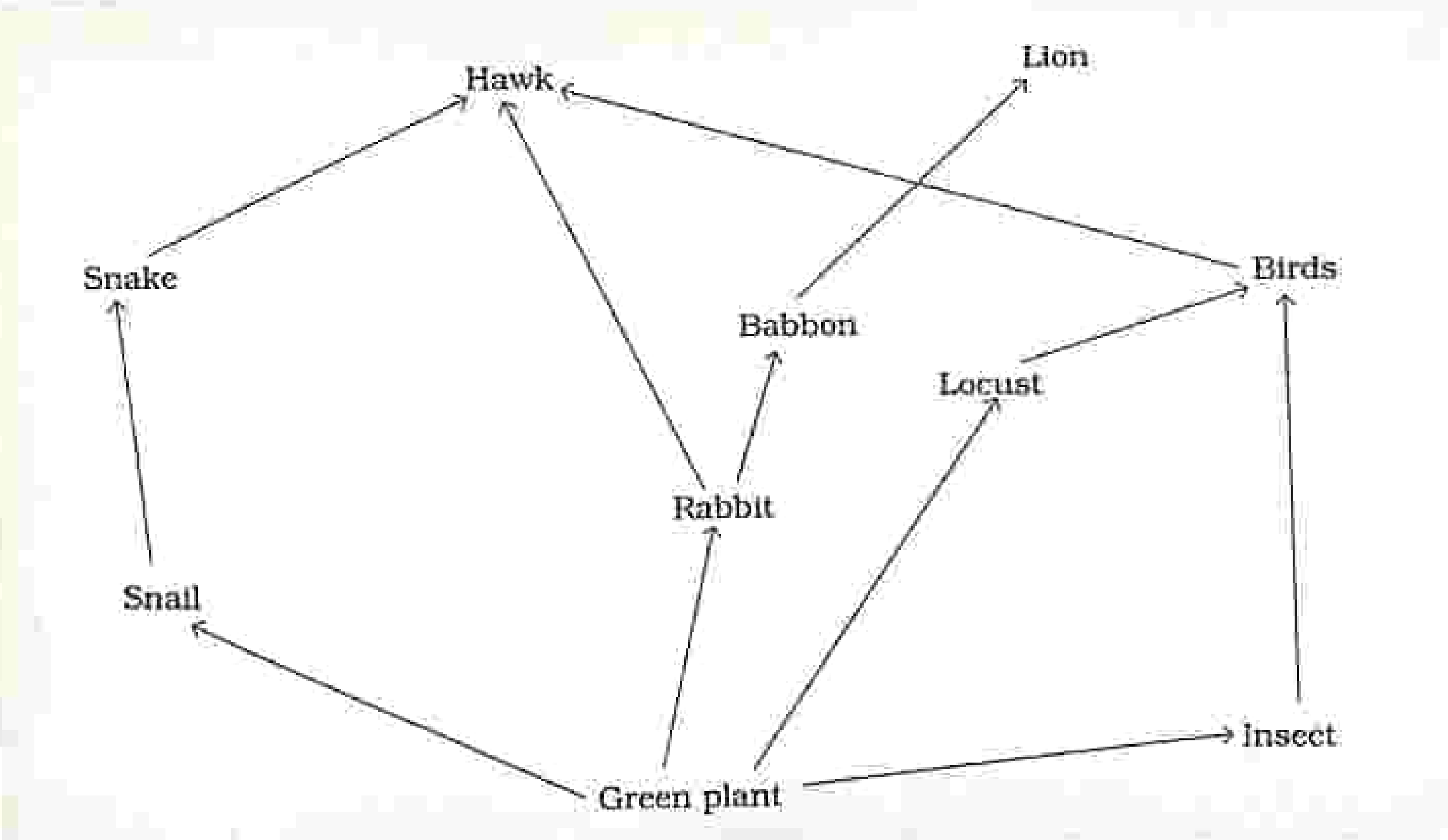


Fig 2.1 Food web

NOTE: The links in the food chains and food webs depend on each other. If one group of organism dies, the other groups are affected.

Pyramid of biomass

- It shows the total mass of organisms in each trophic level. It is a more accurate way of finding the amount of food produced and the amount consumed in an ecosystem.
- In a balanced ecosystem, the biomass of producers should be greater than that of primary consumers, whose biomass must in turn be greater than that of secondary consumers.
- Energy is lost at each level.

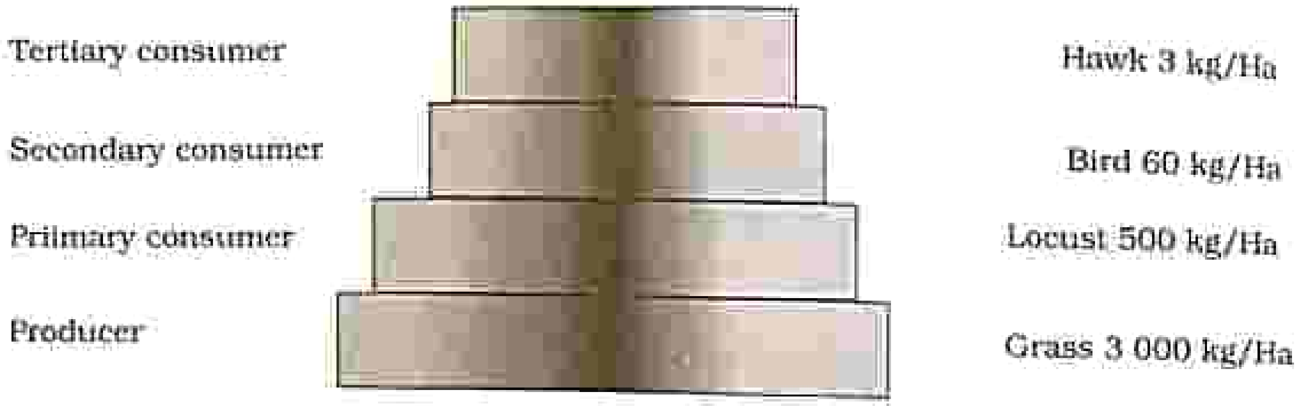


Fig 2.2: Pyramid of Biomass

Carbon cycle

- It is the movement of carbon around the ecosystem. Plants take in carbon in form of carbon dioxide from the air and use it for photosynthesis.
- Carbon is passed on from plants to animals during feeding.
- When plants and animals die, decomposers get the carbon when acting on the dead and decaying plants and animals.
- During respiration, carbon is released into the atmosphere in form of carbon dioxide.
- During burning (combustion) of fossil fuels and plant matter carbon in the form of carbon dioxide is released back into the atmosphere.

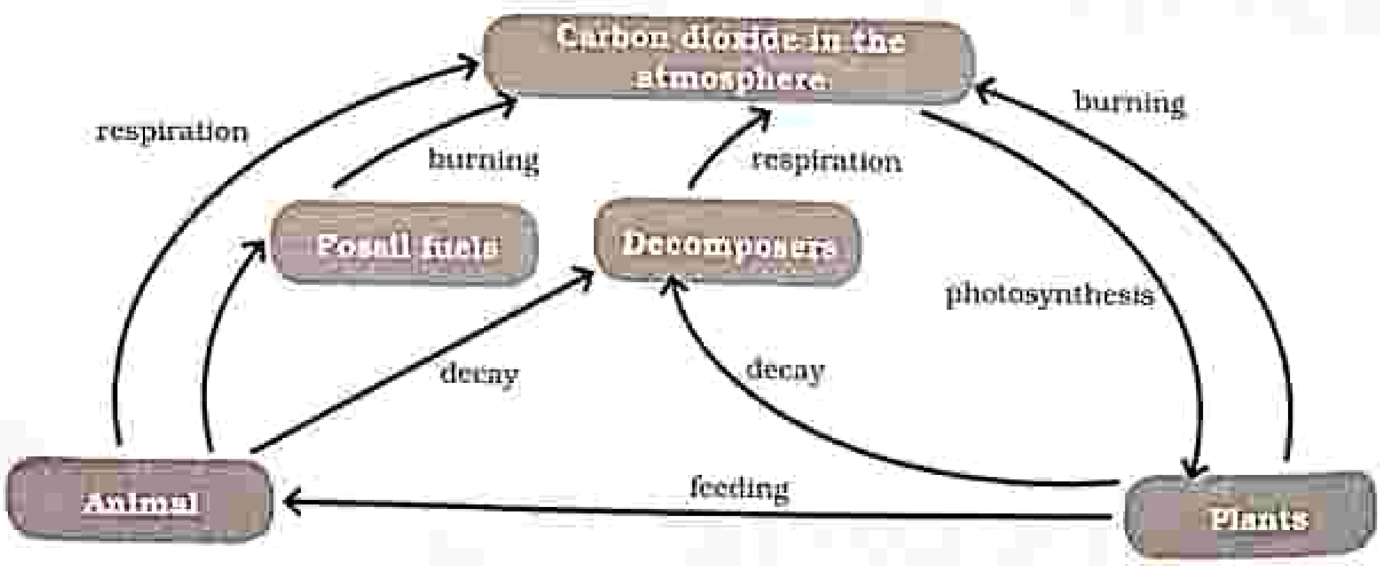


Fig 2.3: Carbon cycle

Nitrogen cycle

- It is the movement of nitrogen between air and living organisms of the earth.
- Air is 78% nitrogen.

Micro-organisms involved in the nitrogen cycle

- **Nitrifying bacteria** – when plants and animals die nitrifying bacteria act on their proteins in the soil to form ammonia and nitrites which are converted to nitrates which plants can absorb from the soil.
- **Decomposers** – when decomposers (bacteria and fungi) act on dead plants and animals and their waste (faeces and urine) ammonia is made which is then converted to nitrates which are added to the soil for plants to absorb. Some of the proteins in the dead plants and animals is also broken down by decomposers and use the nitrogen to make decomposer protein.
- **Nitrogen fixing bacteria** – nitrogen fixing bacteria live in the roots of leguminous plants. Examples of leguminous plants include groundnuts, peas and beans. Nitrogen fixing bacteria convert nitrogen gas from the atmosphere into amino acids in plants. Nitrogen fixing bacteria also convert nitrogen from the atmosphere to ammonia which builds into ammonia compounds. The ammonia compounds are then made into nitrates which can be absorbed by plants.
- **Denitrifying bacteria** – these are harmful bacteria which are found in the soil especially in water-logged soil. Useful nitrates are broken down by the bacteria and nitrogen gas is released into the atmosphere.
  - **NB**: Nitrogen can be added into the soil in form of organic fertilizers like compost and manure or as inorganic fertilizers like ammonium salts and nitrate salts.

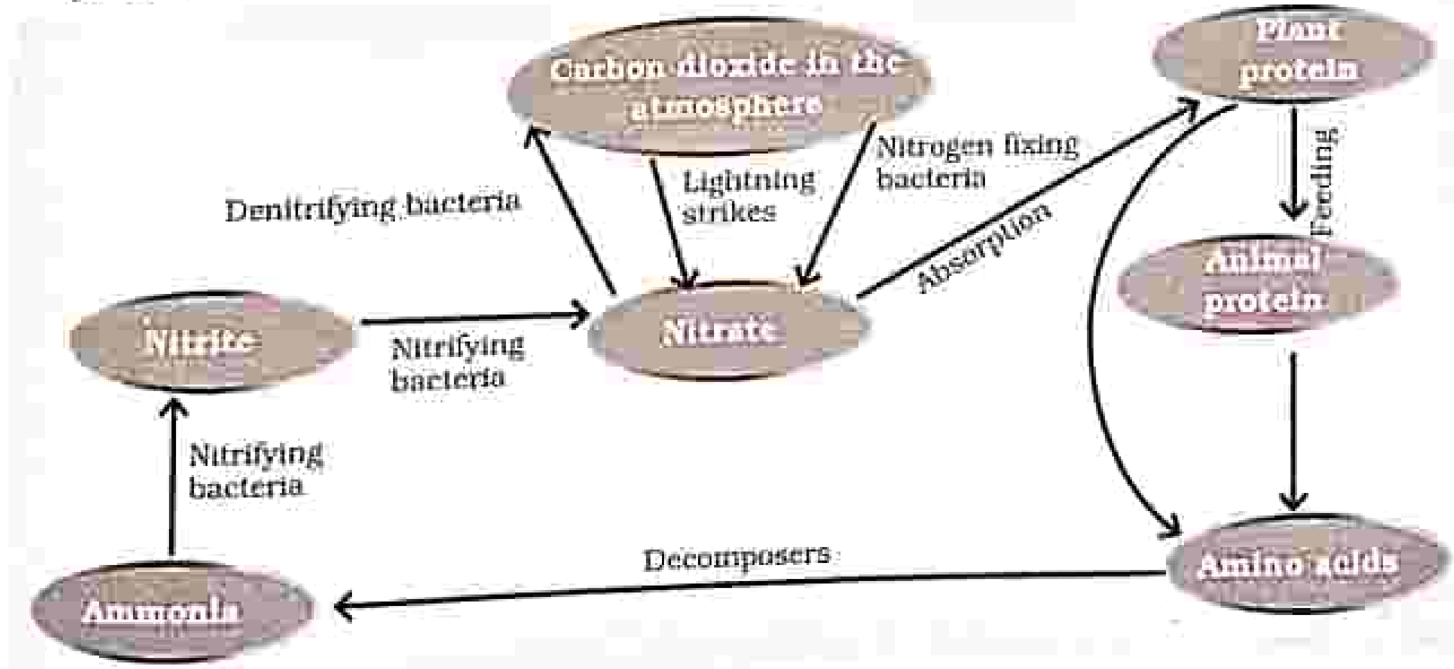


Fig 2.4: Nitrogen cycle

Artificial ecosystem

- The system is not left to develop and control itself naturally.
- There is human interference whereby species are removed and others are added.
- Humans create an artificial ecosystem when they clear land for settlement, agriculture and other activities. This disturbs the balance in an ecosystem and makes it unstable.

Examples of artificial Ecosystems

- Garden
- Pond
- Plantation
- In an artificial ecosystem pests and diseases are controlled.
- Artificial fertilizers like ammonium nitrates are added into the soil to add nutrients.
- Grazing is controlled by the use of paddocks so that the grazing land has a chance to recover.

Bio-diversity

- It refers to a variety of different types of organisms in an area.

Bio-diversity in a natural and artificial ecosystem

Natural ecosystem	Artificial ecosystem
• have greater bio-diversity because there is a lot of different plants and animal species	• less bio-diversity because there are very few animal and plant species
• the extinction of one animal or plant species may not have an effect on other organisms because there are other organisms to eat	• the extinction of one animal or plant species will have an effect on other organisms as they may also become extinct because there will be nothing to feed on

Problems caused by limited bio-diversity

- Soil infertility – caused by one type of plant species growing in an area resulting in the draining of soil nutrients for example in a plantation where the same plant grows in the area each season. All nutrients are drained until the soil becomes infertile.
- Pests and disease – caused by very few animal and plant species because the species which could control the pests by feeding may have become extinct as a result must be controlled artificially by human beings.

Advantages of biodiversity

- Wide variety of food source
- Self-sustenance of an ecosystem
- Interdependence
- Less spread of diseases

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

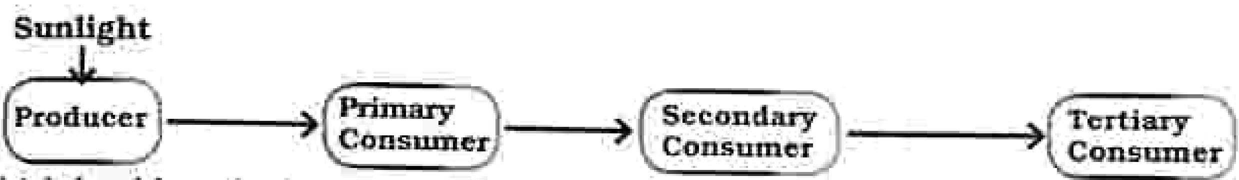
Section B

Structured questions

Answer all the questions.
The number of marks is given in brackets[ ] at the end of each question.

Multiple choice questions

1. The diagram below shows the flow of energy through an ecosystem.



   Which level has the lowest amount of energy?
   A. Tertiary Consumer
   B. Primary Consumer
   C. Producer
   D. Secondary Consumer
2. Which one is an example of the physical components of an ecosystem?
   A. Bacteria
   B. Sunlight
   C. Fungi
   D. Plants
3. Identify a producer
   A. locust
   B. bird
   C. grass
   D. Owl
4. In the carbon cycle, which process removes carbon from the atmosphere?
   A. burning
   B. respiration
   C. digestion
   D. photosynthesis
5. In the carbon cycle, which two processes return carbon to the atmosphere?
   A. burning and photosynthesis
   B. respiration and burning
   C. photosynthesis and respiration
   D. digestion and burning
6. Which one is an example of an artificial ecosystem?
   A. pond
   B. sea
   C. desert
   D. river

7. In the pyramid of biomass, at which level should biomass, be greater?
   A. primary consumer
   B. producer
   C. secondary consumer
   D. tertiary consumer

8. Nitrogen fixing bacteria can be found ______________ in the nitrogen cycle.
   A. in the roots of leguminous plants
   B. in leaves
   C. twigs
   D. branches

9. Biodiversity ....
   A. is a variety of organisms in an area.
   B. is movement of carbon dioxide around the ecosystem.
   C. is the living part of the ecosystem.
   D. shows the total mass of organisms.

10. Green plants make their own food using energy from the sun, so they are called..........
   A. secondary consumers
   B. primary consumers
   C. producers
   D. tertiary consumers

Section B: Structured questions

1 a) Explain the term ecosystem. [2]
  b) List components of an ecosystem. [2]

2 a) Explain
    i) natural ecosystem [1]
    ii) artificial ecosystem [1]
  b) Make the correct sequence to construct food chains from the lists given

i) insects	green plant	hawk	birds	
ii) hawk	snail	green plant	snake	
iii) grasshopper	bird	grass		[3]

c) Complete using arrows to construct a food web

Human being

King fisher

Frog

Small fish

Water beetle

Crab

Worms

Water weed

[5]

3. In the carbon cycle:

i) Which two processes return carbon to the atmosphere? [2]

ii) Which process removes carbon from the atmosphere? [1]

4. In the nitrogen cycle briefly explain the function of:

i) Decomposers [1]

ii) Nitrifying bacteria [1]

iii) Nitrogen fixing bacteria [1]

iv) Denitrifying bacteria [1]

5 a) Explain the term biodiversity. [1]

b) Identify two problems caused by limited bio-diversity. [2]

c) State three advantages of bio-diversity. [3]

CHAPTER 3

Plant nutrition

Key concepts

- Photosynthesis
- Word equation for photosynthesis
- Starch test
- Factors which affect photosynthesis
- Fate of end products of photosynthesis
- Test for oxygen using a glowing splint
- Leaf structure
  - external leaf structure
  - internal leaf structure
- Adaptation of a leaf for photosynthesis

Summary notes

Photosynthesis

- It is the process whereby green plants use light energy to make their own food (carbohydrates) using carbon dioxide and water, releasing oxygen as an end product.

Word equation

$$\text{Carbon dioxide} + \text{water} \xrightarrow[\text{chlorophyll}]{\text{light energy}} \text{Carbohydrates} + \text{Oxygen}$$

Practical Activity 3.1

Aim : Testing a leaf for starch

Apparatus :
1. green leaf
2. iodine solution
3. methylated spirit
4. beaker/boiling tin containing water
5. bunsen burner
6. test tube
7. white tile
8. dropper

Method

1. Boil the leaf in water.
   Reason: to denature the enzymes in the leaf in order to prevent any chemical reactions.

2. Boil the leaf in methylated spirit.

NB: Methylated spirit is inflammable so do not put the test tube containing methylated spirit on direct flame, instead place the test tube in a beaker with water.

Reason – to remove chlorophyll

– to enable the colour changes to be visible

3. Dip the leaf in hot water.

Reason – to soften the leaf

4. Place the leaf on a white tile and using a dropper, drop iodine solution on it.

Reason – iodine is used to test for starch

i) Blue black colour : Starch is present

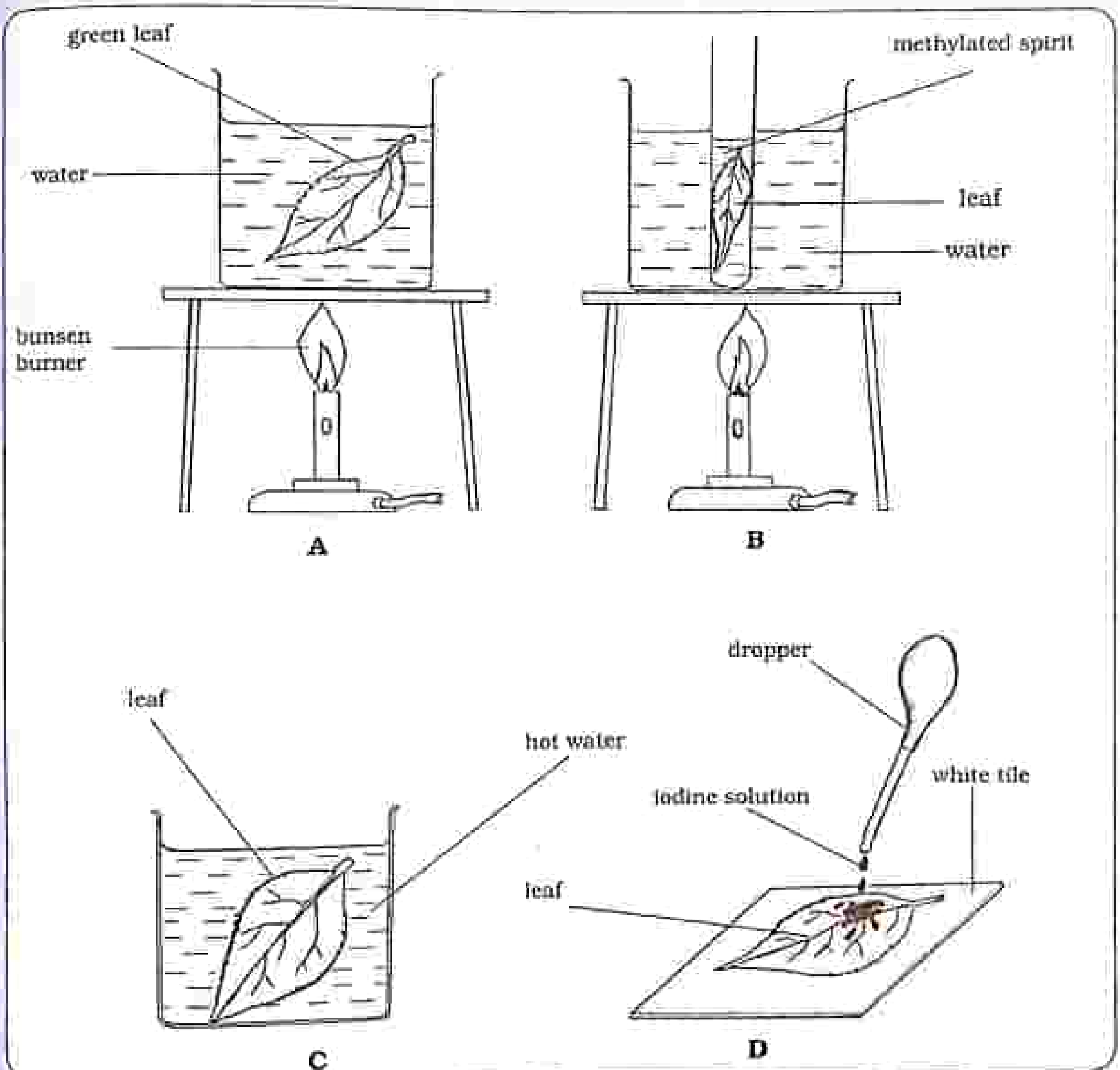


Fig 3.1 Starch test

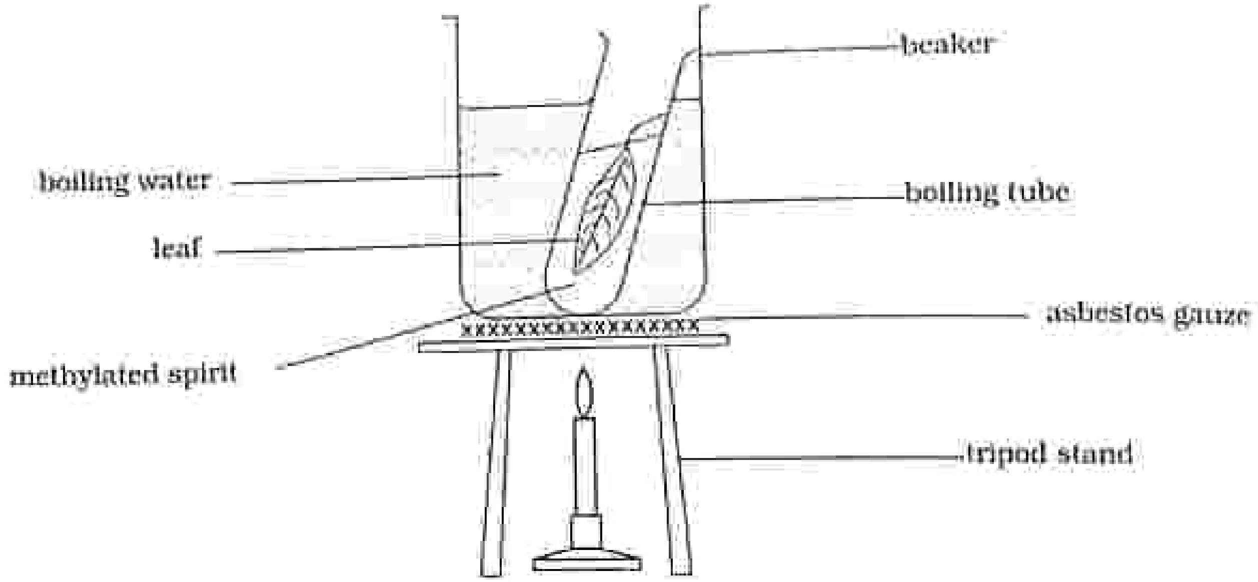


Fig 3.1 Starch test (continued)

Results

A blue-black colour denotes the presence of starch.

Conclusion

Green plants manufacture (make) starch during photosynthesis.

Practical questions

1. What is your observation at stage D in fig 3.1?
2. What conclusion can be drawn from your observation in (1)

Factors which affect photosynthesis

1. **Water**
    - Water is important for the chemical reactions to take place.
        - Carbon dioxide dissolves in water to form carbohydrates.
    - Water also keeps the plant cells turgid in order to support the plant.

2. **Light**

   It is necessary for the chemical reactions which take place in the plant because it provides the plant with energy for photosynthesis.

Practical activity 3.2

Aim : Is light necessary for photosynthesis

Apparatus :
1. one potted plant
2. Cardboard or aluminum foil
3. sellotape
4. Paper clips
5. Scissors
6. Starch testing kit

Method

1. De-starch a green plant.
2. Cover part of the leaf firmly using a small piece of aluminium foil or cardboard ensuring no light is allowed through.
3. Leave the plant in sunlight for a few hours (4-6 hours).
4. Test the covered leaf and the one fully exposed to light for starch.

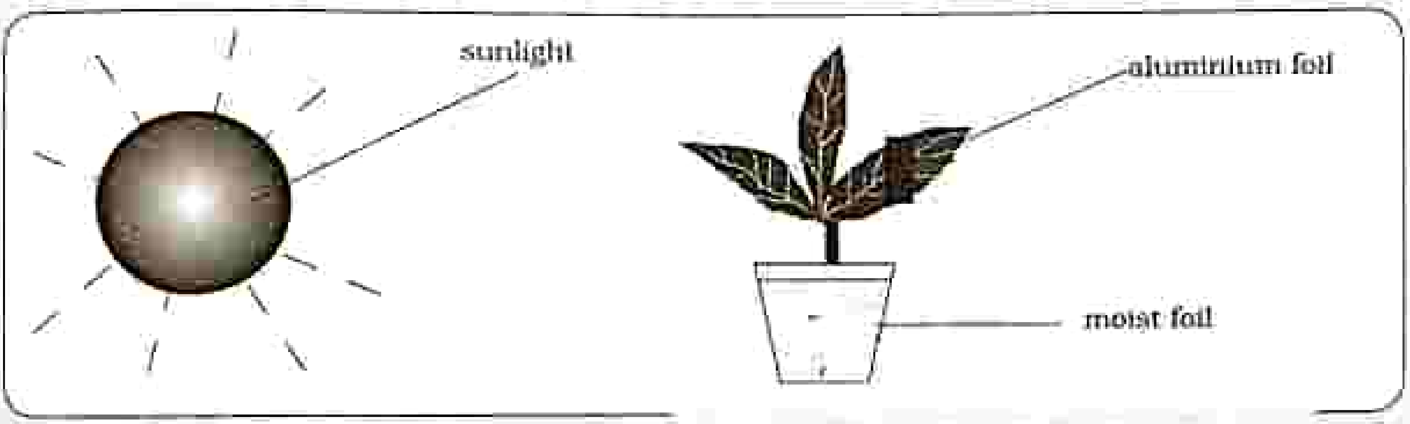


Fig 3.2 Is light necessary for photosynthesis

Results

Leaf fully exposed to light: iodine solution turns blue-black

Leaf covered with aluminium foil: no colour change

Conclusion

Light is necessary for photosynthesis

Practical questions

1. What do you observe on the part covered with the aluminium foil?
2. Explain the starch test results for the part which was covered by the aluminium foil?
3. What is the reason for de-starching the plant before the experiment?

3. **Carbon dioxide**
   - Enters the leaves through the stomata (pores).
   - Dissolves in water to form carbohydrates (glucose).

Practical activity 3.3

Aim : Is carbon dioxide necessary for photosynthesis

Apparatus :
1. Two potted plants
2. Soda lime/sodium hydroxide/sodium hydrogen carbonate
3. Polythene bags
4. String
5. Starch testing kit

Method

1. De-starch the two potted plants by placing the plants in the dark for 48 hours.
2. Cover each potted plant with a polythene bag to ensure no more carbon dioxide enters the plant.
3. Place a container of soda lime inside one of the potted plants.
   NB: Soda lime/sodium hydroxide/sodium hydrogen carbonate, absorb carbon dioxide.
4. Set up both plants in the light for a few hours.
5. Test one leaf from each plant for starch.

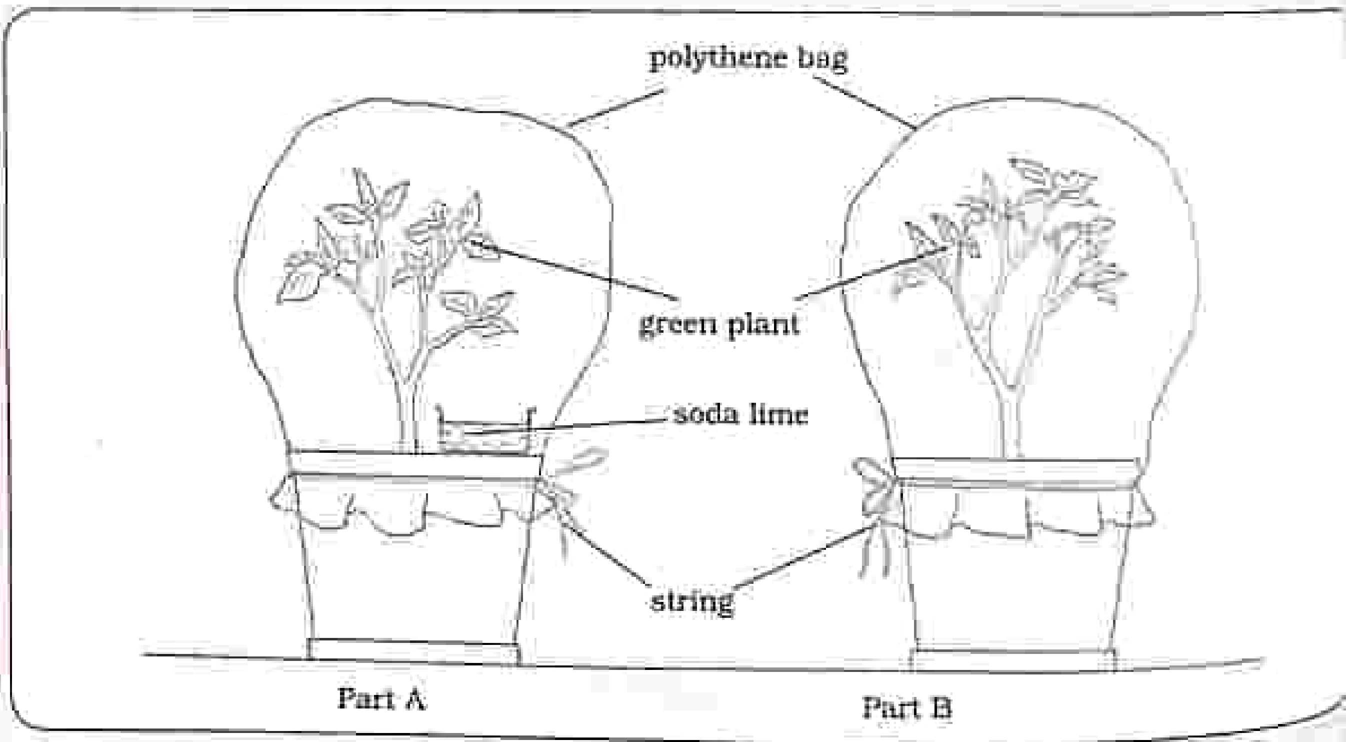


Fig 3.4 Is carbon dioxide necessary for photosynthesis

Result

Leaf from plant A : no colour change on iodine solution.

Leaf from plant B : iodine solution turns blue-black.

Conclusion

Carbon dioxide is necessary for photosynthesis.

Practical questions

1. Identify the control plant.
2. What do you observe after the starch test for:
   a) Leaf from plant A?
   b) Leaf from plant B?

4. Chlorophyll

- is the green pigment found in the chloroplasts which traps light energy which is used for photosynthesis.

Practical activity 3.4

Aim : Is chlorophyll necessary for photosynthesis

Apparatus :
1. Plant with variegated leaves
2. Starch testing kit

Method

1. De-starch a plant with variegated leaves.
2. Place the plant in light for a few hours.
3. Test the leaf for starch.

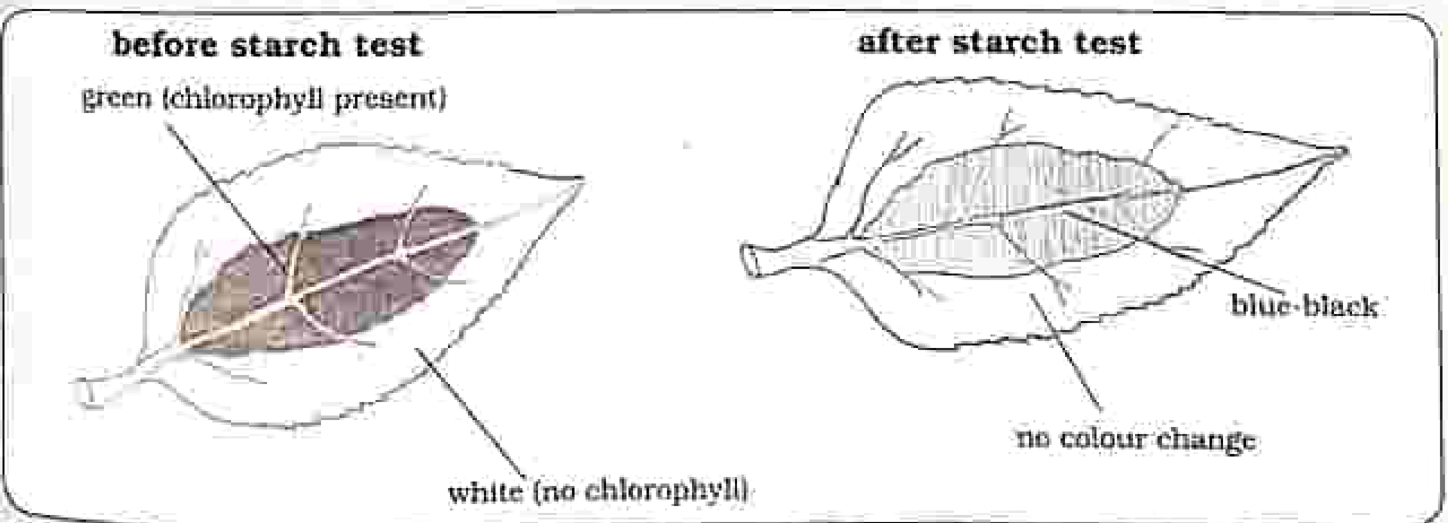


Fig 3.5 Is chlorophyll necessary for photosynthesis

Result

Green part of the leaf: iodine solution turns blue-black.

White part of the leaf: no colour change.

Conclusion

Chlorophyll is necessary for photosynthesis.

Practical questions

1. Explain your observation after the starch test for the:
   a) Green part of the leaf.
   b) White part of the leaf.

Fate of end products of photosynthesis

1. **Carbohydrates**

   Translocation – carbohydrates (in form of sugars) are transported from the leaf to other parts of the plant.

   Storage – carbohydrates may be changed into starch for storage.

   Structure formation – carbohydrates may be changed into cellulose to form cell walls for plant support.

2. **Oxygen**
   a) Respiration – oxygen produced is used for respiration in the plant cell.
   b) Oxygen which is not used by the plant will leave through the stomata.

Practical activity 3.5

Aim : Do green plants produce oxygen

Apparatus :
1. Aquatic plant
2. Beaker
3. Test tube
4. Plasticine/stikistuff

Method

1. Set up the apparatus as shown in the diagram.
2. Leave the apparatus in sunlight for a few hours.
3. Test the gas collected for oxygen.

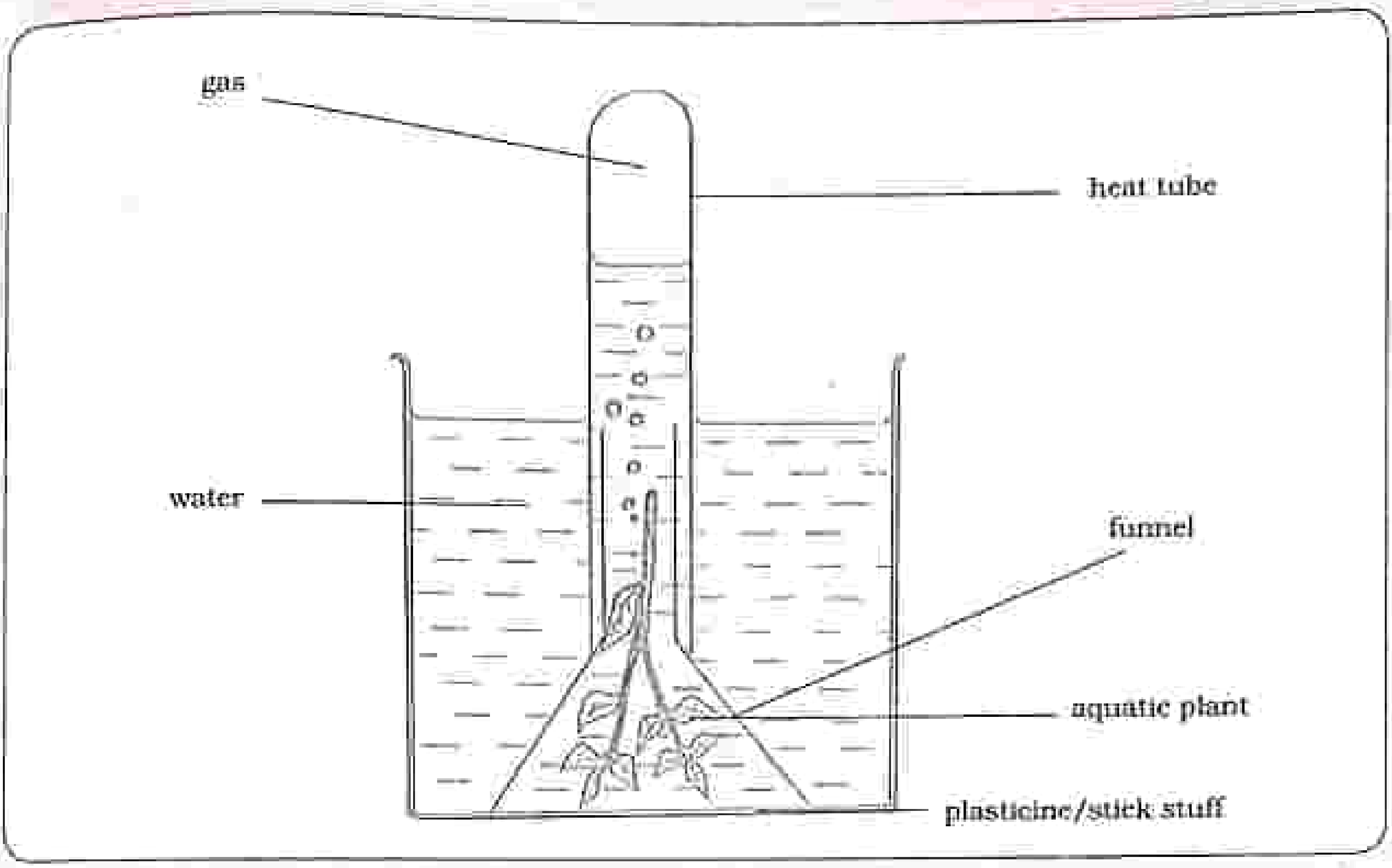


Fig 3.6 Do green plants produce oxygen

Result

A gas collects in the test tube.

Practical activity 3.6

Aim : Testing for oxygen

Apparatus :
1. Gas collected in **Practical activity 3.5**
2. Matches/ lighter
3. Wooden splint

Method

1. Carefully remove the test tube in **Practical activity 3.5** and close the end of the test tube using your finger.
2. Turn the test tube upright.
3. Light the end of a wooden splint and blow out the flame so that it glows.
4. Remove your finger from the end of the test tube and immediately put the glowing splint into the test tube.

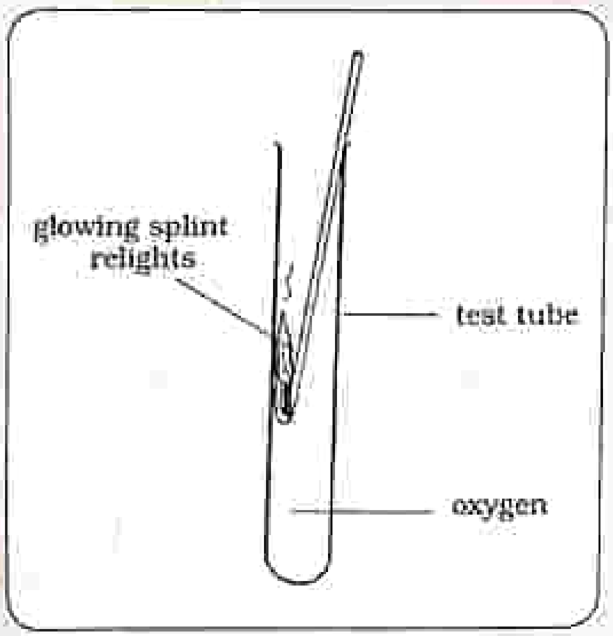


Fig 3.7 Testing for oxygen

Results

The glowing splint burst into a flame (relights).

Conclusion

- In the presence of oxygen, a glowing splint relights.
- Oxygen is produced during photosynthesis.

Practical questions

1. What happens to the glowing splint when put in the test tube?
2. What conclusion can be drawn from your answer in (1)?

Leaf structure

External leaf structure

These are visible parts of a leaf.

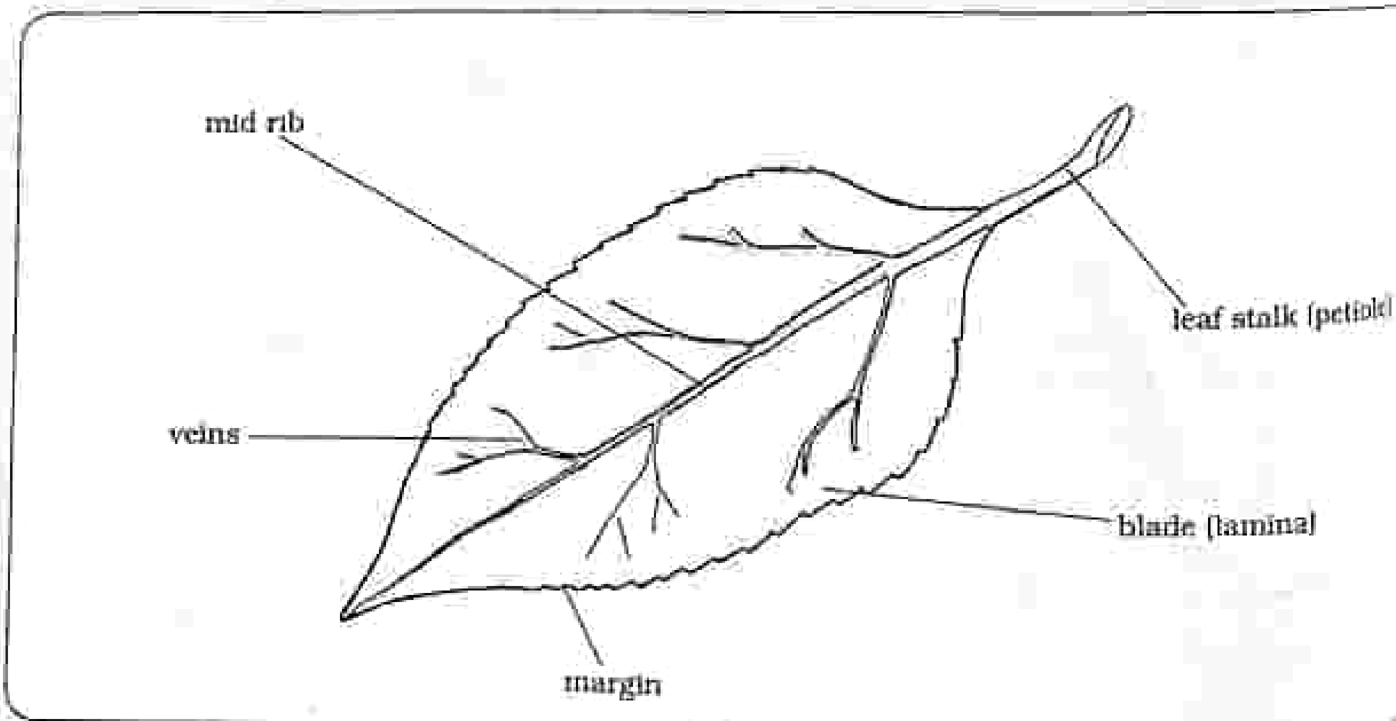


Fig 3.8 External structure of a leaf

Leaf stalk – attach leaf to the stem.
Mid rib – leaf stalk continues into the leaf as mid rib.
Veins – branch from mid rib.
Leaf blade – is broad.

Internal leaf structure

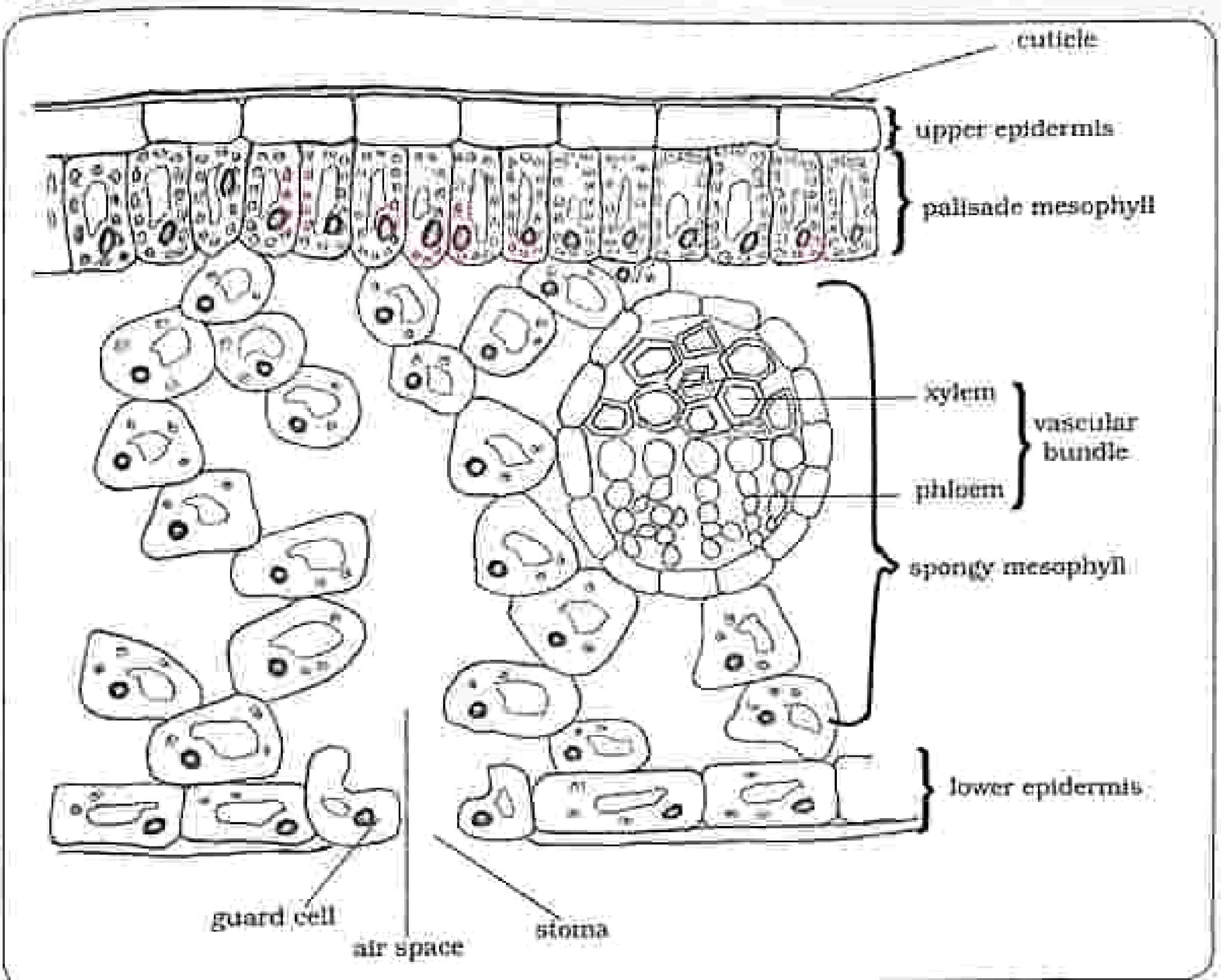


Fig 3.9 Internal structure of a leaf

Leaf adaptation for photosynthesis

Surface area

- Large surface area to absorb maximum light

Stomata

- Allows carbon dioxide to diffuse in and oxygen diffuses out during photosynthesis.
- Allows water vapor to pass out during transpiration.

Palisade cell

- It is packed with chloroplasts which contain chlorophyll to trap light for photosynthesis.

Air spaces

- Allow movement of carbon dioxide, oxygen and water vapour.

Cuticle

- It is a thin, shiny and transparent layer which prevents loss of water.
- It is transparent to allow light to enter.

Vascular bundle

- It is made up of xylem and phloem.
- Xylem brings water and minerals to the leaf.
- Phloem transports sugars and amino acids out of the leaf.

Spongy mesophyll

- Have cells with intercellular spaces to allow gases in and out of the leaf.
- It contains very few chloroplasts for photosynthesis.

Upper epidermis

- It is below the cuticle and allows light to pass through for photosynthesis.

Lower epidermis

- It is the lower surface of the leaf.
- It contains stomata for gaseous exchange.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

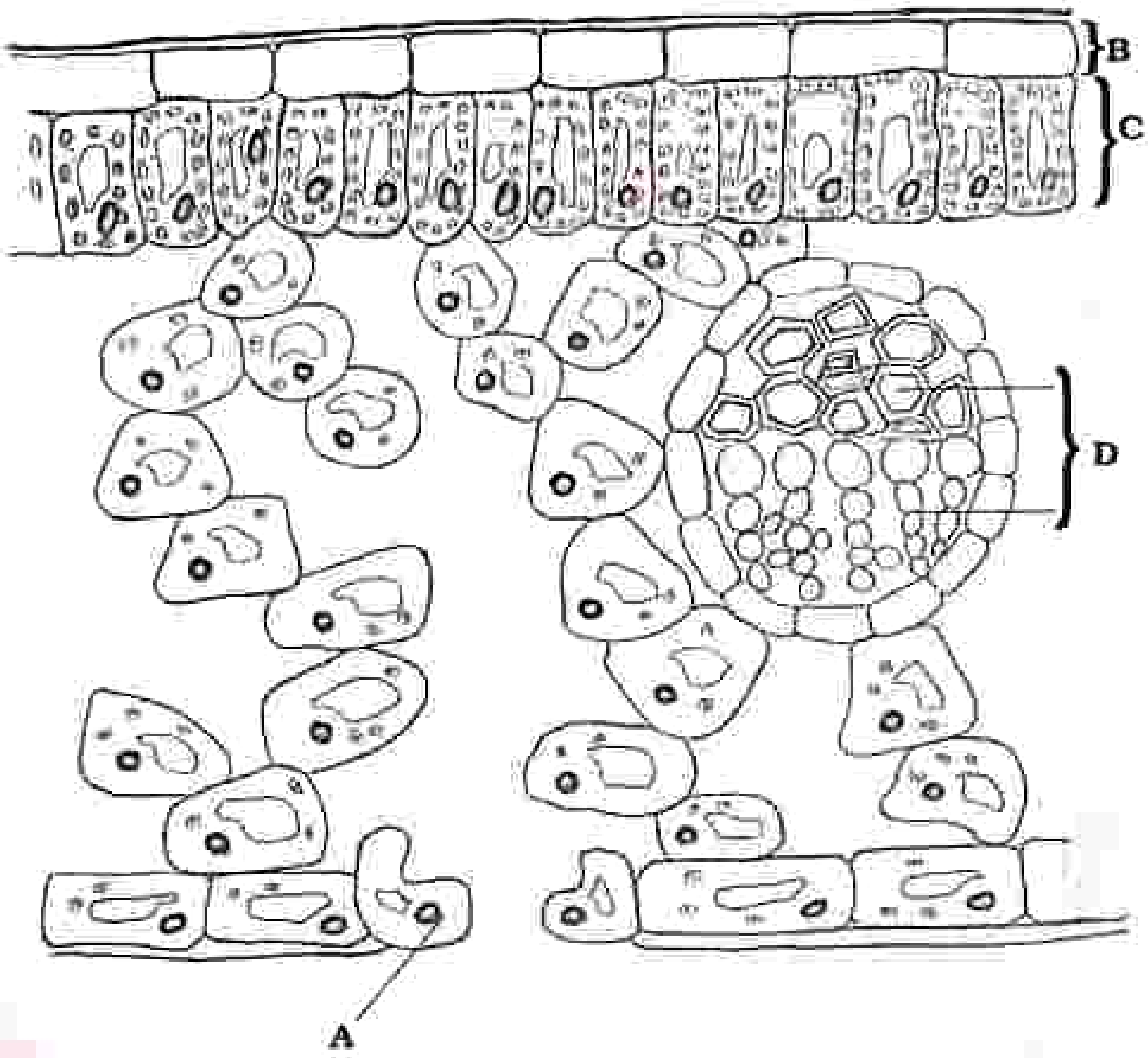
Structured questions

Answer all the questions.
The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Which one is a factor which affects photosynthesis.
   A. water
   B. carbohydrates
   C. oxygen
   D. respiration
2. Identify the correct word equation for photosynthesis
   A. Carbon dioxide + water $\xrightarrow[\text{chlorophyll}]{\text{light energy}}$ Carbohydrates + Oxygen
   B. Oxygen + glucose $\longrightarrow$ Carbon dioxide + water
   C. Oxygen + carbon dioxide $\longrightarrow$ water + glucose
   D. Carbon dioxide + glucose $\longrightarrow$ water + oxygen
3. Identify the two end products of photosynthesis
   A. carbohydrates and carbon dioxide
   B. oxygen and carbon dioxide
   C. carbohydrates and oxygen
   D. water and oxygen
4. What is the purpose of boiling the leaf in methylated spirit when testing it for the presence of starch?
   A. to denature the enzymes
   B. to soften the leaf
   C. to prevent chemical reactions
   D. to remove chlorophyll
5. In the presence of starch iodine solution changes to
   A. brown
   B. yellow
   C. blue-black
   D. brick- red

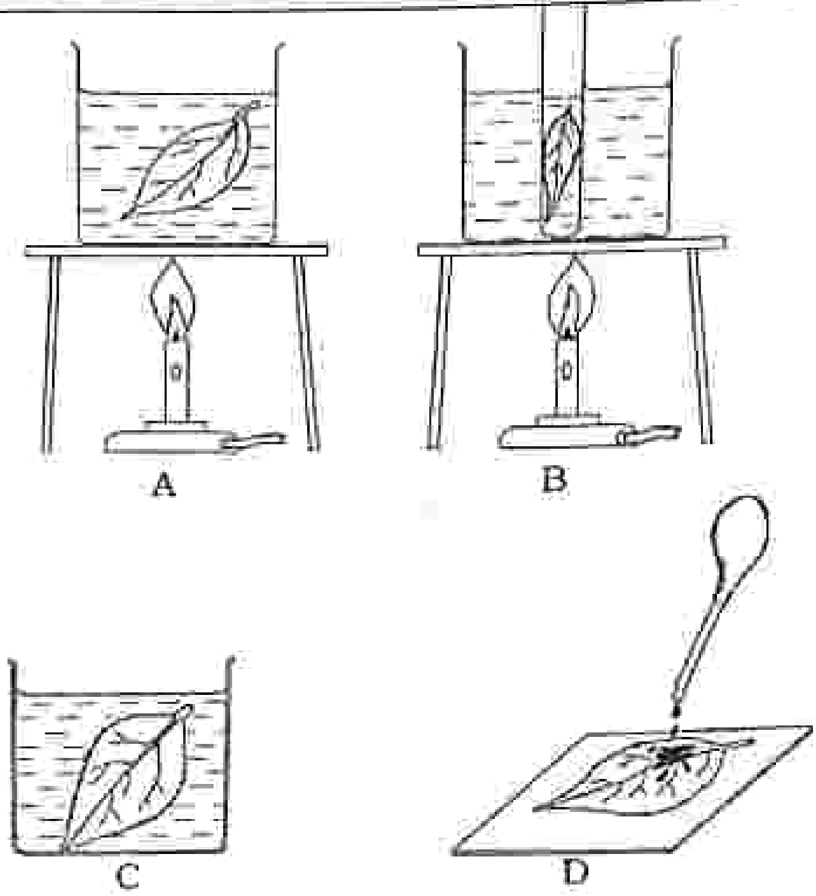
Use the following diagram which shows the internal structure of a leaf to answer questions 6 and 7



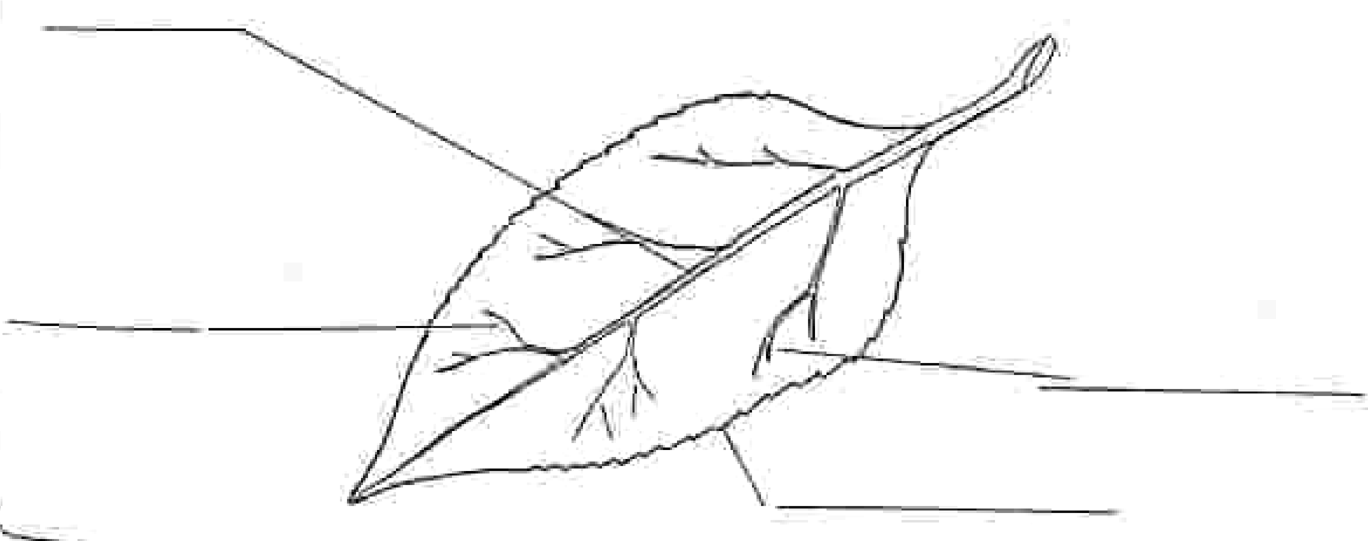
6. Identify the guard cell
   A. A B. B
   C. C D. D
7. Identify the vascular tissue
   A. A B. B
   C. C D. D
8. Identify the part which is packed with chloroplasts to trap maximum light for photosynthesis
   A. Upper epidermis
   B. Palisade mesophyll
   C. Spongy mesophyll
   D. Lower epidermis
9. The following are ways in which a leaf is adapted for photosynthesis **except**
   A. palisade cells
   B. air spaces
   C. stomata
   D. digestion
10. Oxygen test can be carried out us ________.
    A. soda lime
    B. a glowing splint
    C. sodium hydrogen carbonate
    D. sodium hydroxide

Structured questions

1. a) State any three factors which affect photosynthesis. [3]
   b) Name the two end products of photosynthesis.[2]
   c) Describe the fate of the two end products of photosynthesis named in 1(b). [2]
2. Briefly describe each stage A to D on starch test for a leaf.



3. Using the diagram above, describe oxygen test using a glowing splint. [3]
4. Describe any three ways the leaf is adapted for photosynthesis. [3]
5. In the diagram below, label parts of the internal structure of a leaf. [9]



CHAPTER 4

Human nutrition

Key concepts

- Human alimentary canal
  - parts of the alimentary canal and associated organs
  - functions of parts of the alimentary canal of a human
- Types of teeth and their function
- Digestion
  - mechanical digestion
  - chemical digestion
- Importance of digestion
- Function of a typical enzyme
- End products of digestion
- Food tests
- Balanced diet
  - components of a balanced diet and their functions
  - balanced diet for different groups of people
- Malnutrition
  - effects of malnutrition
- Deficiency diseases and their causes

Summary notes

Human alimentary canal

- It is made up of the organs which process food.
- Each organ has its own function in the processing of food.

Parts of the alimentary canal and associated organs

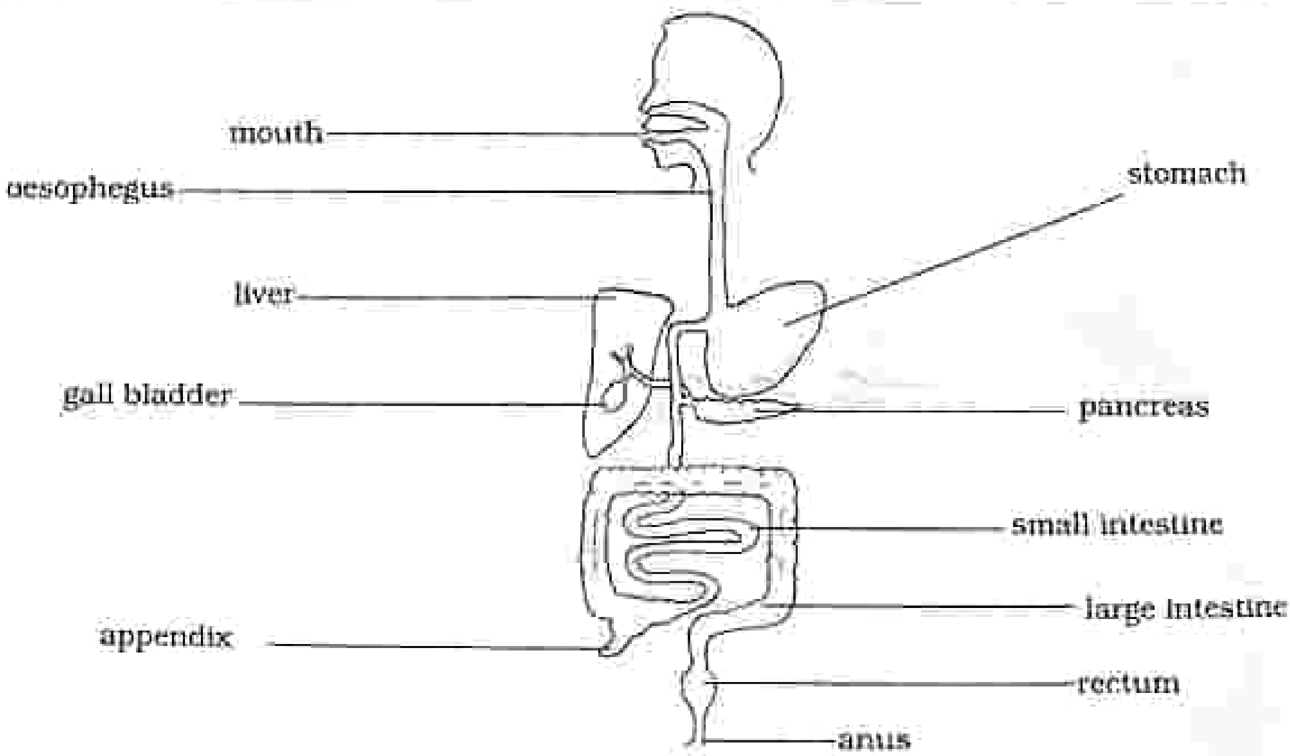


Fig 4.1 Alimentary canal

Functions of parts of the alimentary canal

Table 4.1: Parts of the alimentary canal and their functions

Part of the alimentary canal	Function
Mouth	Ingestion: is the taking in of food through the mouth • saliva helps to moisten the food for easy passage down the alimentary canal • contains teeth which help break down food into smaller pieces • digestion starts in the mouth
Oesophagus	• carries food from the mouth to the stomach • it has muscular walls that contract and relax to push food down to the stomach
Stomach	• produces gastric juice which contains enzymes and acids for digestion of food
Liver	• produces bile which helps to break down fats • stores vitamins and minerals • breaks down toxic substances like alcohol • regulates blood sugar
Gall bladder	• stores bile
Pancreas	• produces pancreatic juice containing enzymes for digestion
Small intestine	• for final digestion of food • absorption of soluble food molecules from small intestine into the blood vessels
Large Intestine	• absorption of water and salts into the blood vessels
Rectum	• store undigested food (faeces)
Anus	• Egestion takes place which is the passing out of undigested material (faeces)

Types of teeth and their function

- Teeth are responsible for breaking down large food molecules into small molecules.

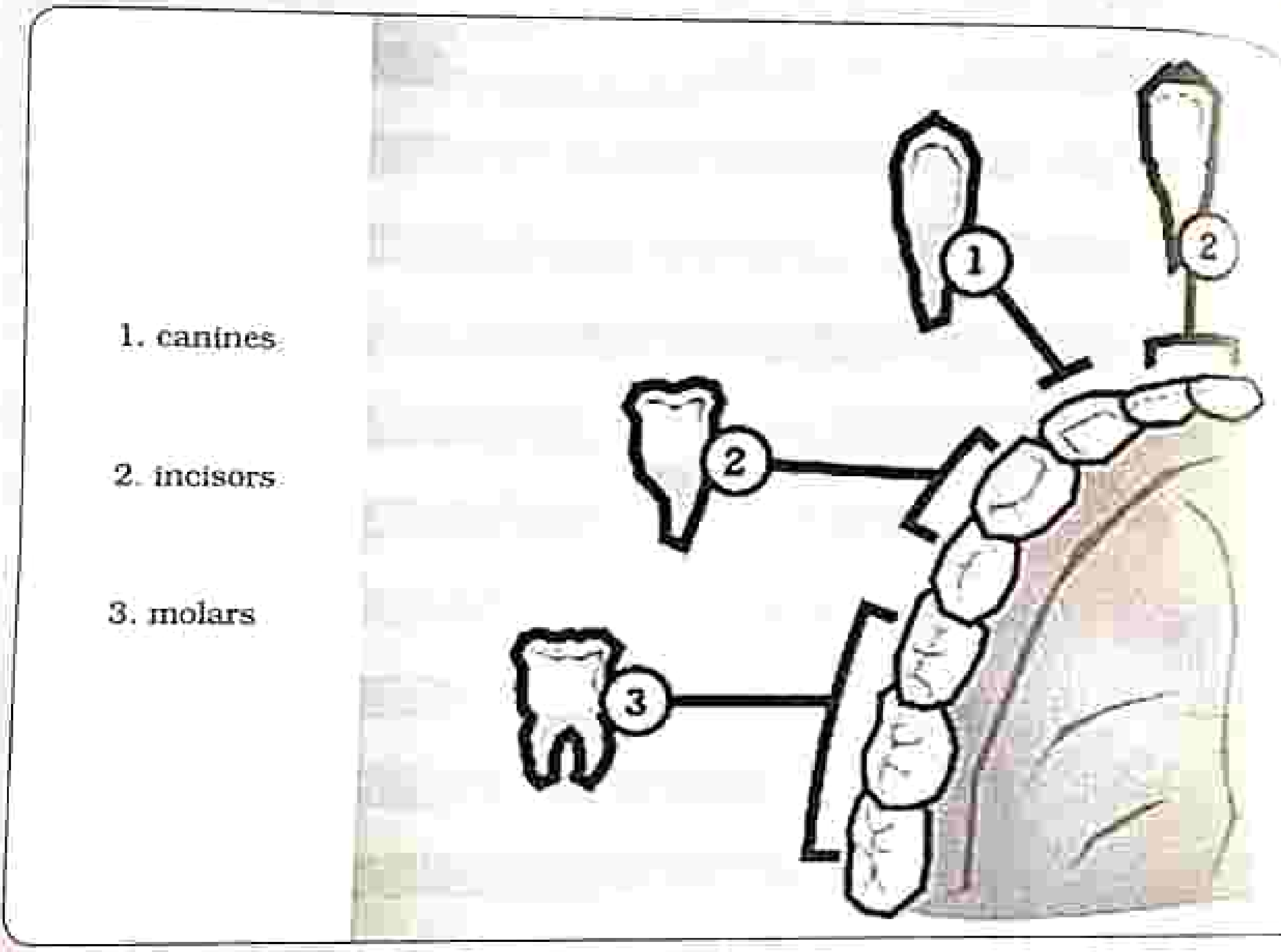


Fig 4.2: Types of teeth

Type of teeth	Functions
Incisor	• cutting • biting
Canines	• tearing • gripping
Premolar	• grinding • chewing
Molar	• grinding • chewing

Digestion

- It is the breaking down of large insoluble food into small soluble food molecules.

Two types of digestion

1. Mechanical digestion

- It is the breaking down of food into smaller pieces.
- Teeth aid in the breaking down of the food into smaller pieces.

2. Chemical digestion

- It is the use of enzymes which act as catalysts to aid in the conversion of food from insoluble to soluble molecules.
- It starts in the mouth when enzymes in saliva act on the food, then proceeds to the stomach where gastric juice which contains enzymes digests food and it ends in the small intestine where enzymes continue to aid in the digestion of protein, starch, sugars, fats and oils.

Importance of digestion

1. **Surface area** – increases surface area for enzyme action

Practical activity 4.1

Aim : Effect of surface area on the rate of a reaction

Apparatus :
1. 2 test tubes
2. Potato or fresh liver
3. Stop watch
4. Hydrogen peroxide

Method

1. Label the two test tubes A and B.
2. Cut two small potato or liver cubes of the same size.
3. Place one cube in test tube A.
4. Cut the other cube into many small pieces and place in test tube B.
5. Add 5ml hydrogen peroxide to test tube A.
   - Using a stop watch, time how long it takes for the bubbles to stop forming
   - Record the time.
6. Add 5ml hydrogen peroxide to test tube B.
   - Using a stop watch, time how long it takes for the bubbles to stop forming
   - Record the time.

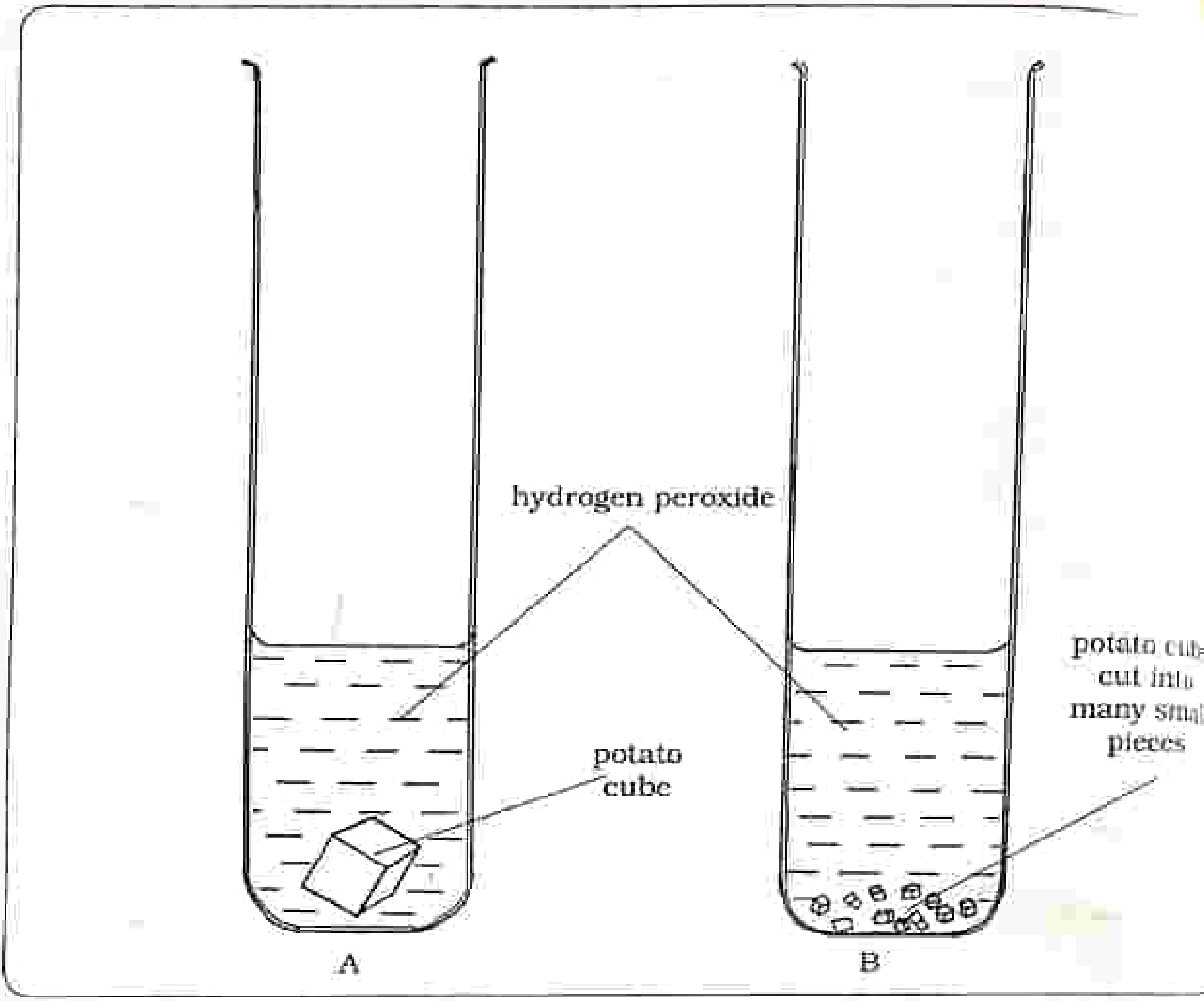


Fig 4.3: Effect of surface area

Results

Potato/liver	Reaction time
Single cube	More time
Many small pieces	Less time

Conclusion

- Many small cubes offer a greater surface area for the reaction than a single cube.
- Therefore a large surface area increases the rate of reaction.

Practical questions

1. In which test tube did the reaction take?
   a) a short time for the bubbles to stop forming?
   b) a long time for the bubbles to stop forming?
2. Solubility - digestion also breaks down large insoluble food molecules into small soluble molecules which can be absorbed into the blood.

Enzyme

- It is a catalyst which speeds up a chemical reaction but it is itself not changed during the reaction.

Function of a typical enzyme

- Amylase is an example of a typical enzyme.
- Amylase is found in pancreatic juice and in saliva.
- Amylase aid in the catalysis for conversion of starch to maltose.
- Enzyme maltase acts on maltose to give the final end product glucose.
- Enzyme protease acts on proteins.
- Enzyme lipase acts on lipids (fats and oils).

Practical activity 4.2

Aim : Using amylase on different food substrates

Apparatus :
1. Three beakers
2. Visking tubing
3. 1% amylase enzyme
4. 2% starch solution
5. Egg white solution
6. Iodine solution

Method

1. Using visking tubing, prepare three model intestines.
2. Model A – add starch solution (control).
   Model B – add starch solution and amylase.
   Model C – add egg white solution and amylase.
3. Place each model in a beaker containing water as shown in the diagram.
4. Test the contents of each model and the surrounding water for starch and reducing sugar at the start of the activity.
5. Leave the models for a few hours.
6. Repeat the test for the contents of each model and surrounding water for starch and reducing sugar.

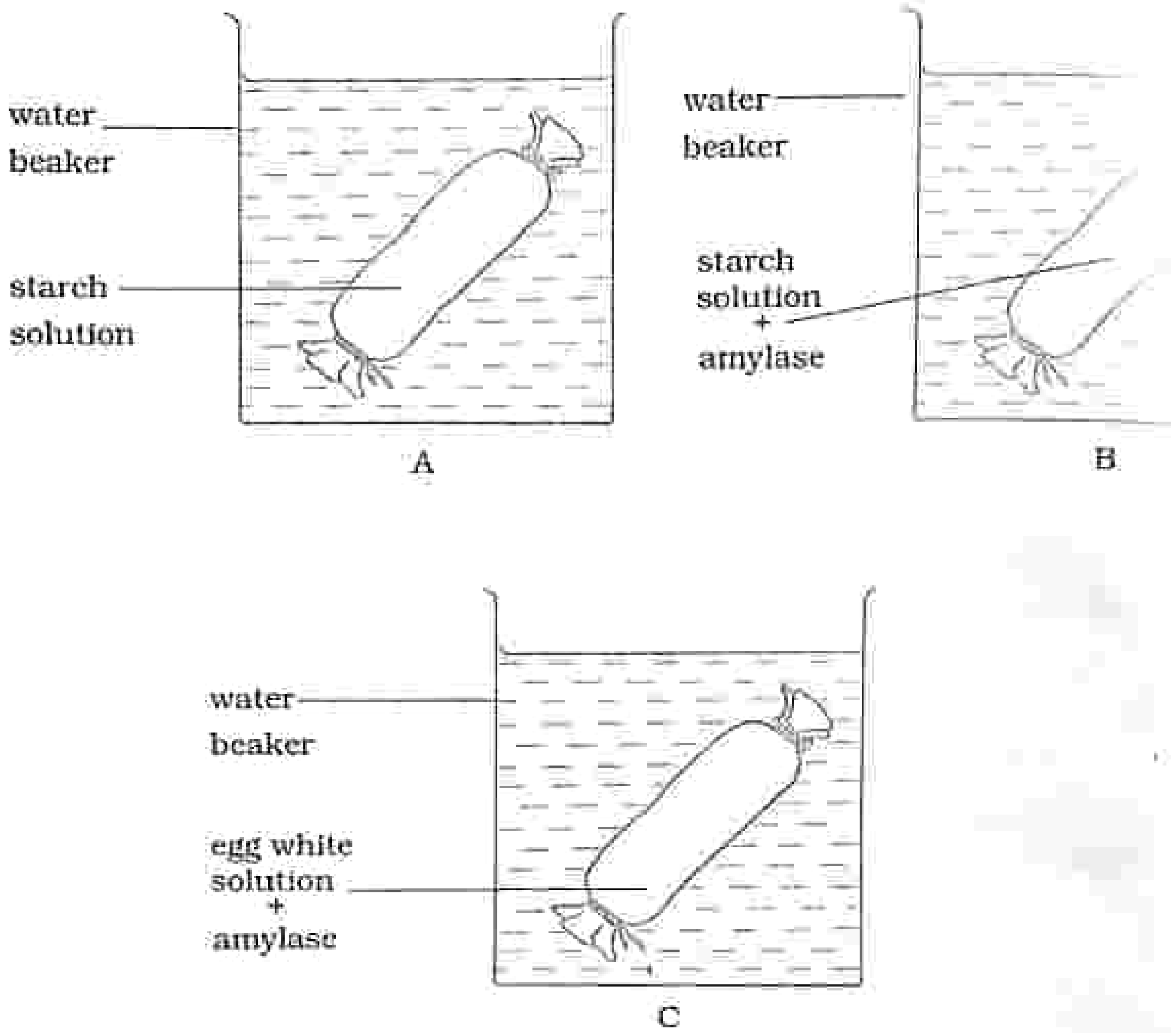


Fig 4.4: Using amylase on different food substrates

Results

Contents in model	Test for sugar (Benedict's solution)
A – starch solution	Negative for maltose
B – starch solution and amylase	Positive for maltose
C – egg white solution and amylase	Negative for maltose

Conclusion

- Digestion took place in model B (starch solution and amylase) only.
- Amylase is an enzyme which can only act on starch and break it down maltose (glucose).

End products of digestion

Food group	End products
Carbohydrates	Glucose
Proteins	Amino acids
Fats	Fatty acids and glycerol

Food tests

Food group	Food test	Positive result
Glucose	**Benedict's test** • put glucose solution in a test tube • add Benedict's solution • heat in a water bath for a few minutes	Brick red precipitate
	Clinistix • dip clinistix into glucose solution	Purple colour
Protein	**Biuret test** • put protein solution 1%(albumen) • add $5cm^3$ Biuret solution • dilute sodium hydroxide and $5cm^3$ 1% copper sulphate solution	Purple colour
	Albustix • dip albustix into the protein solution	Green to dark green colour
Fats	**Ethanol** • add ethanol to a food solution and shake • add water and shake	Cloudy white emulsion
Starch	• add drops of iodine solution to a food sample	Blue black solution

Balanced diet

It is food that has the correct amount of each food nutrient for a healthy living.

Components of a balanced diet and their function

The following are components of a balanced diet and their functions:

a) **Carbohydrates** – Energy giving

b) **Fats** – Energy store and keeping the body warm

c) **Proteins** – growth and repair of worn out tissues

d) **Vitamins**

A – good sight

B – healthy skin

D – aid in the absorption of calcium by the body which help in bone formation and maintenance.

e) **Mineral salts**

Iodine – required by the thyroid gland to regulate the body's metabolism and growth.

Calcium – for strong bones and teeth.

Iron – formation of haemoglobin in the red blood cells.

f) **Water** – for chemical reactions inside the cell.

- blood, urine, saliva and sweat contains water.

g) **Roughage** – helps in the movement of food along the alimentary canal.

– prevents constipation.

Balanced diet for different groups of people

- A balanced diet for different groups of people differs in proportions depending on the amount of activity and gr... taking place.

0-6 months baby

- Exclusive breast milk

6 months -12 months baby

Balanced diet comprising of:

- Breast milk
- A small amount of carbohydrates.
- A small amount of vitamins.
- A small amount of fats.
- A small amount of mineral salts.
- Water as often as the baby can drink

Toddler

Balanced diet comprising of:

- Breast milk up to 24 months
- Additional carbohydrates.
- A small amount of vitamins.
- A small amount of mineral salts.
- A small amount of proteins.
- A small amount of fats.
- Water as often as the baby c... drink.

Adolescent

Balanced diet with

- An increased amount of carbohydrat... because the body is very active.
- An increased amount of vitamins.
- An increased amount of mineral sa... for different body changes taking pla...
- An increased amount of proteins ... body building.
- An increased amount of water ... replace water lost during different bo... activities.

Manual worker

- An increased amount of carbohydrates because the muscles will be using more energy.
- An increased amount of vitamins.
- A small amount of mineral salts.
- A small amount of proteins because manual worker is active but not growing.
- A small amount of fats.
- An increased amount of water to replace water lost through sweating.

Sedentary worker

- A small amount of carbohydrates because of less activity therefore small amount of energy is used.
- An increased amount of vitamins.
- A small amount of mineral salts.
- A small amount of proteins because manual worker is active but not growing.
- A small amount of fats.
- An increased amount of water to replace water lost.

Pregnant woman

- An increased amount of carbohydrates to give the pregnant mother energy.
- An increased amount of vitamins to boost the body's immune system.
- An increased amount of mineral salts for different developments taking place for both the pregnant mother and the unborn baby.
- An increased amount of proteins for body building.
- An increased amount of water to replace water lost during different body activities.

Malnutrition

- Malnutrition can be caused by eating nutrients in wrong proportions (too much or too little).

Effects of malnutrition

- Malnutrition may result in certain conditions such as obesity, anorexia nervosa, diabetes mellitus 2, kwashiorkor, goiter and marasmus.

Obesity

Fig 4.5: An obese person

- Obesity is eating too much food rich in carbohydrates and fats which results in the excess being stored as fat under the skin which increases the risk of health problems.
- Obesity often results from taking in more calories than are burned by exercise and normal daily activities.

NOTE: This condition may damage the heart and the circulatory system.

Anorexia nervosa

Fig 4.6: A person suffering from anorexia nervosa

- Caused by depriving one's body of the essential nutrients and too much exercise.
- It is an eating disorder which causes the person to starve himself or herself due to fear of being overweight.
- The body starts to waste away (weak and thin) because the energy reserves in the body have been used up causing a distorted body image.

Diabetes mellitus 2

- A condition which occurs when the hormone insulin does not function properly.
- The body either doesn't produce enough insulin, or it resists insulin.
- The hormone insulin regulates the concentration of glucose in the blood so that the body functions well.
- Signs and symptoms include increased thirst, frequent urination, hunger, fatigure and blurred vision.

Note: In some cases there may be no symptoms

Treatment

- Following the doctor's prescribed diet which include small amounts of carbohydrates and an increased amount of vegetables
- Exercise
- Medication
- Insulin therapy

Deficiency diseases

These are caused by lack of a specific [illegible] nutrient in the diet.

Kwashiorkor

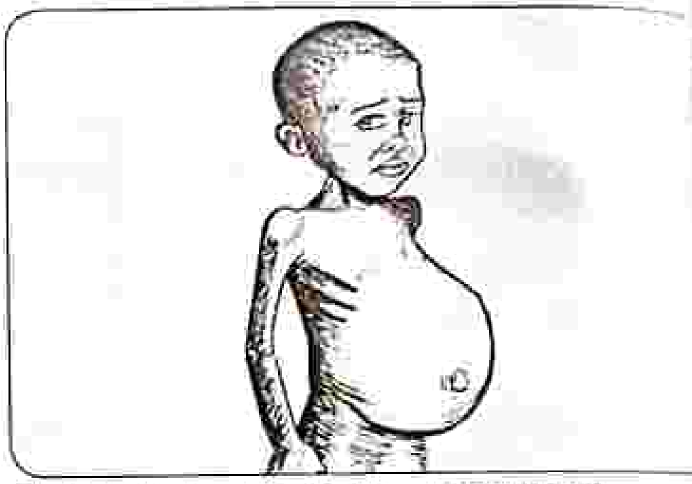


Fig 4.7: Child suffering from kwashiorkor

- It is caused by lack of protein in the diet.
- A child who suffers from kwashiorkor has a swollen abdomen, thin arms and legs, delayed growth in children, frequent infections and sometimes hair turns brown.

Treatment

- Eating a balanced diet with correct proportion of protein.
- Most people recover fully if treated early but if treatment is delayed, there can be serious complications which include coma, shock and permanent mental and physical disabilities.

- If left untreated, kwashiorkor can cause major organ failure and eventually death.

Goitre

Fig 4.8: A person with goitre

- It is when there is lack of iodine in the diet which causes the thyroid gland to enlarge.

Treatment

- Inclusion of the correct amount of iodine in the diet.

Rickets

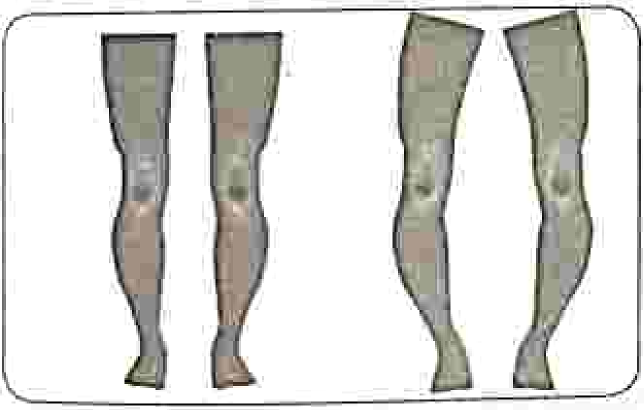


Fig 4.9: A child with rickets

- It is caused by lack of vitamin D in the diet.
- Bones are distorted and bend.

Treatment

- Inclusion of vitamin D in the diet and exposure to sunlight.

Scurvy

- Lack of vitamin C in the diet.
- Leads to gum infection, bleeding which can lead to anaemia, painful limbs and tiredness.

Treatment

- Inclusion of vitamin C in the diet and taking vitamin C supplements.

Anaemia

- It is caused by lack of iron in the diet.
- The person's complexion is pale and is always tired and weak.

Treatment

- Inclusion of iron in the diet and taking iron supplements and sometimes there is need for blood transfusion.

Night blindness

- It is caused by lack of vitamin A in the diet.
- The person cannot see well in darkness.

Treatment

- Inclusion of vitamin A in the diet and taking vitamin A supplements.

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the
you consider correct. Each correct answer will score one mark.

Section B

Structured questions

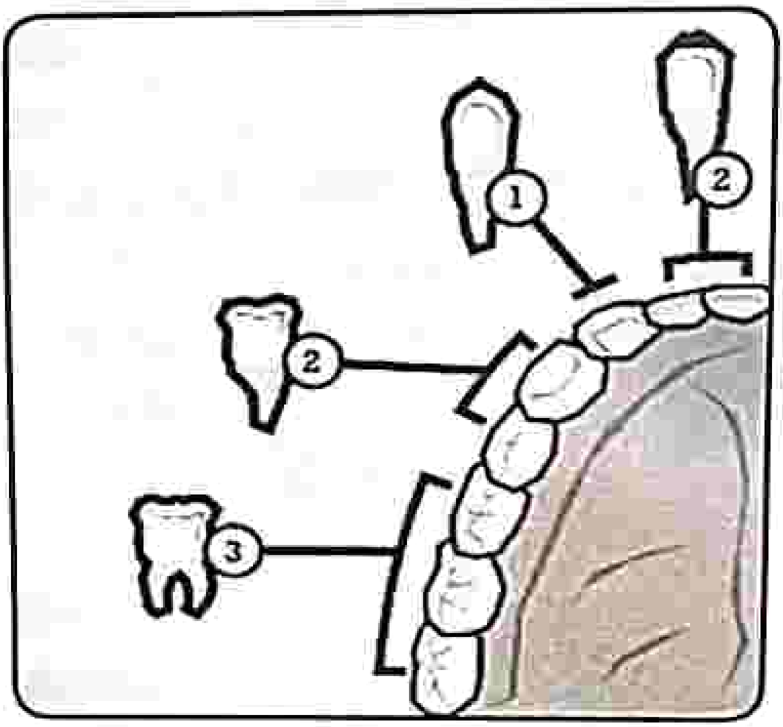
Answer all the questions.
The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. State the function of the mouth in the human alimentary canal.
   A. absorption
   B. photosynthesis
   C. ingestion
   D. assimilation

2. Digestion starts in the ________
   A. small intestine
   B. stomach
   C. large intestine
   D. mouth

3. From the picture below identify canine teeth

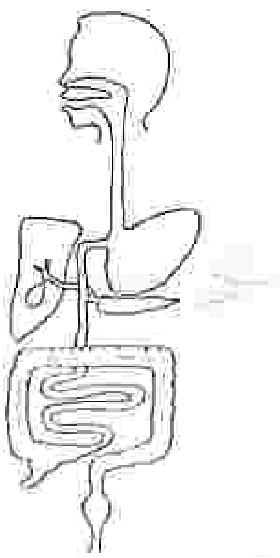


A. 1   B. 2   C. 3   D. 4

4. What is the function of the typical enzyme amylase?
   A. catalyst for the conversion of starch to glucose
   B. catalyst for the conversion of protein to amino acids
   C. catalyst for the conversion of fats to fatty acids
   D. catalyst for the conversion of fats to glycerol
5. Identify the end product of protein digestion
   A. glucose
   B. fatty acids
   C. amino acids
   D. glycerol
6. What causes night blindness?
   A. lack of vitamin A
   B. lack of vitamin C
   C. lack of vitamin D
   D. lack of iron
7. Identify the effect of over eating
   A. anorexia nervosa
   B. diabetes mellitus 2
   C. obesity
   D. kwashiorkor
8. What is the positive result for protein test using albustix?
   A. brick red precipitate
   B. dark green colour
   C. purple
   D. blue-black
9. State the function of vitamin D in the diet
   A. aid in the absorption of calcium
   B. energy giving
   C. keeping the body warm
   D. growth
10. A manual worker's diet must contain more
   A. proteins
   B. carbohydrates
   C. iodine
   D. vitamin D

Section B: Structured questions

1. The diagram shows part of the human alimentary canal and associated organs.



Label on the diagram the following:

a) mouth
b) oesophagus
c) stomach
d) small intestine
e) large intestine
f) gall bladder
g) pancreas
h) liver

[8]

2. Complete the table which shows the types of teeth and their function

Type of teeth	Function
Incisor	
Canine	
Premolar	
Molar	

3. a) Describe;
   i) Mechanical digestion [1]
   ii) Chemical digestion [1]
   b) Explain the importance of digestion. [2]
   c) Complete the table which shows the end products of digestion.

Food group	End product
Carbohydrates	
Proteins	
Fats	

4. a) Explain the terms:
   i) Malnutrition [1]
   ii) Deficiency disease [1]
   b) State causes of the following deficiency diseases
   i) Kwashiorkor
   ii) Goiter
   iii) Rickets
   iv) Scurvy
   v) Anaemia [5]
5. a) Identify components of a balanced diet and describe their function. [14]
   b) Plan a balanced diet for a toddler. [3]
   c) Complete the table of food test

Food group	Food test	Positive result
Glucose	Benedict's test	
Protein	Biuret test	
Fats	Ethanol test	
Starch	Iodine solution test	

CHAPTER 5

Respiratory systems

Key concepts

- Human respiratory system
- Percentage composition of inhaled and exhaled air
- Role of alveoli in gaseous exchange
- Aerobic respiration
- Anaerobic respiration

Summary notes

Human respiratory system

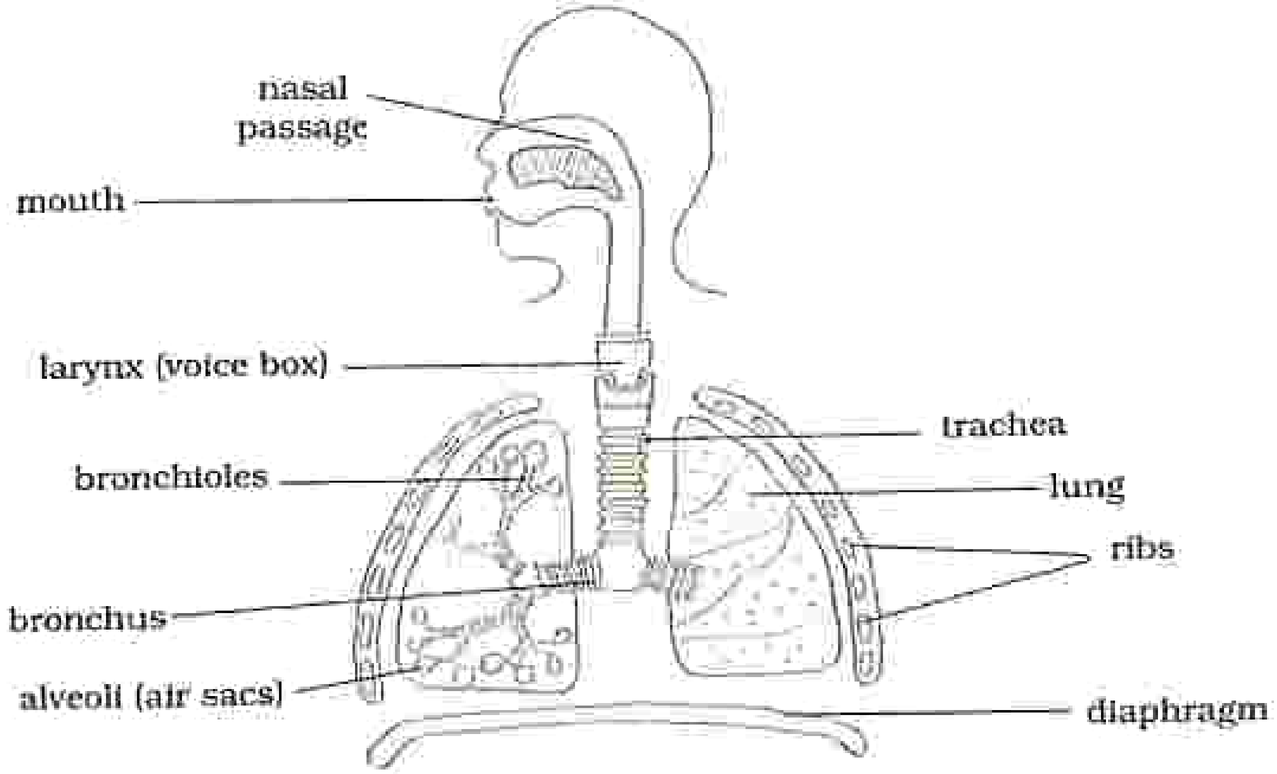


Fig. 5.1: Human respiratory system

- This is a system which is responsible for getting air in and out of the lungs.
- When you breathe in air enters through the nasal passage and mouth.
- In the nasal passage the air is filtered, warmed and moistened.
- The air moves down the trachea.
- From the trachea the air branches into two tubes called bronchi.
- The bronchi further divides into smaller tubes called bronchioles which end at the alveoli (air sacs).

CHAPTER 5

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- Anaerobic respiration

Summary notes

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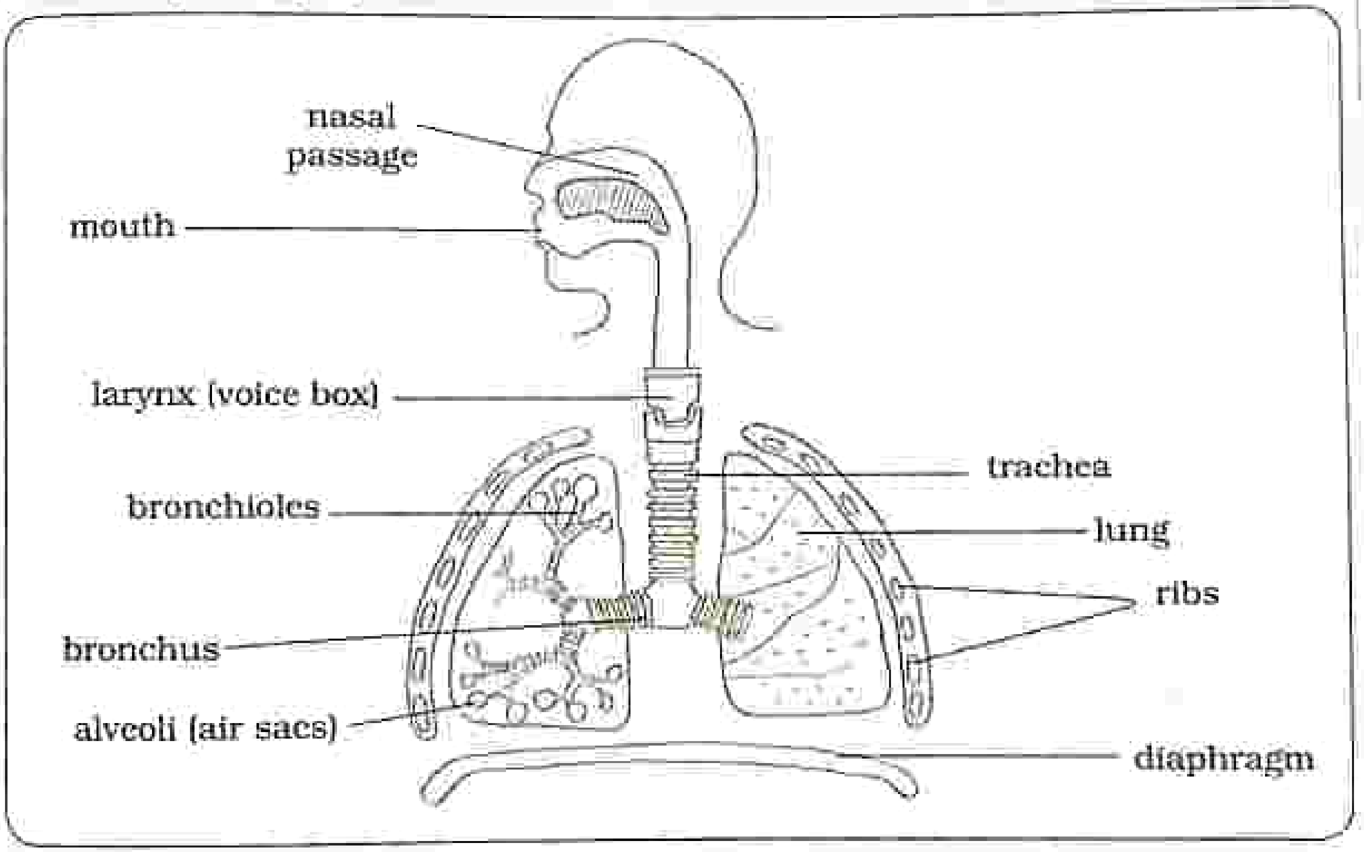

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- The air moves down the trachea.
- From the trachea the air branches into two tubes called bronchi.
- The bronchi further divides into smaller tubes called bronchioles which end at the alveoli (air sacs).

Practical activity 5.1

Aim : To show the change in proportions of carbon dioxide in inha[...] exhaled air

Apparatus : 1. lime water/bicarbonate indicator solution
2. Apparatus set up as shown in Fig 5.2

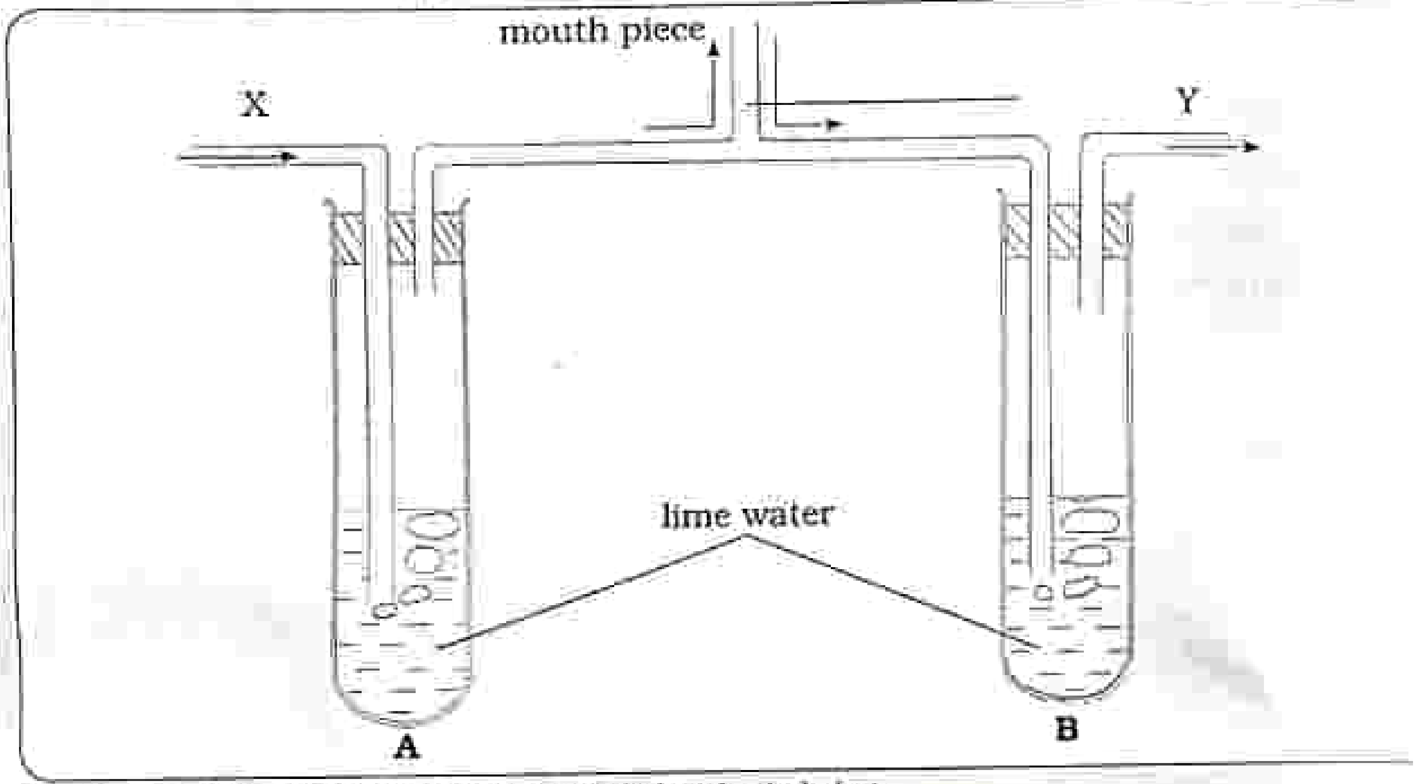


Fig. 5.2: Proportion of carbon dioxide in inhaled and exhaled air

Method

1. Set up apparatus as shown in Fig 5.2 above.
2. Put equal amount of limewater/bicarbonate indicator solution in each container
3. Close X and Y and then breathe in and out of the mouthpiece.
4. Observe any colour changes.

Results

Test tube A - No colour change

Test tube B - Lime water turns milky

Conclusion

- Exhaled air contains more carbon dioxide than inhaled air as noted by the colo[ur] change in the lime water.

Practical questions

1. When carbon dioxide is present, lime water turns from colourless to ________
2. Which air contains more carbon dioxide inhaled or exhaled air?

Practical activity 5.2

Aim : To show the change in proportion of oxygen in inhaled and exhaled air

Apparatus :
1. graduated large transparent container
2. candle on a float
3. large basin
4. rubber tubing

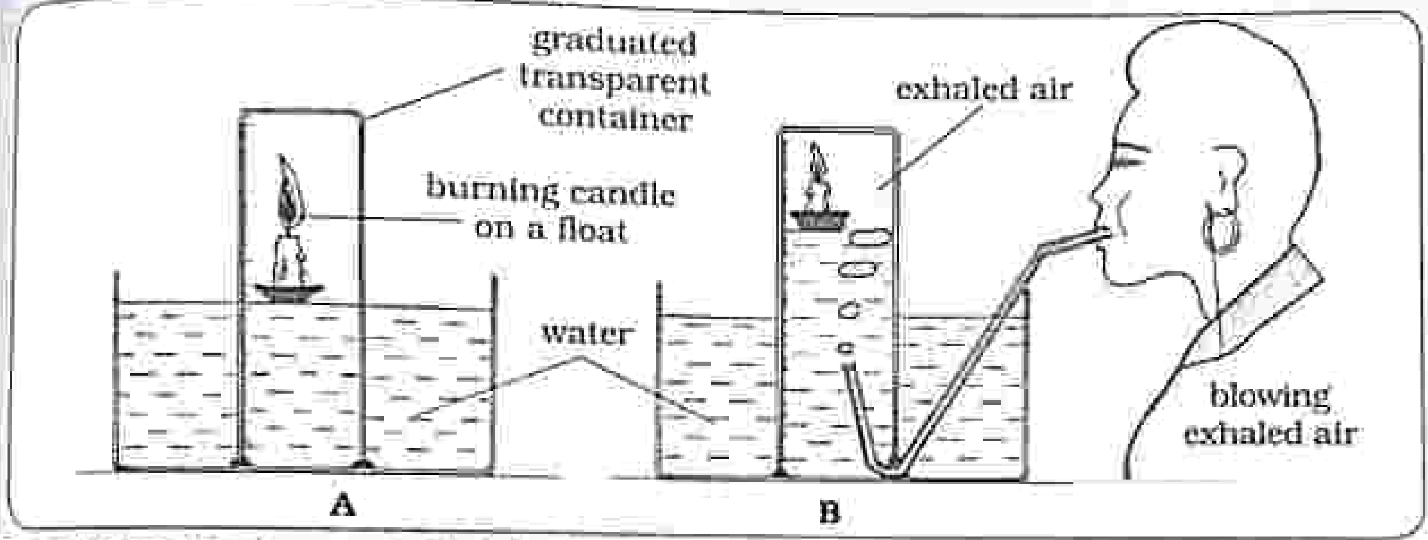


Fig 5.3 Proportion of oxygen in inhaled and exhaled air

Method

1. Set up apparatus as shown in Fig 5.3A.
   - Place a lit candle on a float in a large basin covered by a graduated large transparent container.
2. Take the measurement of how far the water rises up the container.
3. Set up apparatus as shown in Fig 5.3B where exhaled air is blown into the container until all the water is out. Place a lit candle on a float in a large basin covered by a graduated large transparent container.
4. Take the measurement of how far the water rises up the container.

Results

A - Water rises more than in B.
  - Candle takes longer to stop burning.

B - Water rises less than in A.
  - Candle stops burning earlier than in A.

Conclusion

- There is more oxygen in inhaled air compared to exhaled air.

Practical questions

1. Explain why the candle in A takes long to stop burning?
2. Explain why the water in B rises a little?

Percentage composition of inhaled and exhaled air

Air component	Inhaled air	Exhaled air
Oxygen	20	16
Carbon dioxide	0.03	4
Water vapour	varies	higher

Role of alveoli in gaseous exchange

- The role of the alveoli is for the diffusion of carbon dioxide and oxygen.
- During gaseous exchange, oxygen diffuses from the alveolus (air sacs) into the blood capillaries because the capillaries contain less oxygen than contained in the alveolus.
- From the blood capillaries, the oxygen is carried to all parts of the body.
- During gaseous exchange, there is also a higher concentration of carbon dioxide in the blood capillaries than the air in the alveolus, hence by diffusion carbon dioxide moves into the alveolus (air sac) then out through the bronchioles, bronchi, trachea and is exhaled through the mouth or nose.

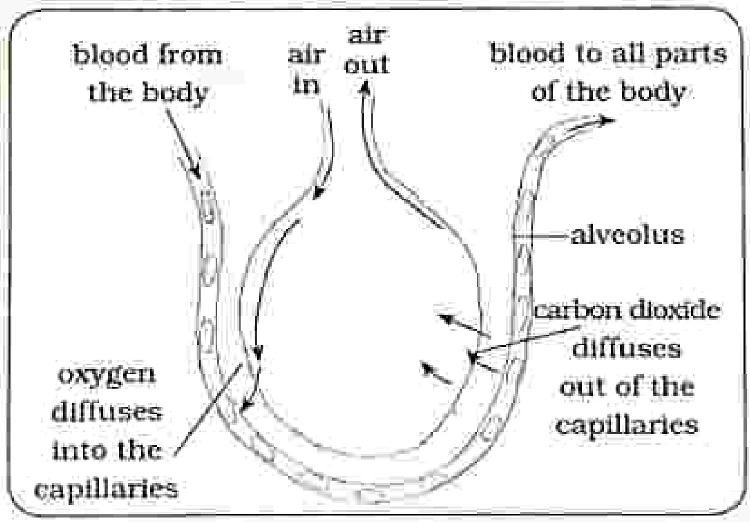


Fig. 5.4: The alveolus

Adaptation of the alveolus for gaseou[s] exchange

- Large surface area – allows mo[re] oxygen to get into the blood capilla[ries] and allows carbon dioxide out of t[he] blood capillaries.
- Network of the blood capillaries – ha[s] a good blood supply because th[ey] are surrounded by numerous blo[od] capillaries.
- One cell thick
- Moist surface

Respiration

- It is the process by which ener[gy] is released by the breakdown [of] glucose.
- It takes place in the cells and th[is is] called cellular respiration.

Aerobic respiration

- It is the process by which energy [is] released by the breakdown of gluco[se] using oxygen.
- Carbon dioxide and water are produ[ced] during the process.

Word equation

Glucose + oxygen $\longrightarrow$ carbon dioxide + water + energy

Anaerobic respiration

- It is a process by which less energy is released by the breakdown of glucose into lactic acid in the absence of oxygen.
- Glucose is not completely broken down so less energy is produced.
- When there is lack of oxygen in our muscles anaerobic respiration occurs and the lactic acid formed will cause muscle stiffness and pain.

Word equation

Glucose ⟶ lactic acid + less energy

Practical activity 5.3

Aim : Is energy released by germinating seeds?

Apparatus :
1. Two insulated containers
2. Wet cotton wool
3. Water
4. Two thermometers
5. Germinating seeds (beans)

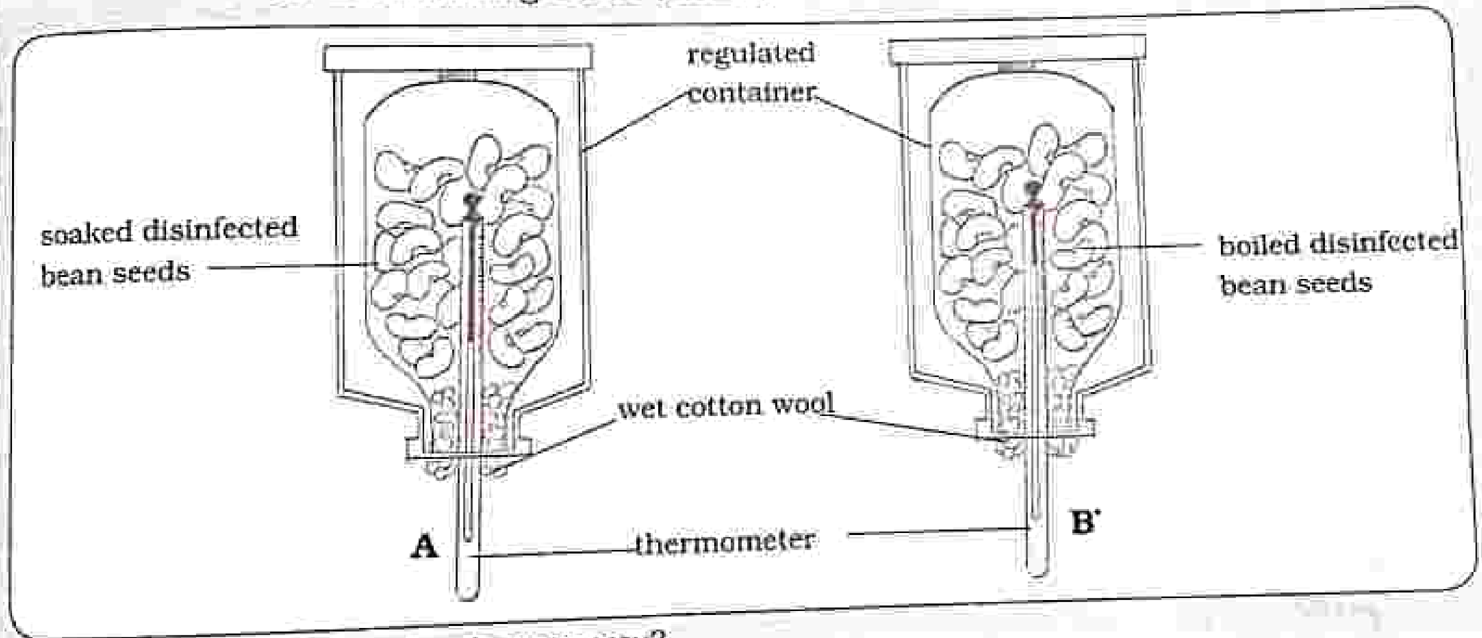


Fig. 5.5: Do germinating seeds release energy?

Method

1. Soak the beans in water to soften them.
2. Place half the soaked bean seeds which have been disinfected in A and set up apparatus as shown in Fig 5.5A.

Note: disinfectant kills any other living organisms in the seeds.

3. Boil the remaining half of the bean seed to kill them (stop any chemical reactions).
4. Place the boiled seeds which have been disinfected and set up the apparatus as shown in Fig 5.5B.
5. Record the initial temperatures in container A and container B.
   - leave the containers in a warm place for a few days because a suitable temperature is needed for germination to take place.
6. Record any temperature changes.

Results

A – There was an increase in temperature.
– Seeds germinated.

B – No temperature change.
– Seeds did not germinate.

Conclusion

- Germinating seeds release energy
- Energy is released during respiration

Practical questions

1. Explain why seeds in B did not germinate.
2. State conditions necessary for germination.

Practical activity 5.4

Aim : Is carbon dioxide released by germinating seeds?

Apparatus :
1. Two test tubes
2. Soaked seeds
3. Disinfectant
4. Carbon dioxide indicator solution (limewater or bicarbonate Indicator
5. Cloth bag

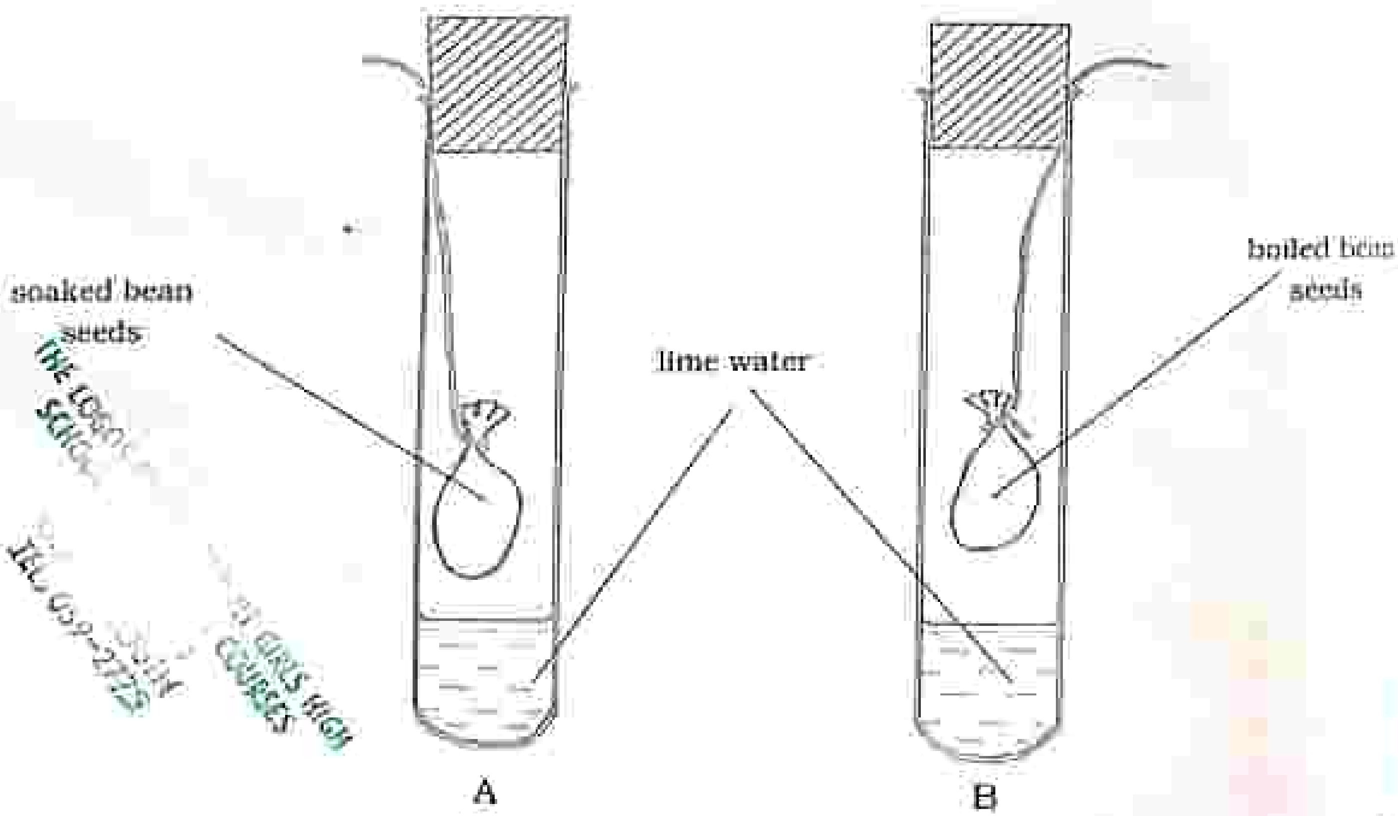


Fig 5.6: Do germinating seeds release carbon dioxide?

Method

- Place half of the soaked and disinfected seeds in a cloth bag.
- Set up as shown in Fig 5.6A.
- Boil the remaining half of the soaked seeds.
- When cool disinfect and place in a cloth bag.
- Set up as shown in Fig 5.6B.
- Examine the lime water after a few hours.

Results

A – Limewater turns milky.

B – Limewater remains colourless.

Conclusion

- Germinating seeds release carbon dioxide.
- Carbon dioxide is released during respiration.

Practical questions

1. Explain why two test tubes are used, one with living seeds and one with boiled seeds.
2. What is the reason for boiling the seeds?

Practical activity 5.5

Aim : Do animals release carbon dioxide?

Apparatus : 1. Small animal (frog)
2. Three test tubes
3. Soda lime
4. Sodium hydroxide

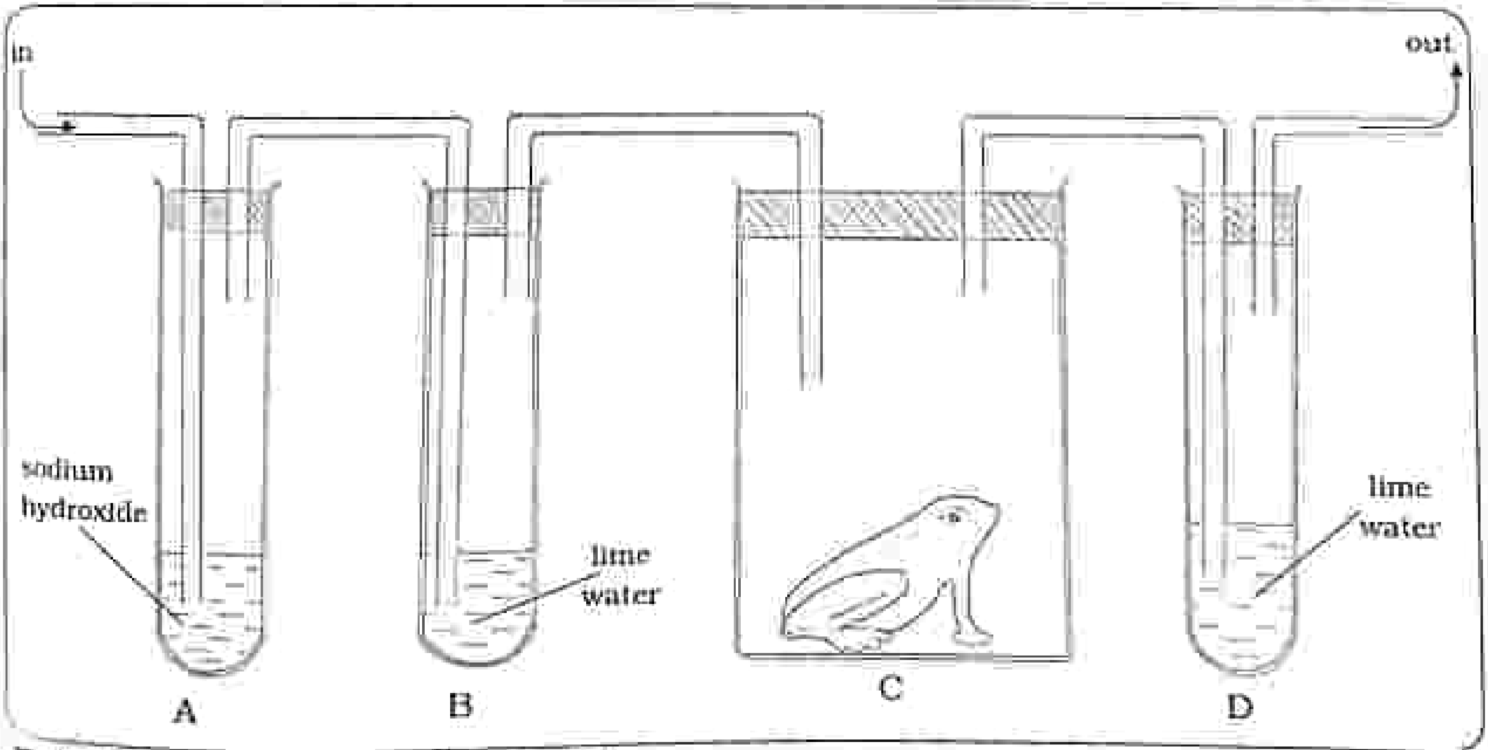


Fig. 5.7: Do animals release carbon dioxide?

Method

1. Set up apparatus as shown in Fig 5.7
2. Contains in:
   A – Sodium hydroxide to remove carbon dioxide
   B – Lime water to test for carbon dioxide
   C – Frog left to respire freely
   D – Lime water to test for carbon dioxide
2. Observe after an hour.

Results

Test tube B – no colour change
Test tube D – lime water turns milky

Conclusion

- Respiring animals release carbon dioxide.

Practical questions

1. What is the purpose of sodium hydroxide in test tube A?
2. Identify the test tube which acts as the control for the experiment.

Differences between aerobic and anaerobic respiration

Aerobic respiration	Anaerobic respiration
Oxygen needed	Oxygen absent
More energy released	Less energy released
Carbon dioxide and water produced	Lactic acid produced

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

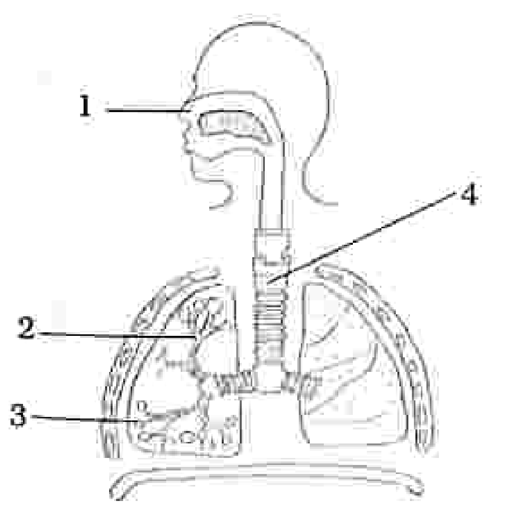
Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. What is the role of alveoli in gaseous exchange?
   A. digestion of food
   B. diffusion of carbon dioxide
   C. photosynthesis
   D. assimilation
2. The diagram shows the human respiratory system. Identify the bronchioles.



   A. 1
   B. 2
   C. 3
   D. 4
3. In which part of the respiratory system is air filtered, warmed and moistened?
   A. nasal passage
   B. trachea
   C. alveoli
   D. bronchioles
4. Identify respiratory gases.
   A. oxygen and carbon dioxide
   B. oxygen and neon
   C. carbon dioxide and nitrogen
   D. oxygen and nitrogen
5. To test for carbon dioxide gas, use
   A. lime water.
   B. glowing splint.
   C. iodine solution.
   D. sodium hydroxide.

6. The table shows the difference between inhaled and exhaled air. Identify the correct combination.

Air component	Inhaled air	Exhaled air
A. Carbon dioxide	4%	0.03%
B. Water vapour	Higher	Varies
C. Oxygen	16%	Varies
D. Oxygen	20%	16%

7. Which one describes how the alveolus is adapted for gaseous exchange?
   A. small surface area
   B. very thick
   C. network of blood capillaries
   D. many chloroplasts
8. Identify the word equation for aerobic respiration
   A. glucose $\longrightarrow$ lactic acid + less energy
   B. glucose + oxygen $\longrightarrow$ carbon dioxide + water + energy
   C. carbon dioxide + water + energy $\longrightarrow$ glucose + oxygen
   D. lactic acid + less energy $\longrightarrow$ glucose
9. Identify the word equation for anaerobic respiration
   A. glucose $\longrightarrow$ lactic acid + less energy
   B. glucose + oxygen $\longrightarrow$ carbon dioxide + water + energy
   C. carbon dioxide + water + energy $\longrightarrow$ glucose + oxygen
   D. lactic acid + less energy $\longrightarrow$ glucose
10. Which process takes place during gaseous exchange?
   A. digestion
   B. photosynthesis
   C. diffusion
   D. osmosis

Section B: Structured questions

1. The table below shows percentage composition of some components of air in inhaled and exhaled air. Complete the table;

Air component	Inhaled air	Exhaled air
Oxygen	....	....
Carbon dioxide	....	....
Water vapour	varies	higher

2. a) Describe the role of alveoli in gaseous exchange. [2]

b) Explain four ways the alveolus is adapted for gaseous exchange. [4]

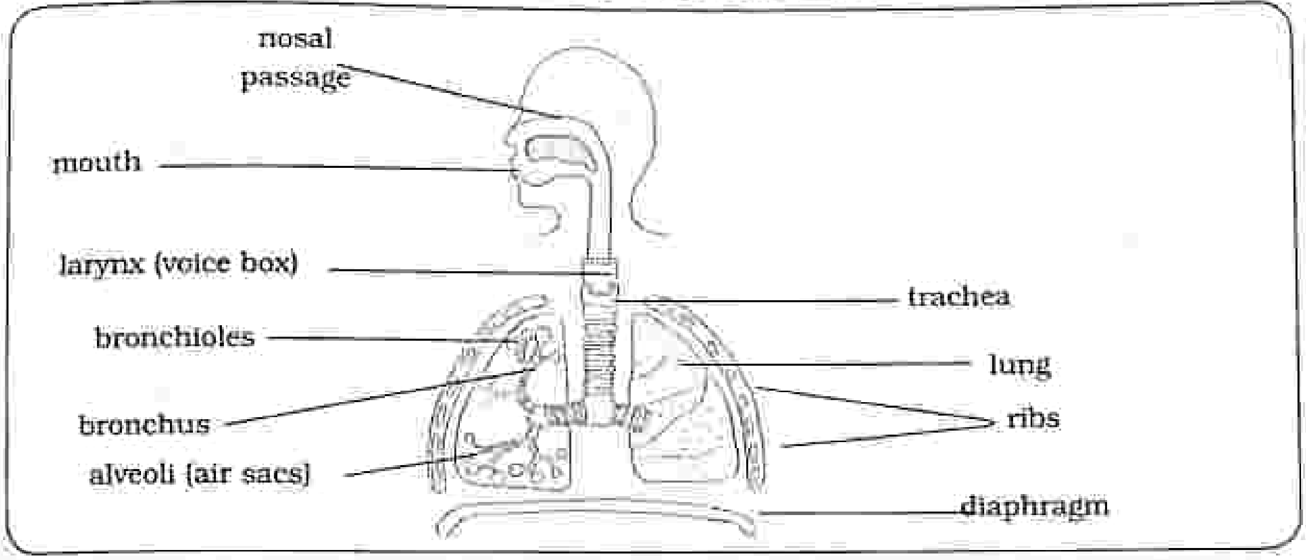
3. a) (i) Describe aerobic respiration. [1]

(ii) State the word equation for aerobic respiration. [1]

b) (i) Describe anaerobic respiration. [1]

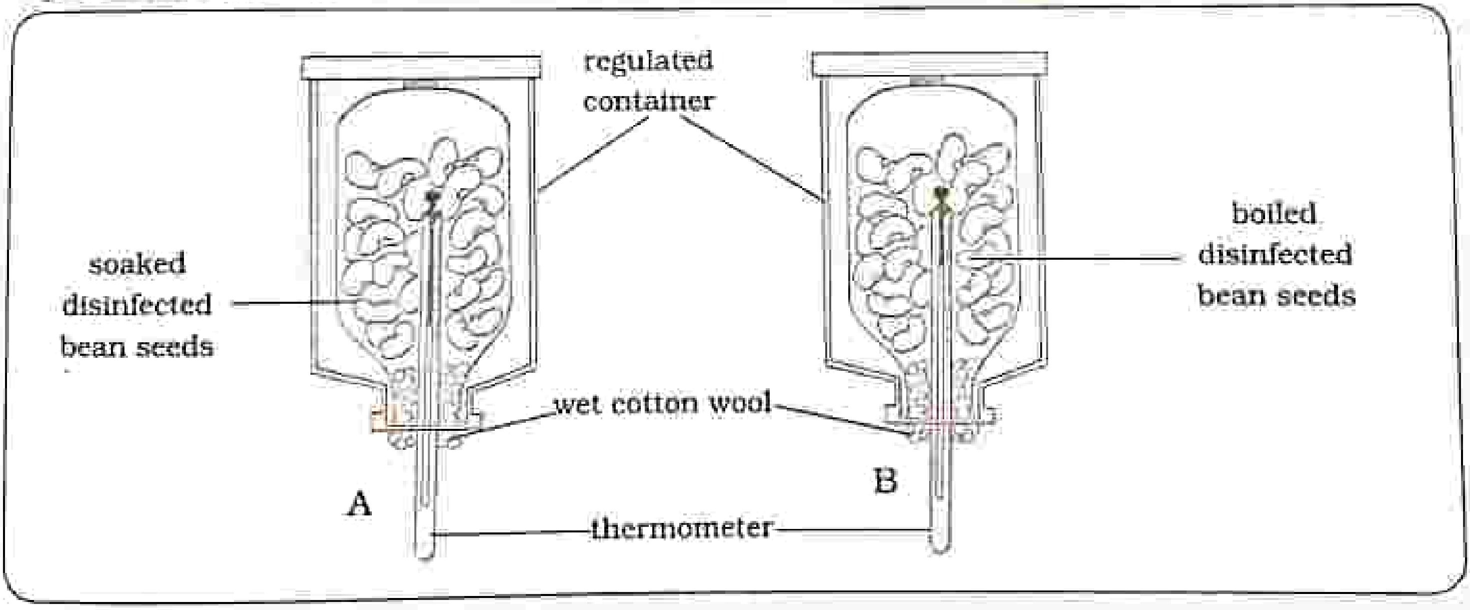
(ii) State the word equation for anaerobic respiration. [1]

4. The diagram shows the human respiratory system. Label the diagram



[5]

5. The setup of apparatus was used to investigate if carbon dioxide is released by germinating seeds.



a) After a few hours explain the observation in

A ....

B .... [2]

b) What conclusion can be drawn from the observation at (b) [1]

CHAPTER 6

Transport systems in plants

Key concepts

- Osmosis
- Diffusion
- Active uptake
- Root structure of a dicotyledonous plant
- Stem structure of a dicotyledonous plant
- Transpiration
- Measurement of transpiration
- Factors affecting rate transpiration
- Adaptation of plant leaves to reduce transpiration
- Importance of transpiration
- Plasmolysis
- Turgidity

Summary notes

Diffusion – is the movement of a substance from an area of high concentration to an area of low concentration.

- Cells obtain materials from their surroundings by diffusion for example oxygen enters plant cells and carbon dioxide leaves the plant cells by diffusion provided there is a diffusion gradient.
- Diffusion stops when the concentrations of the two regions are equal (at equilibrium).

Osmosis – is the movement of water particles from a region of high concentration to a region of low concentration through a semi permeable membrane.

- Root cells absorb water from the soil through osmosis.

Active uptakes – it is when mineral salts are absorbed into the cell against a concentration gradient and the process needs energy which comes from the food.

Root structure of a dicotyledonous plant

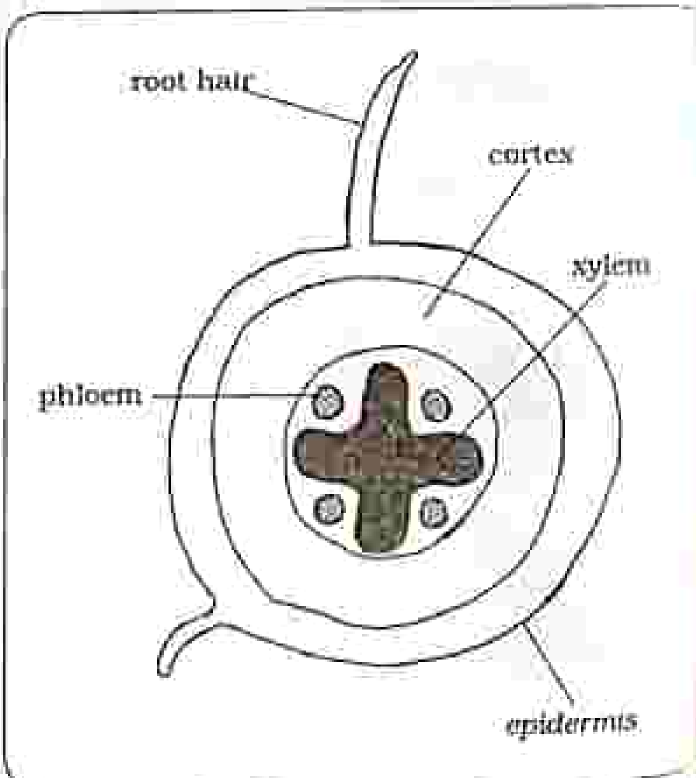


Fig. 6.1: Root structure of a dicotyledonous plant

Stem structure of a dicotyledonous plant

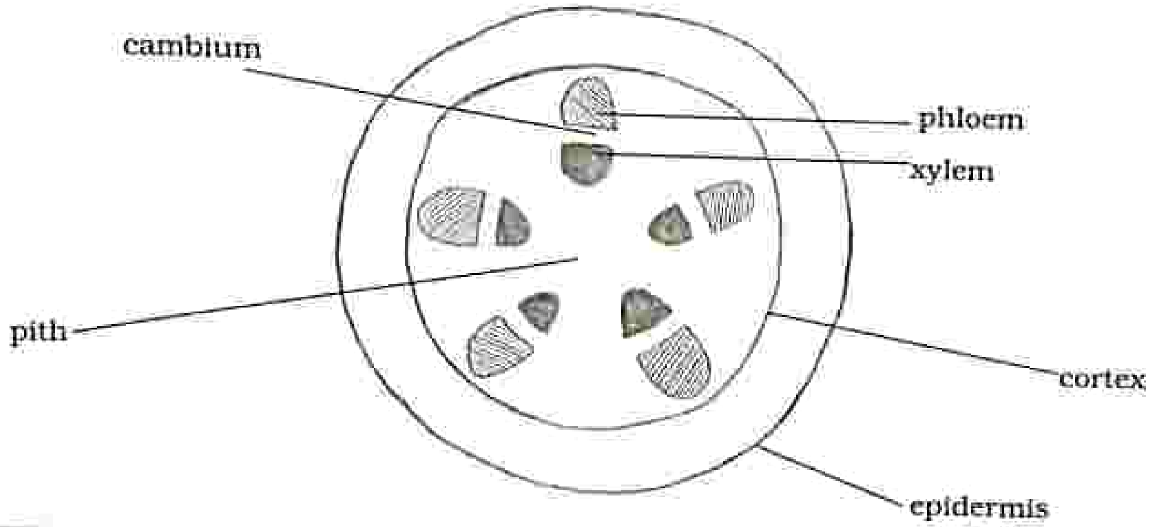


Fig. 6.2: Stem structure of a dicotyledonous plant

Xylem – transport water and dissolved mineral salts from roots to stems and leaves.

Phloem – transport soluble foods to all parts of a plant.

Epidermis – it is the outer layer which protects the stem or root from damage.

- It prevents excessive loss of water.

Cortex – offers stability and strength to the root and stem.

- Water moves across the cortex towards the centre.

Pith – offers stability and strength to the root and stem.

Cambium – produces xylem and phloem vessels.

Transpiration

- It is a process by which plants lose water to the atmosphere in form of water vapour.
- Transpiration takes place through a concentration gradient where water diffuses out of the leaf through the stomata and into the atmosphere.

Measurement of transpiration

Practical activity 6.1

Aim : measuring the rate of transpiration and the factors affecting rate of transpiration

Apparatus :
1. Potometer
2. Water
3. Green plant
4. Vaseline

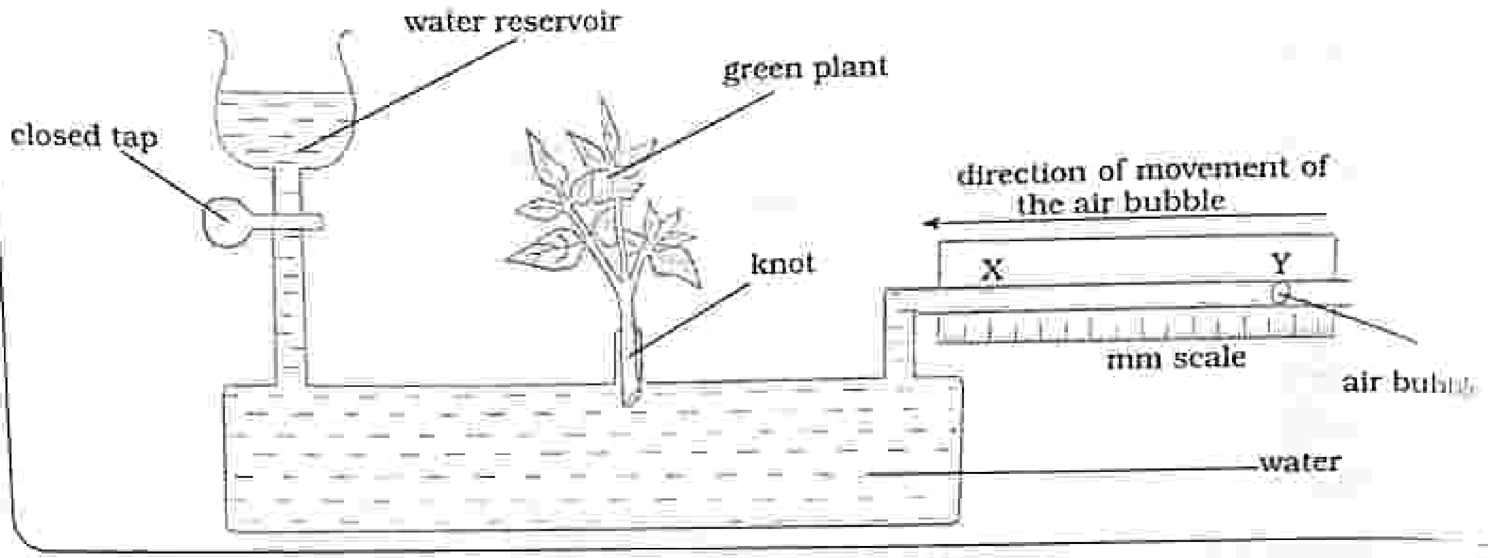


Fig. 6.3: Potometer

Method

1. Place a green plant into the potometer as shown in Fig 6.3.
2. Put vaseline at the knot to prevent water evaporating from the potometer through the knot.
3. Insert an air bubble and label the initial point Y and record the length.
4. Record the time and length the bubble reaches point X.
5. Calculate the rate of transpiration.

Factors affecting rate of transpiration

5. Open the tap and allow water in until it reaches Y again.
6. Repeat the procedure by placing the potometer in the following conditions
   a) Moving air.
   b) Still air.
   c) In the sun (there is more light and high temperature).
   d) In the dark (there is less light and low temperature).
   e) Humid air.
   f) Dry air.
   g) Change plant and put one with narrow leaves.

Results

Initial position of air bubble Y = 10mm

Final position of air bubble X = 22mm

Distance moved by air bubble

= 22mm – 10mm = 12mm

Time taken for air bubble to move from Y to X = 100 seconds

Rate of transpiration.

= Distance X – Y/Time from Y to X

= 12mm/100s

= 0.12mm/s

Conclusion

- The green plant will lose water through transpiration
- This is noted by the movement of the air bubble from point Y to point X.

Practical questions

1. Why is vaseline applied on the knot?
2. Explain why the change of plant to one with narrow leaves?

Factors affecting rate of transpiration

- **Wind speed** – the faster the wind speed, the faster the rate of transpiration.
- **Temperature** – the higher the temperature, the faster the rate of transpiration. Low temperatures result in less transpiration.
- **Humidity** – the less humid the air is, the higher the rate of transpiration. When the air is saturated with water vapour transpiration is reduced.
- **Surface area** – a large surface area (broad leaves) increases the rate of transpiration.
- **Light intensity** – the more the light intensity, the stomata opens therefore the faster the rate of transpiration. In the dark, the stomata are almost closed.
- **Number of stomata** – more stomata increases the rate of transpiration.

Adaptation of plant leaves to reduce transpiration

- Leaves with small surface (narrow leaves and thorny leaves) will transpire less than those with a large surface area.
- Thick cuticle prevents excessive lose of water from the leaves.
- Most leaves have more stomata on the underside and less on the upper side as a result less stomata are exposed to direct light thus reducing the rate of transpiration.
- Hairs cover the leaf surface as a result less water is lost by transpiration.

Importance of transpiration

- When the leaves lose water by transpiration, water and mineral salts are pulled up from the roots and move through the xylem to all parts of the plant.
- During transpiration, water vapour diffuses from the leaf through the stomata into the surrounding air.
  - The water vapour carries with it heat energy from the leaf leaving the leaf cooled.

Plasmolysis

- Is when water moves out of the plant cell.
- Water is highly concentrated inside the plant cells than outside.
- The plant cells will lose water to the surrounding by osmosis.
- When the cells lose water, they become soft, the cell membrane is pulled in, the vacuole and cytoplasm shrinks.
  - The cell is said to be flaccid

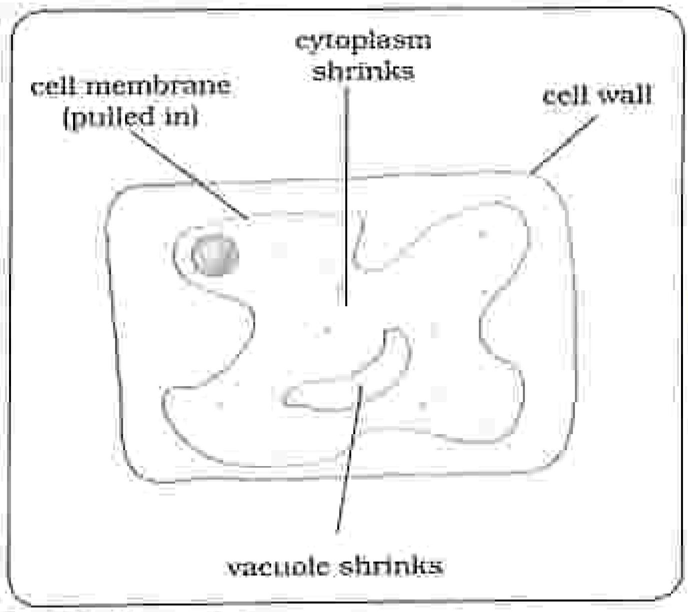


Fig. 6.4: Plasmolysis

Turgidity

- Is when water moves into the cytoplasm and vacuole through the semi permeable membrane by osmosis until the cell cannot take in more water.
- There is a low concentration of water inside the cell and a high concentration of water outside the cell.
- Vacuole gets larger, cytoplasm is pushed out against the cell wall.
- When the cell gains water, the cell becomes firm and provides support to the plant.

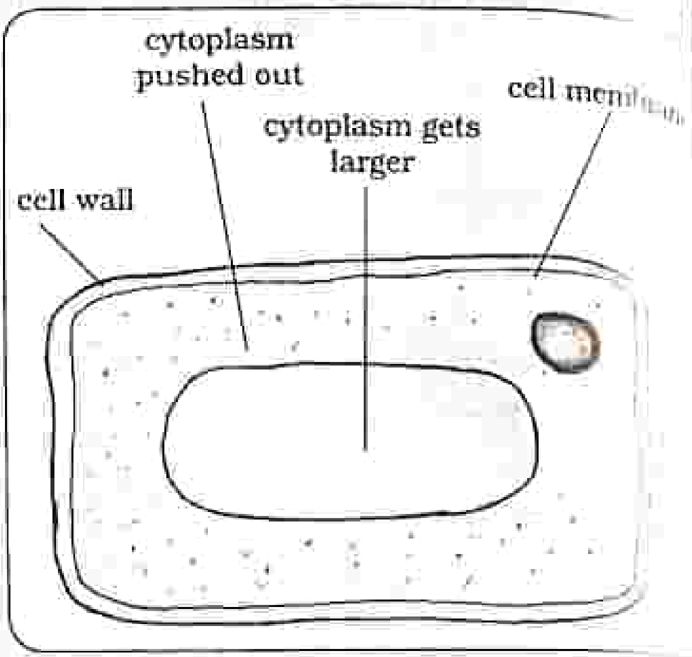


Fig. 6.5: Turgidity

Practical activity 6.2

Aim : To demonstrate plasmolysis and turgidity

Apparatus :
1. Potato
2. Two beakers
3. Salt solution
4. Distilled water

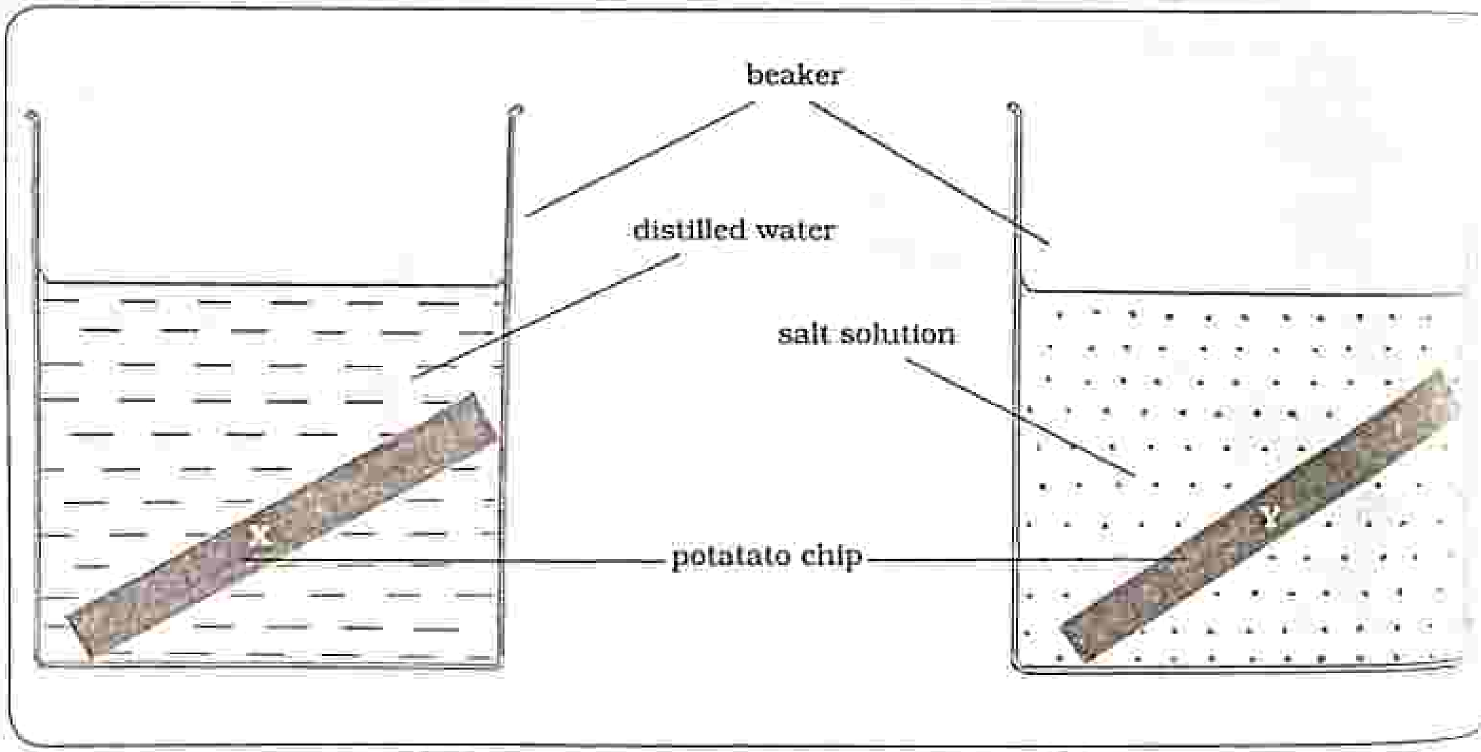


Fig. 6.6: Plasmolysis and Turgidity

Method

1. Cut two identical potato strips and measure the length.
2. Place potato X into the beaker containing distilled water.
3. Place potato Y into the beaker containing salt solution.
4. Leave for few hours and then measure the lengths of the potato strips again.

Results

After an hour

Length of potato strip X = Increased from the original length

Length of potato strip Y = Decreased from the original length

Conclusion

Potato strip X

- Water is highly concentrated in the beaker than in the potato cells.
- Water will move by osmosis from the beaker to the potato cells.
- The potato becomes firm because of turgidity.

Potato strip Y

- Water is highly concentrated inside the potato cells than in the beaker.
- Water will move by osmosis from the potato cells to the beaker.
- The potato strip will shrink because of plasmolysis.

Practical questions

1. When potato strip X is left in distilled water, what happens to the length of the strip after a few hours?
2. When potato strip Y is left in salt solution, what happens to the length of the strip after a few hours?

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

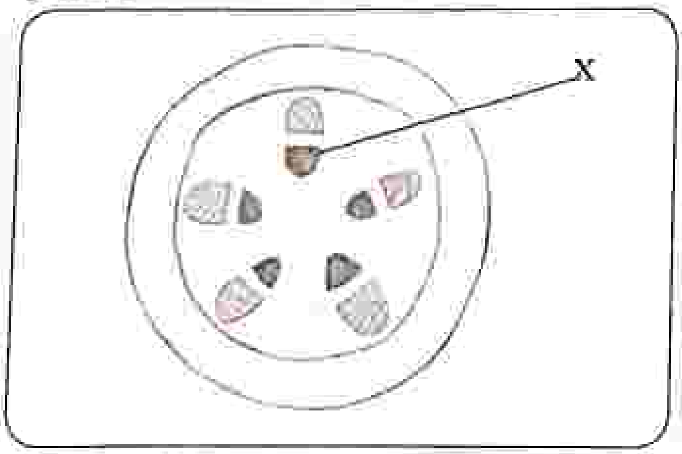
Section B

Structured questions

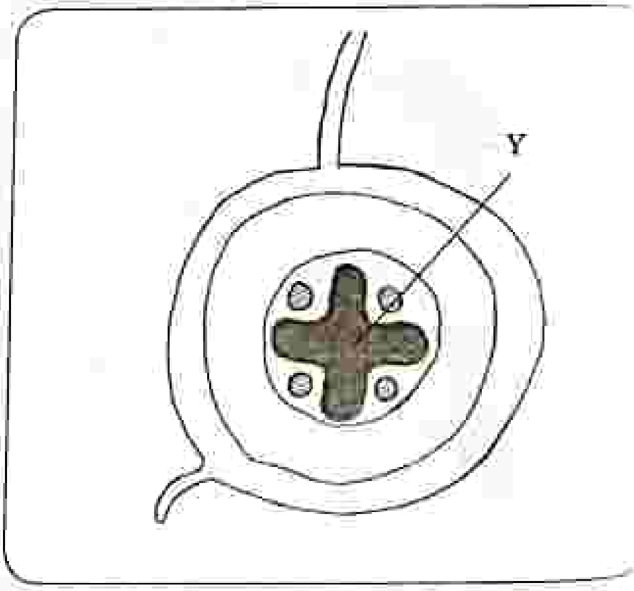
Answer all the questions.
The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. The xylem transport ____________.
   A. water and dissolved mineral salts
   B. soluble food
   C. soil
   D. oxygen
2. The phloem transport __________.
   A. water and dissolved mineral salts
   B. soluble food
   C. soil
   D. oxygen
3. The diagram shows the stem structure of a dicotyledonous plant. Identify part X



   A. epidermis
   B. cortex
   C. xylem
   D. phloem
4. The diagram shows the root structure of a dicotyledonous plant. Identify part Y.

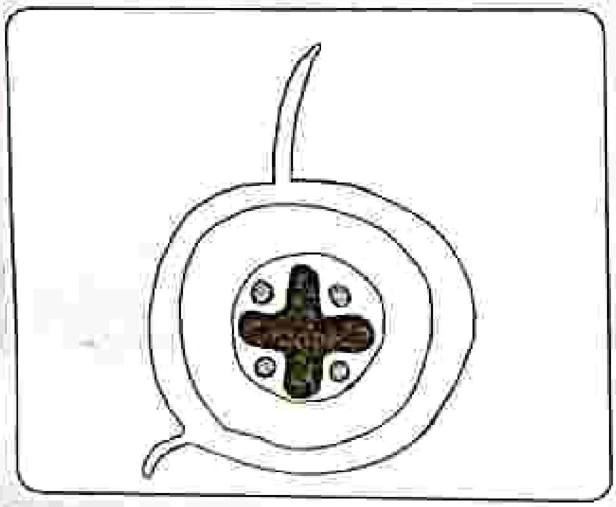


   A. epidermis
   B. cortex
   C. xylem
   D. phloem

5. Which one is a factor which affects the rate of transpiration?
   A. wind speed
   B. oxygen
   C. carbon dioxide
   D. digestion
6. Identify the importance of transpiration.
   A. cooling the plant
   B. increase wind speed
   C. increase surface area
   D. increase light
7. When water moves from an area of high concentration to an area of low concentration through a semi permeable membrane, the process is called ______.
   A. osmosis
   B. diffusion
   C. digestion
   D. active uptake
8. The process responsible for the loss of water from plant leaves and stem is ________.
   A. transpiration
   B. diffusion
   C. digestion
   D. active uptake
9. When minerals are absorbed into the cells against a concentration gradient, the process is called ________.
   A. transpiration
   B. diffusion
   C. digestion
   D. active uptake
10. How is a leaf adapted to reduce transpiration?
   A. presence of hairs
   B. large surface area
   C. more stomata on the upper side
   D. thin cuticle

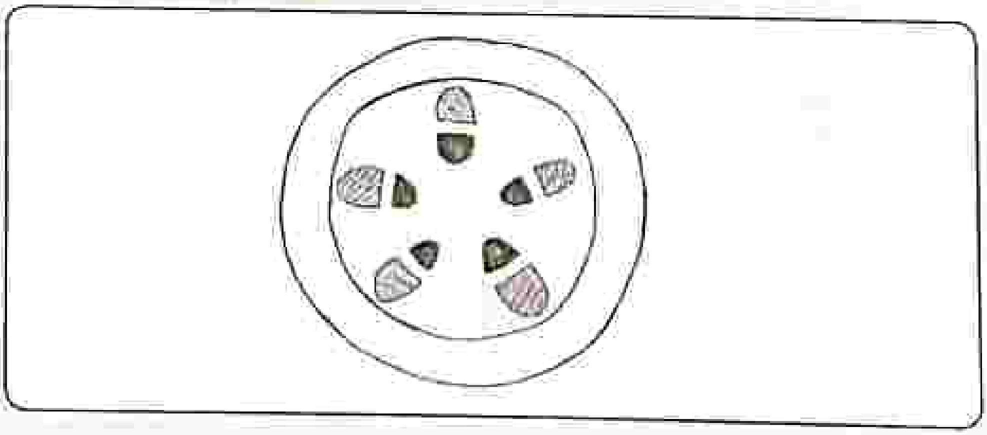
Section B: Structured questions

1. Explain the following terms
   a) Diffusion [1]
   b) Osmosis [1]
   c) Active uptake [1]
   d) Plasmolysis [1]
   e) Turgidity [1]
2. a) The diagram shows the root structure of a dicotyledonous plant. Label the diagram.



[4]

   b) The diagram shows the stem structure of a dicotyledonous plant. Label the diagram.



[4]

3. a) Explain the process of transpiration. [2]
   b) Outline the importance of transpiration. [2]
4. State any four factors which affect the rate of transpiration. [4]
5. Describe four ways plant leaves are adapted to reduce transpiration. [4]

CHAPTER 7

Transport in humans

Key concepts

- Functions of blood
- Structure of blood vessels
- Double circulatory system

Summary notes

Blood

- Blood is a highly specialised connective tissue which continuously supplies body cells with oxygen, and all the nutrients needed.
- It also collects waste materials from the body for excretion.
- Blood plays the defense role as it fights infection.
- It also distributes heat throughout the body and keeps water, temperature and glucose concentration at acceptable levels (homeostasis).

Table 7.1 Components of blood and their functions

Plasma	The liquid part of blood where dissolved nutrients,waste products, gases and physical components of blood are found
Red blood cells	Transport carbon dioxide and oxygen. They do not have a nucleus. They contain hemoglobin which gives blood its red color and carry gases throughout the body.
White blood cells	Help the body in fighting infection. They have a nucleus. Some produce antibodies which destroy pathogens such as bacteria, some engulf the pathogens and destroy them.
Platelets	Help in the clotting of blood.

Blood vessels

- Blood flows in vessels as it circulates around the body.
- The different types of vessels are arteries, veins and capillaries.
- Arteries branch into arterioles and veins branch into venules.
- Capillaries connect arterioles to venules.

Passage of blood from heart and back

Arteries ⟶ arterioles ⟶ capillaries ⟶ venules ⟶ veins

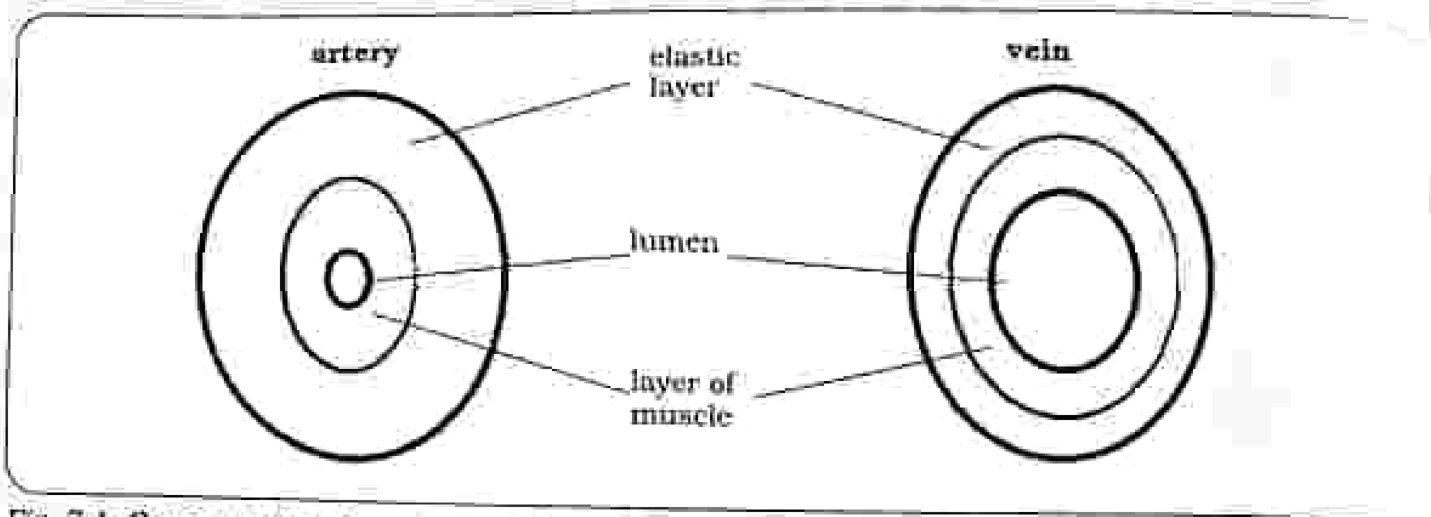


Fig. 7.1: Cross section of an artery and a vein

Table 7.2 Veins and Arteries

Arteries	Veins
Carry blood from the heart	Carry blood to the heart
Carry blood at high pressure	Carry blood at low pressure
Have very thick and strong walls that can stretch	They have thin walls
Have a narrow lumen	Have a wider lumen
Do not have valves	Have valves to prevent back flow of blood
Carry oxygenated blood apart from pulmonary artery	Carry deoxygenated blood apart from pulmonary vein

Capillaries

- Through capillaries blood reaches every cell.
- They are microscopic blood vessels found in the muscles and lungs and, they are one cell thick with very low blood pressure.
- Where gaseous exchange takes place, the thin walls enable nutrients and gases to diffuse between the blood and cells.

Blood vessels and internal organs

- With the exception of the liver which receives blood through two vessels, each internal organ is supplied blood by a large artery and blood is taken away by a large vein.

Table 7.3: Internal organs and blood vessels

Organ	Blood supplied by	Blood taken away by
Kidney	Renal artery	Renal vein
Liver	Hepatic artery (blood with oxygen) Hepatic portal vein (blood with dissolved food)	Hepatic vein
Lungs	Pulmonary artery	Pulmonary vein

The heart

- It has four chambers, two atria and two ventricles.
- Atria force blood into ventricles, which then contract and blood is pushed out of the heart. The function of the heart is to receive and pump blood.
- The right side receives blood from the body cells through the vena cava and pumps it to the pulmonary circuit, through the pulmonary artery.
- The left side receives blood from the lungs through the pulmonary vein and pumps it to the systemic circuit through the aorta.
- The two sides of the heart are separated by the septum.
- The left side of the heart is thicker than the right side.
- The heart has four valves to prevent the back flow of blood.
- Each side has a semi-lunar valve which is found in the pulmonary artery and the aorta.
- Left side has a bicuspid valve (valve with two flaps) and the Right side has a tricuspid valve (valve with three flaps). These are found between the atria and ventricles.

The double circulatory system

- The human circulatory system is a double circulatory system.
- Blood passes through the heart twice and has two separate circuits, pulmonary circuit and systemic circuit.
- Pulmonary circulation takes place between heart and lungs.
- Systemic circulation takes place between heart and body.
- Blood is pumped at high pressure away from the heart and at low pressure as it returns to the heart.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.
The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. The function of hemoglobin is to ______________.
   A. give blood the red colour
   B. help in the clotting of blood
   C. carry oxygen and carbon dioxide
   D. destroy pathogens
2. Which is a function of platelets?
   A. defence
   B. clotting
   C. transport
   D. homeostasis
3. The table shows changes in oxygen and carbon dioxide concentration as blood flows through an organ.

Gas	Change in concentration
Oxygen	Increased
Carbon dioxide	Reduced

The organ has the blood passed through is the ______________.
   A. kidney
   B. liver
   C. lung
   D. stomach
4. The structure in the human heart that separates oxygenated blood from deoxygenated blood is the ______________.
   A. semi lunar valve
   B. bicuspid valve
   C. ventricle
   D. septum
5. Arteries carry blood from the heart. Which combination best describes the pressure in them and the size of the lumen.

	Lumen	Pressure
A	Narrow	High
B	Narrow	Low
C	Wide	High
D	Wide	Low

6. Blood from the lungs enters the heart through the ____________.
   A. left atrium.
   B. right atrium.
   C. left ventricle.
   D. right ventricle.
7. Compared to an artery a vein has ____________.
   A. a thicker wall
   B. A narrow lumen
   C. valves
   D. a wall with more elastic tissue
8. Absorbed food from small intestines is carried to the liver by ________.
   A. coronary artery.
   B. hepatic portal vein
   C. pulmonary artery.
   D. renal vein
9. The table shows the characteristics of blood in one blood vessel in the body

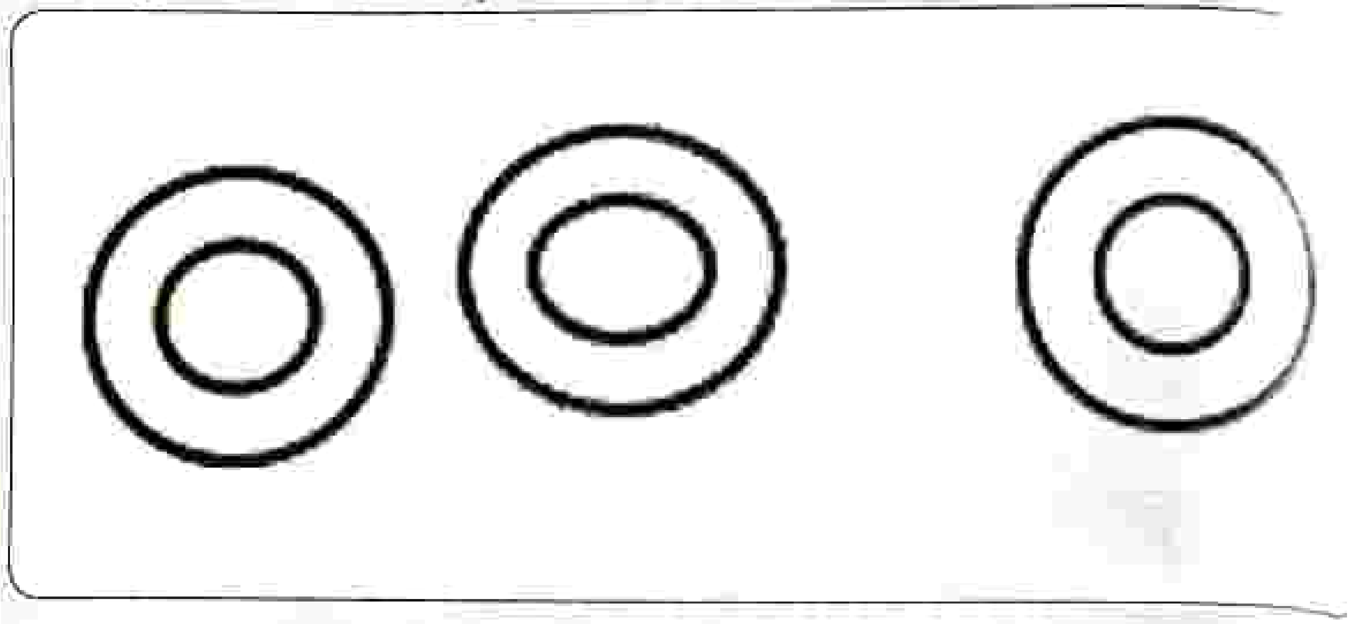
Pressure	Oxygen concentration	Carbon dioxide concentration
Low	Low	High

   Which blood vessel contains blood with these characteristics?
   A. aorta
   B. vena cava
   C. pulmonary artery
   D. pulmonary vein
10 Humans are said to have a double circulatory system because ________.
   A. as blood circulates it passes twice through the heart
   B. each side of the heart has two chambers
   C. the heart has two valves on each side
   D. the heart has two separate sides

Section B: Structured questions

1. Name the component of blood which carries oxygen. [1]
2. a) List any two physical components of blood. [2]
   b) Explain how each component in (a) is adapted for its function. [2]
3. Explain the two ways in which white blood cells fight infection. [2]
4. State the three functions of blood. [3]
5. Describe how capillaries are adapted for material exchange with the body. [2]
6. Distinguish between veins and arteries. [3]
7. State the functions of the heart. [1]
8. State with a reason the difference in structure of the left side and the right side of the heart. [2]

9. The diagram below shows specialised cells



(i) Identify the cells. [1]

(ii) Explain how the cells are adapted for their function. [2]

(iii) Describe how the cells differ from other cells. [2]

10. Distinguish between pulmonary circulation and systemic circulation. [2]

CHAPTER 8

Reproduction in plants

Key concepts

- Sexual reproduction
- Germination
- Asexual reproduction
- Advantage and disadvantages of asexual reproduction

Summary notes

Sexual reproduction in plants

Sexual reproduction in plants is centered on the flower. Each part of the flower performs a function.

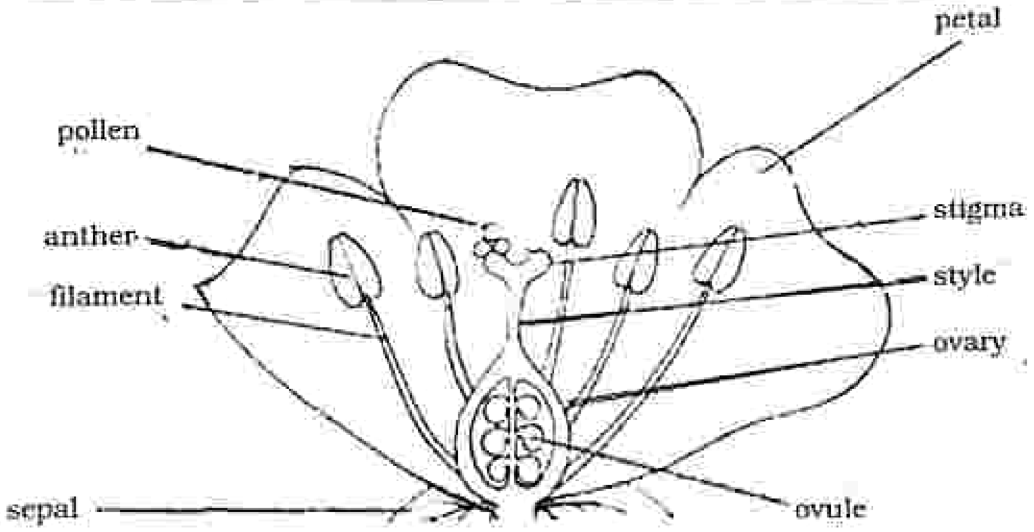


Fig. 8.1: A flower

Table 8.1: Functions of parts of the flower

Part	Function
Stamen	Male part of the flower contains; filament anther with pollen.
Anther	Produces pollen grains.
Filament	Holds the anther in position.
Pollen	Contain male sex cell (gamete).
Pistil/carpel	Female part of the flower contains; stigma, style and ovary.
Stigma	Collects pollen grains, it is usually sticky.

Part	Function
Style	Holds stigma in position, creates a path through which pollen tube grows.
Ovary	Contain ovules and develop into a fruit after fertilisation.
Ovule	Contain the female sex cell (gamete) develops into a fruit after fertilization.
Petals	Scented and brightly coloured to attract insects.
Sepals	Protects the flower whilst in a bud.
Nectary	Produces nectar.
Nectar	Sweet scented juice that attracts insects.

Pollination

- It is the transfer of pollen grains from the anther to the stigma.
- There are two types of pollination namely, self-pollination and cross-pollination.

Self-pollination is the transfer of pollen grain from the anther to the stigma of the same flower or to the stigma of another flower on the same plant.

Advantages of self-pollination

- Ensures continuity of a race or species.
- Helps preserve the parental characteristics.
- Plants do not have to produce pollen grains in large quantities.

Disadvantages of self-pollination

- New varieties cannot be obtained.
- Genetic defects of a breed cannot be removed.

Cross-pollination is the transfer of pollen grains from the anther of a flower on one plant to the stigma of a flower on another plant but of the same type.

Advantages of cross pollination

- Variations are introduced.

Disadvantages of cross-pollination

- Pollen grains have to be produced in large quantities.
- Good parental characteristics may not be preserved.

Agents of pollination

- Insects and wind are the agents of pollination.
- Each flower is adapted to a type of pollination.
- There are wind pollinated and insect pollinated flowers.

Table 8.2 compares wind and insect pollinated flower

Table 8.2: Comparison of wind and insect pollinated flowers

Insect pollinated flowers	Wind pollinated flowers
Have large and brightly coloured petals.	Small and dull coloured petals.
Produce scent.	Not scented.
Produce nectar.	Do not produce nectar.
Pollen grains are sticky and heavier.	Pollen grains are non-sticky and light.
Pollen grains are produced in small quantities.	Pollen grains are produced in large quantities.
Filaments are short.	Filaments long to expose anthers.
Stigmas are sticky.	Stigmas are feathery.
Female and male parts are not exposed.	Female and male parts are exposed to wind.
Example: peas, beans	*Example:* maize, wheat

Fertilisation

- It is the fusion of the male and the female sex cells.
- After pollen grains land on a stigma, they are nourished and develop a tube which moves down the style to the ovary and into an ovule.
- The male sex cell nucleus moves down the pollen tube into the ovule where it will fuse with the female sex cell nucleus, forming a zygote.
- After fertilisation has taken place, sepals, petals, stamens, stigma and style wither away and usually fall off. The ovule develops into a seed and the ovary develops into a fruit.

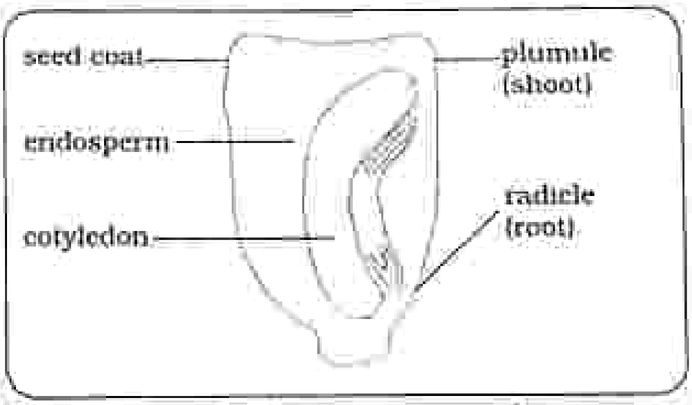


Fig. 8.2:monocotyledonous maize seed

Types of seeds

Monocotyledonous seeds

These seeds have one cotyledon, they store the bulk of their food in the endosperm e.g. maize seed.

Dicotyledonous seeds

These seeds have two cotyledon, food is stored in the two cotyledons.

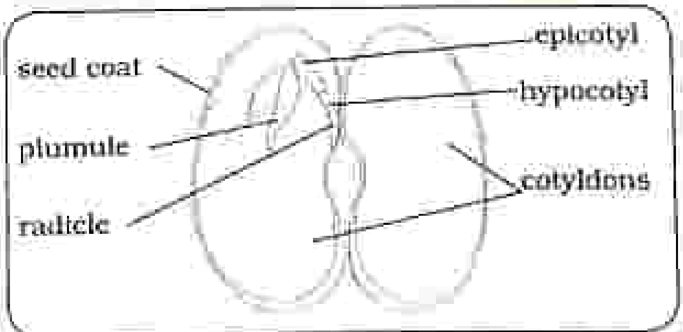


Fig. 8.3: dicotyledonous bean seed

Germination

- It is a process whereby seeds develop into new plants.

Conditions necessary for seed germination

- **Water:** Causes the seed to swell. Seed coats are softened and rapture easily to make it easy for radicle and plumule to come out. It also activates enzymes that begin the process of seed growth.
- **Oxygen:** is used in respiration to produce energy needed.
- **Temperature:** the seed contain enzymes which work faster at optimum temperatures. Optimum temperature also speeds up metabolic activities within the seed.

Asexual Reproduction

- It is a process which results in the production of genetically identical offsprings.
- Only one parent is required and there is no production of seeds.

Advantages of asexual reproduction

- Only one parent is required, reproduces quickly and it takes less energy.
- Good characteristics of parent plant are preserved.
- More food is stored in tubers than in seeds so the young plants grow faster.

Disadvantages of asexual reproduction

- Lack of diversity and prone to extinction due to inability to adapt.
- There is usually overpopu... and overcrowding which mean... chances of survival. Undesirable ... characteristics cannot be elimin...

Methods of asexual reproduction

There are many different types of as... reproduction some of them are;

- **Cuttings:** this is simply taking a ... piece of a plant, plant it and then ... for it to grow.
- **Rhizomes:** are underground ... that grow horizontally and parallel ... the ground. The stem develops ... and shoots.
- **Tubers** are a thickened rhizome ... underground stem or root structur... that has been enlarged for use as ... storage organ. A tuber has tiny sca... leaves with buds that grow on ... surface. Each bud can form a ne... plant. e.g. potato.
- **Runners:** horizontal stems that spread above the ground and outward from the main plant. e.g. strawberry.
- **Bulbs** roughly spherical undergroun... buds (shoots) with fleshy leaves. Eac... bulb contains several other buds whic... can give rise to a new plant.

Differences between asexual reproductio... and sexual reproduction

- Asexual reproduction differs fr... sexual reproduction in a number ... ways.

Table 8.2 shows the differences between the two

Table 8.2 Differences between asexual and sexual reproduction

Asexual reproduction	Sexual reproduction
Only one parent is required.	Requires both male and female.
Lower chances of offspring survival.	Higher chances of offspring survival.
Less energy needed.	More energy needed.
No variation.	There is variation.
No formation or fusion of sex cells.	Formation and fusion of sex cells occur.
Multiplication is very rapid, takes less time to complete.	Multiplication is not rapid, takes longer to complete.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

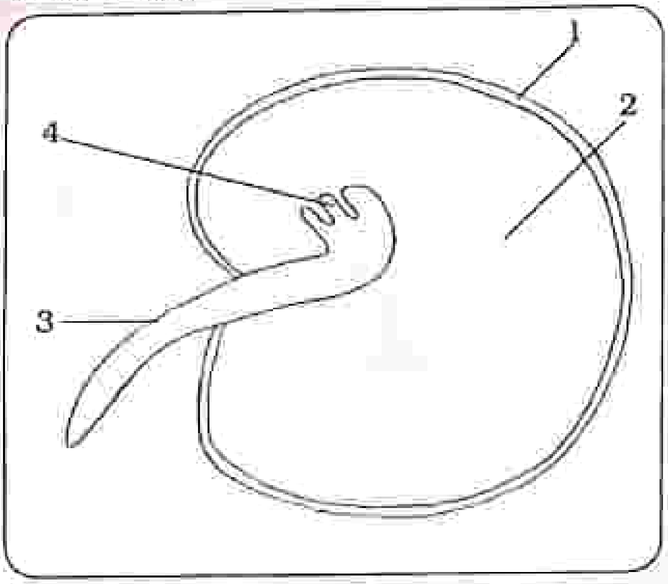
The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. The male part of the flower is called a/an
   A. carpel.
   B. filament.
   C. anther.
   D. stamen.

2. Dull coloured, smaller and non-scented flowers rely on ____________.
   A. insect pollination
   B. wind pollination
   C. self-pollination
   D. cross-pollination

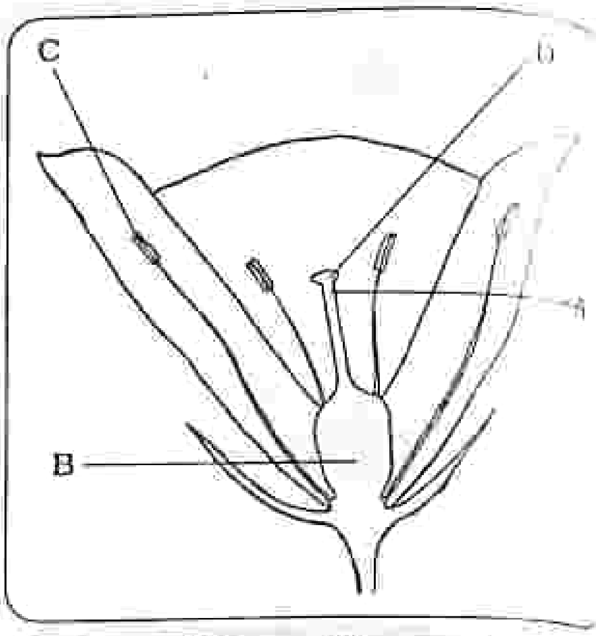
3. Wind pollination requires that the grains are ____________.
   A. heavy and wet
   B. heavy and none-sticky
   C. light and dry
   D. heavy and sticky
4. On which part of the flower will pollen grains be deposited in pollination?
   A. anther
   B. ovary
   C. ovule
   D. stigma
5. The diagram shows a germinating bean seed.



Which label shows the cotyledon and the plumule?

A.	1	3
B.	1	4
C.	2	3
D.	2	4

6. The diagram shows a flower. In wh
   structure do seeds develop?



7. Which conditions will prevent seed fr
   germinating whilst in storage?
   A. High humidity
   B. High oxygen levels
   C. Low light intensity
   D. Low temperature
8. In asexual reproduction, baby pla
   are
   A. Very different from parent plant
   B. Exactly like the parent plant
   C. 50% like parent plant
   D. 80% like parent plant
9. A bulb is a ____________.
   A. horizontal underground stem
   B. modified shoot
   C. thick underground stem
   D. swollen underground stem

10. What are advantages of sexual and asexual reproduction?

	Advantages of sexual	Advantages of asexual
A.	Less population growth	Only one parent required
B.	More energy efficient	Gametes can be transferred by environment
C.	More genetic variation	Faster
D.	No transfer of gametes needed	Does not compete with parent for nutrients

Section B: Structured questions

1. Define pollination. [1]
2. Give two advantages and two disadvantages of self-pollination. [4]
3. Explain the difference between self-pollination and cross-pollination. [2]
4. Distinguish between insect and wind pollinated flowers. [3]
5. What is the difference between pollination and fertilization? [2]
6. Describe the events taking place between pollination and fertilization. [2]
7. Describe the post fertilization changes that take place on the flower. [3]
8. Complete the table below and explain how each part is adapted for wind or insect pollination.

Part of flower	Insect pollinated	Wind pollinated
Petals		
Nectary		
Scent		
Stigma		

[8]

9. Discuss the advantages and disadvantages of asexual reproduction. [4]
10. Distinguish between asexual and sexual reproduction. [2]

CHAPTER 9 # Reproduction in humans

Key concepts

- Male and female reproductive systems
- Male and female sex cells
- Fertilisation
- Pregnancy
- The menstrual cycle
- Methods of contraception

Summary notes

Sexual reproduction in humans

- Humans reproduce sexually through the use of sex cells.
- The male reproductive system produces sperms and the female reproductive system produces ova or eggs.

The female reproductive system

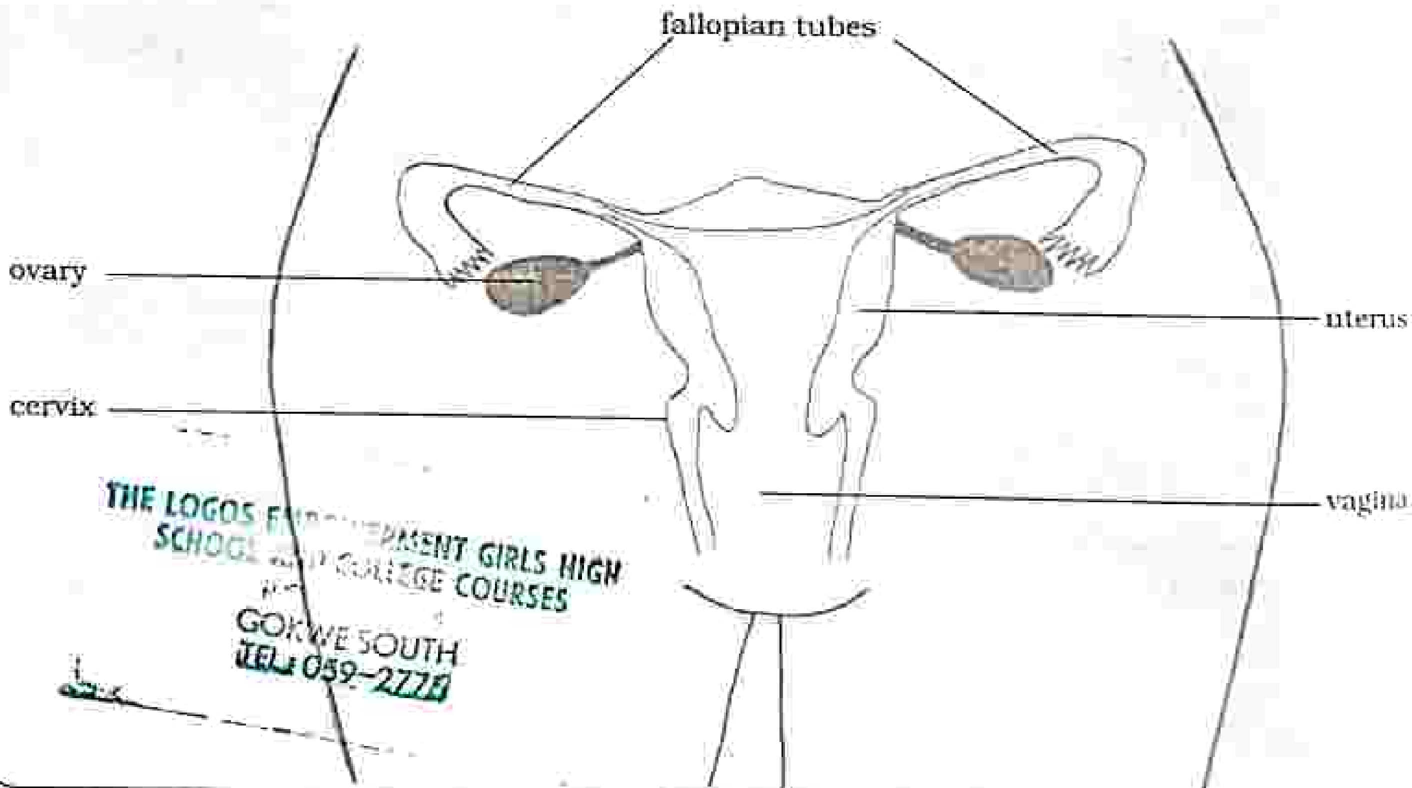


Fig. 9.1: Female reproductive system

- Each part of the reproductive system performs a specific function in reproduction. Table 9.1 shows the functions of the female reproductive organs.

Table 9.1: Functions of female reproductive organs

Organ	Functions
Ovary	Produces ova or eggs.
Fallopian tube/ oviduct	Connects an ovary to the uterus and this is where fertilisation occur.
Uterus	Its where the foetus develops.
Cervix	Separates the vagina from the uterus, closes during pregnancy to keep the baby in place.
Vagina	Accommodates the penis during intercourse.

The male reproductive system

- The main function of the male reproductive system is to produce sperms which are male sex cells.
- Each part of the system performs a specific function. Table 9.2 shows the functions of the male reproductive organs.

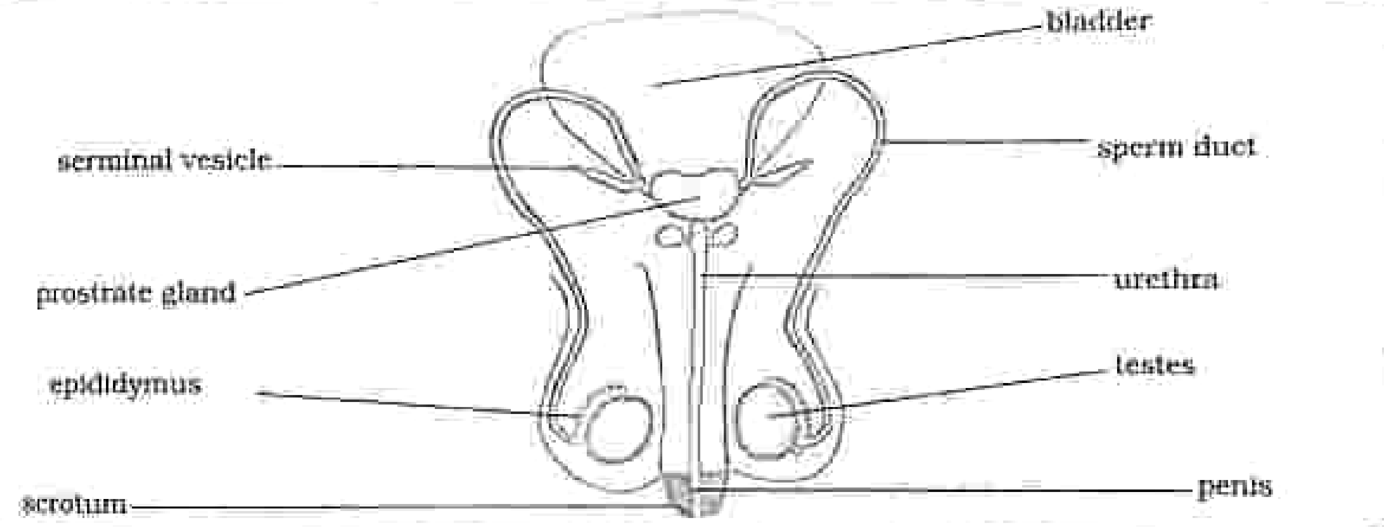
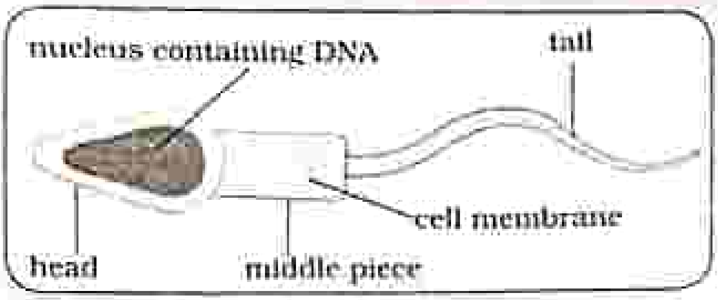


Fig. 9.2: Male reproductive system

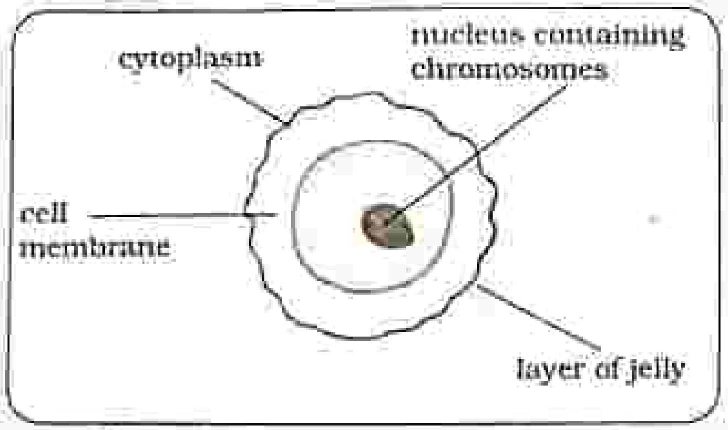
Table 9. 2: Functions of male reproductive organs

Organ	Functions
Scrotum	Contains and regulates temperature of testes.
Testes	Production of sperms and testosterone.
Penis	Deposit semen into the vagina during sexual intercourse.
Prostate gland	Secretes a fluid which combines with sperms to form semen.
Sperm duct	Tube that carries sperms from testes to the urethra.
Urethra	Tube through which urine and semen are excreted from the body.
Epididymus	Temporary storage for sperms.

- **Sex cells** – In humans the sex cells are: the sperm and the ovum
- **Male sex cell** – the sperm is the male sex cell. It has a nucleus containing half set of genetic instructions.



- **Female sex cell** – the ovum or egg is the female sex cell. Its nucleus contains half set of genetic instructions.



Fertilization

- It is the fusion of male and female sex cells to form a zygote.
- During sexual intercourse sperms are deposited into the vagina.
- They swim through the cervix, uterus to the oviduct where the egg will be.
- Only one sperm enters the ovum and fuses with its nucleus.

Pregnancy and the placenta

- Once fertilisation has occurred, the zygote undergoes cell-division (mitosis) to produce many cells which then make up an embryo.
- The embryo moves down to the uterus where it attaches itself to the uterus lining where it will grow. The process is called implantation.
- After implantation, the placenta develops and it allows exchange of substances between the mother an the foetus.
- The placenta also produces hormon such as oestrogen and progesterone.
- Essential nutrients and antibod diffuse from the mother to the foetu
- Waste products diffuse from the foet to the mother for excretion.
- The placenta ensures that the blood the foetus and mother do not mix.
- The foetus is connected to the placen by the umbilical code.
- Amniotic sac produces Amniotic flu which surrounds and protects th foetus during pregnancy.
- In humans, the period of pregnancy 9 months.

Birth

- When the time is due for the bab to be born, the amniotic sac break releasing the amniotic fluid.
- Muscles in the uterus wall contra to push the baby out while the cerv dilates.
- The baby then exits the moth through the vagina and the umbilic cord, which would be still attached t the baby, is cut and tied.

The menstrual cycle

- The cycle happens approximately eve 28 days.
- Around day 14, an egg is released fr the ovaries during each cycle.
- The uterus wall thickens with blood preparation for pregnancy.
- If fertilisation does not occur bo the egg and uterus lining break dow

and are then expelled from the body through the vagina. The process is called menstruation.

- The menstrual cycle is regulated by hormones namely: progesterone and oestrogen.
- These hormones are produced by ovaries.
- Progesterone is responsible for maintaining the thick uterus lining in the cycle and during pregnancy.
- Oestrogen triggers the release of an egg by the ovaries and is also responsible for thickening of uterus lining.
- At the beginning of the cycle oestrogen levels increase and decrease once the egg is released. Progesterone levels increase as soon as the egg is released to maintain the uterus.
- If fertilisation does not occur, progesterone levels decrease and the uterus lining breaks resulting in menstruation.

Methods of contraception

- There are many methods of preventing unwanted pregnancy; these can be put into five groups namely: natural methods, barrier methods, hormonal methods, spermicides and permanent methods and chemical methods.
- The safest way of preventing pregnancy and transmission of sexually transmitted infections is abstinence.

Natural methods – These do not involve the use of hormones chemicals or devices, these may involve making use of knowledge of the menstrual cycle.

Examples of natural contraceptives

- **Rhythm methods** – involve abstaining from sexual intercourse during periods when the woman is likely to ovulate.
- **Withdrawal method** – involves the pulling out of the penis from the vagina before the man ejaculates.

Advantages of natural contraceptives

- Natural methods are cheap, easily available and are most suitable for those who have problems with artificial hormones and chemicals.

Disadvantages of natural contraceptives

- Natural methods are not always reliable.
- Menstrual cycles can be irregular and difficult to predict.
- Semen may also come out of the penis before the man ejaculates.

Barrier methods – these methods prevent the sperm from reaching the egg.

Examples of barrier contraceptives

- **Intrauterine device (IUD)** – it prevents implantation of the fertilized egg.
- **Condoms** – stop sperms from entering the vagina. Condoms also protect against sexually transmitted infections.
- **Diaphragm** – stop sperms from entering the uterus. The device can be left in the uterus for up to ten years.

Advantages of using barrier contraceptives

- With the exception of the IUD which needs to be inserted by a doctor, barrier methods are cheap.

- They are suitable to those who have problems with artificial hormones and chemicals.

Disadvantages of barrier contraceptives

- For barrier methods to be effective they have to be used correctly.
- The IUD is expensive

Hormonal methods – these involve the use progesterone and oestrogen hormones to prevent pregnancy by preventing ovulation. The hormones can be administered in form of injections, pills, implants and patches.

Examples of hormonal contraceptives

- **Contraceptive Injections** – the injection is administered every three months.
- **Contraceptive pills** – one pill is taken every day at the same time. In case of emergency there is the morning after pill which is taken within 72 hours after unprotected sex.
- **Contraceptive implants** – inserted under the skin by a medical practitioner and can last up to 5 years
- **Contraceptive patch** – it is worn on the skin for three weeks and taken off the fourth week to allow menstruation to occur.

Advantages of hormonal contraceptives

- They are effective and easy to use

Disadvantages of hormonal contraceptives

- Hormones could have side effects such as weight gain, headaches and nausea.
- They do not provide protection agains STIs.

Permanent methods – Sometimes calle sterilisation, is a permanent metho because it is difficulty to reverse.

Examples permanent contraceptives

- In females tubal ligation is done, involves cutting the oviducts during surgery. Sperms can no longer reach the egg, and the egg cannot travel to the uterus.
- In males vasectomy is done, it involves cutting of the sperm ducts. This prevents sperms from leaving the testes, as a result semen produced will not contain sperms.

Advantages of permanent contraceptives

- Has no side effects and is effective.

Disadvantages of permanent contraceptives

- Cannot be reversed.
- It is expensive.
- Does not protect against STIs.

Spermicides – These are chemicals that kill sperms, they are inserted into the vagina just before intercourse.

Advantages of spermicides

- They are easy to use, cheap and can be bought without a prescription.

Disadvantages of spermicides

- They do not protect against STIs.
- Some individuals might have allergic reactions to the chemicals.
- They are not very effective.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

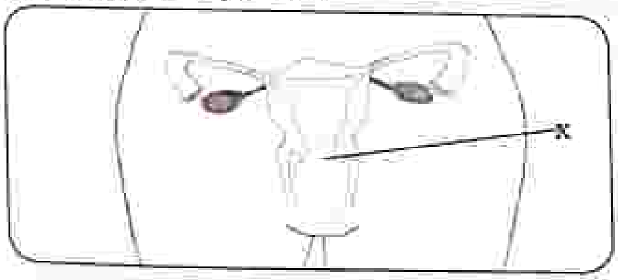
Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. In humans why are sperm cells produced in large numbers?
   A. more sperm cells are needed to fertilise the egg.
   B. sperm cells live for a short period.
   C. sperm cells are smaller in size.
   D. the chance of a sperm cell reaching the ovum is small.

2. The diagram shows a female's reproductive system.



What is the Name of the part labeled X?
A. ovary B. oviduct
C. vagina D. cervix

3. The male sex hormone is called.
   A. testosterone
   B. semen
   C. progesterone
   D. oestrogen

4. Where does fertilization occur in humans?
   A. vagina
   B. uterus
   C. oviduct
   D. ovary

5. The placenta does all of the following except:
   A. supply oxygen to the foetus
   B. supply nutrients to the foetus
   C. supply blood the foetus
   D. secretes hormones

6. The umbilical cord connects the foetus to the
   A. uterus
   B. placenta
   C. cervix
   D. amniotic fluid
7. Identify the combination that correctly describes what happens to the cervix and uterus walls during child birth.

	Uterus	Cervix
A.	Relaxes	Contracts
B.	Contracts and relaxes	Contracts
C.	Relaxes	Dilates
D.	Contracts and relaxes	Dilates

8. What day of a typical 28-day menstrual cycle is a woman likely to ovulate?
   A. 28
   B. 14
   C. 10
   D. 4
9. Which method of birth control is based on understanding a woman's menstrual cycle?
   A. rhythm method
   B. diaphragm
   C. condoms
   D. contraceptive pills
10. Most men are reluctant to have a vasectomy because of the ________
   A. permanency of the procedure
   B. major surgery involved
   C. fear of decreased sexual performance
   D. cost of the procedure

Section B: Structured questions

1. Name the gland that produces testosterone. [1]
2. Name the tube through which sperm are released through the penis. [1]
3. Explain how a sperm cell is adapted for its function. [2]
4. State where implantation occur in humans. [1]
5. Describe the role of the placenta in humans. [2]
6. Explain the importance of separating the mother's and the baby's blood supply during pregnancy. [2]
7. Describe the role of oestrogen and progesterone in controlling the menstrual cycle. [2]
8. Explain the advantages of using condoms in birth control. [2]
9. Explain how contraceptive pills prevent pregnancy. [2]
10. Draw and label the female reproductive system. [4]

CHAPTER 10

Diseases

Key concepts

- Hygiene
- Methods of waste disposal
- Causes, transmission, signs and symptoms of diseases
- Cure and control of pathogenic diseases
- STIs, Typhoid, Malaria, Ebola and, Cholera

Summary notes

Health

- It is state of physical, mental and social well-being.

Hygiene

- It is any practices or activities performed to keep things healthy and clean.
- Hygiene is important in disease prevention in communities.

Levels of hygiene

Table 10. 1 Levels of hygiene

Personal hygiene	Food hygiene	Community hygiene
• bathing regularly • washing hands after using the toilet and before handling food • cleaning homes and environments	• washing hands before handling food • eating food whilst hot • cooking food thoroughly • washing fruit and vegetables thoroughly with clean water • covering food at all times • using clean utensils	• proper refuse disposal methods • proper sewage disposal methods • protecting sources of drinking water

Methods of refuse disposal

- If not properly disposed waste becomes a breeding ground for pathogens and vectors.
- This might result in the spreading of pathogenic diseases.
- Methods of waste disposal include burning, burying and recycling.

Table 10.2: Methods of refuse disposal

Methods	Advantages	Disadvantages
Burning	• Cheap • Amount of land needed is reduced. • Prevents toxins from contaminating ground. • High temperatures will destroy pathogens.	• Pollute the air • Global warming
Burying	• Bad odour from decomposing waste will not pollute air. • Prevents disease carrying animals from breeding on waste.	• Waste may contaminate underground water sources. • Waste may be harmful to soil organisms. • A lot of labour is required. • A lot of space is needed.
Recycling	• Creates jobs • Saves raw materials. • No land and air pollution.	• May not be hygienic.

Diseases

Diseases can be put into four categories namely:

- Pathogenic disease
- Genetic diseases
- Deficiency diseases
- Physiological diseases.

Pathogenic Diseases

- These are caused by organisms called pathogens.
- There are different types of pathogens including; bacteria, viruses, protozoa, fungi and worms.
- Pathogens live on other living organisms called hosts.
- These diseases are transmitted by various methods.

Methods of disease transmission

Table10. 3: Methods of disease transmission

Methods of disease transmission	Examples of diseases
Air – tiny water droplets	TB, influenza, mumps, measles, pneumonia
Contaminated food and water	Cholera, typhoid, bilharzia
Contact (direct or indirect)	Ebola, STIs, chickenpox, leprosy, tetanus, measles
Vectors e.g. mosquitoes, tsetse flies	Malaria, sleeping sickness, bilharzia

Infectious diseases

Malaria

Cause

- Protozoan parasite *Plasmodium*

Transmission

- Vector: female anopheles mosquito

Signs and symptoms

- fever
- headache
- chills
- nausea
- vomiting
- muscle pain
- fatigue
- sweating.

Treatment

- Artemisinin-based combination therapy (ACT)

Malaria control

- Avoid mosquito bite by: using insecticide, mosquito treated mosquito nets, insect repellant, wearing long clothes that cover the whole body.
- Taking anti-malaria/ prophylactic drugs.
- Correct use of malaria medication; take full course.
- Control breeding of mosquito by: draining water logged areas, clearing all tall grass around the homes, applying oil on water surfaces.
- Using insecticides to kill adult mosquitos.

Life cycle of the plasmodium

The life cycle of the plasmodium parasite involves two organisms: humans and female anopheles mosquito. The plasmodium reproduces asexually inside the human body and sexually inside the mosquito.

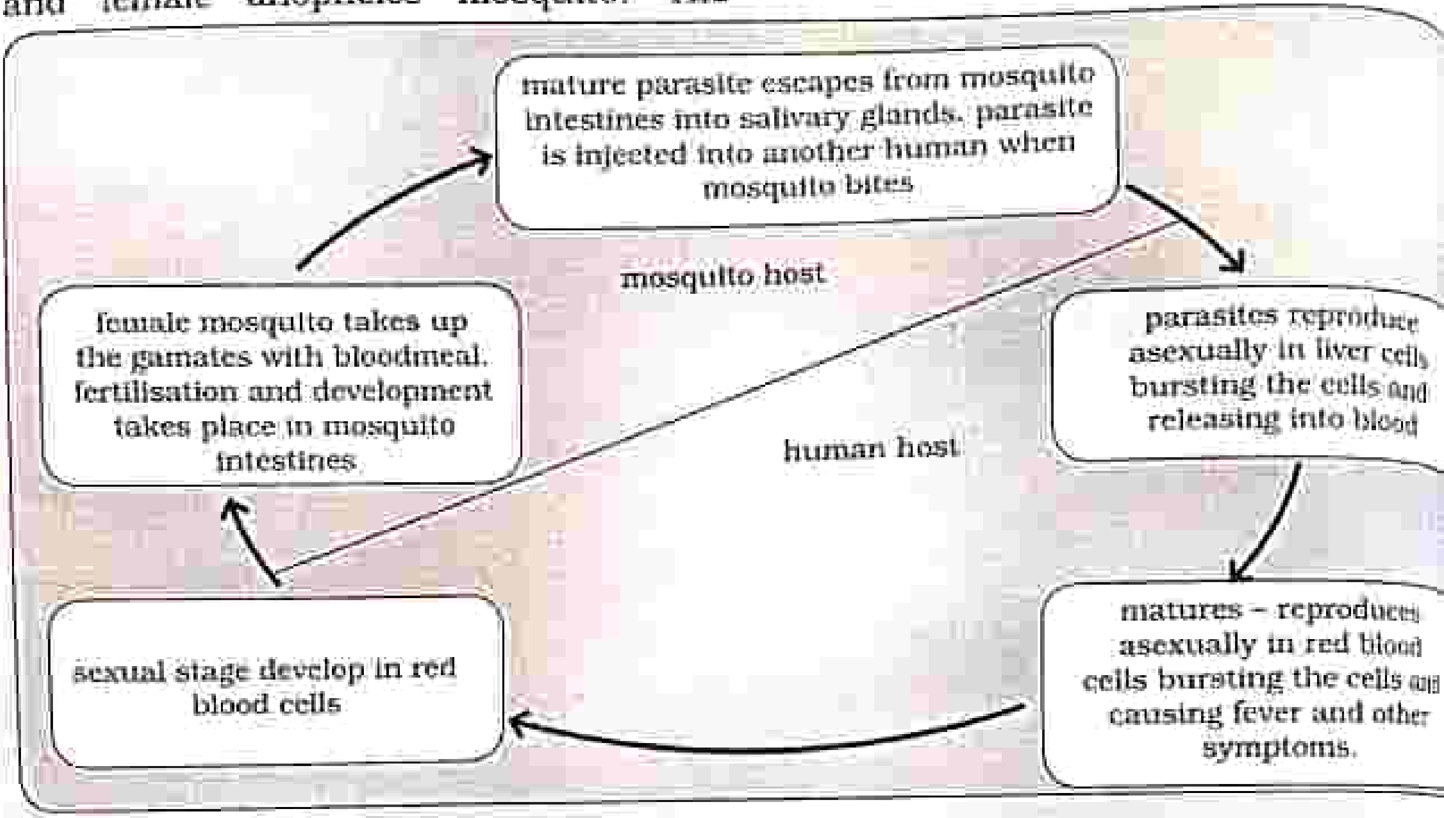


Figure 10.1: Life cycle of the malaria parasite

Life cycle of the anopheles mosquito

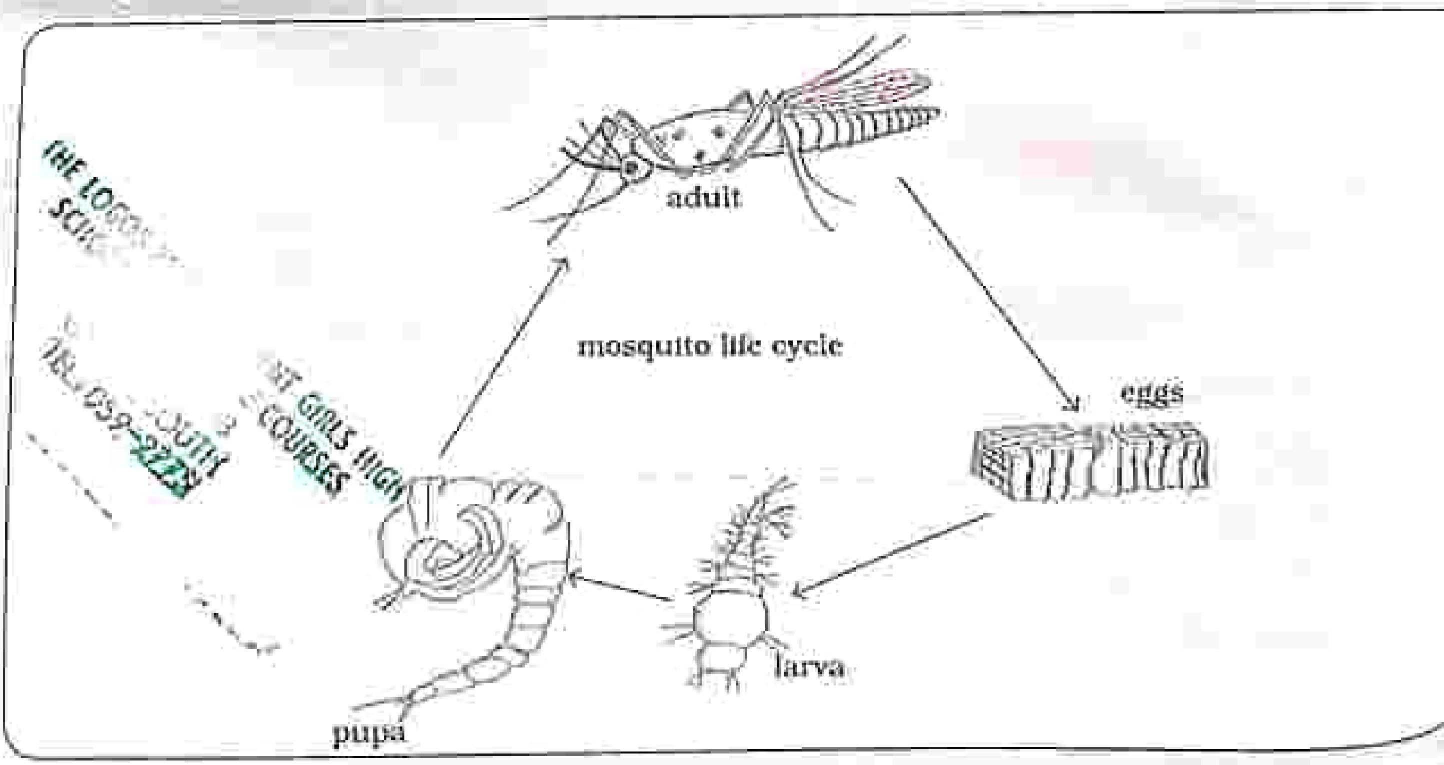


Fig. 10.2: Life cycle of the anopheles mosquito

Bilharzia (Schistosomiasis)

Cause

- Parasitic worms called schistosoma

Transmission

- Contact with fresh contaminated water

Signs and symptoms

- Abdominal pain
- Diarrhea
- Bloody stool
- Blood in urine
- Kidney failure

Treatment

- Taking in anti-parasitic medication

Control

- Taking prevention drugs.
- Take care when swimming, bathing in fresh water.
- Boiling water from unsafe sources.
- Control of the snails.

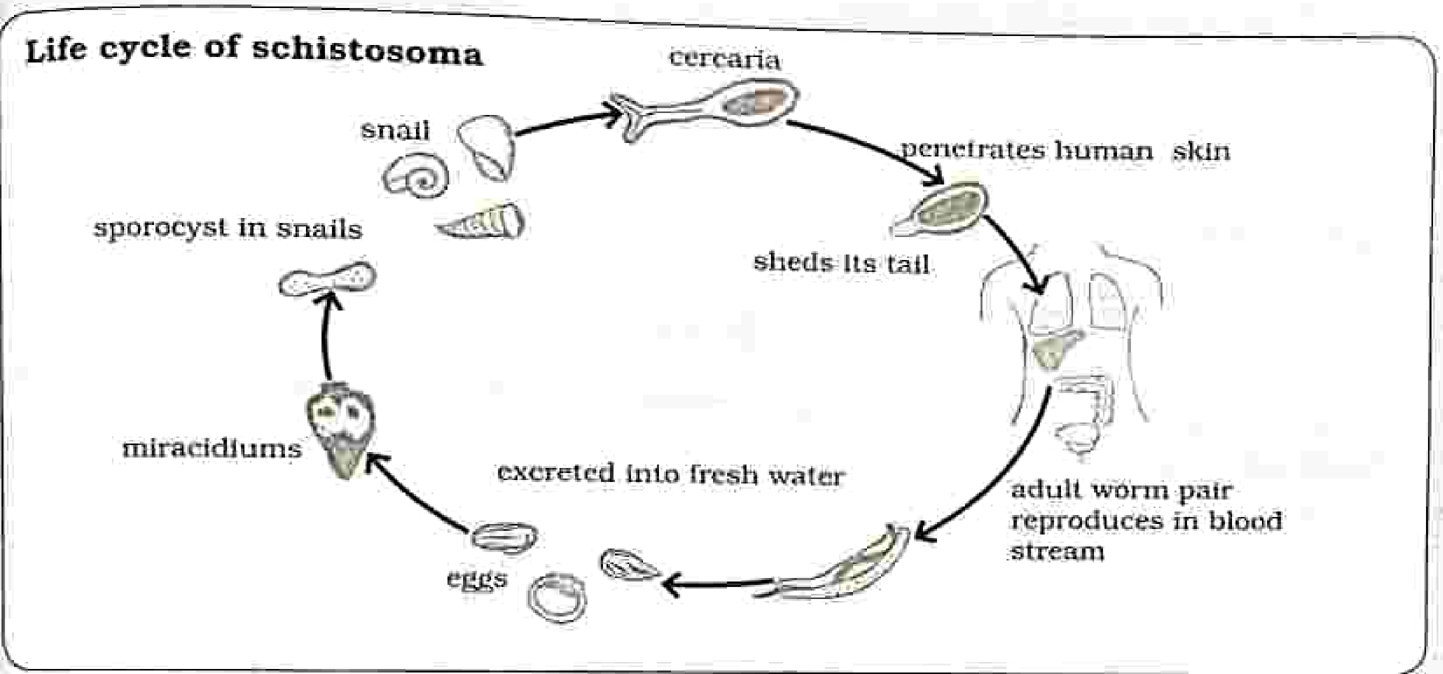


Fig. 10.3: Life cycle of schistosoma

Cholera

Cause

- Bacteria called *vibrio cholera*

Transmission

- Contaminated food and water

Signs and symptoms

- severe watery diarrhea
- vomiting
- dehydration
- fever
- sunken eyes

Treatment

- Taking antibiotics
- Drinking rehydration solution to replace lost fluids.

Control

- Washing hands after using the toilet and before handling food.
- Eating food whilst hot.

- Cooking food thoroughly.
- Washing fruit and vegetables thoroughly with clean water.
- Drinking clean safe water.

Ebola

Cause

Virus

Method of transmission

- Contact with an infected person or body fluid of an infected person.

Signs and symptoms

- fever
- weakness
- muscle and joint pains
- headache, lack of appetite
- diarrhea (sometimes containing blood)
- nausea, vomiting.
- Second stage bleeding from the nose, gums and skin
- bloody vomiting
- stools with blood
- difficulty swallowing

Treatment

- No cure, but there are treatments to relieve symptoms and prevent further complications and side effects.
- Taking in oral rehydration solutions.
- Blood transfusion.
- Control
- Avoid contact with body fluids from an infected person.
- Isolation of infected persons to prevent spreading of the disease.

Typhoid

Cause

- Bacteria

Transmission

- Eating contaminated food or drinking contaminated water.

Signs and symptoms

- diarrhoea
- constipation
- abdominal pain
- nausea
- decreased appetite
- headache, high temperature
- extreme tiredness
- confusion
- rash

Treatment

- Taking in antibiotics.
- Drinking oral rehydration solutions to replace lost body fluids.

Control

- Washing hands after using the toilet and before handling food.
- Eating food whilst hot.
- Cooking food thoroughly.
- Washing fruit and vegetables thoroughly with clean water.
- Drinking clean safe water.
- Vaccination

Sexually transmitted infections

- These are infections that are passed from one person to another through sexual contact.
- There are more than 20 types of

STIs, including gonorrhea, chancroid, syphilis, genital herpes and HIV/AIDS.

- HIV/AIDS can also be spread through sharing of sharp objects such as: needles and razor blades with an infected person and from mother to child during birth.

Table 10.4: Sexually transmitted infections

Disease	Cause	Signs and symptoms	Treatment
Chancroid	Bacteria	Painful ulcers on genitals, swollen lymph nodes, inflammation of the urethra, abnormal vaginal discharge	Taking in antibiotics
Genital herpes	Virus *herpes simplex*	Painful blistering sores on genital area or anus, fever, headache, muscle aches, itching or burning in genital area, pain when urinating	No cure, medicines can heal the sores
Gonorrhea	Bacteria	**In women**: yellowish discharge, vaginal bleeding between menstrual periods, pain during sex **In men**: cause urethritis, leading to pain when urinating, pain during sex, yellowish discharge Sterility in both men and women if left untreated	Antibiotics
HIV/AIDS	Virus	HIV causes no symptoms, some may get flu-like illness AIDS; weight loss, fevers and night sweats, headache, diarrhoea, mouth genital or anal sores, dry cough, rash, swollen lymph nodes, vaginal yeast infections If untreated may lead to death	No cure: anti-retroviral drugs help infected people live longer and healthier
Syphilis	Bacteria	**Primary stage**: painless ulcer called chancre which heals on its own **Secondary stage**: skin rash, fever, swollen lymph nodes, weight loss, headaches, tiredness arthritis. **Tertiary stage:** brain infection loss of sight, heart problems, kidney and liver problems and death.	Antibiotics

STIs prevention control methods

- Abstinence – it is the most effective way.
- Using condoms consistently and correctly.
- Being faithful to one uninfected partner.
- Avoid sexual contact with a new partner until you have both been tested for STIs.
- Early treatment.
- Not sharing sharp objects such as needles and razor blades.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Which of the following is not a component of health?
   A. social
   B. physical
   C. nutritional
   D. mental
2. A pathogen is a__________.
   A. disease
   B. disease transmitting organism
   C. disease causing microorganism
   D. disease causing animal
3. __________ is a droplet infection.
   A. Syphilis
   B. Ebola
   C. Cholera
   D. Common cold
4. Which of the following is transmitted by a vector?
   A. Ebola
   B. HIV/AIDS
   C. Malaria
   D. Tb
5. Another name for schistomiasis is
   A. malaria
   B. parasite
   C. pathogen
   D. bilharzia
6. What type of pathogen is HIV?
   A. virus
   B. fungus
   C. bacterium
   D. protozoa

7. Which list has a matching cause transmission and symptom?

	Cause	Transmission	Symptom
A	Bacteria	Contact	Yellowish discharge
B	Bacteria	Vector	Pain when urinating
C	Virus	Contact	Yellowish discharge
D	Virus	Vector	Pain when urinating

8. In cholera treatment, what is the goal of the oral rehydration therapy?
   A. slows down the heart rate
   B. replaces lost body fluids
   C. prevents transmission of vibrio cholerae
   D. kills vibrio cholerae
9. Syphilis is caused by a ________.
   A. worm B. virus
   C. protozoa D. bacteria
10. ________ is spread by contact.
    A. bilharzia
    B. chicken pox
    C. cholera
    D. malaria

Section B: Structured questions

1. Describe the ways in which pathogens can be transmitted. Give examples in your answer. [4]
2. Describe how cholera is spread. [2]
3. Explain why oral rehydration therapy is effective in the treatment of cholera? [1]
4. Describe how we can prevent the spread of bilharzia. [2]
5. List any two symptoms of gonorrhea in females. [2]
6. List three ways by which HIV can be transmitted. [3]
7. What are the most effective ways of stopping the spread of HIV/AIDS? [2]
8. What treatments are there for people suffering for Ebola? [2]
9. Give one method of preventing the spread of malaria. [1]
10. Draw a labelled diagram showing the life cycle of an anopheles mosquito. [4]

CHAPTER 11

Drugs and substance abuse

Key concepts

- Effects of tobacco smoking
- Effects of excessive alcohol consumption
- Effects of mandrax and cannabis use
- Effects of breathing solvents

Summary notes

Substance abuse

- It is when a drug is used for a purpose for which it is not intended.

Tobacco smoking

- Smoking cigarettes has many adverse effects on the body, which can lead to life threatening complications.
- Tobacco contains nicotine which is addictive and can cause withdrawal symptoms when someone stops using it, making it difficult for smokers to stop.
- Apart from nicotine tobacco also contains a variety of harmful chemicals such as carbon monoxide and tar.
- Smoking affects nearly every organ of the body including: the respiratory system, the circulatory system, reproductive system, the skin, the eyes and increase the chances of getting cancers.

Withdrawal symptoms

- Cravings
- Increased appetite
- irritability

Smoking and respiratory (lung) disease

- Smoking can damage air ways and alveoli in lungs, causing lung disease such as: emphysema chronic bronchitis and lung cancers.

Emphysema

- It is the destruction of air sacs in the lungs causing the person to be short of breath.

Bronchitis

It is the Permanent inflammation of the bronchi in the lungs.

Smoking and Heart disease

- Smokers are at a greater risk for diseases that affect the heart and blood vessels. It makes blood thicker and more likely to clot causing stroke. Smoking also causes coronary heart disease.

Smoking and cancer

- Smoking can cause cancer almost anywhere in your body including liver, kidney esophagus, pancreas, stomach, lungs, bladder, blood cervix, colon and rectum.

Smoking and fsertility problems and complications

In men

- Smoking reduces fertility by affecting sperms and making them less able to fertilise an egg and also increases the risk of birth defects.

In women

- Smoking makes it harder for a woman to become pregnant.
- It affects the health of the unborn child even after birth.
- It increases the risk of premature delivery, ectopic pregnancy, low birth weight, still birth.
- It may cause sudden infant death syndrome.

Unhealthy skin and hair as a result of smoking

- Smoking can cause premature baldness and makes hair grey earlier.
- It also makes skin look older.

Smoking and oral hygiene

- Smoking affects the health of teeth and gums.
- It discolors teeth and causes tooth loss in some cases.

Smoking and the immune system

- Smoking can cause weakened immune system by depressing antibodies that are supposed to fight infection.

Smoking and Diabetes

- Smoking can cause type 2 diabetes mellitus and makes control difficulty.

Smoking and Vision loss

- Smoking increases the risk of cataracts which results in poor vision.

Second hand smoking

- It is the inhalation of tobacco smoke, by non-smokers from smokers.

Effects of exposure to second hand smoke

- heart damage
- high blood pressure
- making asthma worse
- increasing risk of colds and ear infection

Excessive consumption of alcohol

Excessive consumption of alcohol can cause major health problems including liver cirrhosis, brain damage and cancer.

Alcohol and liver cirrhosis

- The liver filters all toxins from the blood and alcohol is toxic to liver cells.
- Cirrhosis is when the liver is heavily scared.
- Each time the liver gets hurt; it repairs itself and forms tough scar tissue.

Signs and symptoms of liver cirrhosis

- fatigue and weakness
- loss of appetite and loss of body weight
- nausea
- skin turns yellow
- intense itching
- blood vessels become visible on skin

Alcohol and brain damage

- Alcohol affects the brain, resulting in blurred vision, memory lapses, difficulty walking, and slowed reaction time.

Alcohol and cancer

- Excessive drinking increases the risk of cancers including: liver cancer, breast cancer, esophagus cancer.

Alcohol and cardiovascular disease

- Heavy drinking makes blood more likely to develop clots which can lead to heart attack and stroke.

Alcohol and the immune system

- Drinking too much alcohol weakens the immune system, resulting in the body being vulnerable to infectious diseases.

Mandrax and cannabis

- Cannabis is also known as, marijuana, ganja, hemp, weed, mbanje and pot.
- Mandrax and cannabis can have negative effects on users.

Effects of Mandrax and cannabis use

Short term effects

- An altered state of consciousness
- Increased pulse
- Impaired coordination, reaction time, and concentration
- Anxiety
- Hallucinations
- Increased appetite
- Paranoia

Long term effects

- Addiction
- Lung cancer
- Reproductive system disorder
- Impaired memory and learning abilities

Breathing solvents

- Some people inhale or sniff substances such as glues, paint, thinners and some cleaning agents.
- Effects of solvent abuse vary greatly depending on the type of product used and the amount taken.

Effects of breathing solvents

Short term effects

- Hallucinations
- Vomiting
- Dizziness
- Slurred speech
- Impaired judgment
- Unconsciousness

Long term effects

- Kidney, liver, heart and brain damage
- Damaged muscles
- Fertility problems
- Memory impairment
- Death from heart failure or suffocation

Social problems

- Disagreements that can affect personal relationships.
- Legal, financial and health problems
- Road accidents when people drive after the use of drugs.
- Accidents at work when workers operate machines under the influence of drugs or alcohol.
- Poor self-control leading to the spread of diseases such as HIV and AIDS.
- Unnecessary absenteeism from work.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question

Multiple choice questions

1. Which term describes the use of a drug for a purpose for which it is not intended?
   A. misuse
   B. abuse
   C. tolerance
   D. addiction
2. Excessive use of any drug constitutes ____________.
   A. drug abuse
   B. drug misuse
   C. drug addiction
   D. drug tolerance
3. Temporary symptoms that occur when the use of an addictive substance is discontinued is known as _____.
   A. withdrawal
   B. cravings
   C. a disease
   D. addiction
4. Which part of the body is most affected by smoking?
   A. lungs
   B. heart
   C. kidney
   D. brain
5. Nicotine in cigarettes causes ________.
   A. cravings
   B. headaches
   C. hallucinations
   D. increased appetite
6. Which of the following is an effect of second hand smoking?
   A. addiction
   B. heart damage
   C. unconsciousness
   D. vomiting
7. Which drug causes liver damage if consumed in excess?
   A. mandrax
   B. cannabis
   C. cigarettes
   D. alcohol
8. How are solvents abused?
   A. smoking
   B. sniffing
   C. spraying directly into nose
   D. drinking

9. Hallucinations result from _____ abuse?
   A. alcohol
   B. cigarette smoking
   C. mandrax and cannabis
   D. solvents

10. Chose the combination that co[rrectly] shows the type of drug abuse a[nd] effect?

	Type of drug abuse	Effect
A	Alcohol	Addiction
B	Cigarette smoking	Liver cirrhosis
C	Mandrax and cannabis	Unconsciousness
D	Solvents	Hallucinations

Section B: Structured questions

1. What causes people to become addicted to cigarettes? [1]
2. Name any three diseases that are caused by smoking. [3]
3. Describe the effects of nicotine on the body. [2]
4. Explain the effects of second hand smoking. [3]
5. Explain how excessive consumption of alcohol lead to a heart attack. [2]
6. State any two symptoms of liver cirrhosis. [2]
7. What are the dangers of using Mandrax and cannabis? [3]
8. Outline the social implications of drug abuse. [3]

CHAPTER 12

Immunity

Key concepts

- Natural immunity
- Acquired immunity
- Breast feeding
- Immunization

Summary notes

Immunity

- It is the ability of a body to protect itself from all types of pathogens.
- There are two major types of immunity namely: natural immunity and acquired immunity.

Natural immunity

- It is that which an individual is born with and is genetic.
- It protects the body throughout life.
- It is the body's ability to fight infections.

Natural immunity involves: white blood cells (leucocytes), the skin, mucus, acids in stomach, saliva, and vaginal secretions.

Table 12. 1: Types of natural immunity

Skin	Its outer tough layer prevents pathogens from entering the body.
Mucus	Traps microorganisms and dust particles.
Stomach acid	Kills most ingested pathogens.
Vaginal secretions	Acidic and help destroy pathogens.
White blood cells Two types: lymphocytes and phagocytes	Destroy pathogens in two ways: • They engulf pathogens (phagocytes). • Produce chemicals called antibodies that destroy pathogens by natural means (lymphocytes).
Tears	Contain antibodies that destroy pathogens.

Acquired immunity

- It refers to the immunity that one develops later in life.
- Acquired immunity is of two different types namely active immunity and passive immunity.

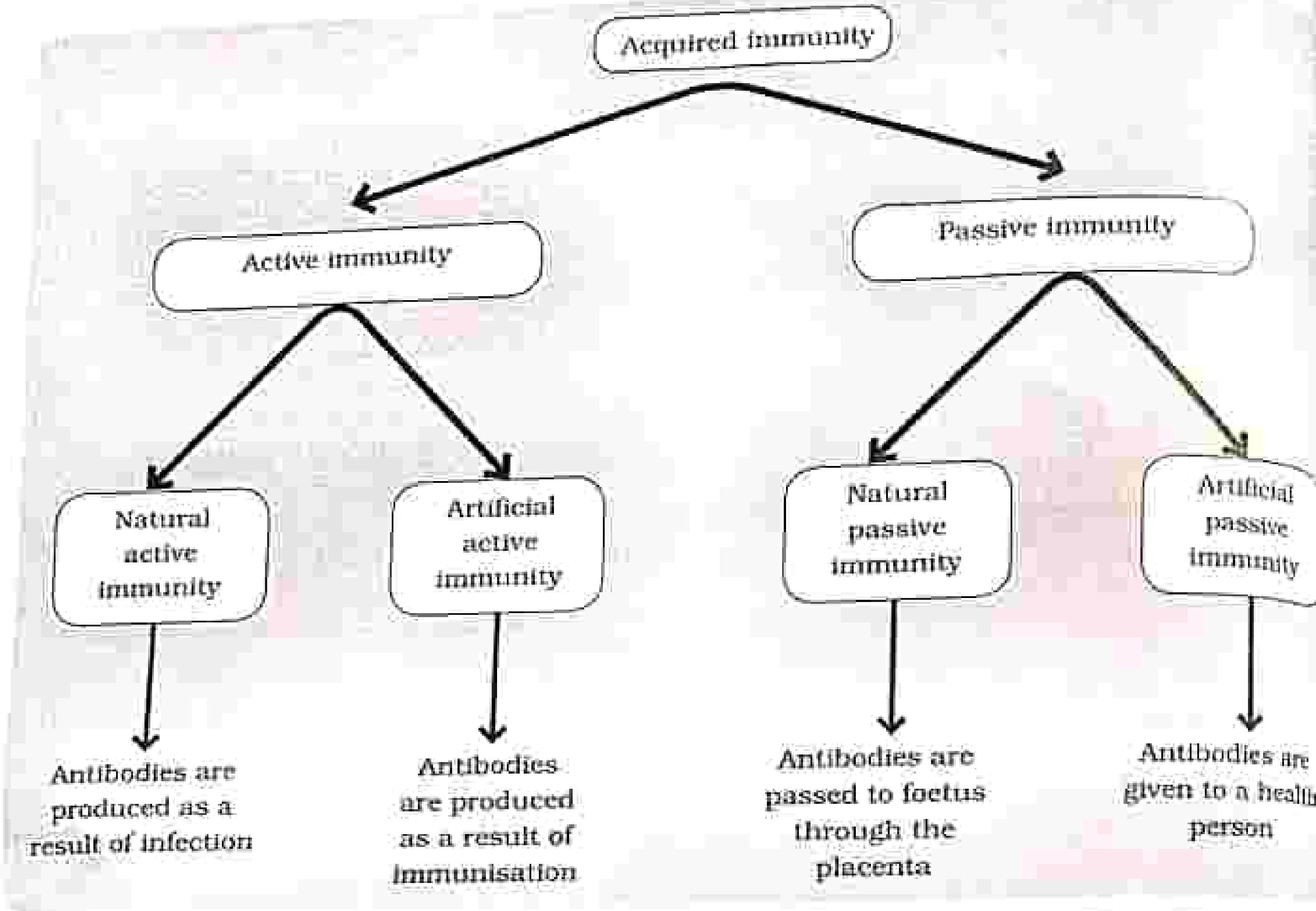


Fig. 12. 1: Types of acquired immunity

Active immunity

- Active immunity is when the body produces antibodies in response to infection.
- The body then develops the ability to fight the pathogens permanently.
- It is slow and takes time in the formation of antibodies.
- It may be natural or artificial.

Natural active immunity

- This is when an infection triggers the production of antibodies to fight the infection.
- A person who recovered from measles develops natural active immunity.

Artificial active immunity

- This is when a person is deliberat exposed to weakened pathogens t cause a particular disease.
- This is done through immunization. response the body produces antibod to destroy the pathogens, which t destroy the pathogens in future.

Passive immunity

- Passive immunity is when rea made antibodies are given to a pers to protect against certain infectio instead of the body producing them.
- This type of immunity provi immediate relief and is not long lasti

- Passive immunity may be natural or artificial.
- Natural passive immunity is when antibodies are transferred from the mother to the foetus through the placenta during pregnancy.
- After birth antibodies are passed through breast milk. This type of immunity is not long lasting.

Artificial passive immunity

- This is when ready-made antibodies are injected into a person who has been exposed to a disease to help fight the infection.
- For example, exposure to tetanus requires one to be injected with antibodies that fight the disease.

Infants and immunity

- Babies get their immunity from their mothers before and after birth.
- This is so because their immune systems will be weak and not fully developed.
- During pregnancy, the foetus gets antibodies from the mother through the placenta.
- After birth the baby gets immunity from their mothers through breast milk.

Breast feeding

- A mother transfers white blood cells and antibodies that fight certain infections to the baby through breast milk.
- This immunity might protect the baby even after breast feeding has stopped.
- This is passive immunity.

Immunization/vaccination

- This is when a weakened pathogen is deliberately injected into a baby.
- In response, the baby produces antibodies, thereby developing active immunity.
- This immunity lasts long and helps fight diseases in future.

In Zimbabwe babies are immunized against a number of killer diseases so as to reduce infant mortality rate.

Table12.2: Vaccination schedule in Zimbabwe

	Age	Vaccinations
Primary course	Birth/ first contact	BCG
	6 weeks	OPV 1, Pentavalent 1 Rotavirus 1 Pneumococcal 1
	10 weeks	Pentavalent 2 OPV 2 Pneumococcal 2 Rotavirus 2
	14 weeks	Pentavalent 3 OPV 3 Pneumococcal 3 IPV
	9 months	Measles rubella 1

	Age	Vaccinations
Boosters	18 months	OPV DPT Measles rubella 2

- BCG – is a vaccine against tuberculosis.
- OPV – oral polio vaccine
- IPV – injected polio vaccine
- Pentavalent is a 5 in 1 vaccine diphtheria, tetanus, whooping cou hepatitis B and Hemophilus influe type B.
- Rotavirus is vaccine for diarrh diseases.
- Pneumococcal is a vaccine f pneumonia and meningitis.
- Measles rubella is a vaccine agai measles and rubella (Germ measles).

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question

Multiple choice question

1. The body combats infection by producing _____.
   A. antibiotics
   B. antibodies
   C. antigens
   D. antiseptics

2. What is the immune system's first line of defence against invading pathogens?
   A. ingestion of pathogens by lymphocytes
   B. ingestion of pathogens byphagocytes
   C. production of antibodies
   D. production of antigens

3. Which examples show different types of immunity?

	Active artificial	Passive natural
A	Immunity to measles after infection	Receiving antibodies to measles from breast milk
B	Immunity to measles after infection	Receiving antibodies to measles by injection
C	Immunity to small pox after vaccination	Receiving antibodies to measles in breast milk
D	Immunity to small pox after vaccination	Receiving antibodies to tetanus by injection

4. The production of antibodies in response to vaccination is an example of which type of immunity?
   - A. natural active
   - B. natural passive
   - C. artificial active
   - D. artificial passive
5. Two types of immunity in humans are ____________.
   - A. natural and acquired
   - B. active and acquired
   - C. natural and passive
   - D. passive and acquired
6. What describes natural passive immunity?
   - A. production of antibodies as a result of pathogen invasion.
   - B. production of antibodies as a result of vaccination.
   - C. injection of antibodies.
   - D. drinking breast milk containing antibodies.
7. Which type of immunity occ... following infection by a pathogen?
   - A. natural active
   - B. natural passive
   - C. artificial active
   - D. artificial passive
8. The physical barriers that form part ... immunity are ____________.
   - A. the brains and the mucus
   - B. the skin and hair in the nose
   - C. the skin and stomach acid
   - D. tears and stomach acid
9. Which vaccine is referred to as th... 5 in 1 vaccine?
   - A. rotavirus
   - B. pentavalent
   - C. pneumococcal
   - D. measles rubella
10. BCG is a vaccine that protects again... ____________.
   - A. tuberculosis
   - B. tetanus
   - C. polio
   - D. measles

Section B: Structured questions

1. Distinguish between natural immunity and acquired immunity. [2]
2. Briefly explain the difference between active and passive immunity. [2]
3. Explain how vaccines stimulate immunity. [2]
4. Give any three ways in which the body prevents pathogens from entering the blood. [3]
5. Name the two types of leucocytes. [2]
6. What is an antibody? [1]

7. Lymphocytes bring about immunity by producing antibodies.
   a) Name this type of immunity. [1]
   b) Besides lymphocytes, another type of white blood cell destroys pathogens. Name this type of white blood cell and describe how it destroys pathogens. [2]
8. Name any three body systems that are responsible for maintaining your health. [3]
9. Suggest two advantages of breast feeding. [2]
10. Explain how the stomach is adapted to reduce the entry of pathogens into blood. [2]

CHAPTER 13 Chemistry

Key concepts

- Paper chromatography
- Distillation of impure water
- Fractional distillation of dilute ethanol and air

Summary notes

Methods of separation

- In a mixture individual substances can be separated using different methods.
- The method used depends on the type of mixture.
- These methods include decanting, evaporation, filtration, magnetism, sieving, distillation and chromatography.

Table13.1: Simple methods of separating mixtures

Method	Description	Application
Decanting	Separating a mixture of two or more liquids having different densities, e.g. water and oil. The less dense liquid floats on top of a more dense liquid. Separation is by pouring off the less dense liquid.	Separate wine from sediments. Some wines give off sediments as they age.
Evaporation	Separating a mixture of a solute and solvent, e.g. salt and water. Heat will evaporate the solvent leaving the solute behind.	Extraction of sugar from sugar cane Extraction of salt from sea water
Filtration	Separating a mixture of an insoluble solid and a liquid, e.g. sand and water. The liquids pass through the filter paper and the solid remains behind.	Water treatment plants
Magnetism	Separating a mixture of substances in which one of the substances is a magnetic material e.g. iron fillings and sulfur powder.	Separating metal objects from grains at grinding mills. Separating metal objects from nonmetal objects at metal scrap plants.

Table13.1: Simple methods of separating mixtures

Method	Description	Application
Sieving	Separating a mixture of fine and coarse solids e.g. fine particles pass through the sieve and the coarse particles remain behind.	Separating particles of different sizes especially in the food industry.
Winnowing	Separating a mixture of lighter particles from heavier particles e.g. maize and chuff. The mixture is thrown into air and the chuff will be blown away by wind.	Separating grain from chuff after harvesting

Simple distillation

- It is a method of separating a solvent from a solution for example water can be separated from a salt solution.
- This method works because water has a much lower boiling point than the solutes.

Distillation of impure water

- When the solution is heated, water evaporates.
- Water vapour is then cooled and condensed and collected in a separate container.
- The solutes stay behind and it enables the collection of both solute and solvent.

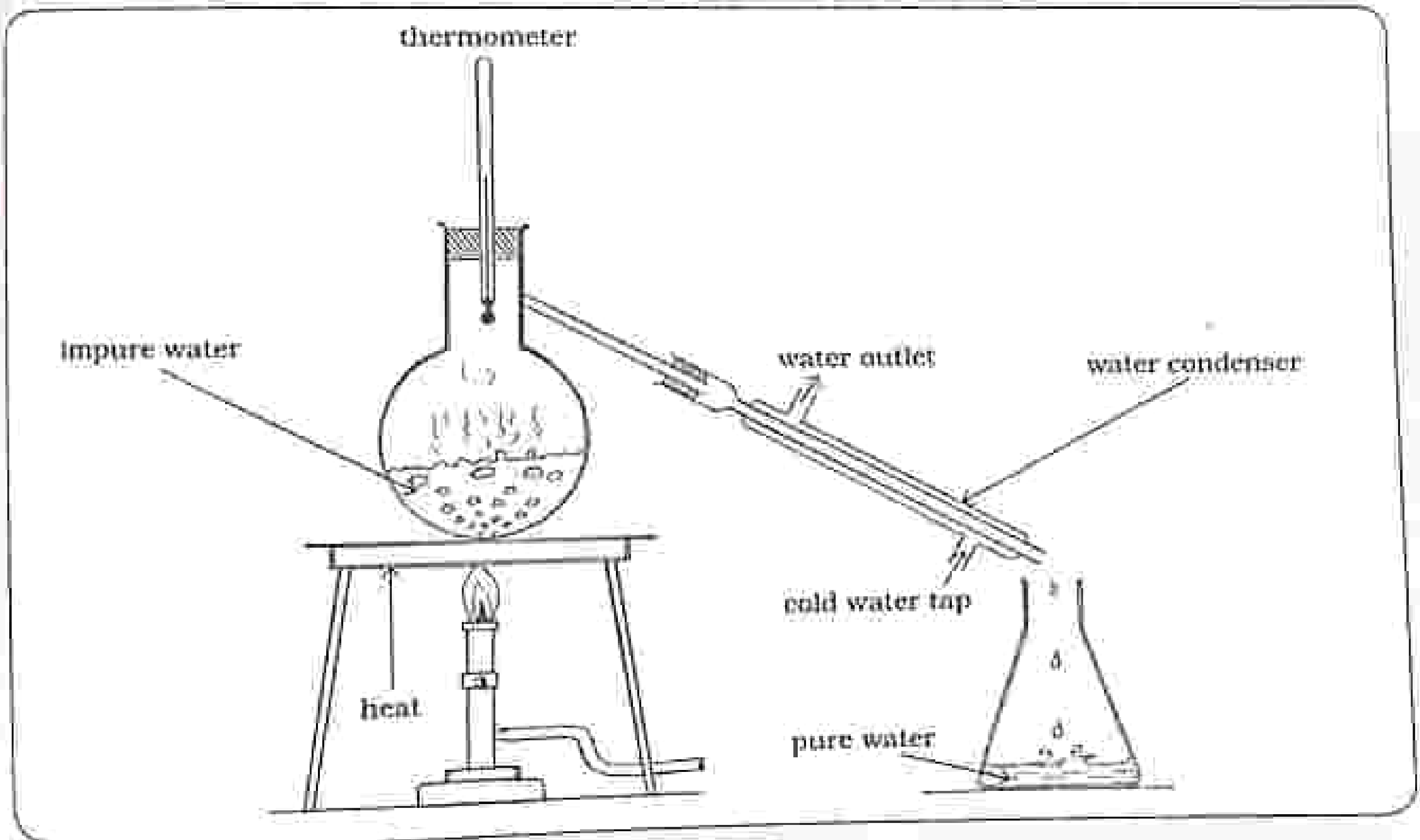


Fig. 13.1: Simple distillation

Fractional distillation

- It is a method of separating a liquid from a mixture of two or more liquids.
- This method uses the fact that liquids have different boiling points, when heated one liquid boils before the other.
- For example gases in liquid air can be separated using fractional distillation.

Fractional distillation of dilute ethanol

- Ethanol and water can be separated by fractional distillation. Ethanol boils at 78°C nd water boils at 100°C.
- When dilute ethanol is heated, ethanol vaporises before water does due to its lower boiling point.
- However the vapour rising will still contain some molecules of water.
- Vapour gets purer as it rises in the column, as heavier molecules (water) will condense back into liquid and fall back down.

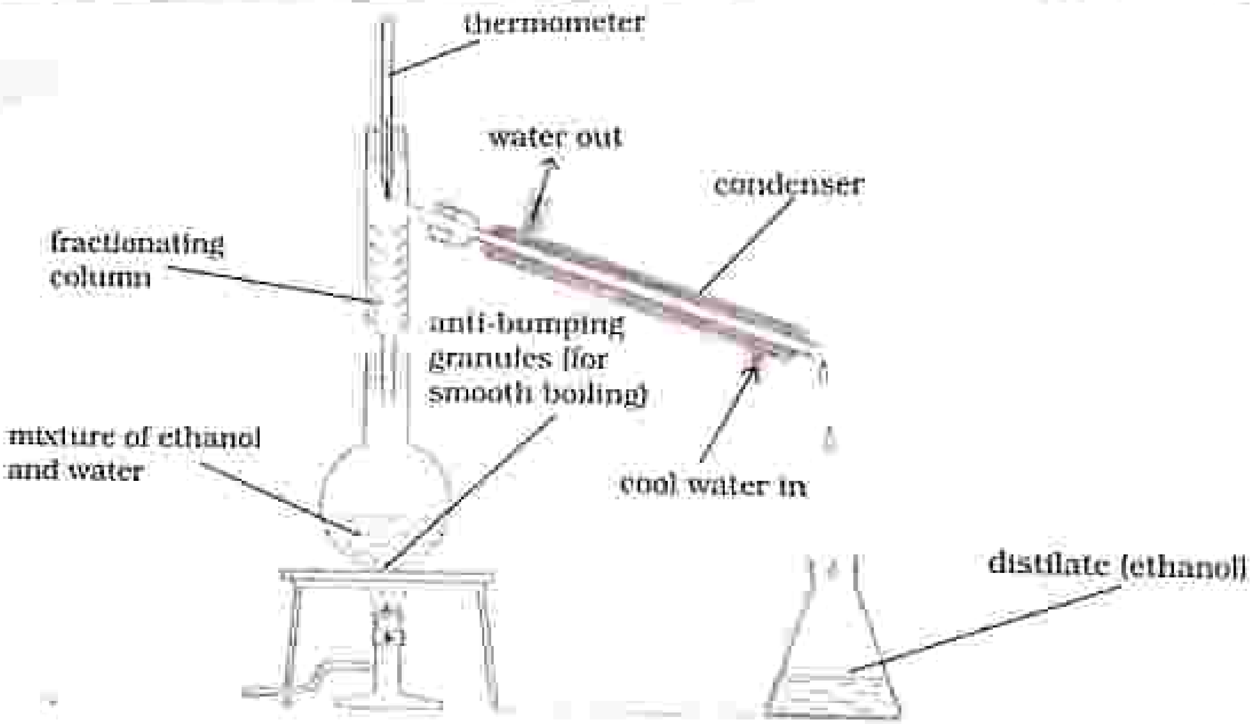


Fig. 13.2: Fractional distillation of dilute ethanol

Chromatography

- It is a method used to separate dissolved substances, especially when the dissolved substances are colored such as ink and plant dyes.
- This method is successful because some of the solutes dissolve in the solvent used better than others and have different molecule size.
- The technique uses two types of substances:
- **Mobile phase** – gas or liquid that transports the solution being tested through the other substance, example water and alcohol.
- **Stationary phase** – liquid or solid through which the substance being tested is carried, for example paper.

The stationery phase will absorb different components of the solution at different rates, creating layers as the components of solution are separated.

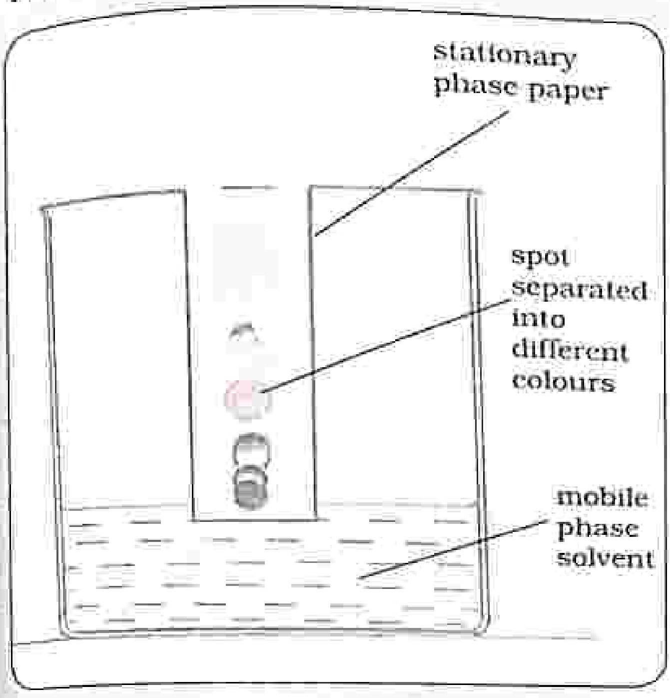


Fig. 13.3: Paper chromatography

Applications of chromatography

- Chromatography plays an important role in pharmaceutical, chemical and food industry.
- It is used for quality analysis.

Food industries

- It is used to separate and analyse additives, vitamins, preservatives.

Pharmaceuticals

- It is used to determine the amount of each chemical in a new product.

Hospitals

- It is used to detect blood or alcohol levels in patients' blood stream.

Manufacturing plants

- It is used to purify chemicals.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question

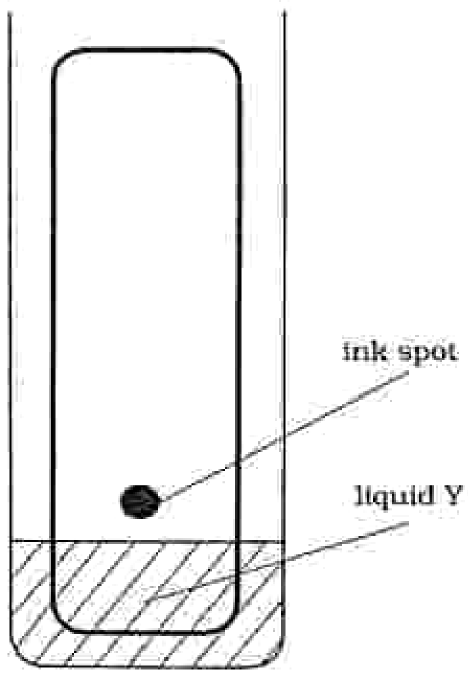
Multiple choice questions

1. What name is given to a change that occurs without changing the chemical makeup?
   A. chemical change
   B. colour change
   C. physical change
   D. substance change
2. Which method is best for separating sodium chloride from a solution of sodium chloride and water?
   A. chromatography
   B. evaporation
   C. filtration
   D. fractional distillation
3. Oil and water may be separated using
   A. chromatography
   B. distillation
   C. decanting
   D. winnowing
4. Filtrate refers to
   A. insoluble solid in the filtrate
   B. solution which dissolves
   C. solution that passes through the funnel
   D. the container in which the solution is collected
5. In a fractionating column of a fractional distillation, higher in the column
   A. the temperature becomes lower
   B. the temperature becomes higher
   C. minimum absorption occurs
   D. risk of sublimation exists
6. Water and alcohol are separated by distillation because they have different
   A. melting points.
   B. boiling points.
   C. densities.
   D. solubility.
7. A technique in which two liquids are separated by heating is known as
   A. chromatography.
   B. distillation.
   C. evaporation.
   D. filtration.

8. Chromatography is generally used by ______
   A. nurses.
   B. magicians.
   C. doctors.
   D. chemists.
9. Which one of the following is a disadvantage of evaporation?
   A. all of the solute is recovered
   B. the solvent is not recovered
   C. It always requires heat
   D. Can only be used for soluble solids
10. Which process can be used to separate oxygen from air?
   A. chromatography
   B. simple distillation
   C. fractional distillation
   D. evaporation

Section B: Structured questions

1. Explain the type of mixtures that can be separated using magnetism. [2]
2. Describe and explain the process of fractional distillation. [4]
3. Describe with the aid of diagrams, how you would separate a mixture of sand and sugar. [5]
4. The diagram shows the separation of dyes in the ink sample



   a) What name is given to this type of separation? [1]
   b) Identify a liquid Y that can be used in this experiment. [1]
   c) What would you expect to notice on the piece of chromatography paper after some time? [1]
   d) Why must the ink spot be placed above the solvent? [1]
5. Name the separation technique that can be used to separate water and alcohol. [1]
6. Distinguish between simple distillation and fractional distillation. [2]
7. Explain any two applications of evaporation. [2]

CHAPTER 14

Atomic structure and the periodic table

Key concepts

- Structure of the atom, nuclide notation and isotopes
- The periodic table
- The mole, Avogadro number and molecular mass
- Empirical and molecular formula
- Concentration of solutions

Summary notes

The atom

- It is the smallest unit of an element that exists. It consists of a nucleus which contains protons and neutrons.
- The nucleus is surrounded by shells which contain electrons.

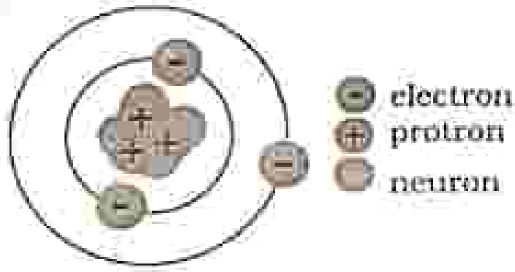


Fig. 14. 1: Atomic structure

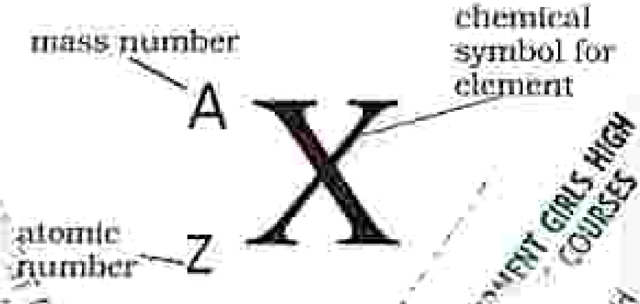
Subatomic particles

Table 14. 1: Relative mass, charge and position of subatomic particles

Subatomic particle	Relative mass	Relative charge	Relative position
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	1/1840	-1	Shells

Nuclide notation

Standard nuclide notation shows the chemical symbol of the element, the mass number and the atomic number



Example

Nuclide notation for nitrogen is:

$^{7}_{14}N$

From the nuclide notation:

- ***N*** is the chemical symbol for nitrogen.
- Atomic number is **7** which is equal to the number of protons.
- Mass number is **14** which is the sum of protons and neutrons.
- Number of neutrons is 14 - 7 = 7.

Mass number

- It is also known as nucleon number.
- It is the sum of protons and neutrons.

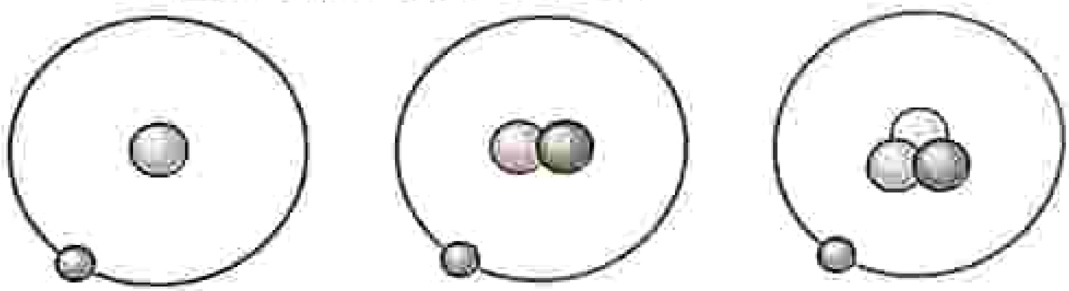
Atomic number

- It is also known as proton number since it gives the number of protons in the atom. In a neutral atom the number of protons is equal to the number of electrons.
- It defines the element's chemical properties since the number of electrons determines the type of bonding.
- It gives the position in the periodic table.

Isotopes

- These are atoms of an element with same number of protons but with different number of neutrons.
- Isotopes have different mass numbers.

Example: Isotopes of hydrogen



Protium (^1_1H) Deuterium (^2_1H) Tritium (^3_1H)

Periodic table

- The periodic table contains information of all elements, such as proton number, mass number, number of shells and the chemical symbol for the element.
- Elements are organised according to their proton number from left to right in an ascending order.
- The periodic table has rows and columns.
- Rows are called periods, the period number corresponds to the number of shells elements in that period have. Elements in period 1 (the first row) have only one shell.
- Columns are called groups, the group number corresponds to the number of electrons in the outer most shell called valance electrons. Elements in group 1 have one electron in the outer most shell.

Bonding

- Matter is made up of different atoms combined to form new compounds.
- In chemical bonding, only valence electrons of an element are involved.
- Many elements follow the octet rule in chemical bonding, which means that an element should have eight electrons in its valence shell. Having eight electrons in the outer most shell ensures stability.

Ionic bonding

- It occurs between a metal and a nonmetal.
- It is a process in which one atom donates electrons and the other atom accepts the electrons.
- Elements on the left side of the periodic table (metals) have few valence electrons, so in order to complete the octet rule, they lose or donate the valence electrons to other elements.
- On the other hand, nonmetals have valence shells that are almost full, so they tend to accept electrons from other elements.
- Those atoms that lose electrons acquire a net positive charge and are called cations.
- Those that have gained electrons acquire a net negative charge and are called anions.

Example

- In the bonding of sodium and chlorine, sodium has one valence electron while chlorine has seven. Sodium will donate its valence electron and becomes a sodium ion Na^+, on the other hand chlorine will accept the electron and become an ion Cl^-.
- The ions will then attract each other and sodium chloride NaCl is formed.

Covalent bonding

- It is a process of sharing valence electrons between two atoms. The bonds are typically between a nonmetal and a nonmetal.
- There are three types of covalent bonds: single bond, double bond and triple bond.

Single bond

- Composed of two bonded electrons.

Example

- In hydrogen gas two atoms share two electrons resulting in a single bond.

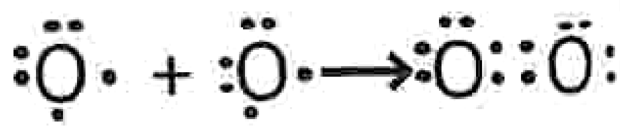
$$H\bullet + H\bullet \longrightarrow H:H$$

Double bond

- It is composed of 4 bonded electrons.

Example

- In oxygen gas two atoms share four electrons.



Triple bond

- It is composed of 6 electrons bond.

Example

- In nitrogen gas; two atoms share six electrons.

Relative atomic mass

- The total mass of protons and neutrons is called relative atomic mass (A_r). It is found on the periodic table.

Example

- Relative atomic mass of oxygen = 16.
- Relative atomic mas of carbon = 12.

Relative molecular mass (M_r)

- When atoms are bonded we get a molecule and the molecule has mass just like atoms.
- We get the relative molecular mass by adding the relative atomic masses of atoms in the molecule.

Example

- Find the relative molecular mass of CO_2.

Solution

- The relative atomic mass (A_r) of C is 12.
- The relative atomic mass (A_r) of O is 16.
- The relative molecular mass of CO_2 = 12 +16 +16 = 44.

The mole and the Avogadro's constant

- The mole is a unit for the amount of substance.
- One mole of a substance contains 6.02×10^{23} particles (atoms, molecules, ions or electrons). This number is known as the **Avogadro's number** or **Avogadro's constant**.

Mole and the molecular mass

- 1 mole of a substance has a mass in grams that is equal to its relative atomic mass or relative molecular mass and is called the molar mass.
- The unit for molar mass is g/mol.

Molar mass for elements

- Molar mass of an element is found on the periodic table, since it is numerically equal to the relative atomic mass.
- One mole of oxygen has a molar mass of 16g/mol.

Molar mass for compounds

- One mole of a molecule has a mass that is equal to the relative molecular mass.
- One mole of H_2O has a molar mass of 1 + 1 + 16 = 18g/mol.

Calculating the number of moles

The equation

n = number of moles

m = mass in grams

M = mass of one mole/molar mass

Example

- How many moles are there in 0.1g of hydrogen gas (H_2)?

Solution

- The molar mass of H_2 = 2 = 0.05 moles
- To find the number of hydrogen gas molecules we multiply the number of moles by the Avogadro's number.

- The number of molecules in 0.1g of hydrogen gas is
  $0.05 \times 6.02 \times 10^{23} = 0.30x^{23}$ molecules.

Empirical formula

- It shows the simplest ratio of elements in a molecular compound.
- Sometime empirical and molecular formula are the same.

Example

- For water, the molecular formula is the same as the empirical formula.
- The molecular formula for water is H_2O.
- The empirical formula for water is H_2O.
- For hydrogen peroxide, the molecular formula and the empirical formula are different
- The molecular formula is H_2O_2
- The empirical formula is HO since the ratio is 1:1.

How to determine the empirical formula

- Step1 – Convert the percentages of each element to mass by assuming we have a 100g sample.
- Step 2 – convert the mass of each element into to moles using the molar mass of each element.
- Step 3 – divide the smallest number of moles calculated into the other numbers.
- Step 4 – round off the mole ratios to the nearest whole number.
- If the numbers are too far from whole numbers, multiply each value by the same factor.

Example

A compound contains 90.0% carbon and 10.0% hydrogen. Find it's empirical formula?

Solution

- Number of moles for carbon
  $= \frac{90.0}{12.0} = 7.5$ mol.
- Number of moles for hydrogen
  $= \frac{10.0}{1} = 10.0$ mol.
- $\frac{7.5}{7.5} = 1.0$ H to C ratio $\frac{10}{7.5} = 1.33$
- 1.33 is not a whole number we multiply each ratio by 3.
- Hydrogen ratio $= 1.33 \times 3 = 4$ carbon ratio $= 1.0 \times 3 = 3$
- The empirical formula $= C_3 H_4$.

Concentration of solutions

- It is a measure of the amount of solute in a solution. The more concentrate the solution, the more the solut dissolves.
- Concentration can be calculated usin
  - Mass or number of moles of solu dissolved
  - The volume of solution
  - Concentration $= \frac{\text{mass of solute in g}}{\text{volume in dm}^3}$
    $= \frac{\text{number of moles}}{\text{volume in dm}^3}$
  - Volume must be in dm^3.
    $1\ dm^3 = 1\ 000\ cm^3$

Example 1

- g of hydrogen chloride is dissol in $100cm^3$ of water. Calculate concentration.
- $\frac{100cm^3}{1000} = 0.10\ dm^3$
- $C = \frac{m}{v} = \frac{1.0}{0.10} = 10g/dm^3$

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice

1. A proton has approximately the same mass as
   A. a neutron.
   B. an alpha particle.
   C. a beta particle.
   D. an electron.
2. Atomic mass of an element is equal to
   A. mass of protons.
   B. number of protons.
   C. the sum of mass of protons and neutrons.
   D. the sum of mass of protons and electrons.
3. All isotopes of a given element have the same ______________
   A. atomic number.
   B. atomic mass.
   C. neutron number.
   D. protons and neutrons.
4. Bromide ion (Br^-) have same configuration as ________.
   A. chlorine B. argon
   C. nitrogen D. helium
5. Relative molecular mass of H_2O is
   A. 8g. B. 16g.
   C. 18g. D. 34g.
6. How many moles are present in 12g of carbon?
   A. 1 mole
   B. 6.02×10^{23} moles
   C. 12 moles
   D. 24 moles
7. How many atoms are in 5 moles of hydrogen?
   A. 6.02×10^{23} atoms
   B. 6.02×10^{22} atoms
   C. 3.01×10^{23} atoms
   D. 3.01×10^{22} atoms
8. The number of molecules present in one mole of any substance is called
   A. Avogadro number.
   B. atomic number.
   C. mass number.
   D. Proton number.

9. The formula that shows the simplest ratio of atoms in a compound is the
   A. molecular formula.
   B. chemical formula.
   C. compound formula.
   D. empirical formula.

10. What name is given to a solid that completely dissolves to give a solution?
   A. solute B. solvent
   C. salt D. precipitate

Section B: Structured questions

1. An atom of magnesium has the symbol $^{24}_{12}Mg$
   a) State the number of protons, electrons and neutrons in this atom. [3]
   b) Why is magnesium positioned in group 2 of the periodic table? [1]
2. Chlorine is a halogen.
   a) In which group is chlorine on the periodic table. [1]
   b) State any three uses of chlorine. [3]
3. Explain why group VIII elements have an unreactive nature. [1]
4. State the three types of covalent bonds [3]
5. Choose an element from group 2 and draw its electronic configuration. Then draw its electronic configuration once it becomes an ion. [4]
6. What is an isotope? [1]
7. Determine the relative molecular mass for water (H_2O). [3]
8. Find the mass of one mole of CO_2. [3]
9. 10g of sodium hydroxide is dissolved in 3 dm^3 of water. Calculate the concentration of the solution. [3]

CHAPTER 15 Metals

Key concepts

- Reaction of metals with oxygen, steam water and dilute acids
- Equations for reactions of metals with oxygen, water and dilute acids
- Reactivity series
- Predicting reactivity from position on reactivity series

Summary notes

Metals and non-metals

Elements on the periodic table can be grouped as mainly metals and non-metals. Metals are on the left hand side and non-metals are on the right hand side of the periodic table.

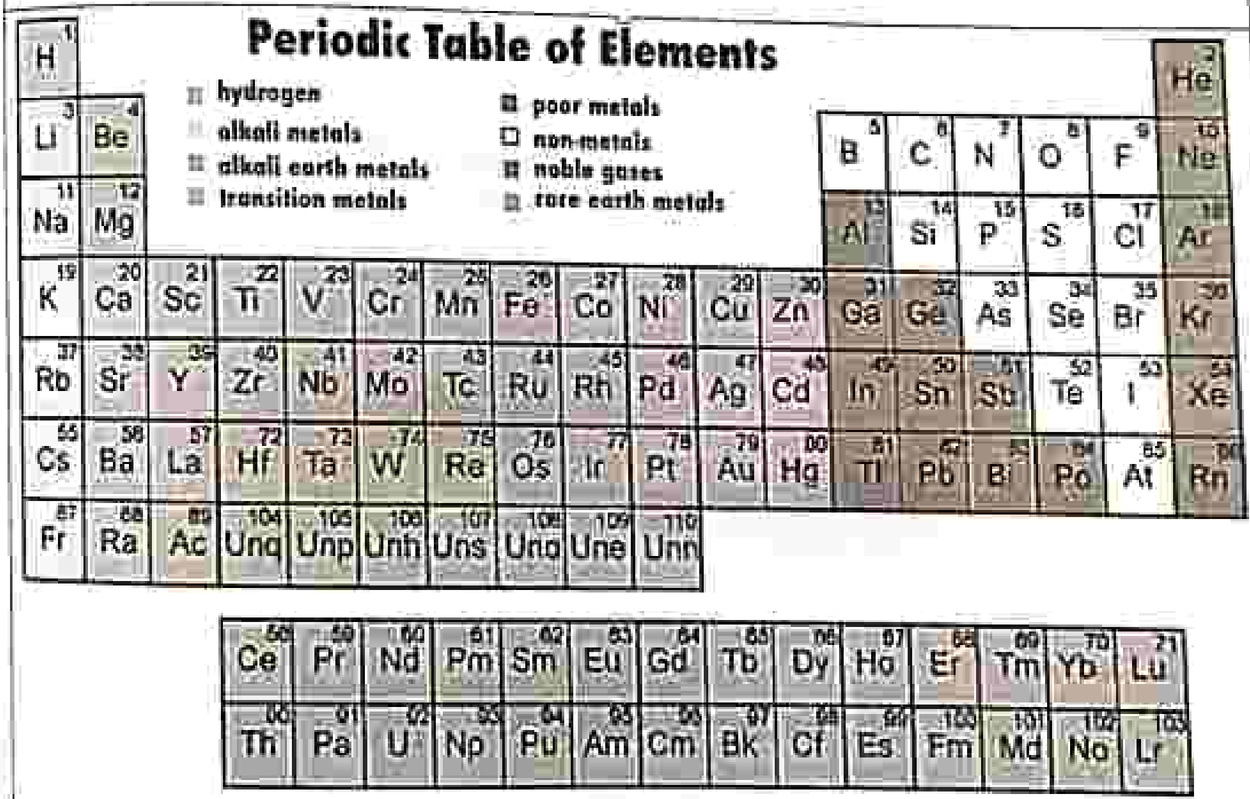


Fig. 15.1: The periodic table (see colour table at the back)

Metals

- Different metals react at different rates with oxygen, water and diluted acids.

Reaction of metals with oxygen

Table 15. 1: Reaction of metals with oxygen

Metals	Heat required	Observation
Potassium	Gentle heat	Burns violently with lilac flame.
Sodium		Burns violently with golden yellow flame.
Calcium	Strong heat	Burns violently with brick red flame.
Magnesium		Burns with a bright light.
Aluminum		White powder oxide is produced.
Zinc		White powder oxide is produced.
Iron		Burns with sparks.
Lead		Yellow powder oxide on surface.
Copper	Very strong heat	Black powder oxide on surface.
Mercury		Red powder oxide on surface.
Silver		No reaction
Gold		No reaction

Equations for the reaction of metals with oxygen

General equation

Metal + oxygen ⟶ metal oxide

Example

Word equation

Sodium + oxygen ⟶ sodium oxide

Chemical equation

$4Na_{(s)} + O2_{(g)} \longrightarrow 2Na_2O_{(s)}$

Reaction of metals with water

- Some metals react with water to give metal hydroxide and hydrogen gas.
- The most reactive metals such as potassium and sodium react vigorously with water.
- Some metals only react with steam to form metal oxide and hydrogen gas.

Table 15. 2: Reaction of metals with water and steam

Metals	Water or steam	Observation
Potassium	Water	Burns with lilac
Sodium		Burns with golden yellow flame
Calcium		Colorless bubbles produced
Magnesium	Steam	White oxide
Aluminium		White oxide
Zinc		White oxide
Iron		Brown oxide
Lead	No reaction	No change
Copper		
Mercury		
Silver		
Gold		

Equations for the reaction of metals with water

General equation

Metal + acid ⟶ salt + hydrogen gas

Example

Word equation

Magnesium + hydrochloric acid ⟶ magnesium chloride + hydrogen

Chemical equation

$Ca_{(s)} + 2H_2O_{(l)} \longrightarrow Ca(OH)_{2(aq)} + H_{2(g)}$

Equation for Reaction of metals with steam

General equation

Metal + steam ⟶ metal oxide + hydrogen gas

Example

Word equation

Zinc + steam ⟶ zinc oxide + hydrogen

Chemical equation

$Zn_{(s)} + H_2O_{(g)} \longrightarrow ZnO_{(s)} + H_{2(g)}$

Reaction of metals with acids

- Most metals react with acids to give a salt and hydrogen gas.

Table 15.3: Reaction of metals with acids

Metal	Reaction rate	Observation
Potassium	Explosively	Explosion
Sodium		
Calcium	Very fast	Gas bubbles produced
Magnesium	Fast	
Aluminium	Moderate	
Zinc	Slow	
Iron	Slow	
Lead	Very slow	
Copper	No reaction	No change
Mercury		
Silver		
Gold		

General equation

Metal + acid ⟶ salt + hydrogen gas

Example

Word equation

Magnesium + hydrochloric acid ⟶ magnesium chloride + hydrogen

$Mg_{(s)} + 2HCl_{(l)} \longrightarrow MgCl_2 + H_2$

Chemical equation

The reactivity series of metals

It lists metals in order of their reactivity, from the most reactive to least reactive.

The reactivity series

Fig. 15.2: The reactivity series

Predicting the reactivity of metals from its position in the reactivity series

- A more reactive metal will displace a less reactive metal in a metal salt solution containing a less reactive metal, this is called displacement.
- Any metal that displaces another in a chemical reaction is above it on the reactivity series.

Example

$Mg_{(s)} + CuCl_{2(l)} \longrightarrow MgCl_{2(aq)} + Cu_{(s)}$

Copper is below magnesium on the reactivity series, therefore magnesium will easily displace copper.

If a less reactive metal is added to a metal salt solution containing a more reactive metal no reaction will occur.

Example

Magnesium + Calcium Chloride $\longrightarrow$ no reaction

$Mg + CaCl_2 \longrightarrow$ no reaction

- Magnesium is below calcium on the reactivity series, therefore it cannot displace calcium. The reaction will not occur.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. The valence electrons for metals are generally
   A. 1 or 2.
   B. 4 or 5.
   C. 6 or 7.
   D. 8 or 9.
2. To become stable metals
   A. gain electrons.
   B. donate electrons.
   C. lose or gain electrons.
   D. gain protons.
3. Which gas is produced when a metal reacts with an acid?
   A. nitrogen
   B. oxygen
   C. hydrogen
   D. carbon dioxide
4. ______does not react with acids.
   A. gold
   B. magnesium
   C. tin
   D. zinc
5. Name a metal whose oxide is yellow when hot and white when cold.
   A. zinc
   B. lead
   C. iron
   D. copper
6. Metals react with steam to produce ____________.
   A. metal oxide.
   B. metal hydroxide.
   C. metal salt.
   D. metal chloride.
7. Which of the following equations correctly show the reaction of magnesium with steam?
   A. $Mg + H_2O \longrightarrow MgO + H_2$
   B. $Mg + 2H_2O \longrightarrow MgO_2 + 2H_2$
   C. $Mg + 2H_2O \longrightarrow Mg(OH)_2 + H_2$
   D. $2Mg + 2H_2O \longrightarrow 2Mg(OH) + H_2$

8. Between which elements would you place zinc in the reactivity series?
   A. aluminium and iron
   B. copper and silver
   C. iron and tin
   D. lead and mercury

9. The most reactive metal is ____
   A. potassium B. lithium
   C. calcium D. sodium

10. Which metal can be displaced by lea[d] from its salt solution?
   A. Copper B. Iron
   C. Potassium D. Zinc

Section B: Structured questions

1. Write the general equation for the reaction of metals with:
   a) Oxygen
   b) Water
   c) Acids [3]
2. List the following in order of reactivity, starting with the most reactive, lead, potassium, copper, calcium, zinc. [2]
3. Sodium reacted with hydrochloric acid.
   a) Complete the word equation.
      Sodium + hydrochloric acid $\longrightarrow$ [2]
   b) Write the balanced chemical equation. [2]
4. Describe what you would observe when potassium reacts with water. [2]
5. When metals react with other substances which type of ions do they form? [1]
6. Write a word equation to show a displacement reaction. [2]
7. Describe the test for the gas produced when a metal reacts with water. [2]
8. Describe what would happen if you add copper to zinc sulphate solution? [2]

CHAPTER 16

Acids bases and salts

Key concepts

- Properties of acids and bases
- The pH scale
- Indicators: universal indicator solution and litmus paper
- Neutralisation
- Reaction of acids with: metals, metal oxides, metal hydroxides and metal carbonate
- Word and chemical equation for reactions
- Acid base titration

Summary notes

- Substances can be acidic, basic, or neutral.
- When a base dissolves in water an alkaline solution is formed.

Examples of acids

- Hydrochloric acid (HCl)
- Sulphuric acid (H_2SO_4)

Examples of bases

- Sodium hydroxide (NaOH)
- Magnesium carbonate ($MgCO_3$
- baking soda and bleach

Table 16. 1: Properties of acids and bases

Properties of acids	Properties of bases
Sour taste	Bitter
pH less than 7	pH more than 7
Corrosive	Caustic
Turn blue litmus paper red	Turn red litmus paper blue
Turn universal indicator from green to red	From green to blue or purple
Proton donors	Proton acceptors
Contain hydrogen ion	Contain hydroxide ion
Sticky	Slippery soapy feel
Examples: vinegar, lemon,	Examples: baking soda and bleach

To test whether a solution is acidic, basic or neutral indicators are used namely: litmus paper and Universal indicator.

Litmus paper

It only shows whether the substance is alkaline or acidic. Litmus paper has two colours red and blue.

Table 16. 2: Litmus colour changes

	Red litmus	Blue litmus
Acids	Turns blue	Remains blue
Bases	Remains red	Turns red
Salts	Remains red	Remains blue

Universal indicator

- It shows how acidic and alkaline a solution is.
- It will change from green to different colours. It comes as a solution or paper.
- Acids change universal indicator from green to red.
- Bases change universal indicator from green to blue/purple.

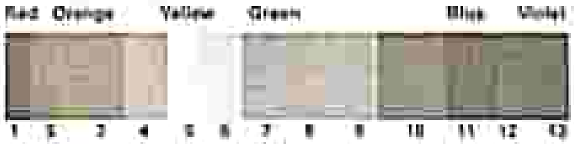


Fig. 16.1: Universal indicator

The pH scale

- It is a measure of acidity or alkalinity of substances.
- The numbers go from 0 – 14. Acidic solutions have ph less than seven, alkaline solutions have a ph greater than seven and neutral solutions have ph of seven.

Reaction of acids with bases

- Reaction between an acid and a base is called neutralisation reaction. When an acid reacts with a base, a salt is formed.
- A salt is a substance that is neutral; is formed when acids react with the following:
  - metals, e.g. sodium, calcium
  - metal oxide e.g. sodium oxide, magnesium oxide
  - metal hydroxide e.g. calcium hydroxide, potassium hydroxide
  - metal carbonates e.g. calcium carbonate, zinc carbonate

The general neutralisation reaction

Acid + base $\longrightarrow$ Salt and water

Reaction of acids with metals

General reaction

Acid + metal $\longrightarrow$ Salt and hydrogen

Example

Word equation

Sodium + Hydrochloric Acid $\longrightarrow$ Sodium Chloride + Hydrogen

Chemical equation

$2Na_{(s)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(aq)} + H_2$

Reaction of acids with metal oxide

General equation

Acid + Metal oxide $\longrightarrow$ Salt and Water

Fig. 16. 2: The pH scale

Example

Word equation

Magnesium + Hydrchloric acid $\longrightarrow$ Magnesim Chloride + hydrogen

Chemical equation

$Mg_{(s)} + 2HCl_{(aq)} \longrightarrow 2MgCl_2(aq) + H_{2(g)}$

Reaction of acids with metal hydroxides

General equation

Acid + Metal hydroxide $\longrightarrow$ Salt and Water

Example

Word equation

Hydrochroric acid + Potassium hydroxide $\longrightarrow$ Potassium chloride + Water

Chemical equation

$HCl_{(aq)} + KOH_{(aq)} \longrightarrow KCl_{(aq)} + H_2O_{(l)}$

Reaction of acids with metal carbonates

General equation

Acid + Metal carbonate $\longrightarrow$ Salt + Carbon dioxide + Water

Example

Word equation

Magnesum carbonate + hydrochloric acid $\longrightarrow$ Magnesium chloride + Water + Carbon dioxide

Chemical equation

$MgCO_{3(aq)} + 2HCl_{(aq)} \longrightarrow MgCl_{(aq)} + H_2O_{(l)} + CO_2$

Acid base titrations

Titration is a technique used to determine the concentration of an unknown acid or base by neutralising it with an acid or base of known concentration.

- titrant – a solution of known concentration
- analyte – a solution of unknown concentration

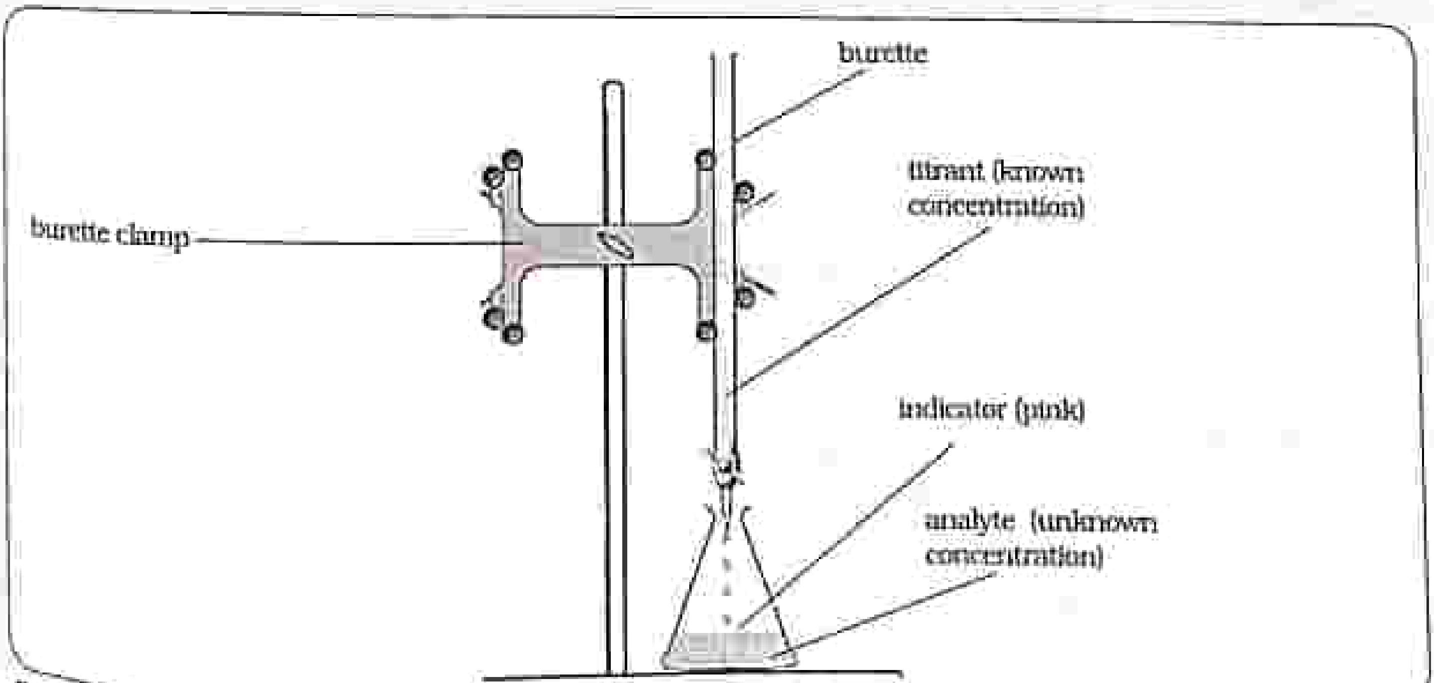


Fig. 16.3: Acid base titration set up

- The titrant is added through a burette to a known volume of analyte until the reaction is complete. An indicator will be added to analyte. An indicator will experience a colour change (end point) when reactions reach equivalent point.
  - end point refers to the point at which the indicator changes colour
  - equivalent point refers to point where equal amounts of acid and base would have reacted at equivalent point moles of acid = moles of base.

Common acid base indicators

- phenolphthalein – colourless in acid and pink in base
- methyl orange – pink in acid and yellow in base

Example

$HCl + NaOH \xrightarrow{\text{at equivalent point}} NaCl + H_2O$

Titration curve

- Titration curve – acid base titrations are monitored by the change of ph as titration progresses.
- A titration curve – is a plot of the ph of analyte versus the volume of titrant.

Titration of strong acid with strong base

- Before adding titrant (base) the ph of analyte (acid) is very low.
- As the base is added the ph of acid starts to increase until the equivalent point is reached and the ph becomes 7.
- If addition of base continues the ph starts to become basic.

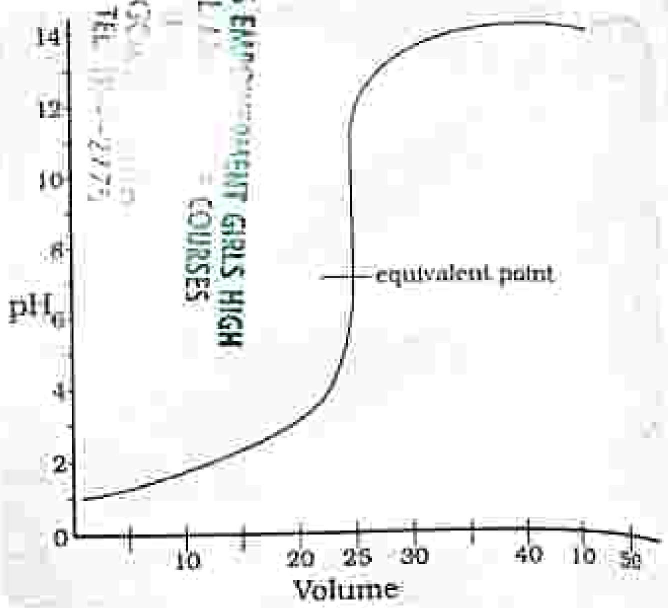


Fig. 16.4: Graph showing strong acid and strong base titration

Titration of strong base with strong acid

- Before adding the titrant (acid) the ph of analyte (base) is very high.
- As the acid is added the ph of the base starts to decrease until the equivalent point is reached and indicator changes colour, at this point the ph becomes 7.
- If addition of acid continues the ph starts being acidic.

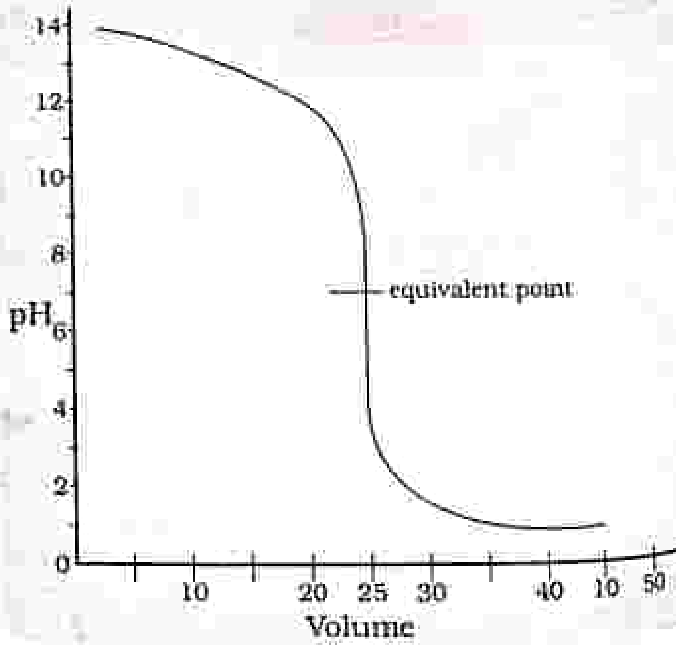


Fig. 16.5: Graph showing strong base and strong acid titration

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. The process in which acids and bases react to form salts is called
   A. titration.
   B. neutralisation.
   C. distillation.
   D. crystallization.
2. Which of the following is neutral?
   A. HCl
   B. NaCl
   C. $MgCO_3$
   D. NaOH
3. Identify the correct match
   A. alkali - pH below seven
   B. acid - pH above seven
   C. acid pH seven
   D. alkaline - pH above seven
4. Which gas is produced when hydrochloric acid reacts with zinc?
   A. hydrogen
   B. oxygen
   C. nitrogen
   D. carbon dioxide
5. When a base is added to an acid the product is
   A. hydrogen gas.
   B. carbon dioxide.
   C. salts.
   D. hydroxides.
6. What is used to measure the exact pH of a solution?
   A. litmus paper
   B. universal indicator
   C. pH meter
   D. methyl orange
7. ______ will turn red litmus blue?
   A. HCl
   B. H_2SO_4
   C. NH_3
   D. NaOH
8. What is the correct formula for calcium chloride?
   A. $CaCl_2$ B. CaCl
   C. Ca_2Cl D. Ca_2C_{12}

9. Which compound is formed when sodium hydroxide reacts with hydrochloric acid?
   A. sodium oxide
   B. sodium hydroxide
   C. sodium chloride
   D. sodium carbonate

10. Phenolphthalein in acidic solution is ________.
   A. colourless
   B. orange
   C. pink
   D. yellow

Section B: Structured questions

1. What is a neutralisation reaction? [1]
2. Distinguish between acids and bases. [3]
3. What is produced when hydrochloric acid reacts with sodium hydroxide? [2]
4. Explain giving examples how an acid reacts with a metal carbonate. [3]
5. Name any two acid base indicators used in the lab and their colour changes. [4]
6. a) Which gas is produced when an acid reacts with a metal. [1]
   b) Describe how you would test for the gas in the laboratory. [2]
7. Write the general equation for the reaction of an acid with a base. [2]
8. Why is sulphuric acid unable to change the colour of red litmus paper? [2]
9. Name one substance that can change red litmus to blue. [1]
10. With the aid of a diagram describe an acid base titration. [5]

CHAPTER 17

Industrial processes

Key concepts

- Production of peanut butter
- Production of oil from peanut butter
- Uses of oil
- Production of soap
- Fractional distillation of liquid air
- Uses of oxygen, carbon dioxide and nitrogen
- Electrolysis
- Electrolytic cell
- Electrolysis of molten lead bromide
- Electrolysis of water
- Uses of oxygen and hydrogen
- Electroplating an iron nail with copper
- Reasons for electroplating materials

Summary notes:

Production of peanut butter

Pre-cleaning – that is removing sticks and stones

Shelling of nuts – is the removal of shells from the peanut seed ensuring the seed is not damaged

Grading – selection of the desired quality of nuts while removing rotten and premature ones

Roasting – destroys enzymes which produce bad flavour

- improves flavour, texture and colour of the nuts

Blanching – removing the outer skin of the nuts leaving them with a pale colour

Grinding – crushing the nuts to provide a smooth texture.

- The peanuts should be ground twice, firstly the peanuts alone and then other ingredients like salt(to add flavour) and stabilizers (to avoid separation of oil from peanuts) are added when the peanuts are ground for the second time.

Cooling – the peanut butter is left to cool

Packaging – peanut butter is packed into clean containers of desired sizes which are then capped and sealed.

Production of peanut oil

Peanut oil can be extracted using two methods :

- Using an oil presser – the oil presser squeezes the oil out of the peanuts.
- Using decanting method – if less or no stabilizer is used during the production of peanut butter and left to set, oil in the peanut butter will separate and move to the top because it is less dense. The oil can now be easily poured from the mixture.

Uses of peanut oil

- For making cooking oil
- Used in baking
- Used in making baby care products
- Used in making skin moisturisers

Production of soap

Saponification – is the action alkalis on fatty acids (fats and oils) make soaps

Fatty acid + sodium hydroxide ⟶ sodium salt of fatty acids (soap) + glycerol (glycerine)

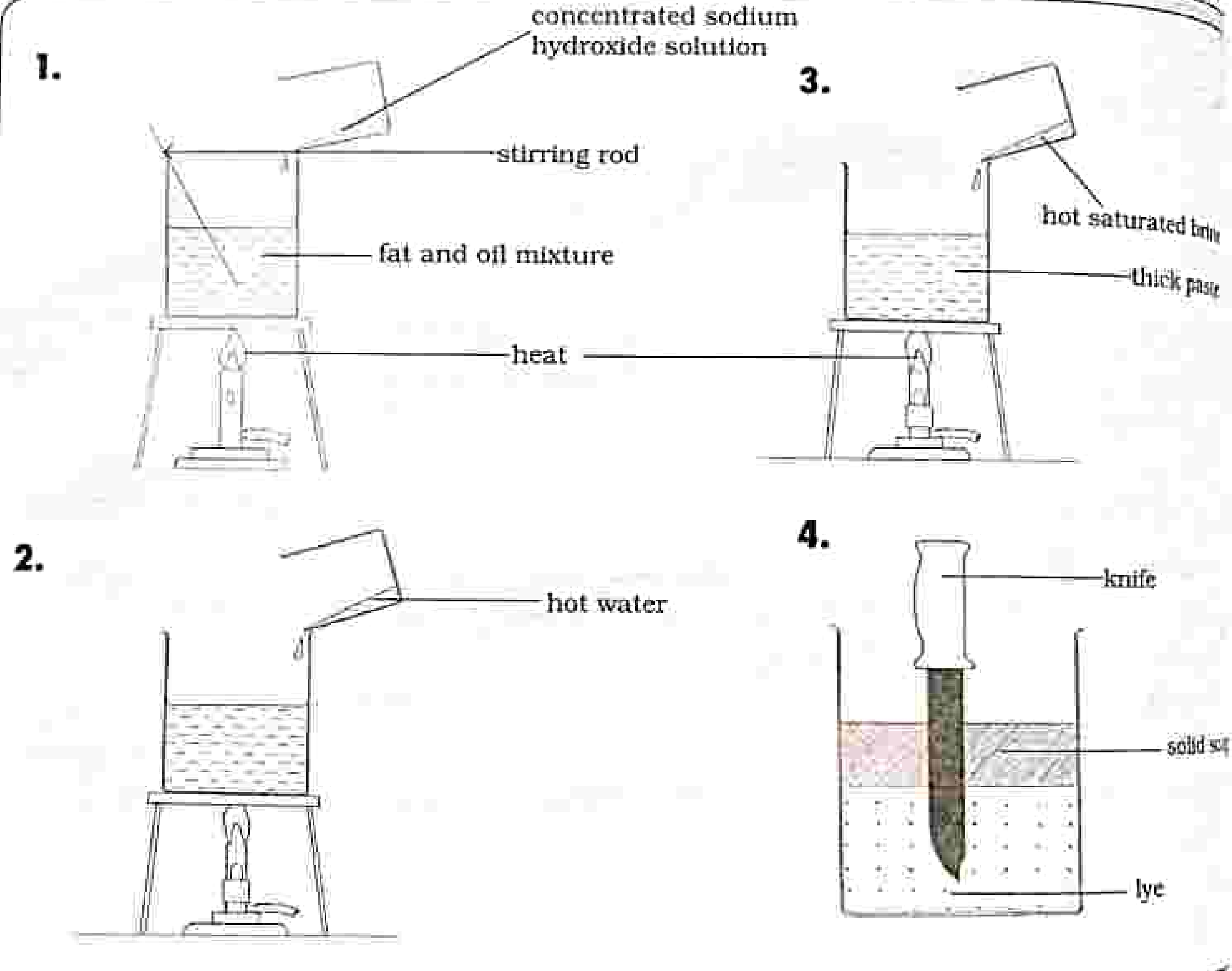


Fig 17.1: Soap making process

Soap making process

1. Heat the fat mixture over a low flame and slowly add concentrated sodium hydroxide solution while stirring using a stirring rod until thick.
2. Cool the mixture and remove the solid. Into the solid add hot water and continue stirring until a thick paste is formed.
3. Into the thick paste, add hot saturated brine. Heat and stir until the mixture no longer stick on the stirring rod.
4. Remove the stirring rod and leave the mixture to cool until the solid soap separate from the lye. Rinse the soap with water.

Fractional distillation of liquid air

Oxygen and nitrogen are produced from the fractional distillation of liquid air. Fractional distillation is used to separate liquids with different boiling points.

Process of liquefaction of air

1. Air from the atmosphere is cooled to -78°C.
   - Water and carbon dioxide turns to solid at this temperature and are removed.
2. Remaining gases are compressed and cooled.
   - Compressed gases are allowed to expand rapidly thus further cooling them.
3. The process of compression and expansion is repeated until temperature drops to -190°C and the gases turn to liquid except for rare gases.
4. The liquid air is pumped into a distillation column and the temperature is allowed to rise.
5. Nitrogen evaporates first and escapes from the top of the tower because it has a boiling point of -196°C.
6. Oxygen remains as a liquid because its boiling point is -183°C and is removed from the bottom of the column.

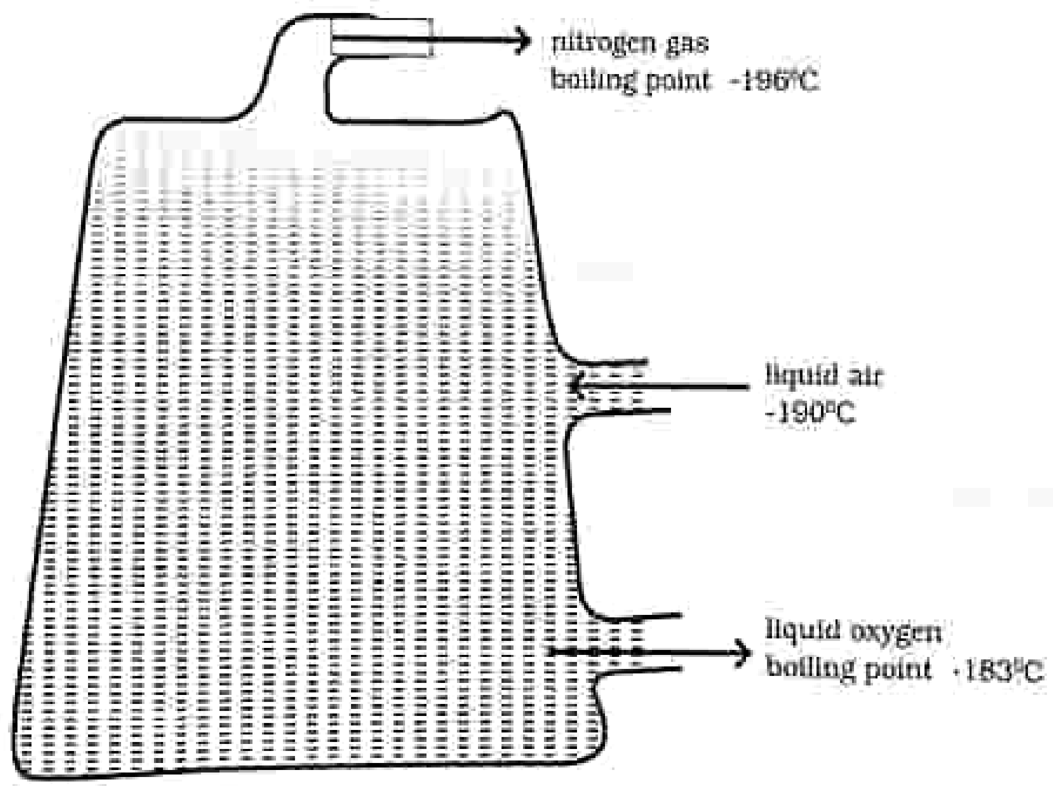


Fig 17.2: Fractional distillation of liquid air

Uses of oxygen

- Medical purposes
- Manufacturing of steel
- Welding

Uses of carbon dioxide

- Used as a refrigerant
- In fire extinguishers
- Used in fizzy drinks

Uses of nitrogen

- Medical use
- Manufacture of nitric acid and ammonia
- Freezing vegetables

Electrolysis

- It is the process of breaking down a compound which is either molten or in aqueous solution when an electric current is passed through it.

Electrolytic cell

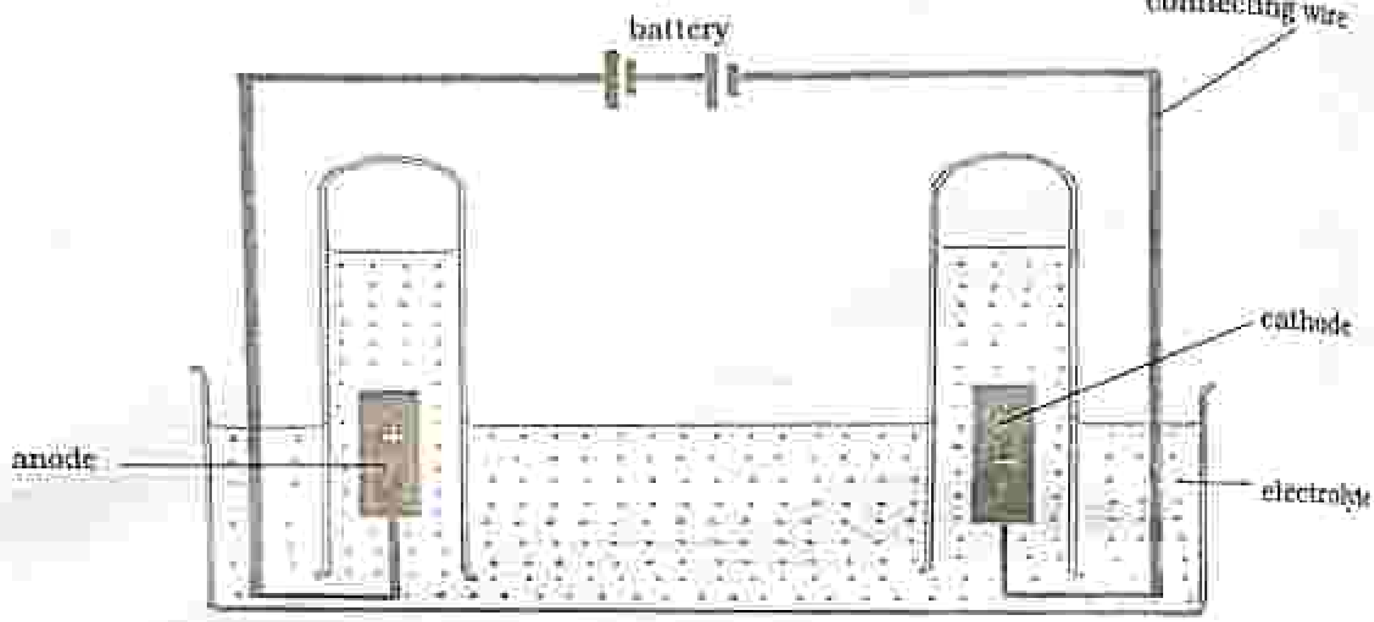


Fig 17.3: Electrolytic cell

- An electrolytic cell converts electrical energy into chemical energy. An electrolytic cell consists of :

1. **Two electrodes**

   These are conductors which allow current to enter or leave a substance. The two electrodes are :

   a) **Anode** – which is connected to the positive terminal of the battery

   b) **Cathode** – which is connected to the negative terminal of the battery

2. **Electrolyte**
   - it is the solution (liquid or aqueous) which contains ions that carry the current. The electrolyte can be broken down by electrolysis.

3. **Connecting wires**
   - connects the positive terminal of the battery to the positive electrode (anode)
   - connects the negative terminal of the battery to the negative electrode (cathode)

Electrolysis of molten lead bromide

Practical activity 17.1

Aim : To investigate the electrolysis of lead (II) bromide

Apparatus : 1) Carbon electrodes
2) Battery
3) Connecting wires
4) Beaker
5) Lead(II)bromide

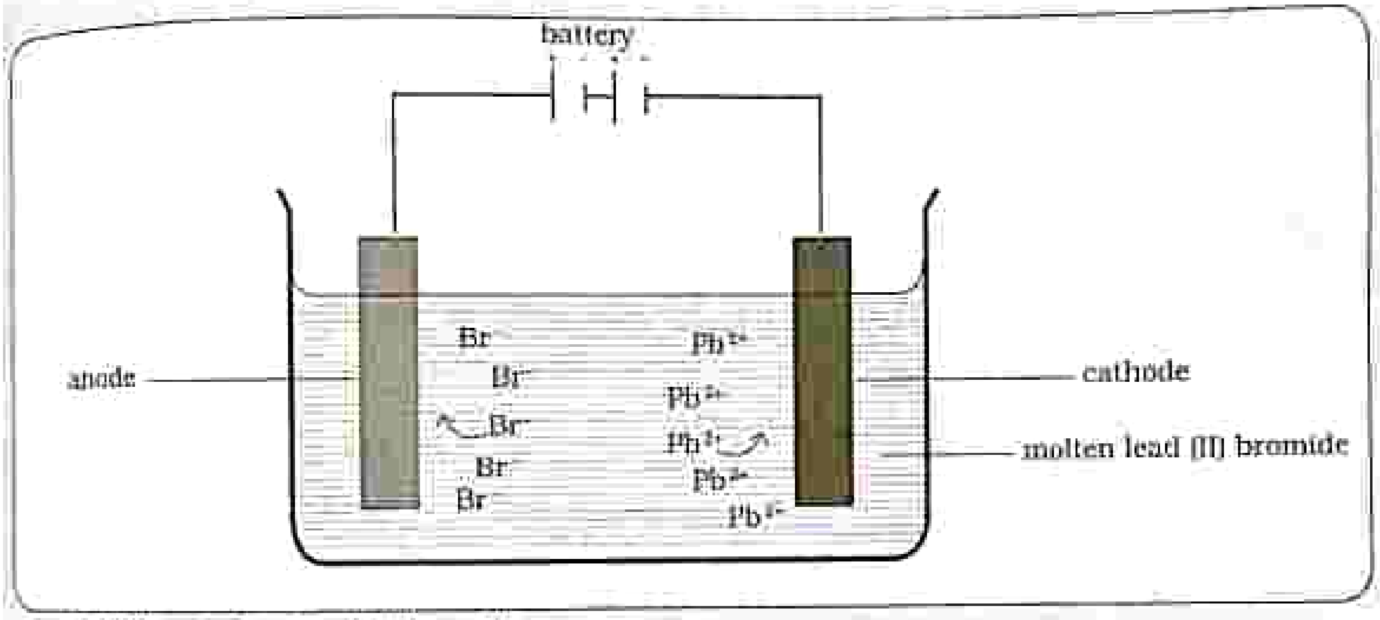


Fig. 17.4: Electrolysis of lead (II) bromide

Method

Caution: Bromine gas is poisonous. This experiment should be done in a fume cardboard or well ventilated room.

1. Heat the lead (II) bromide until it melts
2. Set up the apparatus as shown in Fig 17.4
3. Using the connection wires, place the two electrodes into the molten lead (II) bromide and pass a current through it.
4. Observe what happens at each electrode. To observe the cathode product, switch off the current and remove the electrode.

Results

Anode – bubbles of brown gas
Cathode – silver metal

Conclusion

- lead (II) bromide contains positive lead ions (Pb^{2+}) and negative bromide ions (Br^-)
- opposite charges attract therefore the lead ions (Pb^{2+}) are attracted to the cathode and the bromide ions (Br^-) are attracted to the anode during electrolysis molten lead (II) bromide is broken down into elements that is:
  - Anode : Bromine (Br) fumes
  - Cathode : Solid lead (Pb)

Word equation for the electrolysis of molten lead (II) bromide

lead (II) bromide ⟶ lead + bromine

Practical questions

1. Bromide ions (Br^-) are attracted to the ________ electrode
2. Lead ions (Pb^{2+}) are attracted to the ________ electrode

Electrolysis of water

Practical activity 17.2

Aim : To investigate the electrolysis of water

Apparatus :
1. Beaker
2. Carbon electrodes
3. Two test tubes
4. Battery
5. Connecting wire
6. Splints
7. Burner
8. Water
9. Concentrated sulphuric acid

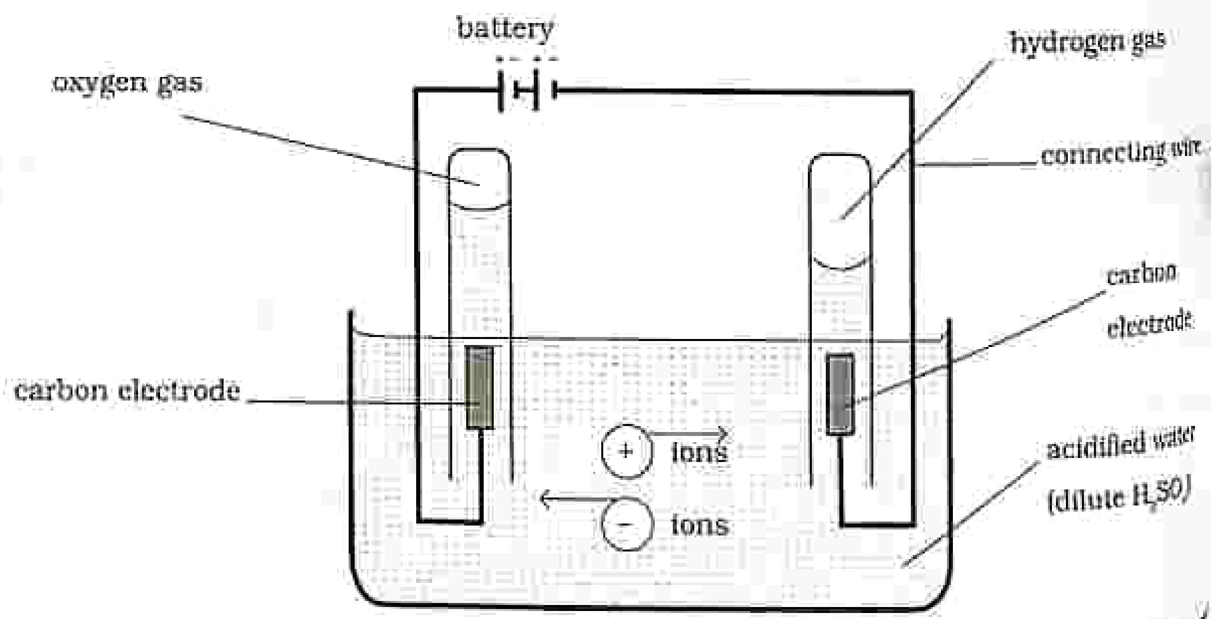


Fig. 17.5: Electrolysis of water

thod

- Set up the electrolytic cell as shown in Fig. 17.5.
- Add a few drops of sulphuric acid and stir.
- Let an adjusted current flow through the water.
  - The current should produce a gentle evolution of gases at each electrode.
- Carefully remove the test tubes one at a time and quickly placing the thumb over the mouth of the test tubes to ensure the gas collected will not escape.
- Test the gas collected.
  Anode – Test with a glowing splint
  Cathode – Test with a lighted splint

Results

Volume	:	Cathode		Anode
		2	:	1

Testing for the gas

Anode – the glowing splint lighted to produce a flame.

Cathode – the lighted splint produced popping sounds.

Conclusion

- During the electrolysis of water, water is broken down into hydrogen and water.
- Hydrogen is produced at the cathode.
- Oxygen is produced at the anode.
- Ratio of gas – Hydrogen 2 : Oxygen 1.

Practical questions

1. Identify the gas produced at the anode.
2. Identify the gas produced at the cathode.

ses of Hydrogen

- Used in the manufacture of ammonia
- Making of margarine
- Processing of petroleum products
- Lifting balloons in air

Electroplating

- Coating a metal using another metal which is less reactive using the process of electrolysis.

Practical activity 17.3

Aim : Electroplating an iron nail using copper

Apparatus :
1. Beaker
2. Copper, sulphate solution
3. Iron nail
4. Copper electrode
5. Dilute sulphuric acid
6. Sand paper

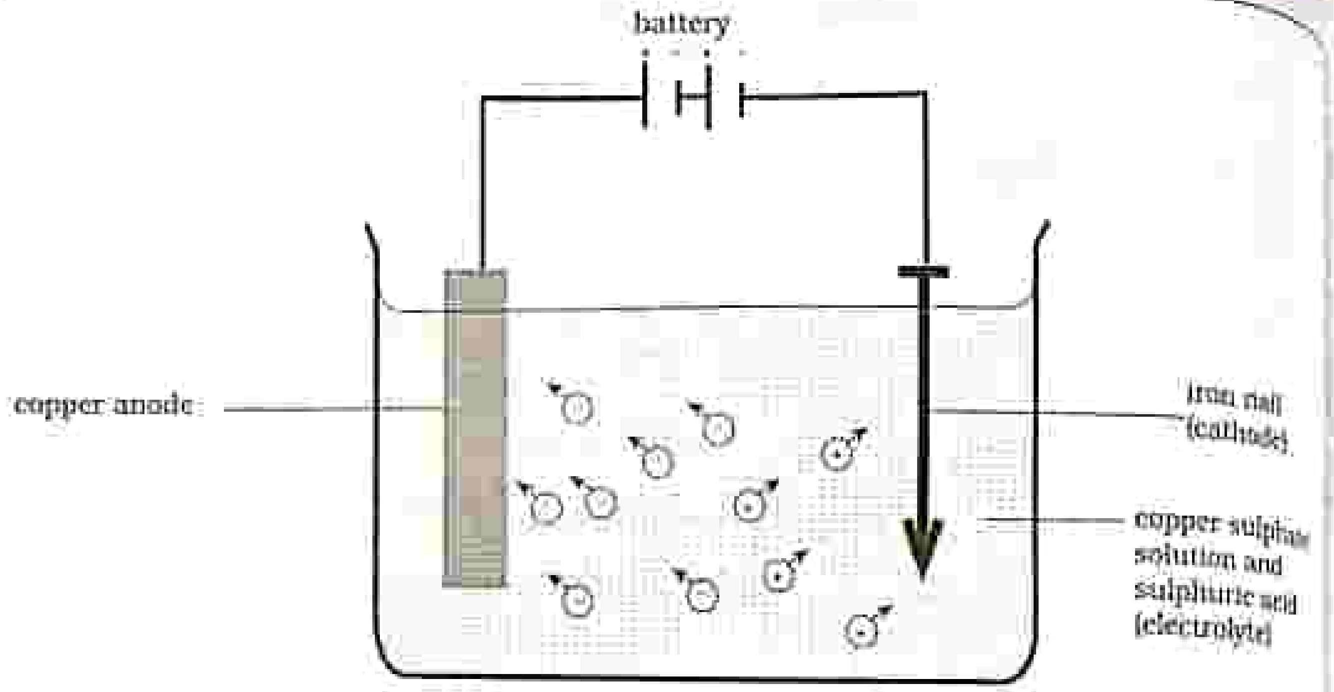


Fig. 17.6: Electroplating an iron nail using copper

Method

- Make the electrolyte using copper sulphate solution and dilute sulphuric acid.
- Using a sand paper clean the iron nail.
- Set up apparatus as shown in Fig 17.6.
- Allow a current to flow for about 20 minutes.
- Switch off the current and disconnect the cathode.

Results

- Copper anode dissolves in the copper (II) sulphate solution and loses weight.
- Iron nail is coated brown.

Conclusion

Cathode process

- positively charged copper ions are attracted to the negatively charged nail and are deposited to the nail.
- Over time the iron nail is copper plated.

Practical questions

1. Explain what happens to the iron nail after some time.
2. Explain what happens to the copper anode after some time.

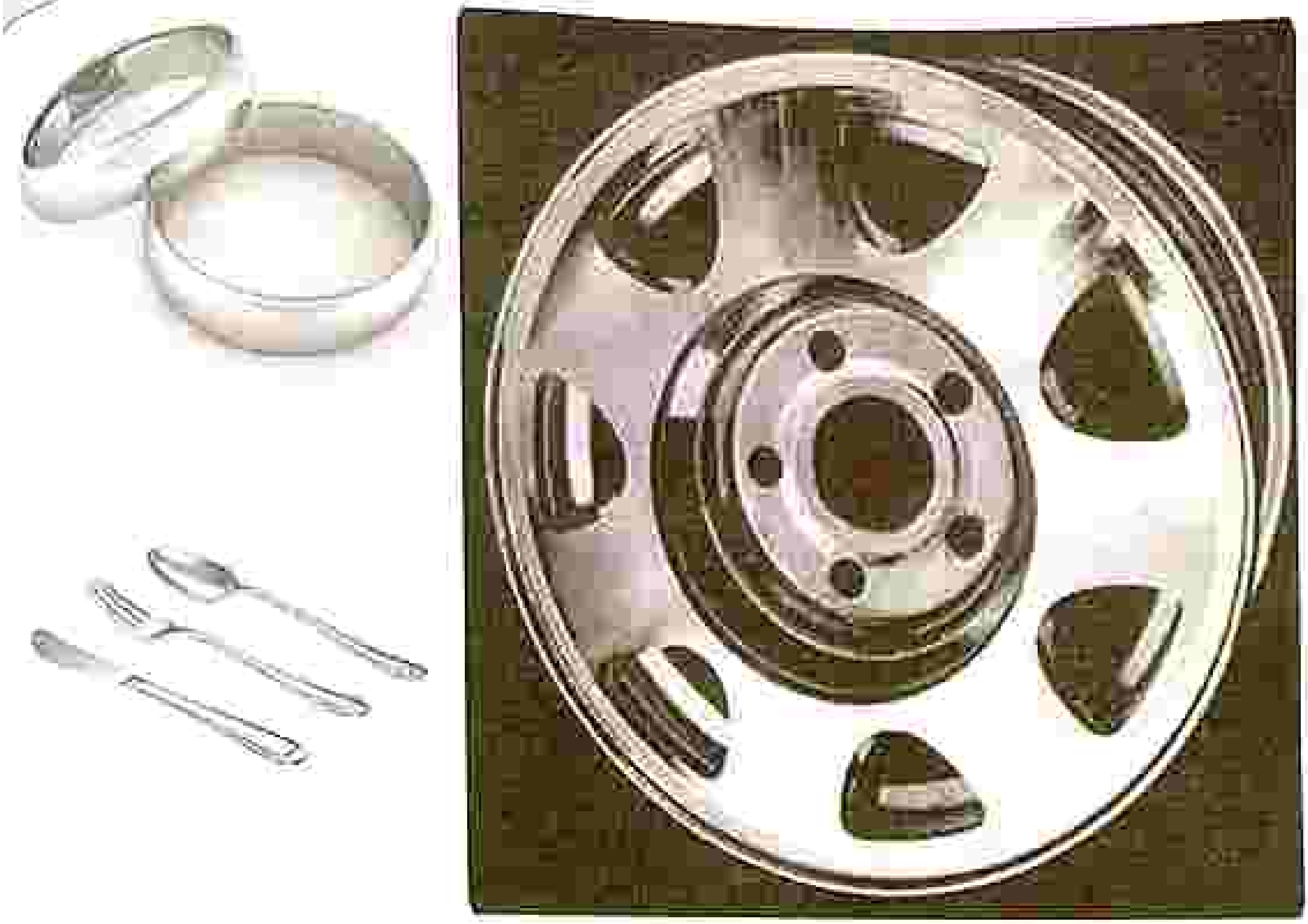
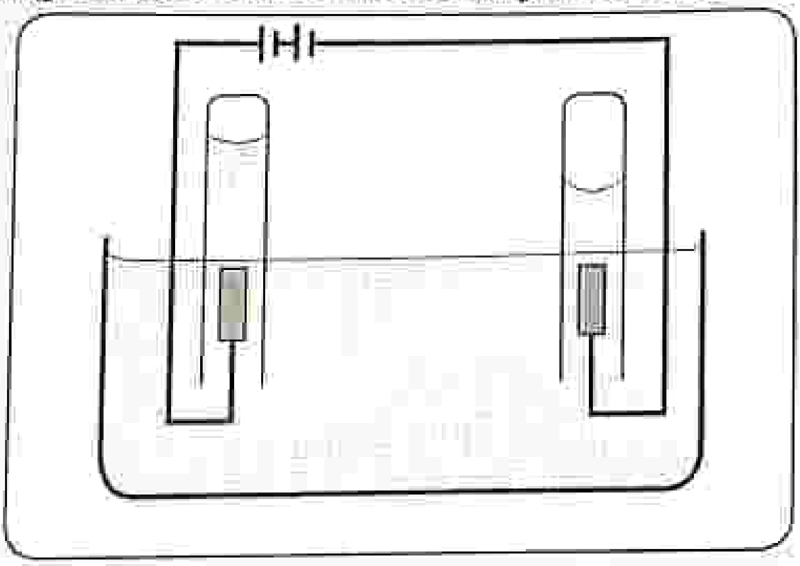


Fig. 17.7: Electroplated objects

Reasons for electroplating materials

- Prevent corrosion
- Decoration

3. The diagram beow shows the electrolysis of water



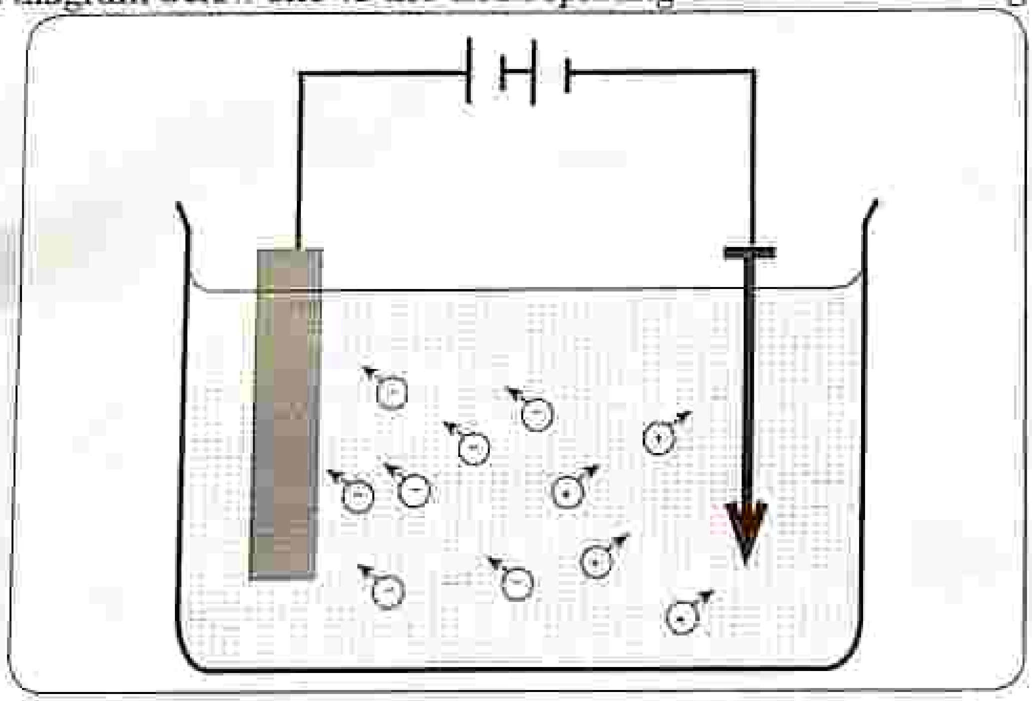
a) Name the gas produced at the:

i) Anode [1] ii) Cathode [1]

b) Briefly explain the test for the gases produced at;

i) Anode [2] ii) Cathode [2]

4. The diagram below shows the electroplating of an iron nail using copper



a) Name the:

i) Anode used [1]

ii) Cathode used [1]

iii) Electrolyte used [1]

b) State reasons for electroplating material. [2]

5. List any three uses of the following gases

a) Hydrogen [3]

b) Oxygen [3]

c) Nitrogen [3]

CHAPTER 18

Harber and contact processes

Key concepts

- Haber process
  - Raw materials used to manufacture ammonia
  - Conditions needed for the manufacture of ammonia
  - Manufacture of ammonia
- Contact process
  - Raw materials used to manufacture sulphuric acid
  - Conditions needed for the production of sulphuric acid
  - Manufacture of sulphuric acid
  - Uses of sulphuric acid
- Production of ammonium nitrate through acid-base titration

Summary notes

Haber process – is an industrial process that uses nitrogen from air and hydrogen from water to manufacture ammonia.

Raw materials needed for the manufacture of ammonia

- Hydrogen from electrolysis of water
- Nitrogen from fractional distillation

Conditions needed for the production of ammonia

- Pressure: 200atm
- Catalyst: iron
- Temperature: 450°C – 500°C

Manufacture of ammonia

- Ammonia (NH_3) is a compound of nitrogen and hydrogen.

Stages in the manufacture of ammonia

- **In a converter** – nitrogen gas (N_2) and hydrogen gas (H_2) are mixed in the ratio 1:3.
- The mixture is passed into the reaction chamber where the temperature is 400°C – 450°C and the mixture is compressed to a pressure of 200atm and passed through an iron catalyst, which helps to speed up the chemical reaction.
  - the mixture will produce ammonia

$$\text{Nitrogen} + \text{hydrogen} \xrightarrow[\text{iron catalyst, 200atm}]{450°C} \text{ammonia} + \text{heat}$$

$$N_2 + 3H_2 \longrightarrow 2NH_3 + \text{heat}$$

- Ammonia is removed from the mixture by condensation.
- The unreacted gases are mixed with hydrogen and nitrogen again and undergo the same process until more ammonia is collected.

Uses of ammonia

- Manufacture of fertilizers such as ammonium nitrate
- Making dyes
- Manufacture of nitric acid
- Manufacture of household cleaners
- Used as refrigerant

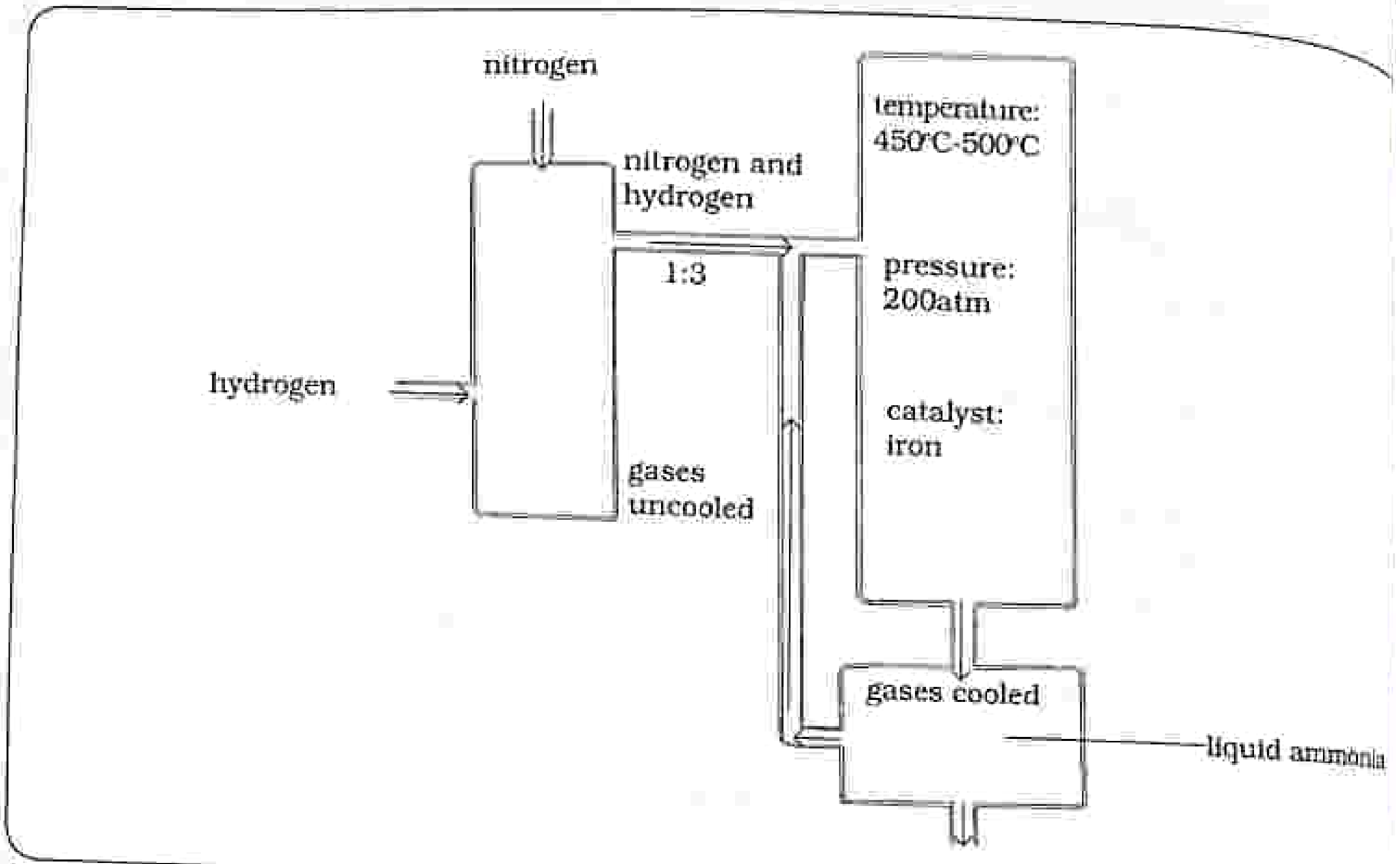


Fig 18.1: Manufacture of ammonia

Contact process – is an industrial process in which sulphur dioxide is oxidized to sulphur trioxide to manufacture sulphuric acid.

Raw materials needed

- Sulphur dioxide from burning iron pyrites or sulphur.
- Oxygen from air.

Conditions needed for the production of sulphuric acid

- Pressure : 1atm
- Catalyst : vanadium(v)oxide
- Temperature: 450°C - 500°C

Manufacture of sulphuric acid

- Sulphur dioxide is produced by burning in air or burning iron pyrites

Iron pyrites + oxygen → sulphur dioxide + iron oxide

$2 FeS + 3O_2 \longrightarrow 2 SO_2 + 2 FeO$

- Sulphur dioxide and oxygen are mixed in a reaction chamber.
- Temperature inside the reaction chamber is maintained at 450°C – 500°C.
- Vanadium (v) oxide is added as a catalyst to speed up the chemical reactions inside the reaction chamber.

- Sulphur dioxide is oxidized to sulphur trioxide and the reaction is reversible

$$\text{Sulphur dioxide} + \text{oxygen} \xrightarrow[\text{450°C, 1atm}]{\text{Vanadium (v) oxide}} \text{sulphur trioxide}$$

$$2SO_2 + O_2 \longrightarrow 2SO_3$$

- In the absorption tower, sulphur trioxide is cooled and absorbed into concentrated sulphuric acid to produce oleum

$$\text{Sulphur trioxide} + \text{sulphuric acid} \longrightarrow \text{oleum}$$

$$SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$$

- To produce sulphuric acid, oleum is added to water, that is the dilution process

$$\text{Oleum} + \text{water} \longrightarrow \text{sulphuric acid}$$

$$H_2S_2O_7 + H_2O \longrightarrow 2H_2SO_4$$

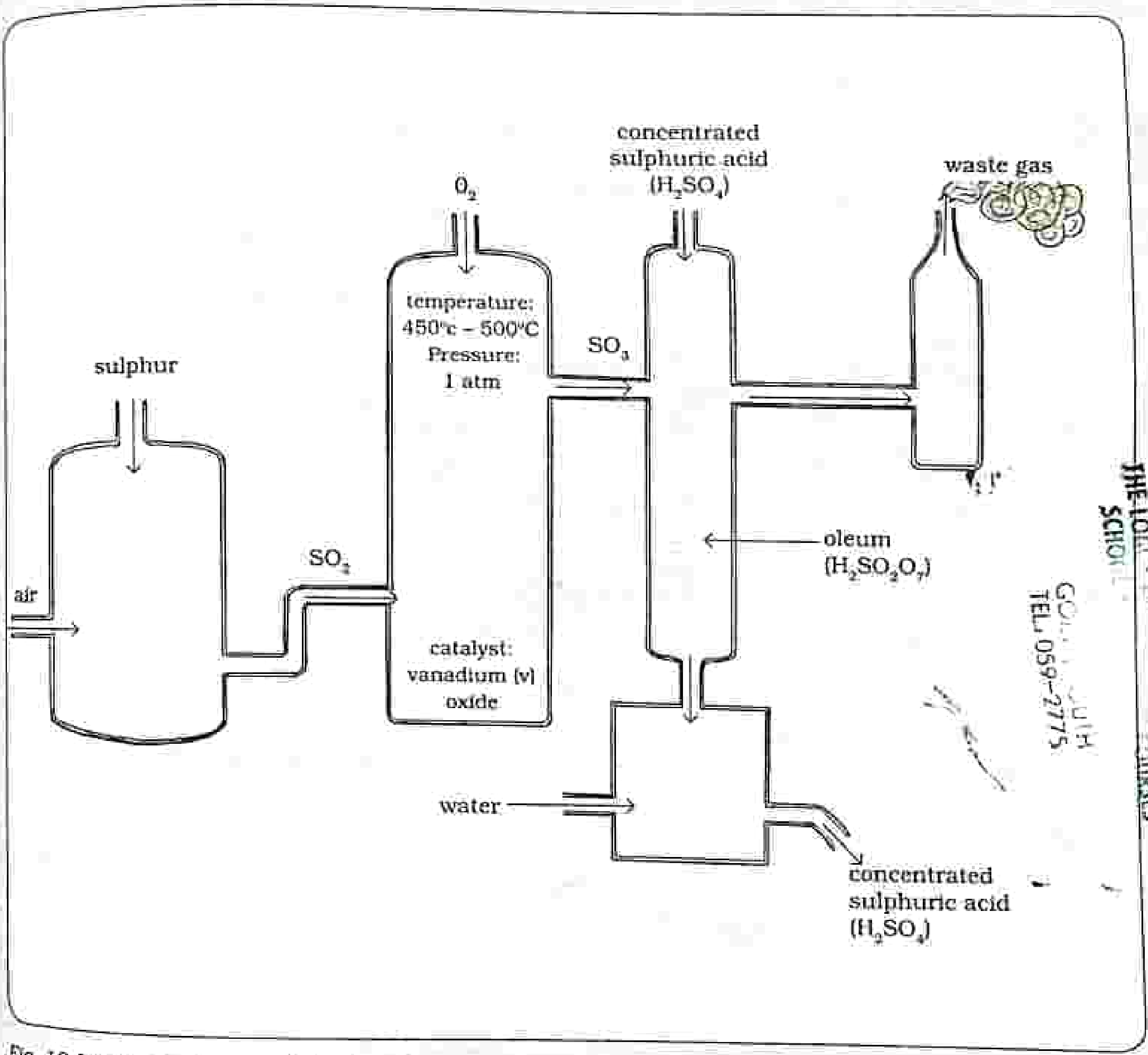


Fig. 18.2: Manufacture of sulphuric acid

Uses of sulphuric acid

- Battery acid
- Manufacturing plastics and paints
- Cleaning materials before electroplating
- Used in the manufacturing detergents
- Drying agent which absorb water
- Making paper

Production of ammonium nitrate through acid-base titration

Practical activity 18.1

Aim : **Production of ammonia through acid-base titration**

Apparatus :

1. burette
2. pipette
3. pipette filler
4. conical flask
5. ammonia solution
6. nitric acid
7. methyl orange pH indicator
8. bunsen burner
9. beaker
10. glass rod
11. clamp
12. white tile

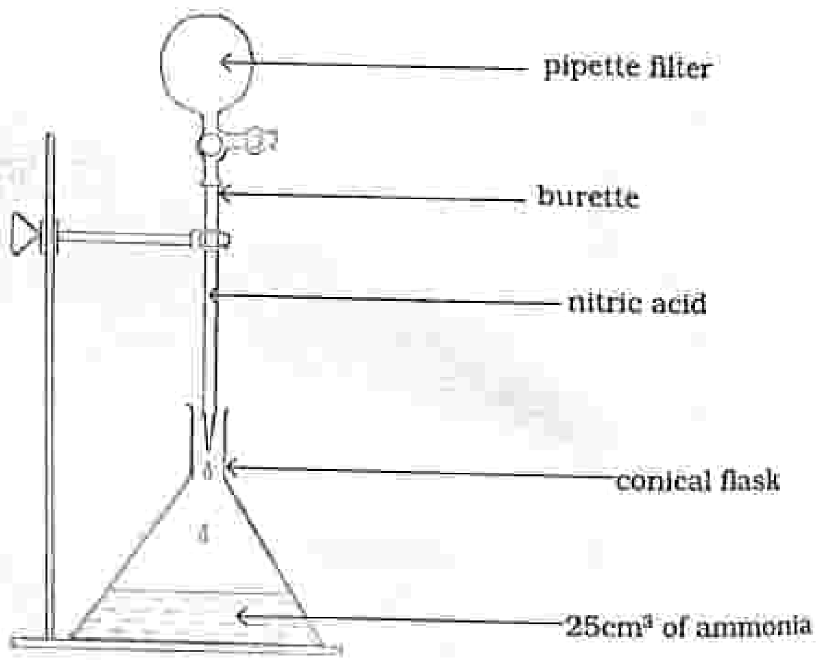


Fig 18.3: Production of ammonium nitrate

Method

1. Set up apparatus as shown in Fig 18.3.
2. Using the pipette and pipette filler, add $25cm^3$ of ammonia to the conical flask.
3. Put the conical flask on a white tile and add to the flask a few drops of methyl orange pH indicator.
4. Fill the burette with nitric acid and record the volume.
5. Slowly add nitric acid to the alkalis in the conical flask ensuring they mix well.
6. Stop adding the nitric acid when the indicator changes colour permanently (reaching the end point) to orange that is an indication of neutralization.

   Note: The volume required to neutralize the ammonia.
7. Mix a new solution of $25cm^3$ of ammonia and the volume of nitric acid noted in step 6.
8. Heat the solution in the beaker, stirring with a glass rod.
   - leave to cool

Results

- Water evaporates and a white substance remains in the beaker.

Conclusion

- The neutralisation reaction results in the formation of salt and water.
- After heating, the water evaporates and the white substance which remains in the beaker is ammonium nitrate.

Practical questions

1. What remains in the beaker when the water evaporates?
2. Name the type of reaction which forms ammonium nitrate?

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Identify raw material used to manufacture ammonia.
   A. hydrogen and nitrogen
   B. oxygen and nitrogen
   C. carbon dioxide and hydrogen
   D. sulphur dioxide and oxygen
2. Identify raw material used to manufacture sulphuric acid.
   A. hydrogen and nitrogen
   B. oxygen and nitrogen
   C. carbon dioxide and hydrogen
   D. sulphur dioxide and oxygen
3. What is the catalyst used in the manufacture of ammonia?
   A. Iron
   B. Vanadium (v) oxide
   C. Oxygen
   D. Sulphur
4. What is the catalyst used in the manufacture of sulphuric acid?
   A. Iron
   B. Vanadium (v) oxide
   C. Oxygen
   D. Sulphur
5. During the manufacture of ammonia, nitrogen and hydrogen are mixed in the ratio ________.
   A. 1:3
   B. 2:1
   C. 3:2
   D. 1:2
6. When manufacturing ammonia, a pressure of _______ is used.
   A. 20atm
   B. 1atm
   C. 200atm
   D. 15atm
7. When manufacturing sulphuric acid, a pressure of ____________ is used.
   A. 20atm
   B. 1atm
   C. 200atm
   D. 15atm

8. Production of ammonium nitrate is a ________ reaction.

A. Neutralisation
B. Oxidation
C. Cold
D. Hot

9. In the Contact process, sulphur trioxide is made from a ________.

A. Endothermic reaction
B. Neutralisation reaction
C. Reversible reaction
D. Cold reaction

10. In the neutralisation reaction of ammonia and nitric acid the salt produced is ________.

A. Ammonium
B. Ammonium nitrate
C. Ammonium sulphate
D. Sulphuric acid

Section B: Structured questions

1. List the raw materials used to manufacture ammonia. [2]
2. List the raw materials used to manufacture sulphuric acid. [2]
3. State the conditions needed for the production of
   a) Ammonia [3]
   b) Sulphuric acid [3]
4. State the uses of
   a) Ammonia [3]
   b) Sulphuric acid [3]
5. Briefly describe the manufacture of
   a) Ammonia [5]
   b) Sulphuric acid 5]

CHAPTER 19

Oxidation and reduction

Key concepts

- Rusting
  - Conditions necessary for rusting
  - Methods of preventing rusting
- Physical and chemical changes
- Oxidation and reduction
  - Reaction of copper oxide with hydrogen
- Iron extraction in the blast furnace
  - Raw materials used in the extraction of iron, their sources and function
  - Reactions in the blast furnace
  - How iron and slag separates
  - Heating iron (III) oxide on a charcoal block
- Alloys of iron
  - Percentage composition of alloys of iron
  - Uses and properties of alloys of iron

Summary notes

Rusting is the corrosion of iron in the presence of oxygen and water to give iron oxide.

Word equation

iron + oxygen + water ⟶ iron oxide (rust)

Conditions for rusting to occur:

1. Oxygen
2. Water (moisture)

Methods of preventing rusting

- **Painting** – prevents direct contact of iron with oxygen and water hence preventing rusting.
- **Galvanising** – is coating of iron (or steel) with a thin layer of another metal that does not react with oxygen for example zinc.
- **Plating** – coating a metal using another metal which is less reactive using the process of electrolysis.

Table 19.1: Physical and chemical changes

Physical change	Chemical change
No new substance is formed	New substance is formed
Properties do not change	New substance has different properties
Easy to reverse	Not easy to reverse
Mass of new substance is usually the same as starting mass	Mass of new substance is usually different from the starting mass
Heat changes do not usually occur	Usually heat changes occur
A mixture is formed	A compound is formed

Oxidation and reduction

Oxidation

- It is loss of electrons.
- It is gain of oxygen.
- It is loss of hydrogen.
- The reactant that loses electrons is being oxidized and the reducing agent is the substance that supplies electrons.

Reduction

- Is gain of electrons.
- Loss of oxygen.
- Gain of hydrogen.
- The reactant that gains electrons is being reduced and the oxidizing agent is a substance that accepts electrons.

Redox reactions

- These are **reduction-oxidation** reactions.
- Redox reactions are reactions where one reactant gains electrons whilst the other reactant loses electrons.
- Oxidation and reduction occur at the same time.

Example

Oxidation of iron (II) chloride

$$2FeCl_{2(aq)} + Cl_{2(g)} \longrightarrow 2FeCl_{3(aq)}$$

OR

$$2Fe^{2+}_{(aq)} + Cl_{2(g)} \longrightarrow 2Fe^{3+}_{(aq)} + 2Cl^{-}_{(aq)}$$

- During the oxidation of iron (II) chloride, iron (II) ions change to iron (III) by loss of electrons and at the same time chlorine changes to chlorine ions by gain of electrons.

$Fe^{2+} - e \longrightarrow Fe^{3+}$ (oxidation)

$Cl^{2} + 2e \longrightarrow 2Cl$ (reduction)

Reaction of copper oxide with hydrogen

Practical activity 19.1

Aim : To show the reaction of copper oxide with hydrogen

Apparatus : 1. Copper (II) oxide
2. Bunsen gas which contains hydrogen
3. Test tube with a small hole near the end
4. Spatula
5. Cork with a glass tube inserted
6. Rubber tube

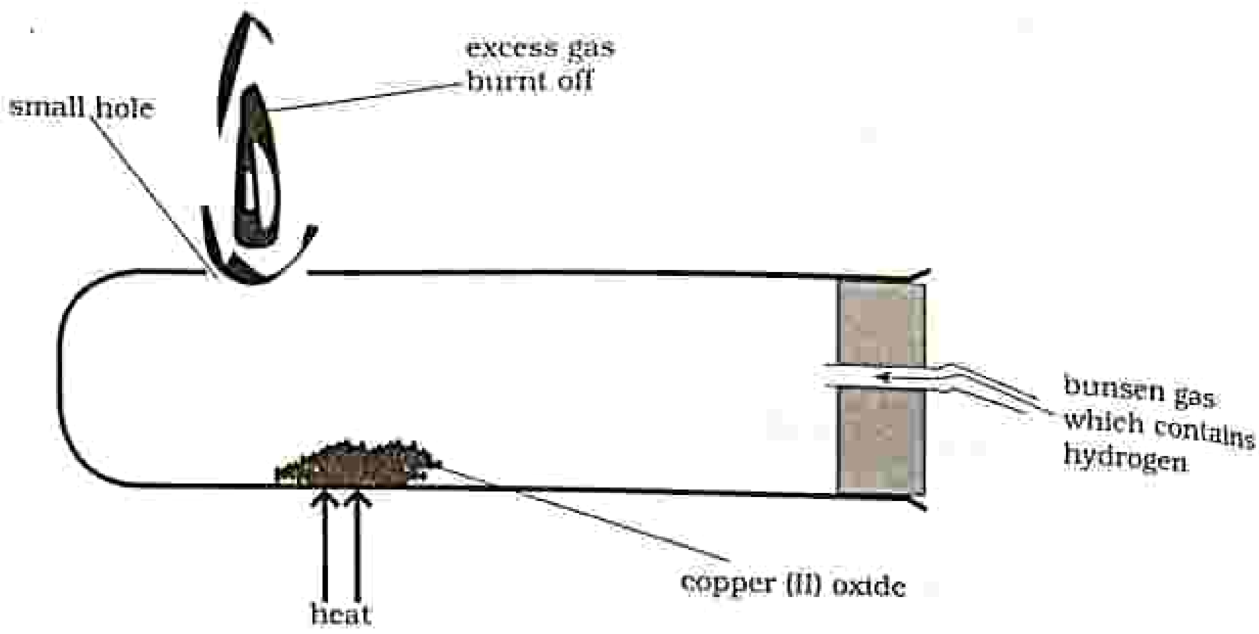


Fig 19.1: Reaction of copper (II) oxide by hydrogen

Method

- Set up apparatus as shown in Fig 19.1.
- Using a spatula, place copper (II) oxide in the test tube.
- Insert the cork on the test tube and using a rubber tube, allow a very slow stream of gas to flow into the test tube.
- Burn the excess gas at the small hole.
- Heat the copper (II) oxide until there is a colour change.
- Remove the heat and turn off the gas after a few more minutes.

Results

- Copper (II) oxide reacts with hydrogen gas to form copper metal and water.

Conclusion

- Hydrogen removes oxygen from copper oxide.
- The copper oxide is reduced and hydrogen is oxidized to form water.
- Hydrogen is the reducing agent, while copper oxide is the oxidizing agent.

$CuO_{(s)} + H_{2(g)} \longrightarrow Cu_{(s)} + H_2O_{(g)}$

Practical questions

1. Copper (II) oxide is ________ and hydrogen is ________. (Oxidized; Reduced)
2. Identify:
   a) the reducing agent
   b) the oxidizing agent

Iron extraction in the blast furnace

Table 19.2: Raw materials used in the extraction of iron and their function

Raw materials	Component	Function
Coke (carbon) C	Carbon C	For producing carbon monoxide which reduces iron (III) oxide
Limestone (calcium carbonate) $CaCO_3$	Calcium carbonate $CaCO_3$	To remove impurities by reacting with them to form slag
Hot air	Oxygen	Contains oxygen for the burning of coke to produce heat and carbon monoxide
Iron ore (haematite) Fe_2O_3	Iron (III) oxide	Source of iron

Reactions in the blast furnace

- **Formation of CO_2**: coke burn in air forming carbon dioxide at very high temperatures of about 1700°C.
  $C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} + \text{heat}.$

- **Formation of CO:** the carbon dioxide is reduced to carbon monoxide further up in the furnace.
  $CO_{2(g)} + C_{(s)} \longrightarrow 2CO_{(g)}.$

- **Reduction of Fe_2O_3**: carbon monoxide reduces iron oxide to molten iron
- This is a reversible reaction.
  $Fe_2O_{3(s)} + 3CO_{(g)} \rightleftharpoons 2Fe_{(l)} + 3CO_{2(g)}.$

- **Decomposition of $CaCO_3$**: limestone decomposes into calcium oxide
  $CaCO_{3(s)} \longrightarrow CaO_{(s)} + CO_{2(g)}.$

- **Formation of slag**: calcium oxide combines with impurities such as sand (silicon (iv) oxide, SiO_2) to form slag (calcium silicate)
  $CaO_{(s)} + SiO_{2(s)} \longrightarrow CaSiO_{3(l)}.$

NOTE: both iron and slag are molten so will drop to the bottom of the furnace.

- molten slag is less dense therefore will float on top of the iron and is tapped off separately.

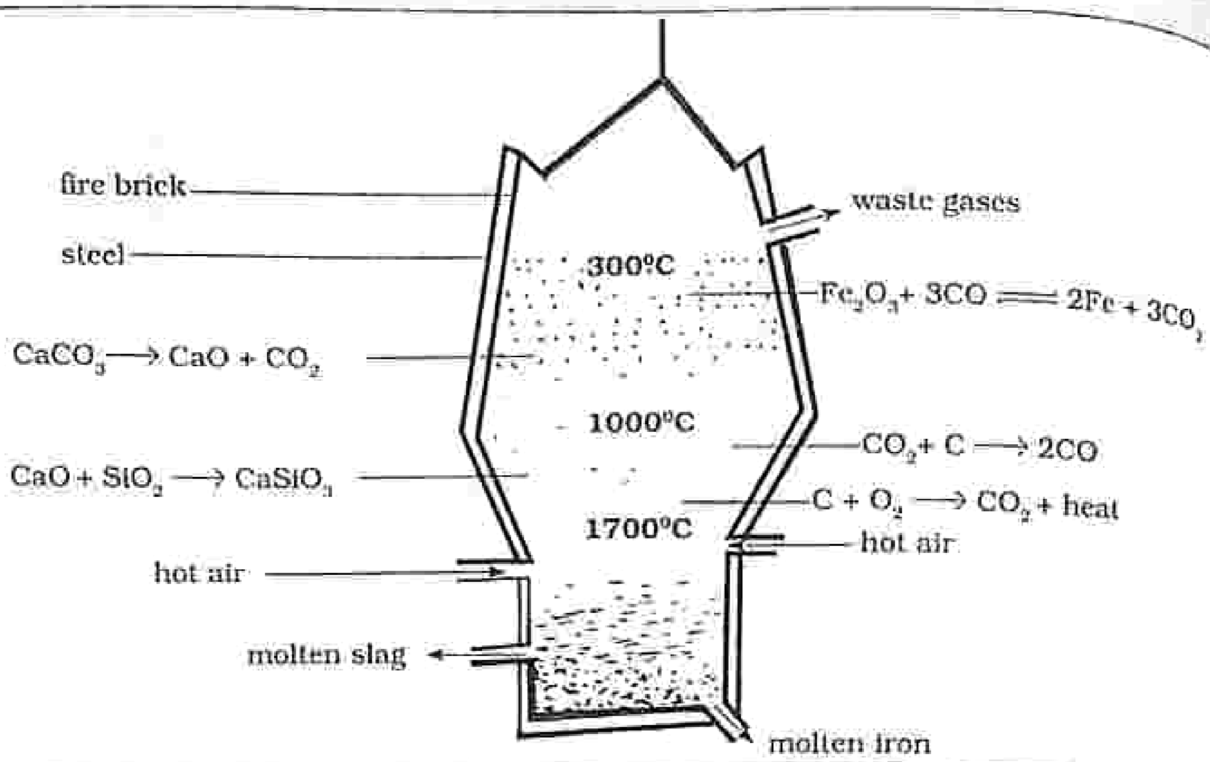


Figure 19.2: The blast furnace

How iron and slag separates

- Iron from the blast furnace is impure.
- Purification of iron is done in the oxygen furnace using the oxygen lance process.
- Impurities in iron are oxidised.
- Oxygen is blown over molten iron and impurities burn into gaseous oxides which escape easily.
- Pure iron is obtained.

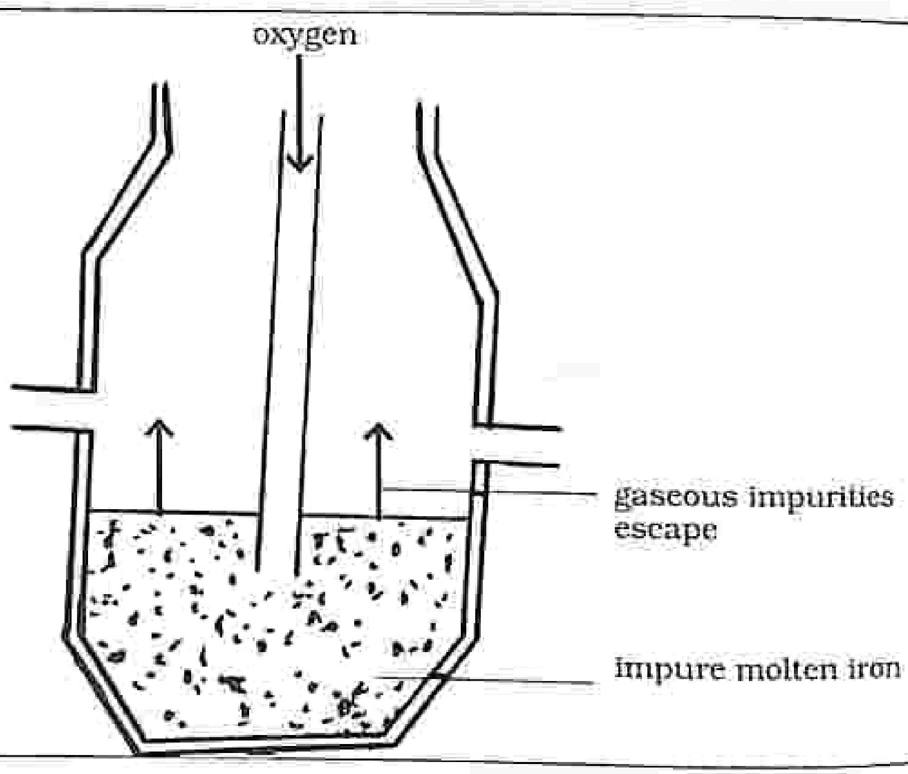


Fig. 19.3: Oxygen furnace

Heating Iron (III) oxide on a charcoal block

Practical activity 19.2

Aim : Heating iron (III) oxide on a charcoal block

Apparatus : 1. Charcoal block
2. Bunsen burner
3. Iron (III) oxide

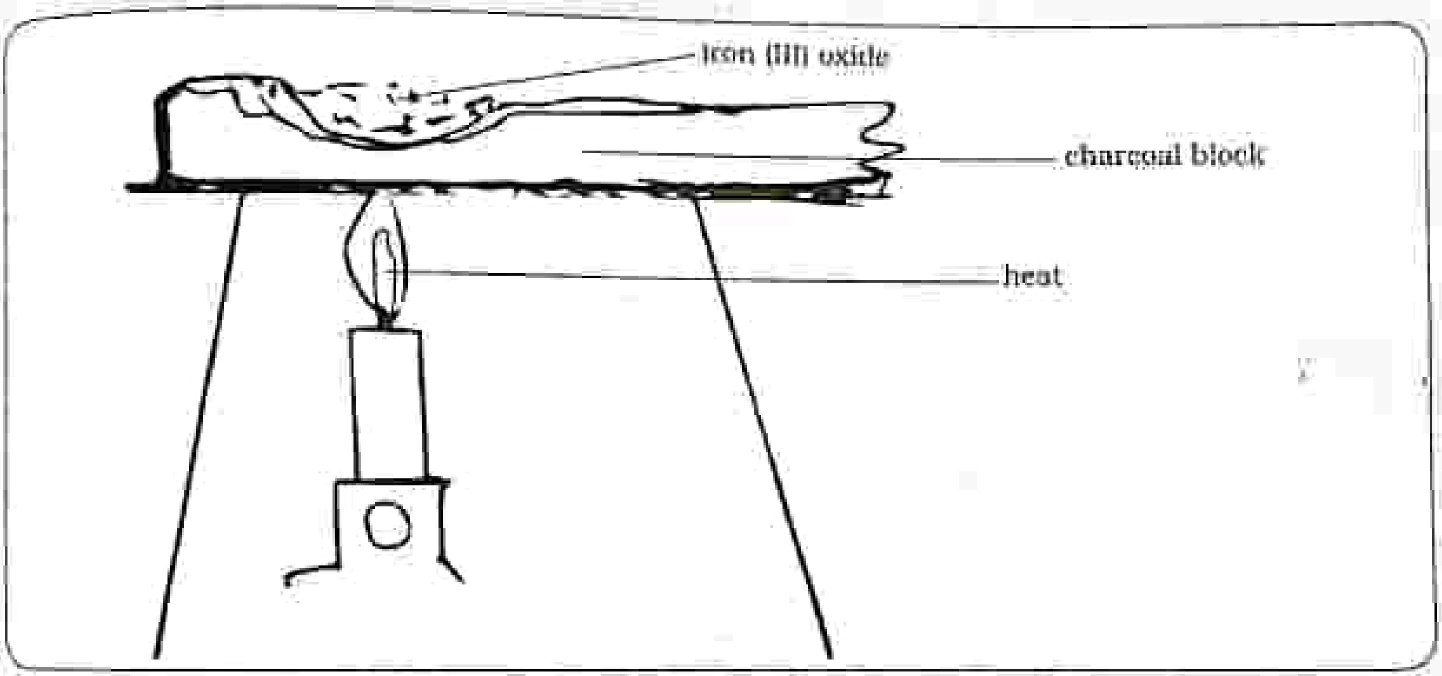


Fig 19.4: Heating iron (III) oxide on a charcoal block

Method

- Add iron (III) oxide to the charcoal block.
- Heat the charcoal block and iron (III) oxide until they glow orange.
- Allow to cool.

Results

- A silvery grey metal is produced which can be attracted to a magnet.

Conclusion

- Charcoal is a form of carbon.
- Carbon is more reactive than iron so it can displace iron from iron (III) oxide.
- Carbon reduces iron (III) oxide to iron and carbon is oxidized to carbon dioxide.

iron (III) oxide + carbon $\longrightarrow$ iron + carbon dioxide

$$2Fe_2CO_{3(s)} + 3C_{(s)} \longrightarrow 4Fe_{(s)} + 3CO_{2(g)}$$

Practical questions

1. Carbon reduces iron (III) oxide to _____.
2. Carbon is oxidized to ______.

Alloy

- It is a mixture of one metal with other metals and or with carbon.
- Each alloy has different properties.

Table 19.3 Alloys of iron

Alloy of iron	Composition	Properties	Uses
Mild steel	0.2% carbon	• malleable • ductile • can be welded • strong under both tension and compression	• railway line • beams • bolts and tools
Stainless steel	18% chromium 6% nickel	• resists heat and acids • resists rusting	• cutlery • sink units • stainless steel objects
Cast iron	up to 4% carbon	• cheaper than steel • hard but brittle • easy to mould • expands when it cools	• cast iron objects like gas stoves and railing • steel manufacture • drain pipes • domestic boilers

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. The two conditions necessary for rusting are ____________.
   A. soil and water
   B. water and hydrogen
   C. moisture and carbon dioxide
   D. oxygen and moisture
2. Plating of a metal to prevent rusting is done using the process of
   A. oxidizing.
   B. galvanizing.
   C. electrolysis.
   D. reduction.
3. Which one is not a way of preventing rusting?
   A. plating
   B. galvanising
   C. electrons
   D. painting
4. Which statement best describes oxidation?
   A. loss of electron, loss of oxygen and loss of hydrogen
   B. loss of electrons, gain of oxygen and loss of hydrogen
   C. loss of electrons, gain of oxygen and gain of hydrogen
   D. gain of electrons, loss of oxygen and gain of hydrogen
5. What is the function of limestone in the extraction of iron?
   A. reducing iron (III) oxide
   B. burning of coke
   C. removing impurities
   D. adding heat
6. Several reactions take place in the blast furnace; write the equation to show the formation of slag/
   A. $C(s) + O_{2(g)} \longrightarrow CO_{2(g)}$ + heat
   B. $CaO_{(s)} + SiO_{2(s)} \longrightarrow CaSiO_{3(l)}$
   C. $CO_{2(g)} + C_{(s)} \longrightarrow 2CO_{(g)}$
   D. $Fe_2O_{3(s)} + 3CO_{(g)} \longrightarrow 2Fe_{(l)} + 3CO_{2(g)}$
7. ____________ is an alloy of iron.
   A. cast iron
   B. limestone
   C. copper
   D. rust

8. Which alloy of iron is malleable and ductile?
   A. stainless steel
   B. cast iron
   C. mild steel
   D. steel
9. Purification of iron is done using ______________.
   A. electrolysis
   B. oxygen lance process
   C. blast furnace
   D. painting
10. When iron (III) oxide is heated on a charcoal block ______ reduces iron (III) oxide to iron.
   A. carbon
   B. oxygen
   C. carbon dioxide
   D. hydrogen

Section B: Structured questions

1. a) State the conditions necessary for rusting. [2]
   b) Explain methods of preventing rusting. [3]
2. a) Define:
      i) oxidation [1]
      ii) reduction [1]
      iii) redox reaction [1]
   b) In reduction, the substance that accepts electrons is known as ______ [1]
   c) In oxidation, the substance that supplies electrons is known as ______ [1]
3. List the function of each raw material used in the extraction of iron.
   a) limestone
   b) coke
   c) iron ore
   d) air [4]
4. a) Outline the reactions that take place in a blast furnace. [6]
   b) How is iron and slag separated? [3]
5. Complete the table on the alloys of iron

Alloy	Percentage composition	Properties	Uses
Mild steel			
Stainless steel			
Cast iron			

[9]

CHAPTER 20

Organic chemistry

Key concepts

- Forms of fuel
- Efficiency of different fuels
- Global warming
- Causes of global warming
- Deforestation
- Veld fire
- Combustion :
  - complete and
  - incomplete
  - combustion
  - lighting burners
- Hydrocarbons
- Members of the homologous series with 3 carbon atoms
  - Alkanes
  - Alkenes
- Drawing displayed structures
- Uses of methane, ethane, propane, ethane, propene
- Alcohols
- Ethanol
- Homologous series
- Structural formula of ethanol
- Production of concentrated ethanol
- Uses of ethanol
- Biogas production
- Factors affecting the production of biogas
- State the use of biogas

Summary notes

- **Fuel** is a compound, element or mixture which is burnt to obtain heat energy. Most fuels contain carbon and hydrogen as the important components.

Forms of fuels

- **Liquid fuel**
  - examples include diesel, petrol, paraffin, ethanol
  - fuels are obtained by fractional distillation of petroleum except for ethanol which is produced by the fermentation of sugar in Zimbabwe
- **Solid fuel**
  - examples include wood, coal, charcoal
  - used as a very cheap fuel
  - it is an efficient fuel because it has a low energy content and burns quickly
- **Gaseous fuel** – is a better fuel for use in the home and industries
  - it causes less pollution
  - Example is cooking gas

Efficiency of different fuels

- The efficiency of different fuels differs depending on how effective heat energy is produced during combustion.

- When a lot of carbon dioxide is released into the atmosphere this causes a rise in the atmospheric temperatures resulting in global warming.

Note : When hydro carbons burn there is need for the presence of plenty oxygen to prevent incomplete combustion.

Lighting Bunsen burner

Wide sleave

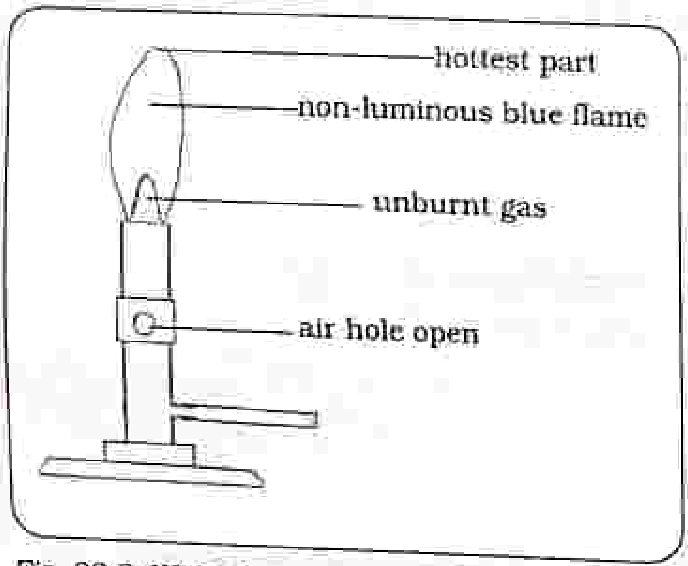


Fig. 20.2: Wide sleeve

- Blue flame which is very hot and clean

Narrow sleeve

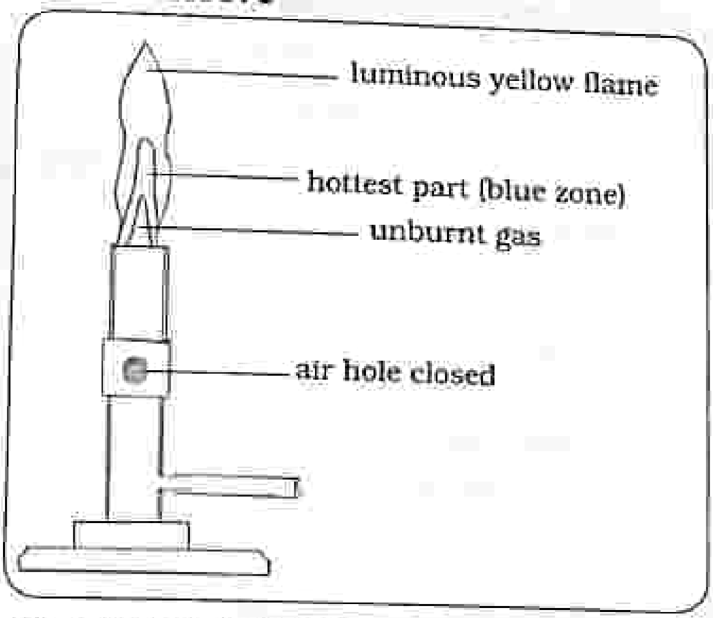


Fig. 20.3: Narrow sleeve

- Yellow flame
- Produces a lot of soot
- Not very hot

Lighting a methylated spirit burner

Long wick

- Produces more soot as it will draw more fuel but lasts long.

Short wick

- Burn more efficiently giving a clean flame but lasts for a short time.

Hydrocarbons

- These are organic compounds which contain hydrogen and carbon only.

Members of the homologous series with 3 atoms

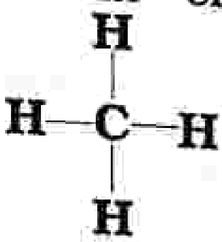
- A homologous series is a series of compounds with the same functional group and chemical properties but differ on the number of repeating units they contain.
- **Alkanes:** are organic compounds that consist of a single bond between their carbon and hydrogen atoms.
- Are hydrocarbons with the formula. C_nH_{2n+2}

Examples are :

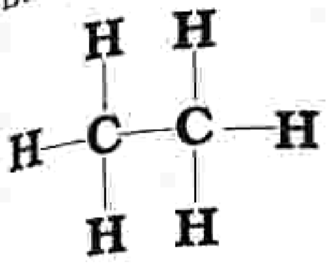
Methane	CH_4
Ethane	C_2H_6
Propane	C_3H_8

Displayed structure

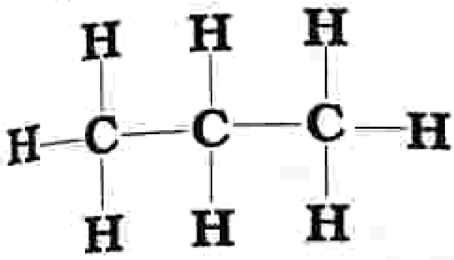
Methane CH_4



Ethane C_2H_6



Propane C_3H_8



Alkenes

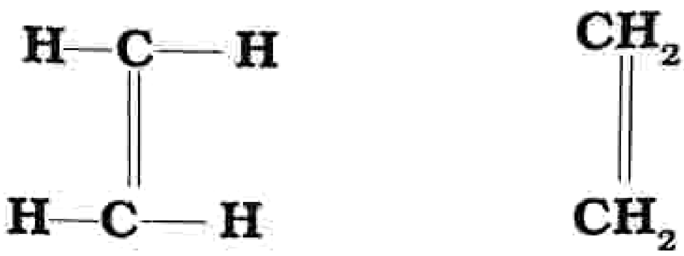
- They contain at least one double bond between the two carbon atoms.
- They are unsaturated compounds.
- Are hydrocarbons with the general formula C_nH_{2n}
- n is 2 or more

Examples are :

1. Ethene C_2H_2
2. Propene C_3H_6

Displayed structure

Ethene C_2H_2



Propene C_3H_6

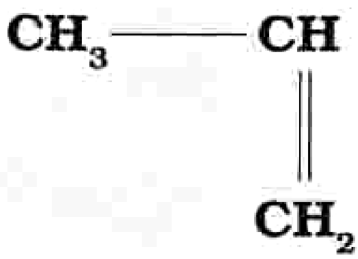


Table 20.1 Uses of hydrocarbons

Hydrocarbon	Use
Methane	• is a component of natural gas which is used as a fuel for heating both in the industries and homes • methane decomposes when heated at high temperatures to hydrogen and carbon black • used to make organic chemicals • carbon black is used to make inks, paints and carbon papers
Ethane	• fuel used in heating • used for the production of ethylene which is used for making plastics and detergents • used as a fuel for residential and commercial heat applications
Propane	• also used as vehicle fuel • used as a refrigerant

Hydrocarbon	Use
Ethene	• when ethene is heated at very high pressure, polythene is formed which is a very good electrical insulator used in making cables • polythene is used for making bottles and plastic bags because it resists acids and alkalis • used in the production of ethylene gycol which is used as an automotive antifreeze agent • used for the production of polypropylene
Propene	• polypropylene is used in the production of furniture and textile

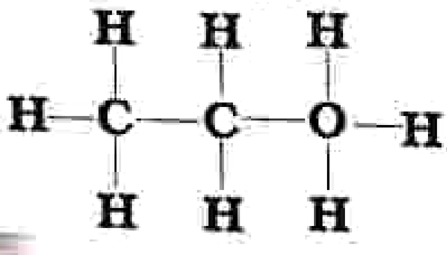
Alcohol

- Alcohols belong to the hydroxyl group (-OH) joined to carbon and hydrogen
- General formula for alcohols is $C_nH_{2n+1}OH$

  Ethanol is an alcohol which belongs to the hydroxyl group joined to carbon and hydrogen.

Displayed structural formula of ethanol

Ethanol: $\mathbf{C_2H_5OH}$



Production of ethanol

- Ethanol is a product of the fermentation process of green plants.
- Green plants produce sugars which can be converted to ethanol using yeast.

Word equation

Glucose $\xrightarrow{\text{yeast}}$ ethanol + carbon dioxide

Practical activity 20.2

Aim : Fermenting sugar solution or maize meal solution (mahewu)

Apparatus :
1. Conical flask
2. Test tube
3. Thermometer
4. Limewater
5. Cork
6. Glass tube
7. Maize meal
8. Yeast
9. Glucose solution

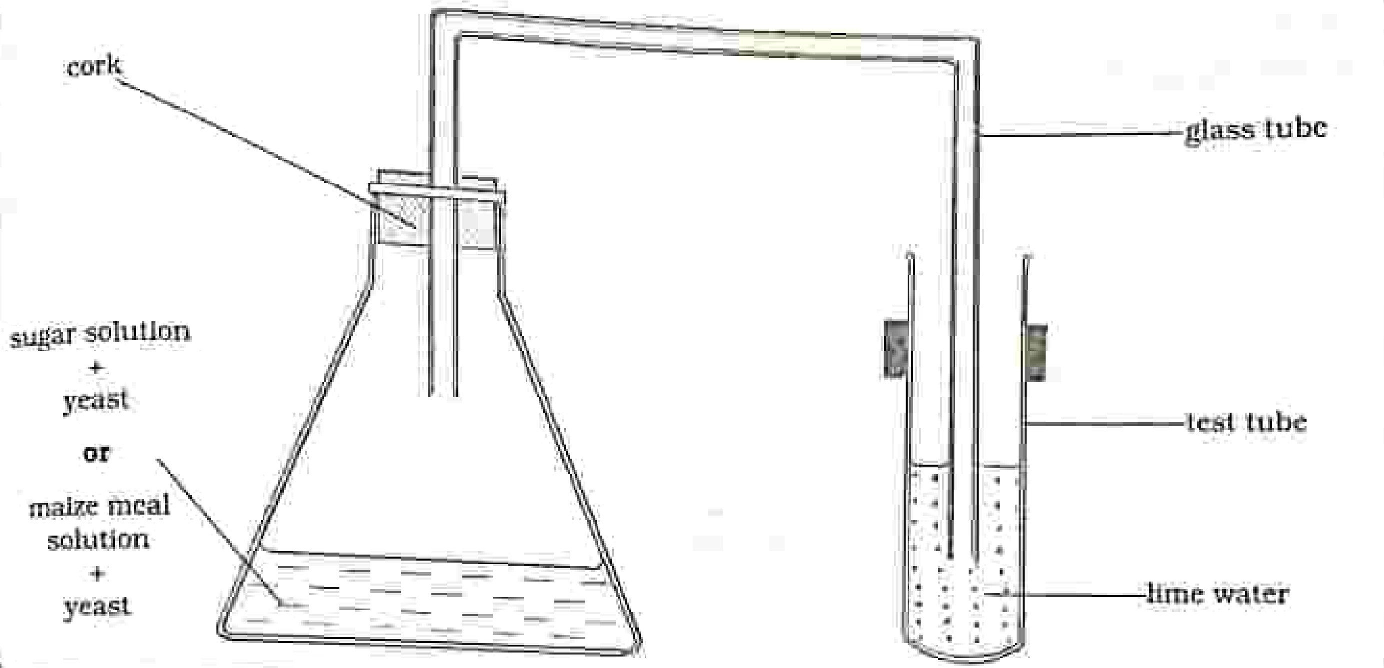


Fig. 20.4: Fermentation of sugar solution or maize meal solution (mahewu)

Method

Part A

1. Set up apparatus as shown in Fig 20.4
2. **Using glucose solution**
   - In the conical flask add about 50cm^3 of 5% glucose to the conical flask and then add one spatula measure of yeast.

Or

Using maize meal

- Mix a few teaspoons of mealie meal with water and put in a conical flask and then add one spatula measure of yeast.

3. Cork the flask and allow the end of the glass tube to dip in the limewater as shown in Fig 20.4.
4. Leave in a warm place with temperatures of 30°C-35°C for a few days.
5. Observe any changes in the limewater.
6. Smell the contents of the flask.

Result

- Limewater turns milky.
- Contents have a strong smell.

Conclusion

- When glucose solution or maize meal solution is acted upon by the enzyme zymase in yeast, it is converted to ethanol and carbon dioxide is also produced.

Role of yeast

- Yeast contains an enzyme, zymase which acts as a catalyst to aid in the breakdown of glucose or maize meal into smaller molecules.

Word equation

Glucose $\xrightarrow{\text{zymase}}$ Ethanol + Carbon dioxide

OR

Maize meal solution $\xrightarrow{\text{zymase}}$ Ethanol + Carbon dioxide

PART B

Fractional distillation of ethanol

- Fractional distillation is used to separate ethanol from the fermented sugar or maize meal solution.
- Concentrated ethanol is obtained from fractional distillation.

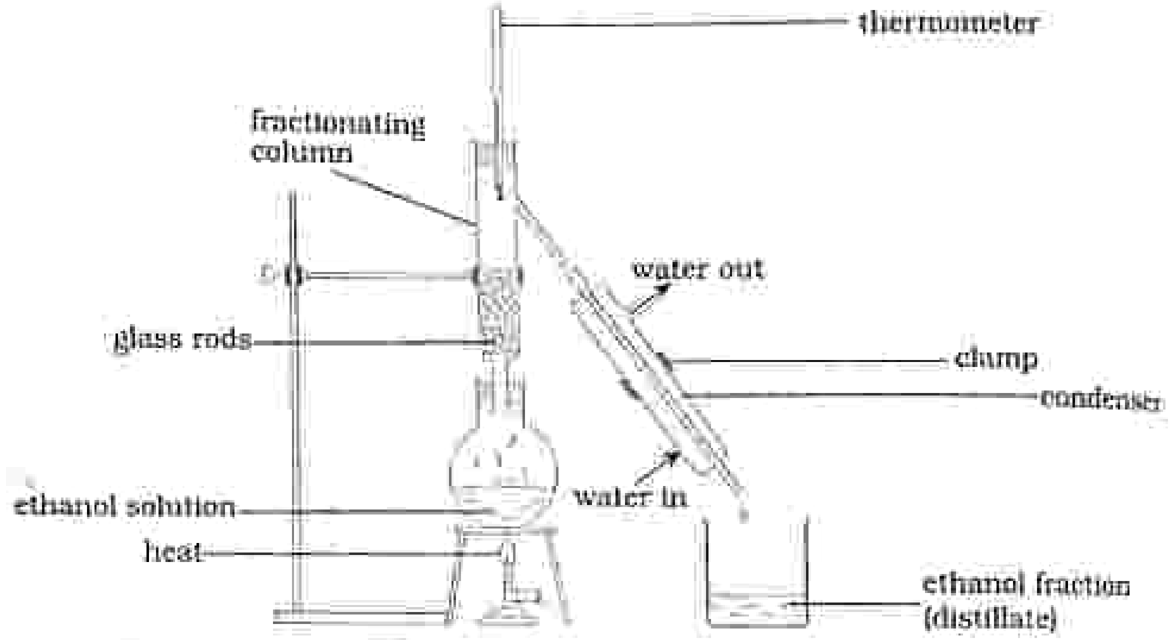


Fig. 20.5: Fractional distillation of ethanol

Method

1. Set up apparatus as shown in Fig 20.5.
2. Decant any solid residue from the ethanol obtained in part A.
3. Pour the solution into a large round bottomed flask fitted with a fractionating column.
   - The solution should not be more than a third full in the flask.
4. To ensure even heating, add a few pieces of broken porcelain or a little sand.
5. Try to ignite a few drops of the solution on a watch glass before heating.
   - If the ethanol concentration is strong enough, it will burn.
6. Heat the solution and collect the fractions of ethanol coming off at temperatures 78°C, 82°C, 86°C, 90°C and 94°C.
7. Ignite each fraction collected at (6) on a watch glass or tin lid to see if it will burn.
8. Record your results in the table.

Results

Temperature	78°C	82°C	86°C	90°C	94°C
Easy to ignite	Yes	Fairly	Fairly	No	No

Conclusion

- Concentrated ethanol is collected at 78°C.
- Concentrated ethanol is easier to ignite and burn.

Table showing proportions of water and alcohol at different temperatures

Boiling point (°C)	Percentage of ethanol (%)
78	86
82	79
86	72
90	62
94	44

Practical questions

1. What is the purpose of lime water?
2. Which fraction burns very well?

Uses of ethanol

- Beverage
- It is used to manufacture alcoholic drinks.
- Medical purpose
- Used as a solvent in the production of medical drugs.
- Fuel
  - Ethanol is blended with petrol for use as a fuel.
  - It is also used in methylated spirit which is a fuel used in spirit burners.
- Solvent
  - Ethanol is used as a solvent for organic compounds for example in dyes and plant pigments.

Biogas

- Biogas is the gas produced from the fermentation of organic waste that is animal and human waste.
- It is a mixture of gases such as methane, hydrogen, carbon dioxide, carbon dioxide and sulphide.

Biogas production

- A biogas digester as shown in Fig 20.6 is constructed.

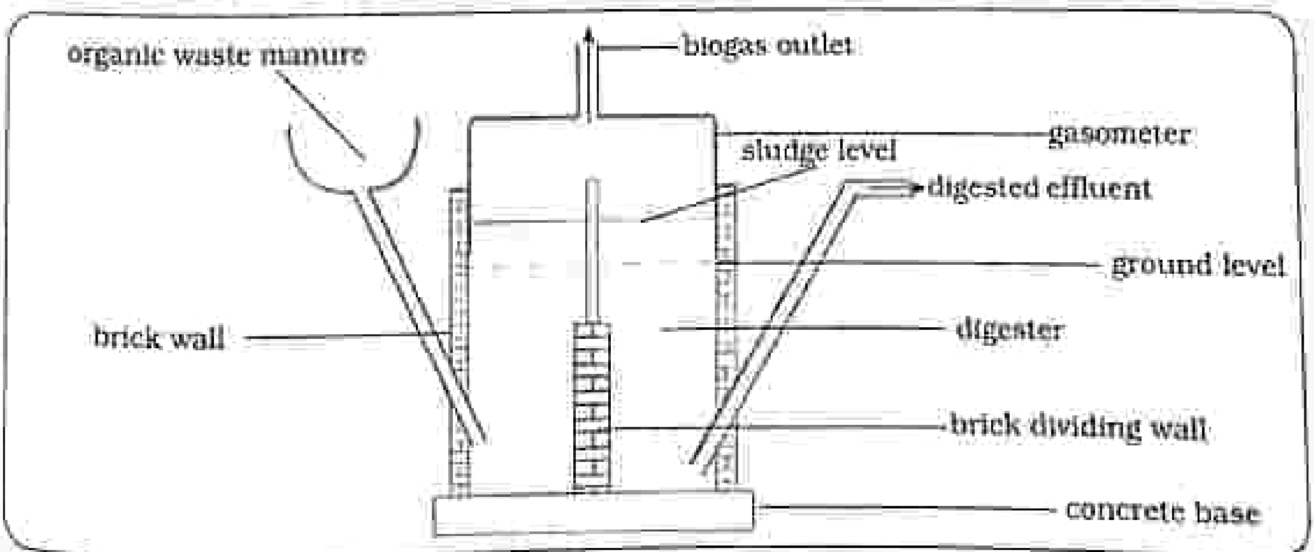


Fig. 20.6: Bio digester

- Put manure in a sealed pit.
- Manure is digested by anaerobic bacteria (in the absence of oxygen).
- The floating gasometer tank is rotated several times a day in order for the sludge in the pit to be agitated.
- Build up sludge is removed and the process is started again.
- A gas is given off which is collected and piped to burners.

Word equation

Organic waste $\xrightarrow{\text{fermentation}}$ methane + carbon dioxide

Note:

- Sludge from the production of biogas can be used as a fertilizer.

Factors affecting the production of biogas

- **Role of yeast**
  - The bacteria will function in an oxygen free environment (anaerobic bacteria) so that organic waste is fermented to produce biogas.
- **Temperature**

An optimum temperature of 35°C to 55°C is used.

- **pH**
  - Solution should not be too acidic or too alkaline.
- **Type of organic waste**
  - The type of organic waste used will affect the biogas produced.

Uses of biogas

- Biogas is used for lighting.
- Biogas is used for cooking.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Identify the forms of fuel
   A. Carbon, Oxygen, Hydrogen
   B. Liquid, Gaseous, Solid
   C. Liquid, Oxygen, Carbon
   D. Gas, Hydrogen, Carbon
2. The most efficient form of fuel is
   A. Gaseous B. Solid
   C. Liquid D. Oxygen
3. Complete combustion is ___________.
   A. the burning of fuel in the presence of plentiful hydrogen to give energy
   B. the burning of fuel in the presence of plentiful water to give energy
   C. the burning of fuel in the presence of plentiful carbon dioxide to give energy
   D. the burning of fuel in the presence of plentiful oxygen to give energy
4. Identify the products of complete combustion
   A. Carbon dioxide + water vapour + Heat
   B. Carbon monoxide + water vapour + Heat
   C. Oxygen + water vapour + Heat
   D. Hydrogen + water vapour + Heat
5. Hydrocarbons are organic compounds which contains
   A. Hydrogen and Carbon only
   B. Oxygen and Carbon only
   C. Hydrogen and Oxygen only
   D. Nitrogen and Carbon only
6. Identify the gas produced in the fermentation process
   A. Carbon dioxide
   B. Oxygen
   C. Hydrogen
   D. Nitrogen

7. What is the role of yeast in the fermentation process?
   A. Give a sweet taste
   B. Add colour
   C. Contains an enzyme zymase which act as a catalyst
   D. Change the smell of the contents
8. Biogas is produced by fermenting
   A. Organic waste B. Oxygen
   C. Yeast D. Carbon dioxide
9. Identify an alkane
   A. Propene
   B. Ethene
   C. Ethane
   D. Ethanol
10. Ethanol belongs to the
   A. Alkenes
   B. Alkanes
   C. Hydroxyl group
   D. Ethane

Section B: Structured questions

1 a) Identify factors affecting the production of biogas. [4]
  b) State the uses of biogas. [2]
2. a) Define the term hydrocarbon. [1]
  b) Draw the displayed structures of
    i) methane
    ii) ethane
    iii) propane [6]
3. a) Define global warming. [1]
  b) List the causes of global warming. [3]
4. a) Draw the displayed structural formula of ethanol. [2]
  b) Describe the production of concentrated ethanol. [5]
  c) List uses of ethanol.
5. a) Draw the displayed structures of
    i) ethene
    ii) propene [4]
  b) State the uses of
    i) ethene
    ii) propene [2]

CHAPTER 21 Data presentation

Key concepts

- Tallies
  - make a tally and interpret data
- Tables
  - make a table and interpret data
- Bar graphs
  - construct a bar graph and interpret data
- Pie chart
  - construct a pie chart
  - interpret and analyse data
- Line graph
  - construct a line graph
  - interpret and analyse data
- Straight line graph
  - construct a straight line graph and interpret data

Summary notes

- Data is information and the information obtained from experiments and investigations must be presented in a way which makes it easy to understand.
- Data can be presented in different forms.

Tallies

- A tally is a quick and accurate method of counting and recording data.
- It shows how often something happens and that is the frequency.
- It can be recorded in form of a table with a heading.

Making a tally

1. Make four vertical lines and then one horizontal or diagonal line which cuts across the four vertical lines. This shows five counts.
2. Leave a space and repeat step (1) until all the data has been captured.

Table 21.1: Making a tally of shoe size for learners in Form 1Y

Shoe size	Tally	Frequency
4	III	3
5	IIII̶	5
6	IIII̶ IIII	9
7	IIII̶ IIII̶ IIII	14
8	IIII̶ I	6
9	II	2

Table

- A table makes data easy to read and understand.

Making a table

1. Give a heading to the table.
2. Make labeled rows and columns.

Table 21.2: Favourite sports for learners in Form 2X

Favourite sport	Number of learners
Hockey	2
Tennis	9
Soccer	15
Netball	7
Rugby	3

Bar graph

- It is an easy way to show how often something happens.
- Bar graphs are used when the data collected is in groups or categories.
- They can be constructed from information presented in a table.

Constructing a bar graph

1. Draw the horizontal axis(X-axis) and the vertical axis(Y-axis).
2. Label the axis.
3. Groups or categories are always plac on the X-axis.
4. Find a suitable scale.
5. Use equal spaces for each category group and draw bars.
6. Using information from Table 21 construct a bar graph.

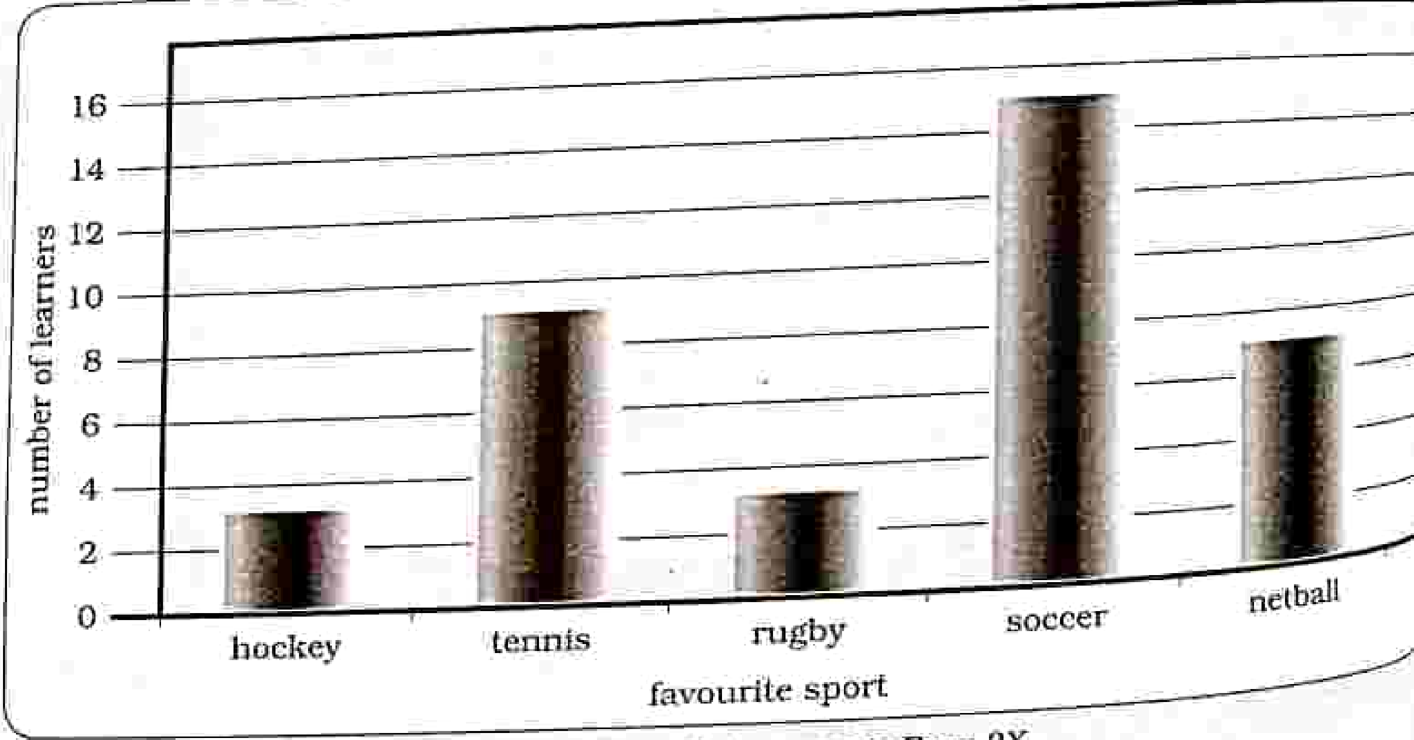


Fig 21.1: Bar graph showing the favourite sport for learners in Form 2X

Pie chart

- It is a type of graph in which a circle is divided into sectors.
- Each sector represents a portion of a whole.

Constructing a pie chart

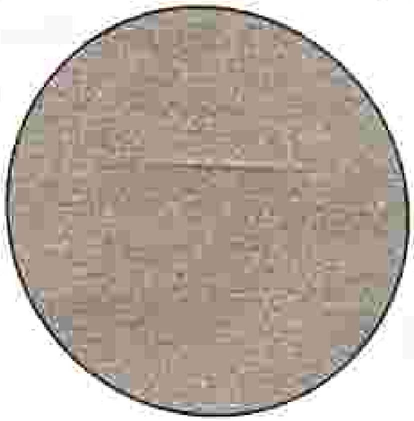
Use table 21.2 to:

1. Find the central angle for each component and the percentage of each learner.
2. Construct a pie chart

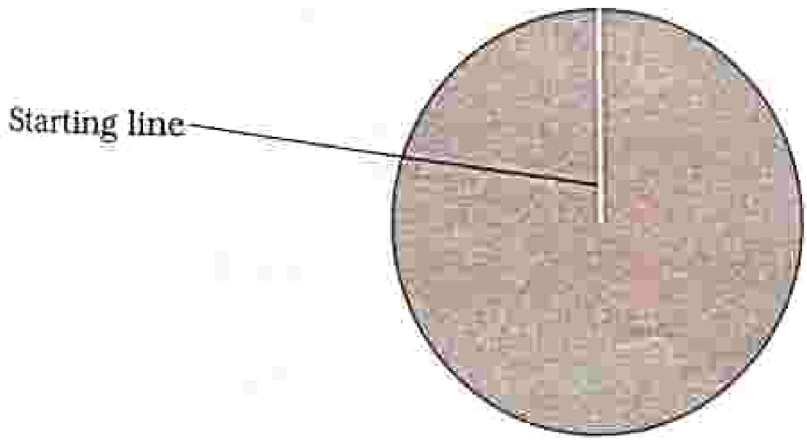
Table 21.3: Central angle and percentage of each learner

Favourite sport	Number of learners	Measure of central angle(°)	Percentage (%)
Hockey	2	2/36 × 360 = 20	2/36 × 100 = 6
Tennis	9	9/36 × 360 = 90	9/36 × 100 = 25
Soccer	15	15/36 × 360 = 150	15/36 ×100 = 42
Netball	7	7/36 × 360 = 70	7/36 × 100 = 19
Rugby	3	3/36 × 360 = 30	3/36 × 100 = 8
Total	36	360	100

2. a) Draw a circle of any radius using a mathematical campus.



b) Draw a radius line from the centre of the circle upwards to intersect the circle.



c) Starting from the radius line drawn in 2(b), draw radii using a protractor making central angles corresponding to the values of the respective components in a clockwise direction.

d) Repeat the process for all the components of the data.
The end of one sector is the start of the next sector.

NOTE:

- At the beginning of each new sector, the protractor must be at 0°.
- Start with the sector which has the largest central angle and end with the sector which has the smallest central angle.

e) Shade each sector with different colours.

f) Use a key to explain the category represented by each colour.

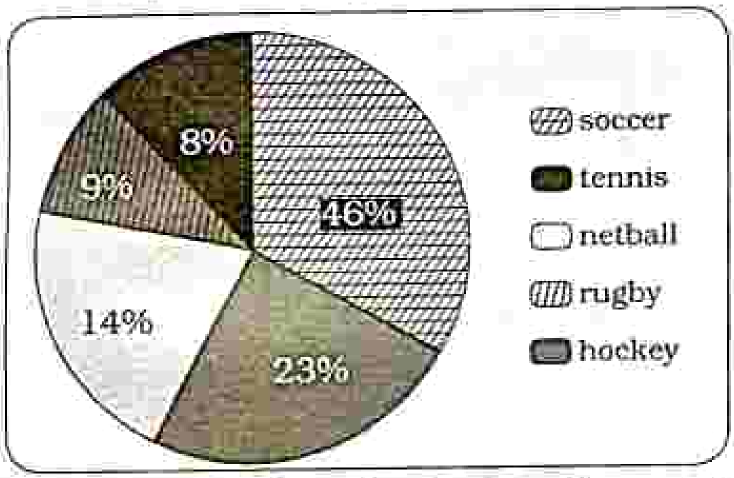


Fig. 21.2: A pie chart showing favourite sport of learners in Form 2X

- The most popular sport is soccer (42%) and the least popular sport is hockey (6%).
- 25% of the learners like tennis and 8% like rugby.

Line graph

- It is a graph with points which are connected by lines.
- A line graph shows the relationship between two variables showing how things change over time.
- The independent variable is plotted on the horizontal axis (X-axis) and the dependent variable is plotted on the vertical axis (Y-axis).

Constructing a line graph

1. Give the graph a title.
2. Identify the independent and dependent variable.
3. Find a suitable scale for the axis.
4. Draw a line for the horizontal axis (X-axis) and the vertical axis (Y-axis).
5. Label the axis
6. Plot the data points and join them using straight lines from left to right.

Table 21.4 below shows the results collected by learners after measuring the height of a pea plant for two weeks everyday.

Table 21.4: Favourite sports for learners in Form 2X

Day	Height (cm)
1	–
2	–
3	–
4	–
5	0.5
6	1
7	1.5

Table 21.4: Favourite sports for learners in Form 2X

Day	Height (cm)
8	2.0
9	3.0
10	4.0
11	5.5
12	7.0
13	10.0
14	14.0

Line graph to show growth of the pea plant

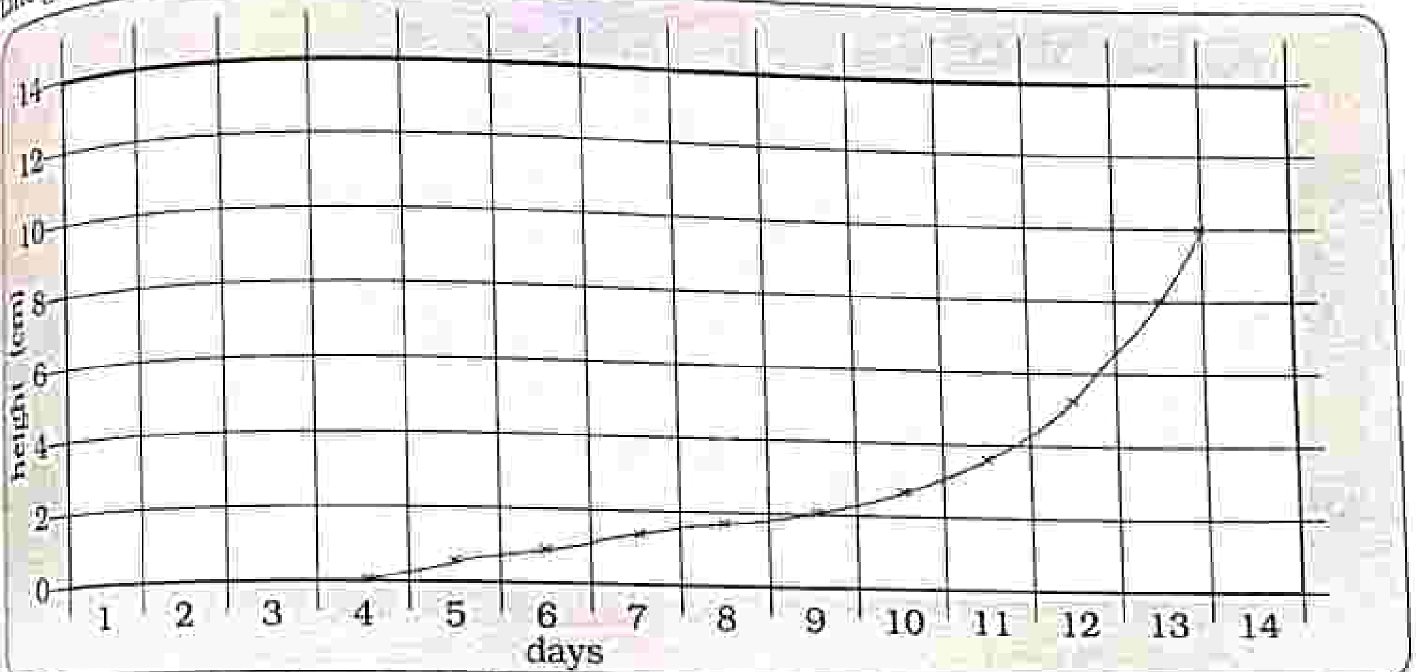


Fig. 21.3: Line graph to show growth of the pea plant

Scale: X – axis 1cm represent 1 unit

Y – axis 2cm represents 2 units

Straight line graph

- It is a graph with points which are connected by one straight line.

Constructing a straight line graph

- Give the graph a title.
- Identify the independent and dependent variable.
- Find a suitable scale for the axis.
- Draw the horizontal axis (X – axis) and the vertical axis (Y – axis)
- Label the axis.
- Plot all the points and join them using a clear plastic rule.
  - Make a thin and clear line.
  - Draw a line of best fit taking note of a balance of points about the line.
- Construct a straight line graph from the results on the table below showing temperature changes.

Table 21.5: Time and temperature changes

Time (minutes)	Temperature (°C)
0	20
2	30
4	40
6	50
8	60
10	70
12	80
14	90
16	100

Straight line graph showing temperature changes

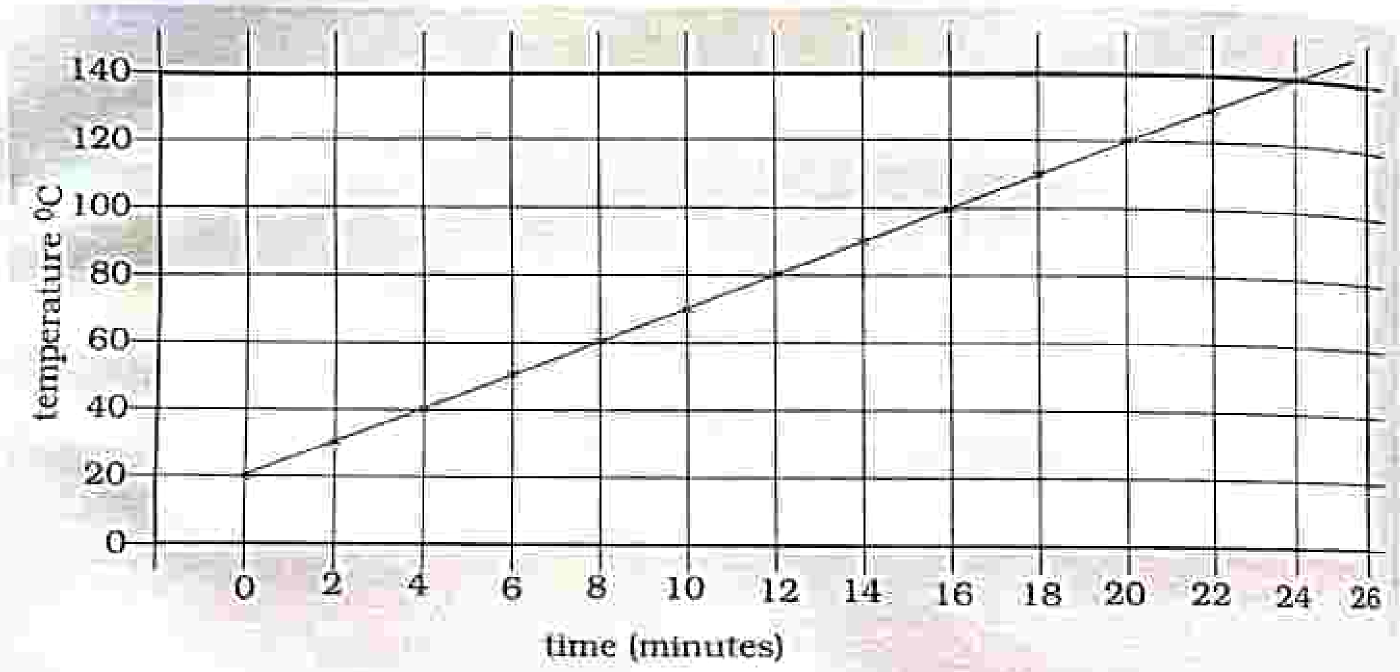


Fig. 21.4: Straight line graph showing temperature changes

Scale: X – axis 2cm represents 2 units

Y – axis 2cm represents 20 units

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Which one is a method of presenting data?
   A. Neutralisation
   B. Pie chart
   C. Physical change
   D. Chemical reaction

2. When making five counts in a tally, you make how many vertical lines and horizontal lines?

	Vertical lines	Horizontal lines
A.	4	2
B.	5	0
C.	4	1
D.	3	2

3. A table consists of
   A. Rows and columns
   B. Y-axis and X-axis
   C. Plotted points
   D. Bars

4. Use the information from the table below to find the number of learners who wear shoe size 4 to 6.

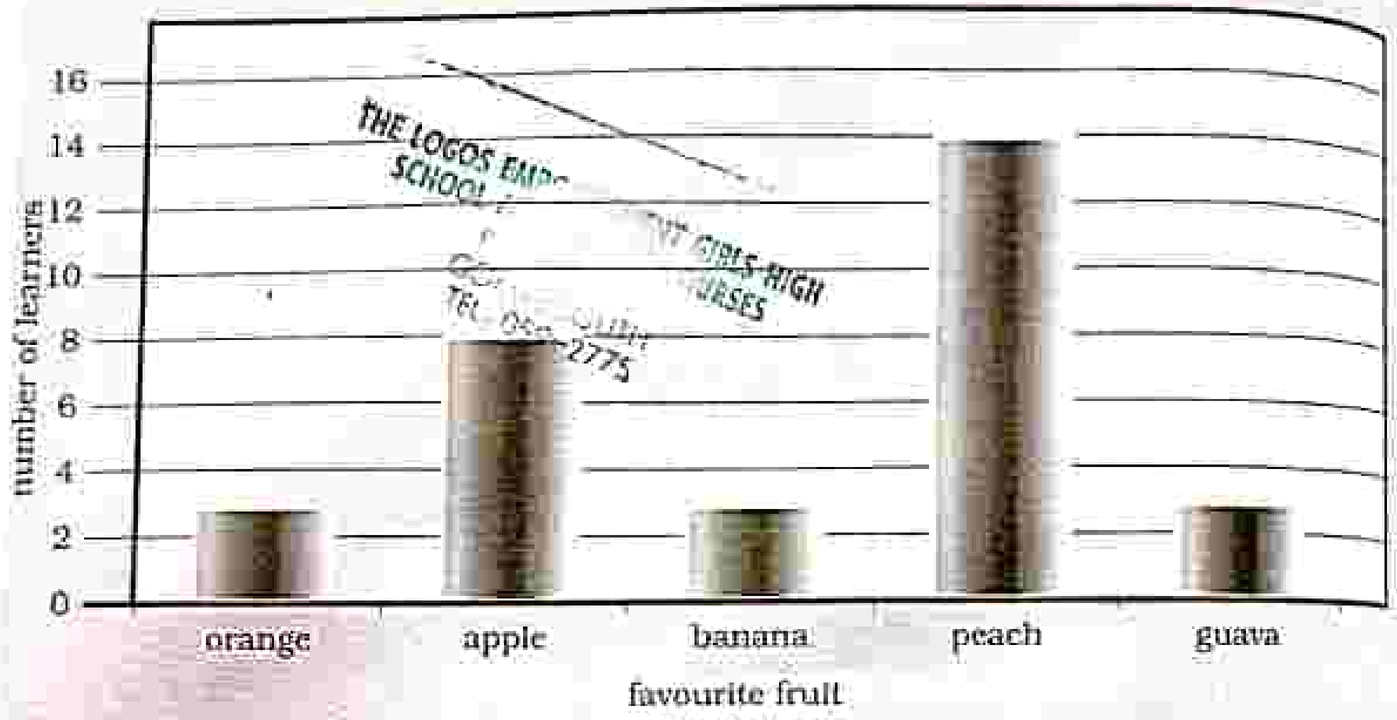
Shoe sizes for learners in Form 2P

Shoe size	Tally	Frequency
4	IIII	4
5	IIII II	7
6	IIII IIII	9
7	IIII IIII IIII	14
8	IIII II	7
9	I	1

A. 15
B. 10
C. 20
D. 12

5. Identify the type of method which uses plotted points for presenting data
   A. Line graph
   B. Pie chart
   C. Tally
   D. Straight line graph

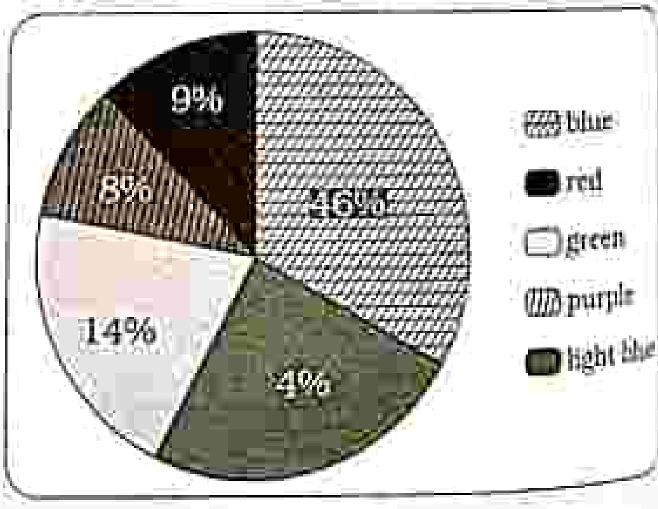
6. The diagram below is a bar graph showing the favourite fruits for learners in Form 4Z. Identify the most favourite fruit for the learners?
   A. Orange B. Banana
   C. Apple D. Peach



7. Using Bar graph in question (6), how many learners are in Form 4Z?
   A. 40
   B. 35
   C. 36
   D. 39

8. A pie chart ....
   A. consists of labeled rows and columns.
   B. is a type of graph in which a circle is divided into sectors.
   C. is a graph with points which are connected by lines.
   D. is a graph with points which are connected by one straight line.

9. Identify the type of method used to present data in the diagram below.



A. Line graph
B. Pie chart
C. Tally
D. Straight line graph

). Using the graph in question 9 identify the most favourite colour?

A. Purple

B. Blue

C. Red

D. Green

Section B: Structured questions

Use the information given below to answer the questions.

Shoe sizes for learners in Form 1Y

Shoe size	Tally	Frequency
4	-	3
5	-	5
6	-	9
7	-	14
8	-	6
9	-	2

a) Complete the table by making a tally. [3]

b) Use the information from the table to construct a bar graph. [4]

Construct a straight line graph from the results on the table below showing the length of a growing plant which was measured daily.

Number of days	Leaf length (mm)
1	5
2	10
3	15
4	20
5	25
6	30
7	35
8	40
9	45

[5]

The table below shows results obtained when a group of students were asked why they chose the university they are enrolled in.

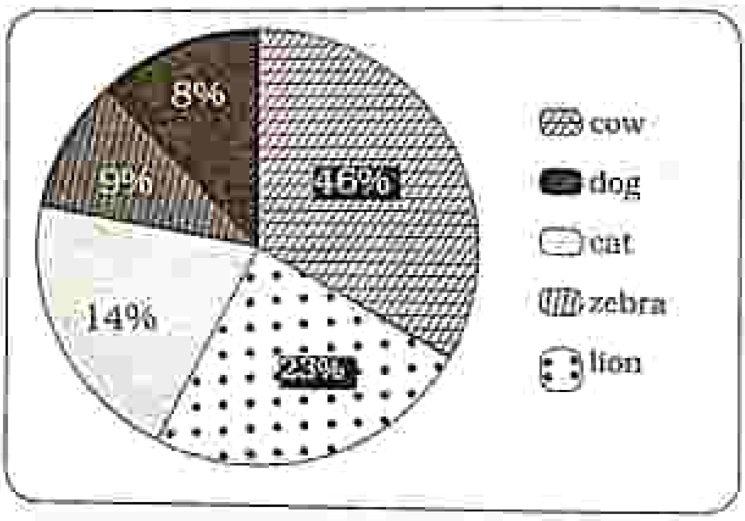
Reasons for enrolling at the university	Number of students
Campus close to home	10
Offered the course wanted	8
Fees a bit cheap	15
Other	7

a) Present the results in the form of a pie chart . [5]

b) Which is the most common reason? [1]

c) Which is the least common reason? [1]

4. The pie chart below shows the favourite animals for Form 4Z learners.

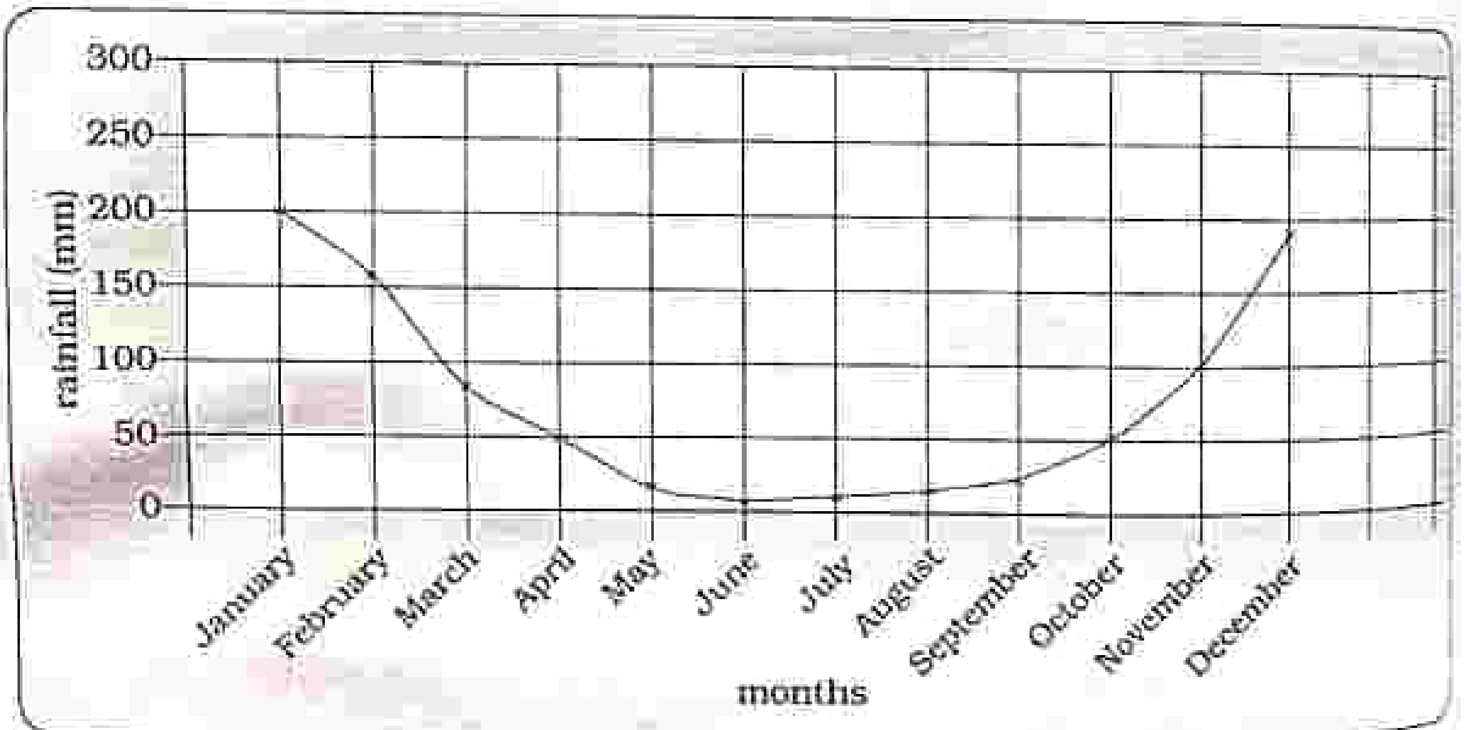


a) What percentage of the learners:

i) Favour a cat [1]

ii) Favour a lion [1]

5. The following diagram is a line graph to show rainfall pattern.



a) In which month was most rainfall recorded? [1]

b) What was the rainfall recorded in April? [1]

CHAPTER 22

Measurements

Key concepts

- Physical quantities
  - derived quantity in terms of base units
  - convert units
  - read scale to the nearest fraction
  - prefixes of SI units
  - conversion of units
  - estimations
- Types of errors in measurement
  - Parallax error
  - Zero error
- Measurement of physical quantities
  - Length : using metre rule
    : using vernier calipers
    : using micrometer screw gauge
  - Time
  - Temperature
  - Measuring current
  - Measuring voltage
  - Mass : of small objects
    : of a liquid
  - Volume : regular objects
    : irregular objects
    : small objects
  - Density : calculate density
    : determine density of liquid

Summary notes

Physical quantities

A physical quantity can be measured to determine its magnitude (size) and has units.

Internationally agreed units are called the SI units (Systeme Internationale).

Physical Quantity	SI Unit	Symbol
Length	Metre	m
Mass	Kilogram	kg
Time	Second	s
Temperature	Kelvin	K
Electric current	Ampere	A

CHAPTER 22

Measurements

Key concepts

- Physical quantities
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  - Temperature
  - Measuring current
  - Measuring voltage
  - Mass : of small objects
    : of a liquid
  - Volume : regular objects
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    : small objects
  - Density : calculate density
    : determine density of liquid

Summary notes

Physical quantities

A physical quantity can be measured to determine its magnitude (size) and has units.

Internationally agreed units are called the SI units (Systeme Internationale).

Physical Quantity	SI Unit	Symbol
Length	Metre	m
Mass	Kilogram	kg
Time	Second	s
Temperature	Kelvin	K
Electric current	Ampere	A

- Other quantities can be derived from the base quantities and these are called derived quantities thus will have derived units.

Derived quantity	Derived SI unit	Symbol	Expression in terms of SI base
Force	Newton	N	$\frac{Kg.m}{s^2}$
Work done Energy	Joule	J	$\frac{Kg.m^2}{s^2}$
Power	Watt	W	$\frac{Kg.m^2}{s^3}$
Potential difference	Volt	V	$\frac{Kg.m^2}{As^3}$

Prefixes of SI units

- Prefixes are used to simplify either very big or very small numbers.

Prefix	Symbol	Factor
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	K	10^{3}
hecto	h	10^{2}
deca	da	10^{1}
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	µ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

Types of errors

a) Parallax error

May be caused by positioning the eye incorrectly.

Parallax error may also be due to the object not being on the same level as the markings of the scale.

To avoid parallax error eye must be vertically looking on the mark.

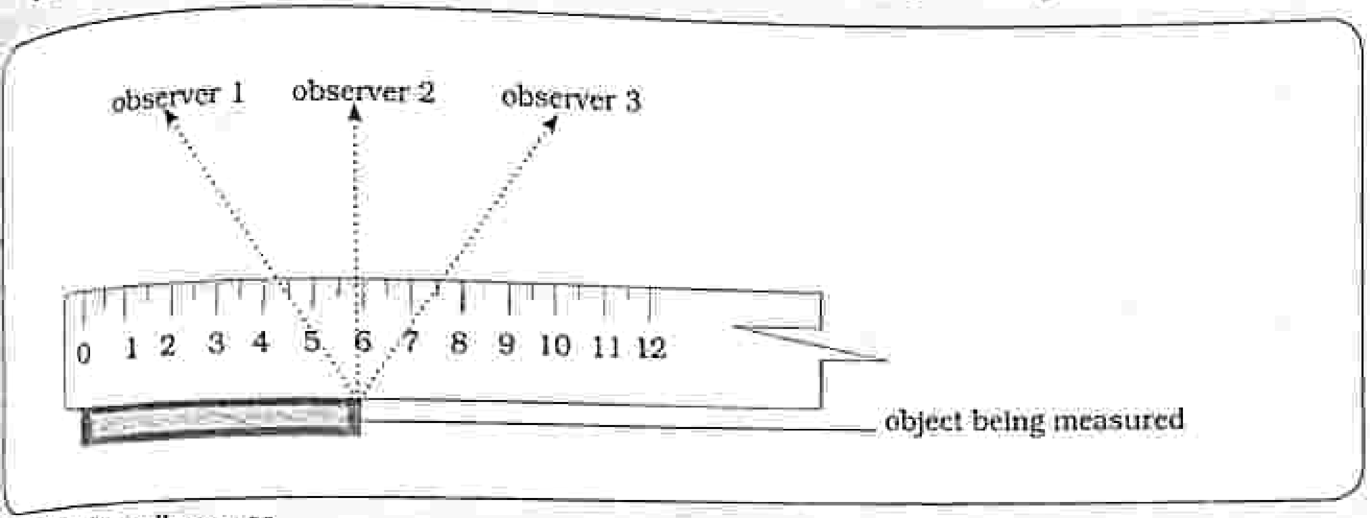


Fig. 22.1: Parallax error

b) Zero error

- It is an error showing how far away the reading of the instrument is from zero.
- The pointer of an instrument will not be pointing at zero before the instrument is used.
- Zero error can be positive or negative.

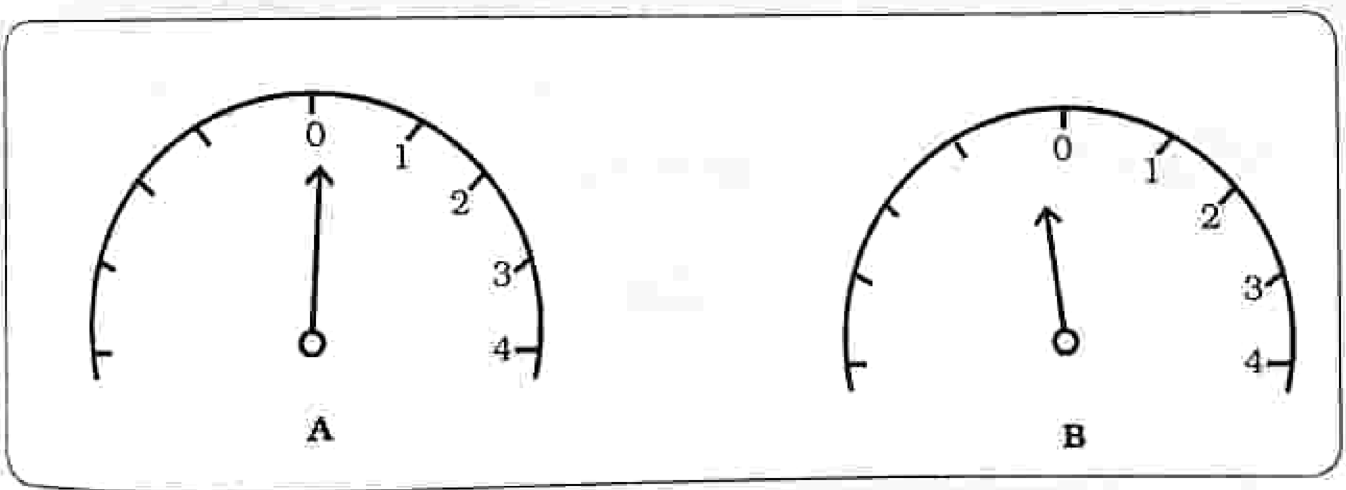


Fig. 22.2: Zero error

If the pointer is not initially on the zero mark then a zero error is committed.

a) Positive Zero error

b) Negative Zero error

Avoiding zero error

1. Ensure pointer is adjusted to point at the zero mark before use of the instrument
2. If adjusting the pointer fails and the instrument cannot be replaced:

a) **Positive zero error** – subtract the error from the final pointer mark.

b) **Negative zero error** – add the zero error to the final pointer mark.

Measuring instruments

Physical quantity	Measuring instrument
length	• metre rule • vernier caliper • micrometer screw gauge
mass	• balance
time	• stop watch
temperature	• thermometer
current	• ammeter
voltage	• voltmeter

Estimation of physical quantities

An estimate is a rough calculation of a physical quantity.

An estimate is usually done before an actual measurement is taken.

Estimate

a) Length of an exercise book
b) Mass of a brick
c) Time taken to reach the ground by a falling orange from an orange tree
d) Temperature of tap water

Object	Estimation of its length (cm)	Actual measurement (cm)
pencil	13	12
Exercise book	30	29.5
table	90	100

Accuracy

- It is defined as the closeness of measured value to a standard value.

Precision

- It is a measure of the degree of agreement within a group of measurements.
- It is the closeness between two or more measured values to each other.
- It is a measure of the consistency of the measurements.

Instrument	Range of measurement	Smallest division
Metre rule	0 - 1m	0.1cm
Vernier caliper	0 - 15cm	0.01cm
Micrometer screw gauge	0 - 2.5cm	0.01mm
Ammeter	0 - 1A	0.02A
Milliameter	0 - 100mA	2mA
Voltmeter	0 - 5V	0.1V
Digital stop watch		0.1s 0.01s
Analogue stopwatch		0.1s
Electronic balance		0.1g 0.01g
Thermometer	-10°C to 110°C	1°C

Measuring length

Length can be measured using:

a) Metre rule

b) Vernier callipers

c) Micrometer screw gauge

Metre rule

It is used to measure length.

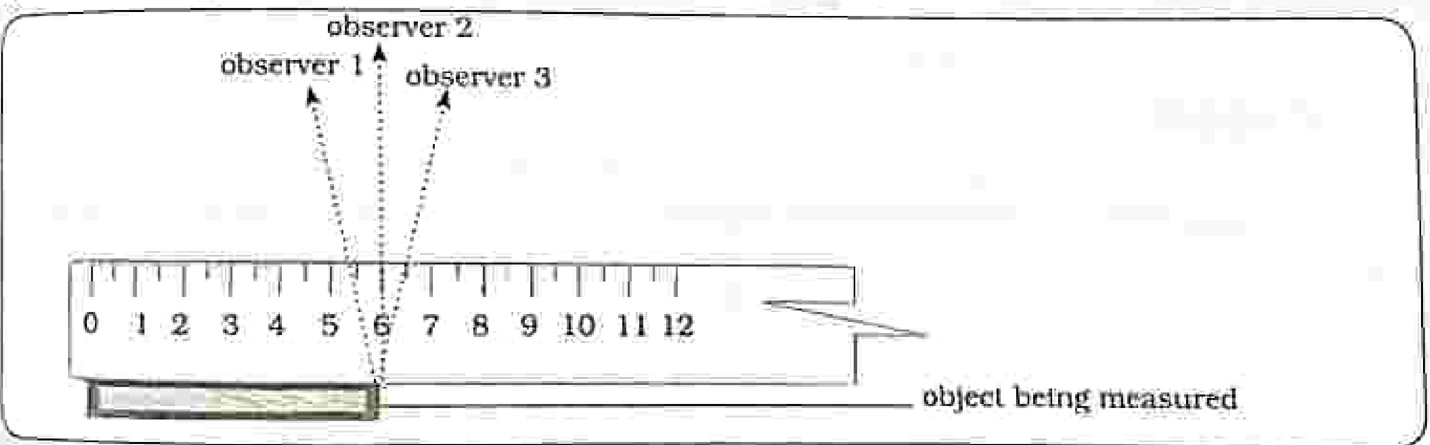


Fig. 22.3: Metre rule

Observer 1 = 5.4cm

Observer 2 = 6.0cm

Observer 3 = 6.5cm

- When taking a reading from a metre rule the eye must be placed vertically above the mark to be read to **avoid parallax error.**
- From Fig 22.3 only observer 2 is taking the reading on the correct position.
- Results obtained by observer 1 and 3 are not accurate due to the wrong positions of the observers resulting in parallax error.

Vernier calipers

It is used when seeking to obtain accurate measurements of small objects.

It consists of:

1. Main scale which is fixed.
2. Vernier scale which slides along the main scale.
   - It gives reading to one tenth of a mm which is 0.1mm or 0.01cm.
3. A pair of internal jaws to measure inside diameters.
4. A pair of external jaws to measure outside diameters.

Taking a reading from a vernier calipers

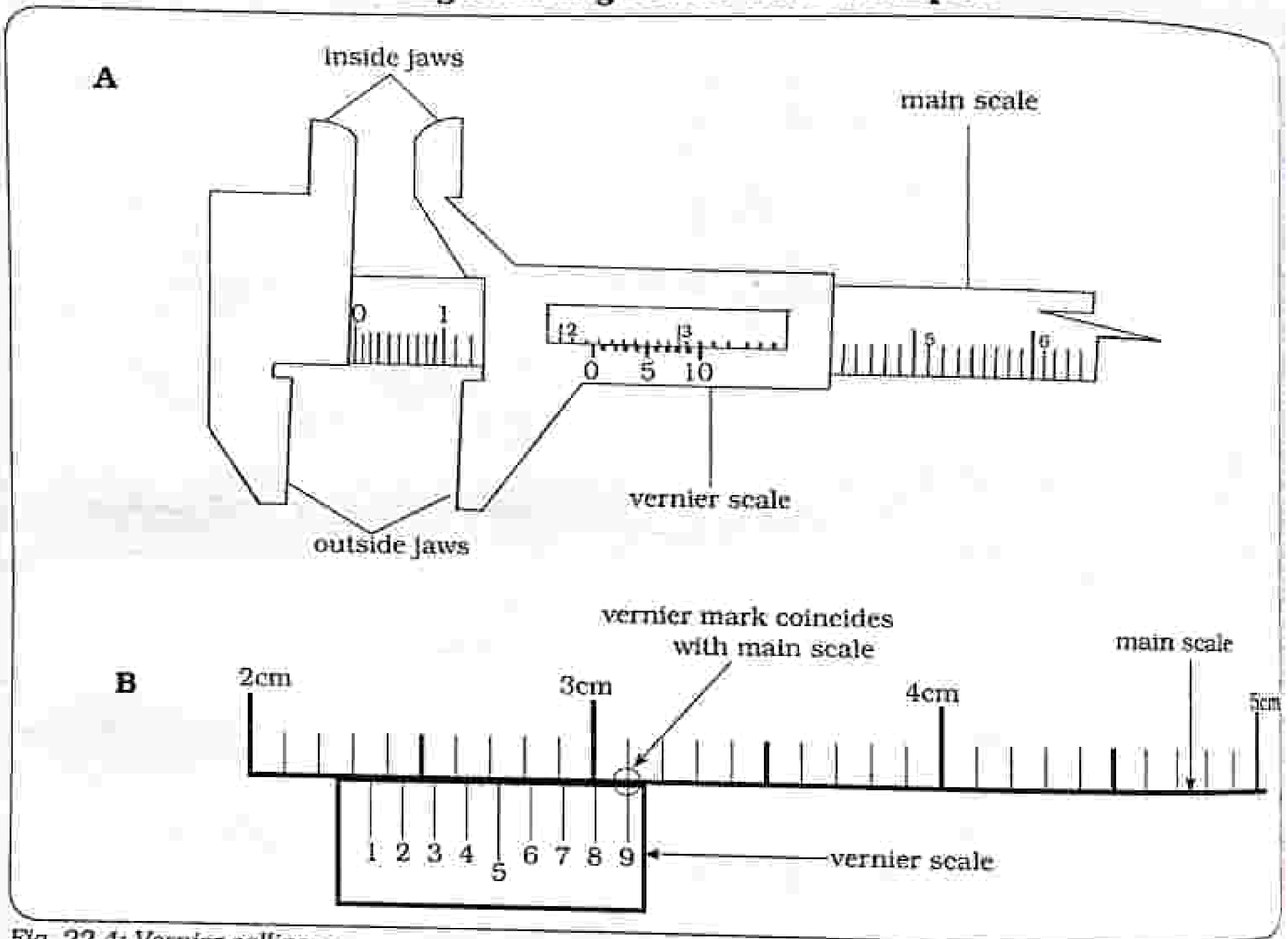


Fig. 22.4: Vernier callipers

- Main scale reading = 2.2cm
- Check a mark on the vernier scale which coincides exactly with a mark on the main scale.

  = 0.9mm or 0.09cm

  Reading shown in Fig 22.4B

  2.2cm + 0.09cm = 2.29cm

Micrometer screw gauge

- It is used to measure very small objects
- It can measure the width, diameter and thickness of any object which fits between its anvil and spindle.

Micrometer consists of:

1. Main scale on the sleeve.

c) Rotating vernier scale on the thimble

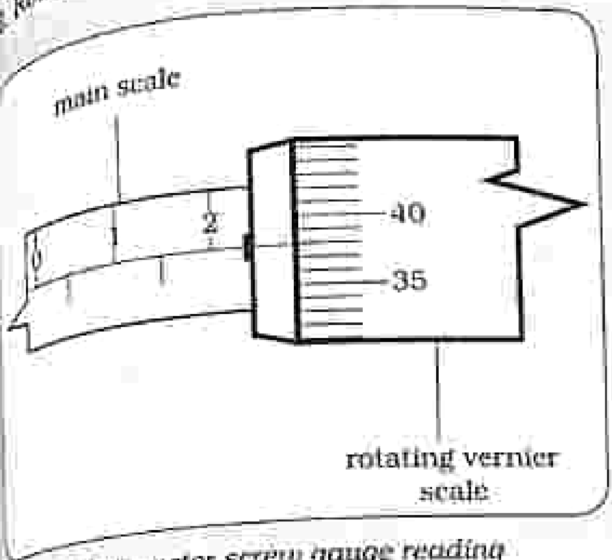


Fig 22.5 Micrometer screw gauge reading

Taking a reading

1. Read and record the reading of the main scale
= 2.5mm
2. Read and record the reading on the rotating vernier scale which coincides with the datum line on the sleeve
= 0.38mm

Reading shown in Fig 22.5

Reading of main scale = 2.5mm
Reading of vernier scale = + 0.38mm
2.88mm

Time

- Time is measured using a stop watch.
- Stop watches can be analogue or digital as shown in Fig 22.6

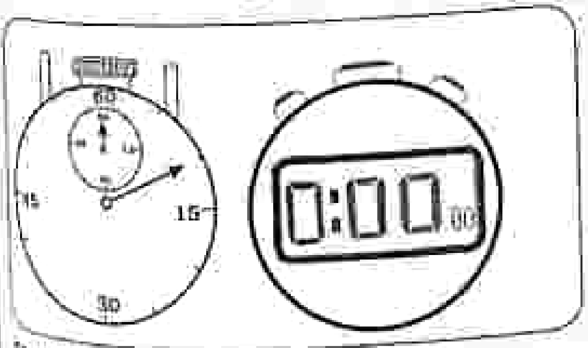


Fig. 22.6: Stop watch

Taking a reading from the stop watch

1. Press reset button to ensure the stopwatch starts at zero.
2. Press the start button to start taking a reading and press the same button to stop then record time.

Note : for more accurate results repeat the readings and calculate the average because the user's reaction time has an effect on the results obtained.

Temperature

- It is a measure of the hotness or coldness of a substance.
- **A thermometer** is used to measure temperature.
- There are different types of thermometers and the most common are the:

a) **Liquid in glass thermometer**
- The liquid can either be mercury or alcohol.

Usually liquid in glass thermometers are laboratory thermometers and clinical thermometers.

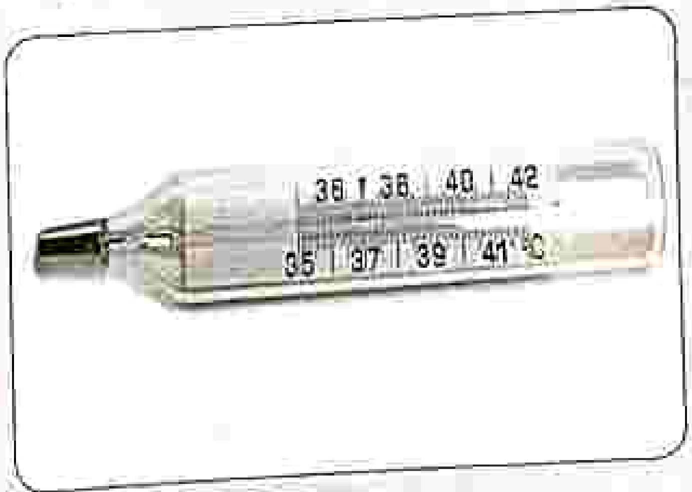


Fig. 22.7: Glass thermometer

Note: The alcohol or mercury in the bulb expands when heated.

Taking a reading

i) **Thermometer using mercury**

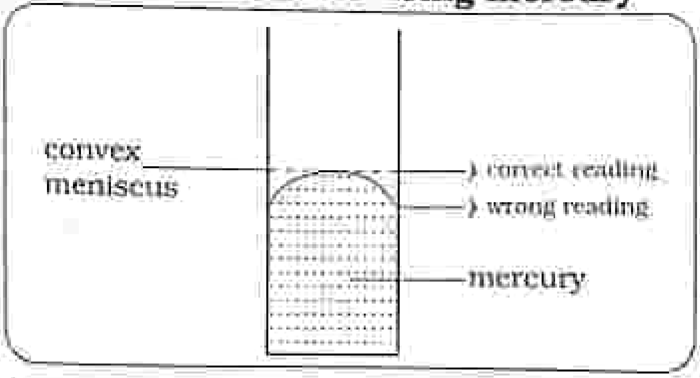


Fig. 22.8a: Thermometer using mercury

- A thermometer containing mercury has a convex meniscus.

(ii) **Thermometer using alcohol**

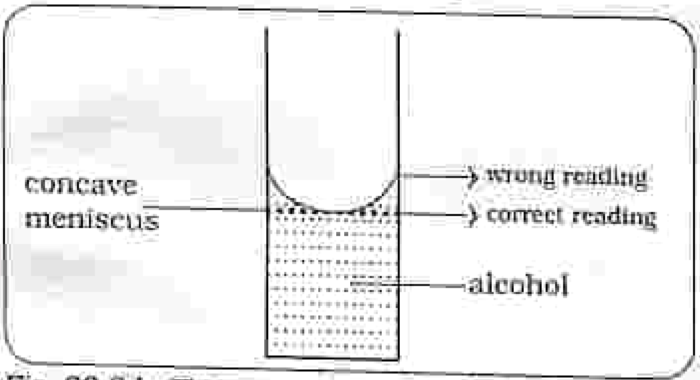


Fig. 22.8 b: Thermometer using alcohol

- A thermometer containing alcohol has a concave meniscus.

b) Digital thermometer

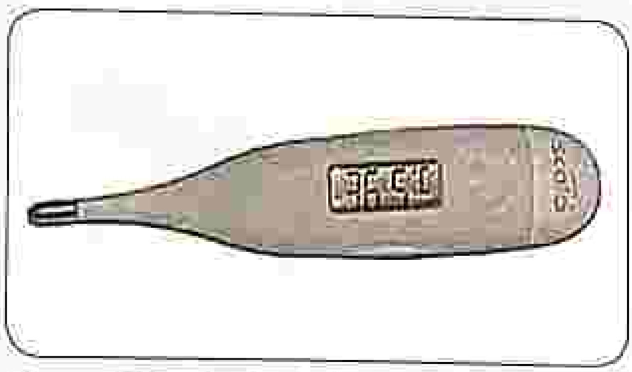


Figure 22.9: A digital thermometer

- A digital thermometer has temperature sensors inside and temperature recorded on the display screen.

Electric current

- It is the flow of charge in an electric circuit.
- It is measured using an ammeter.
- SI unit for current is ampere (A).
- An ammeter is connected in series in a circuit.

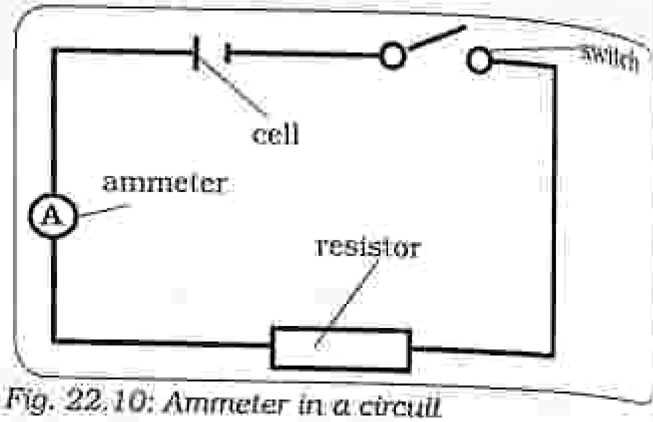


Fig. 22.10: Ammeter in a circuit

- The ammeter can be digital or analogue.
- When using an analogue ammeter deduce the smallest division so that you take an accurate reading.
- Ensure the pointer is pointing at zero before closing the switch in the experiment to avoid zero error.

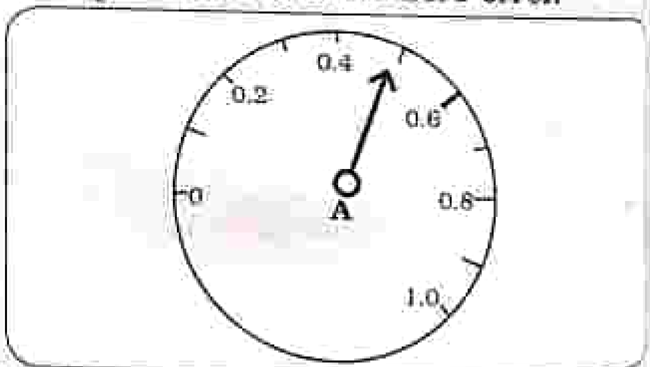


Fig. 22.11: Reading the ammeter

- Reading shown is 0.5A

Voltage

- Voltage or potential difference (pd) is the work done per unit charge.
- Potential difference exists between every positively charged point and a negatively charged point.

- A voltmeter is used to measure voltage or potential difference across an electric circuit.
- A voltmeter is connected in parallel in a circuit.

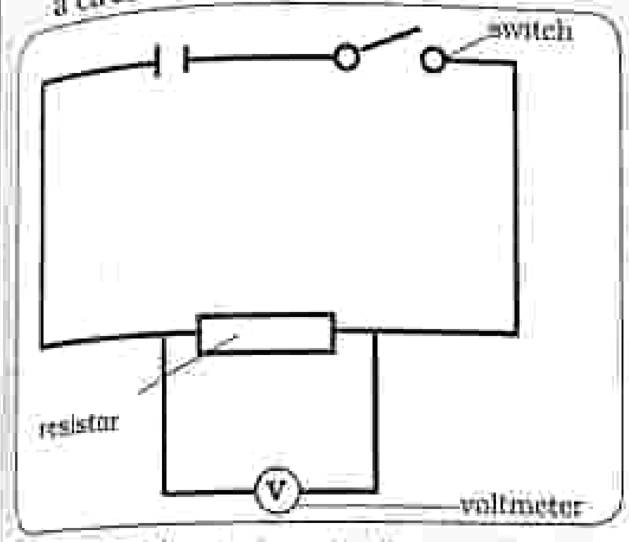


Fig 22.12: Voltmeter in a circuit

- Voltmeters can be digital or analogue.
- When using an analogue voltmeter, deduce the smallest division so that you take an accurate reading.

Reading the voltmeter

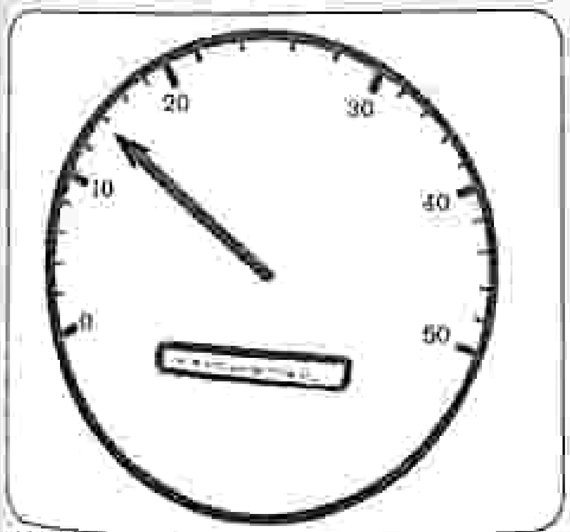


Fig. 22.13: Reading the voltmeter

- Ensure the pointer is pointing at zero before closing the switch.

Reading shown is 14V.

Mass

- It is the amount of matter in an object.
- The SI unit for mass is kilogram (kg).
- Mass is measured using a balance.
- A balance can be digital or analogue.
- Some examples of balances include :
  a) triple beam balance
  b) digital balance

Triple beam balance

It has three scales a) 10g scale
b) 100g scale
c) 500g scale

How to take a reading from the triple beam balance

1. Move the sliding masses until the pointer lines up with the zero mark.
2. Add all the readings from the three scales.

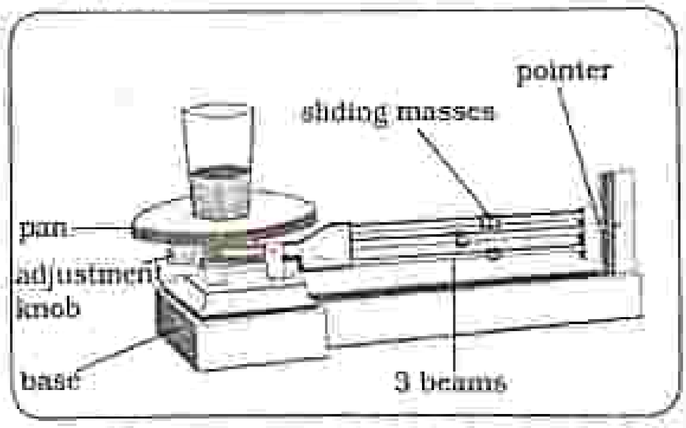


Fig. 22.14a: Triple beam balance


Fig. 22.14b: Digital balance

Practical activity 22.1

Aim : Measuring mass of very small objects

Apparatus : 1. small objects (dried beans)
2. balance
3. small container

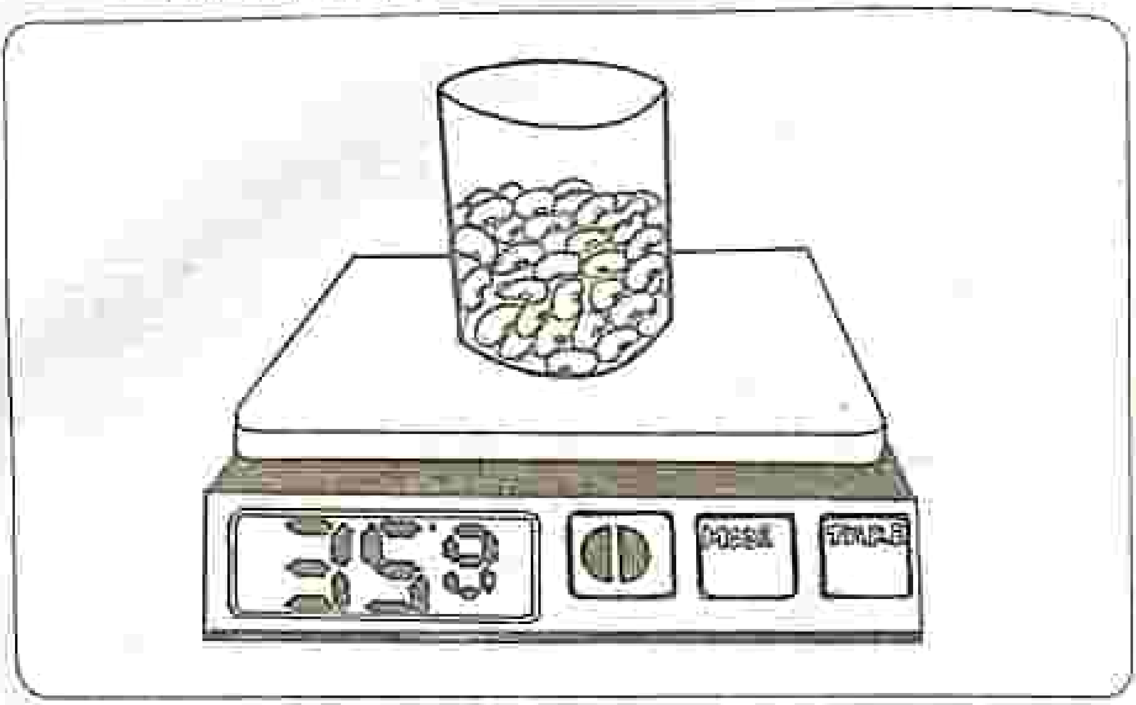


Fig. 22.15: Weighing dried beans

Method

1. Zero the balance
2. Measure the mass of an empty container
   - Take down the reading
3. Add 50 small seeds (dried beans) into the container
   - Measure the mass
   - Record the new reading

Results

Mass of container + 50 dried beans	= 35g
Mass of empty container	= 20g
Mass of 50 dried beans	= 35g − 20g = 15g
To find average mass of one bean	= 15g ÷ 50 bean seeds
	= 0.3g

Conclusion

- The mass of a small object can be found by first finding the mass of a l number of the same small objects and then calculate the average of one s object.

Practical activity 22.2

Aim : Measuring the mass of a liquid

Apparatus :
1. Balance
2. Small container
3. Water

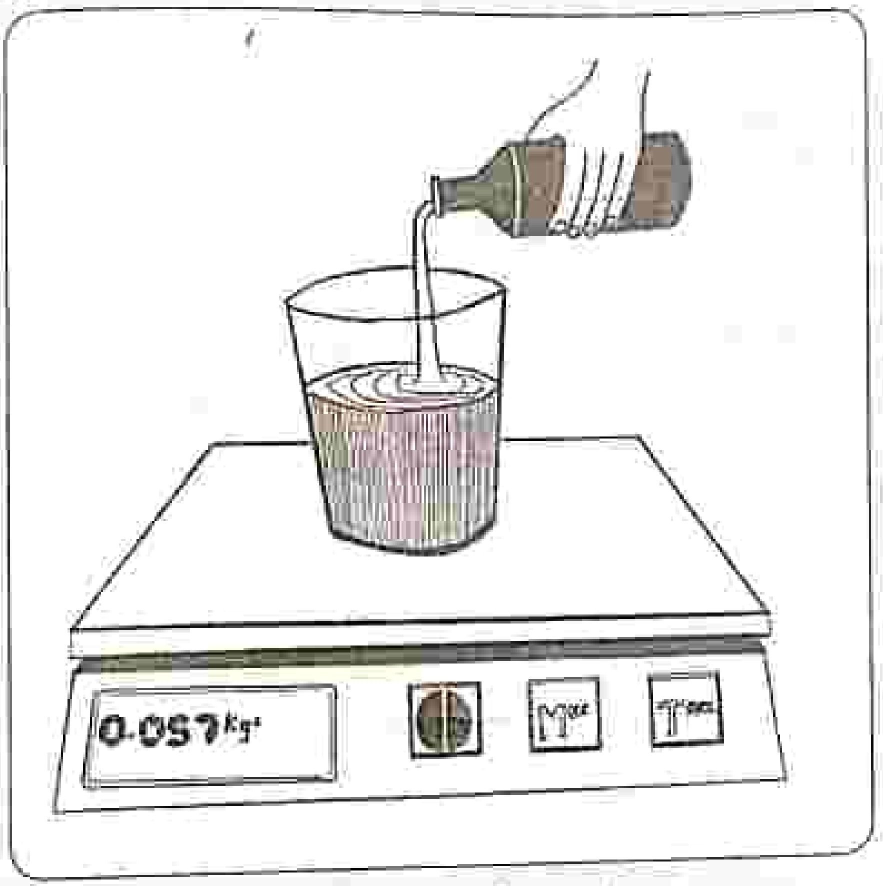


Fig. 22.16: weighing water

Method

1. Zero the balance
2. Measure the mass of an empty container
   - Record the reading
3. Add water to the container until it is half full
   - Measure the mass
   - Record the reading

Results

Mass of container + water = 20g
Mass of empty container = 5g

Mass of water = 20g - 5g
= 15g

Conclusion

- Mass of a liquid can be found by first finding the mass of the empty container, then the mass of container + water. Subtract the mass of empty container from mass of container + water.

Volume

- It is the amount of space occupied by an object.
- Volume of a liquid is measured using a measuring cylinder.
- It is measured in cubic metres (m^3).
- Volume of a regular shape can be calculated.
- Volume of an irregular solid can be found using the displacement method.

Volume of a regular object

To find the volume of a regular object use the formula: Length × Width × Height.

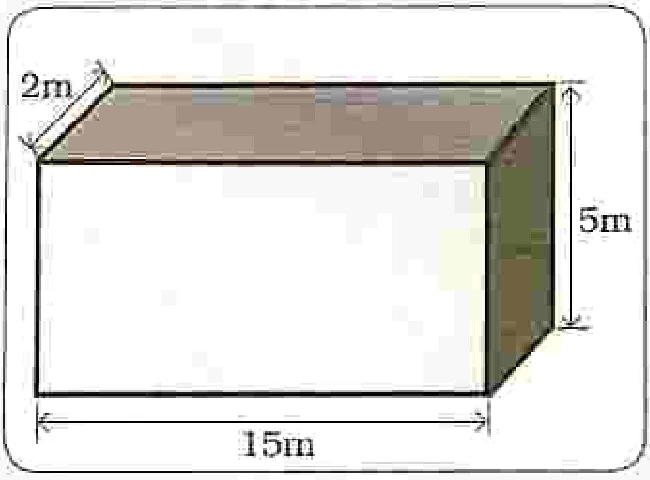


Fig. 22.17: A cuboid

$$\begin{aligned} \text{Volume} &= L \times W \times H \\ &= 15m \times 5m \times 2m \\ &= 150m^3 \end{aligned}$$

Practical activity 22.3

Aim : Measuring the volume of small objects

Apparatus : 1. Measuring cylinder
2. Water
3. Small objects (bean seeds)

Taking a reading from a measuring cylinder

1. Place the measuring cylinder on a flat surface and wait for the liquid to stabilize before taking a reading.
2. To avoid parallax error, read from the meniscus.
3. Take note of the units used on the measuring cylinder for example
   $1ml = 1cm^3$
4. Reading shown below is $80cm^3$.

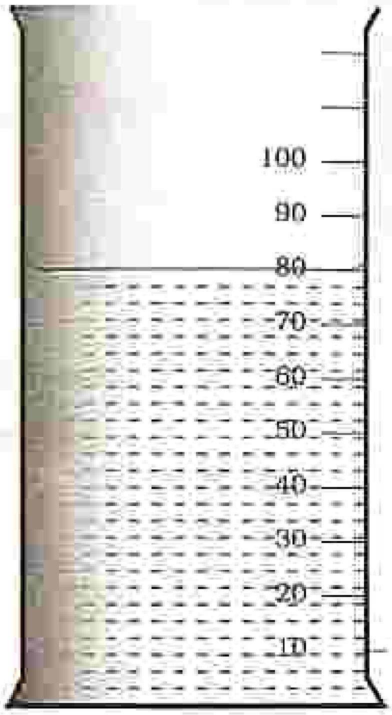


Fig. 22.18: Measuring cylinder

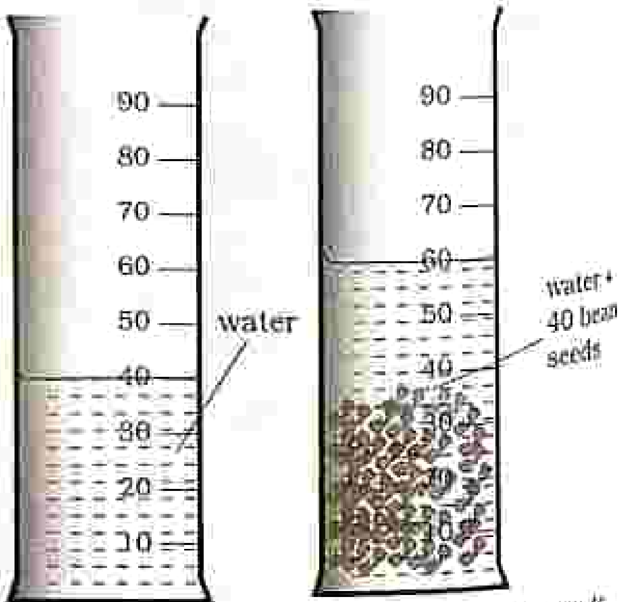


Fig. 22.19: Measuring volume of small objects

Method

1. Half fill the measuring cylinder with water.
2. Take the reading from the meniscus.
   - Record the reading.
3. Count 40 bean seeds and carefully drop them into the measuring cylinder with water.
4. Read and record the new volume.

Results

Volume of bean seeds + water

= $60cm^3$

Volume of water only

= $40cm^3$

Volume of 40 bean seeds

= $20cm^3$

To find the average volume of one bean seed = $20cm^3 \div 40$ bean seeds

= $0.5cm^3$

Conclusion

- The average volume of small object can be calculated by first finding th volume of many small objects and the divide to find the average volume c one small object.

Practical Activity 22.4

Aim : Measuring the volume of an irregular object

Apparatus :
1. Measuring cylinder
2. Overflow can
3. Water
4. An irregular object (stone)

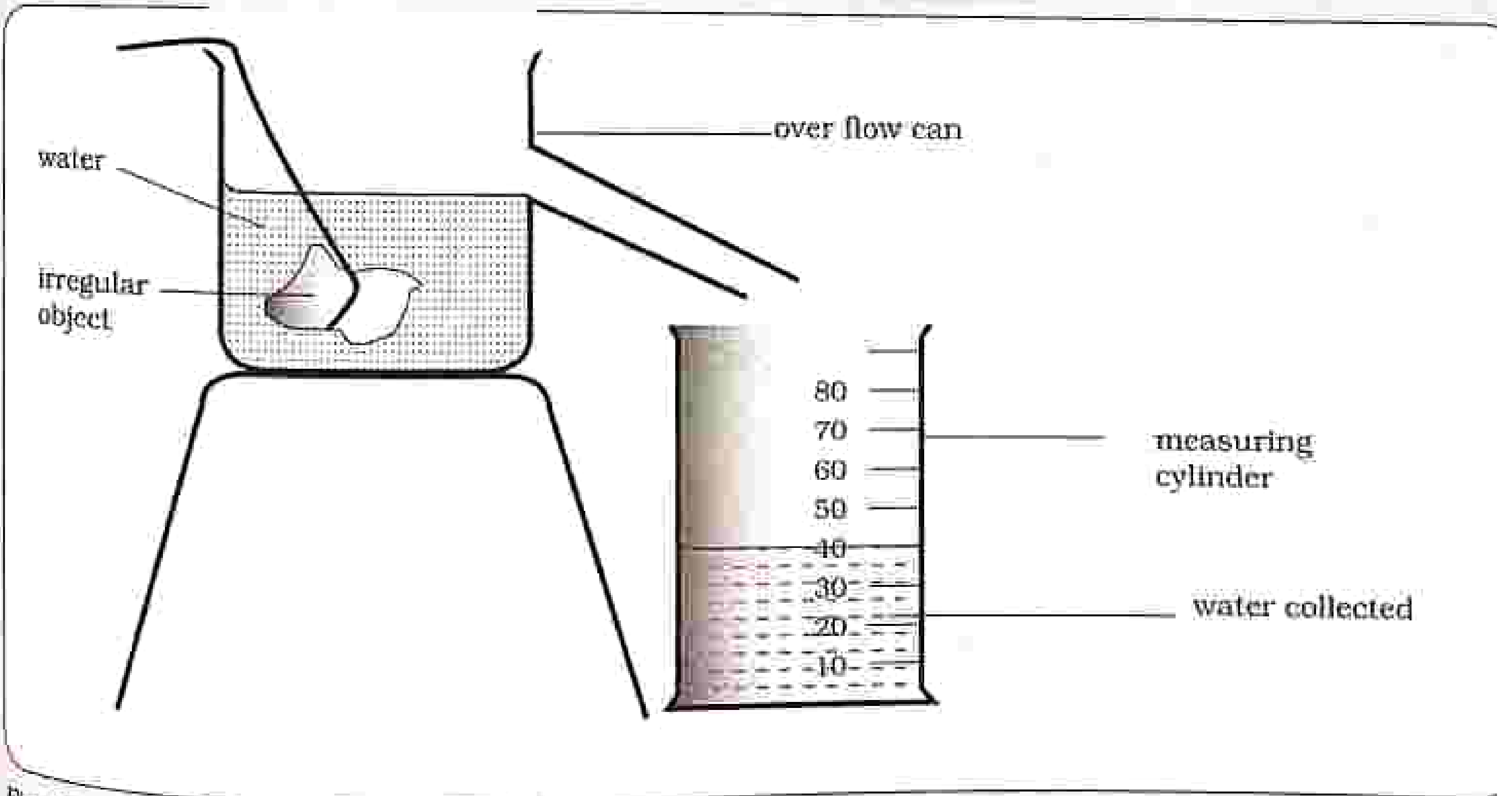


Fig. 22.20: Volume of an irregular object

Method

- Set up apparatus as shown in Fig 22.20.
- Put water into an overflow can just below the spout.
- Carefully place the irregular object into the overflow can.
- Take a reading from the measuring cylinder.

Results

- Volume of water collected in the measuring cylinder = $40cm^3$.

Conclusion

The volume of the irregular object is the volume of the displaced water collected in the measuring cylinder.

Density

It is the mass per unit volume.

It is measured in kg/m^3 or g/cm^3.

Density $= \frac{mass}{volume}$

Calculate the density of an object with a mass of 5kg and volume of $10m^3$

Density $= \frac{mass}{volume} = \frac{5kg}{10m} = 0.5kg/m^3$

Practical activity 22.5

Aim : Density of liquids

Apparatus :
1. measuring cylinder
2. balance
3. water

Method

1. Measure the mass of an empty measuring cylinder using a balance. Record the reading.
2. Add $50cm^3$ of water to the measuring cylinder.
   Measure on the balance the measuring cylinder containing water. Record the reading.
3. Calculate mass of water only.

Results

- Mass of measuring cylinder + water = 30g
- Mass of empty measuring cylinder = 10g
- Mass of water = 30g − 10g = 20g
- Volume of water = $50cm^3$

Conclusion

$$\text{Density} = \frac{mass}{volume} = \frac{20g}{50m^3} = 0.4g/cm^3$$

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Identify the SI unit for measuring mass.

   A. second B. kilogram

   C. Kelvin D. ampere

2. What is the expression for force in terms of SI base units?

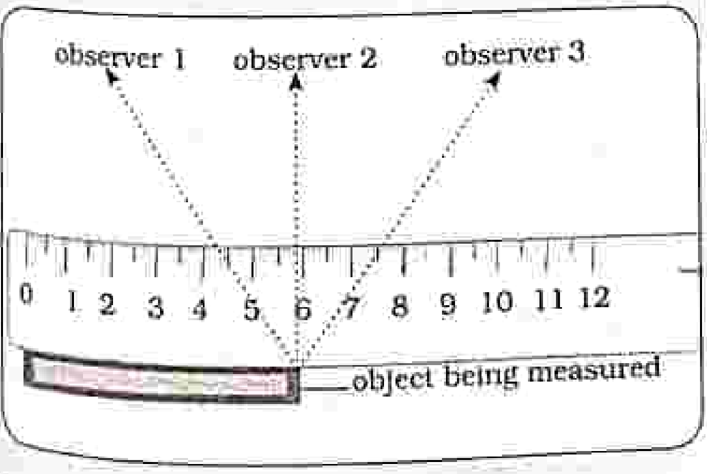
   A. $\frac{kg.mg}{s^2}$

   B. $\frac{kg.m^2}{s^2}$

   C. $\frac{kg.m^2}{s^3}$

   D. $\frac{kg.m^2}{As^3}$

4. Identify the type of error shown in the diagram below.



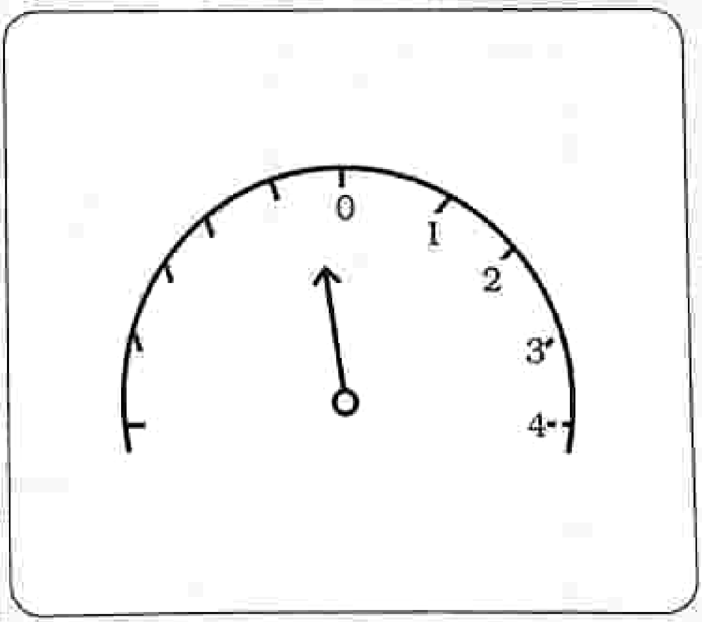
   A. one error

   B. negative one error

   C. parallax error

   D. zero error

5. Name the measuring instrument shown in the diagram below.



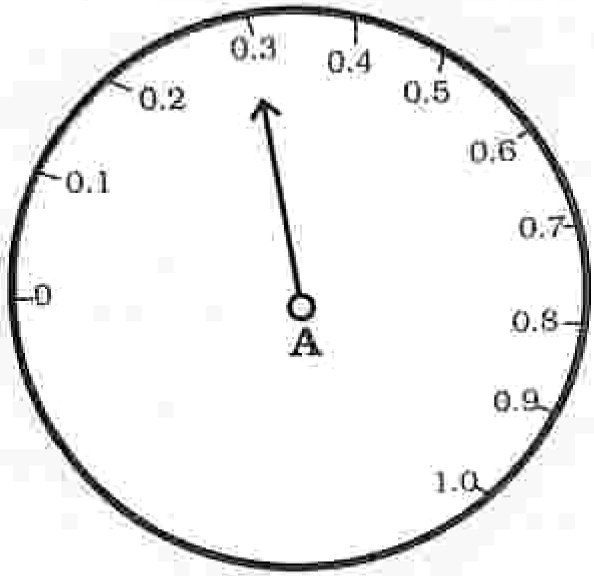
   A. vernier calipers

   B. micrometer screw gauge

   C. metre rule

   D. stop watch

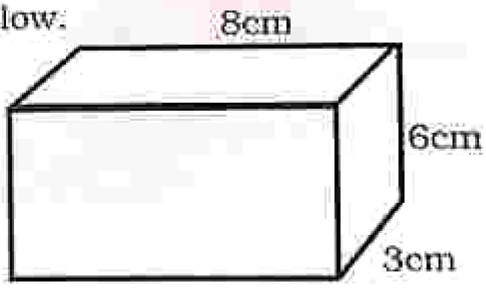
6. What is the reading shown in diagram shown below?



A. 0.1A B. 0.3A
C. 0.4A D. 0.2A

7. How do you find density of an object.
   A. mass/volume
   B. volume/mass
   C. mass × volume
   D. mass × volume × length

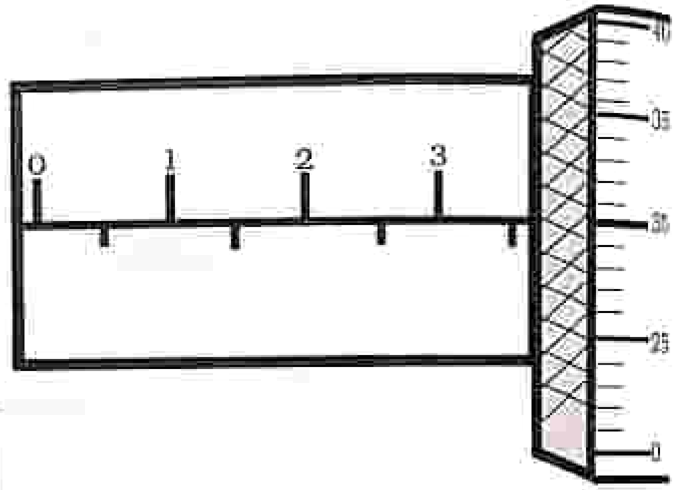
8. Find the volume of the diagram shown below.



A. $144cm^3$ B. $150cm^3$
C. $48cm^3$ D. $18cm^3$

9. Current is measured using____
   A. voltmeter B. vernier calipers
   C. ammeter D. stop watch

10. What is the reading shown in the diagram below?



A. 3.5mm
B. 3.80mm
C. 30mm
D. 3.30mm

Section B: Structured questions

1. Complete the table below.

Physical quantity	SI unit	Symbol
length	-	-
mass	-	-
time	-	-
temperature	-	-
Electric current	-	-

[10]

2. Convert the following

a) 10m = ...cm
b) 3000cm = ...mm
c) 2.5kg = ...g
d) 63kg = ...g
e) 3000g = ...kg
f) 3 hours = ... minutes
g) 3600 seconds = ... minutes
h) 120mins = ...seconds [8]

3. a) Explain parallax error and how the error can be avoided. [2]
   b) Explain zero error and how the error can be avoided. [2]
4. With the aid of a diagram explain how to find the volume of an irregular object. [5]
5. Explain how to find the mass of very small objects. [5]

CHAPTER 23

Force

Key concepts

- SI Unit
- Measurement of force
  - Spring balance
  - Forcemeter
- Effects of force
  - Position
  - Shape
  - Size
  - Speed
- Types of force

  Contact
  - mechanical
  - friction

  Non contact
  - electrostatic
  - magnetic
  - gravitational
  - weight
- Distinguish between weight and mass
- Moments of force
  - Principle of moments
- Resultant force
  - In line forces
  - Balanced and unbalanced
- Momentum
- Inertia
- Newton's laws of motion

Summary notes

Force

- It is a push or a pull.
- The SI unit for force is newton (N).

Measurement of force

- Force can be measured using :
- Forcemeter
- Spring balance

Force meter

- It consists of a spring which is attached to a hook.
- The spring stretches when a force is applied.
- The larger the force applied, the longer the spring stretches and the greater the reading on the forcemeter.

Spring balance

- It has a spring.
- It measures how much the spring stretches when an object is hung.
- The mass of the object is measured in grams or kilograms and then converted to Newtons.

  1kg = 10N

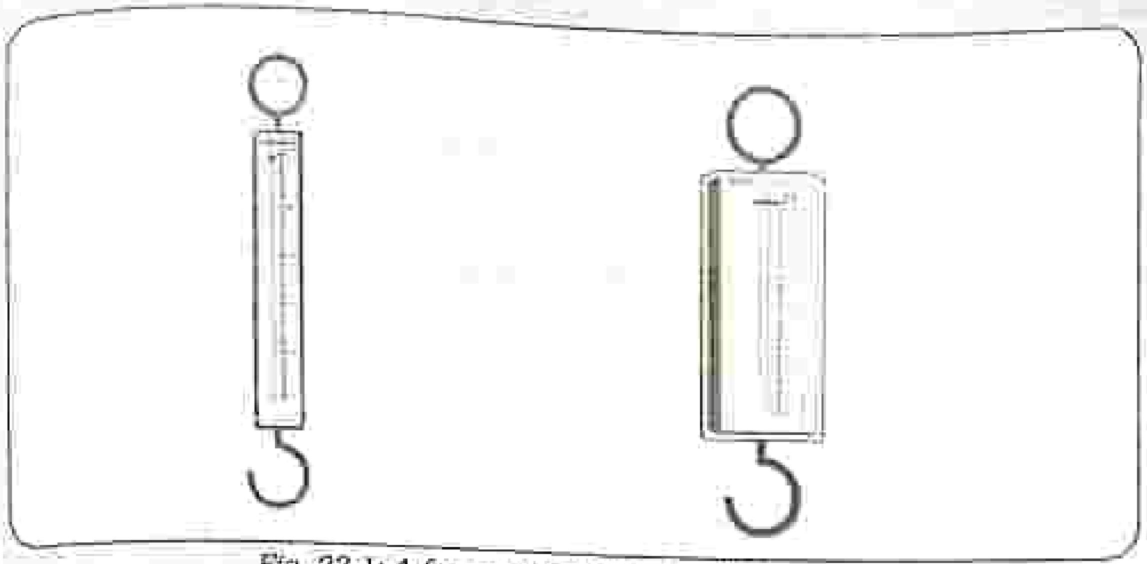


Fig. 23.1: A forcemeter and spring balance

Effects of force

- Force applied to an object can cause the object to :
  a) Change speed
  b) Change direction
  c) Change position
  d) Change shape

Change in speed

- A force can cause an object to change speed.

Example

Fig. 23.2: A rolling ball which changes speed when blocked

Change in size

- A force can cause an object to move from position **A** to position **B**.

Example

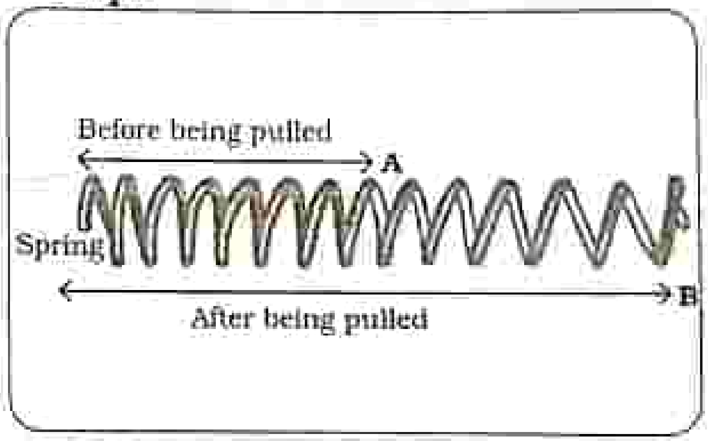


Fig. 23.3: A pulled spring

Change in shape

- A force can change the shape of an object.

Example

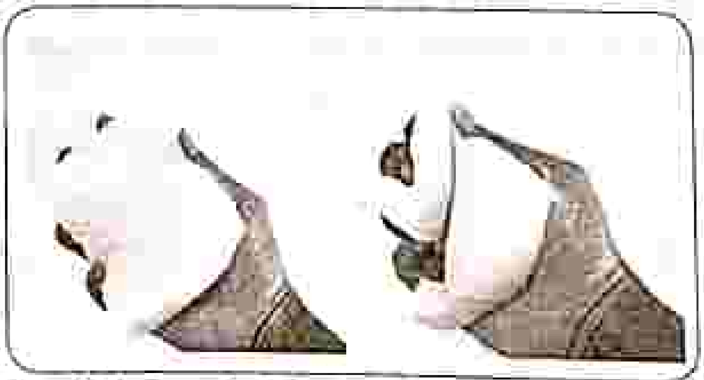


Fig. 23.4: Squashed foam rubber

Change direction

- A force can cause an object to change direction.

Fig. 23.5: A ball which has been bounced back

Types of forces

- The two main types of forces are contact and non contact.

Contact forces

- Two objects are in direct interaction with each other such that they exert a force on each other by touching.
- Contact forces include mechanical force and friction.

Mechanical force

- It is a push or a pull exerted on an object.

Fig. 23.6: A man pulling a cart

- The man in Fig 23.6 is applying a pulling force on the cart so that it moves forward and can apply a pushing force by moving it backwards.

Friction

- It is a force which opposes movement.
- The force of friction is always on the opposite direction to movement.
- Friction causes a moving object to slow down or to stop.
- It occurs between two objects in contact.

Practical activity 23.1

Aim : Measuring friction

Apparatus : 1. Forcemeter

2. Wooden block

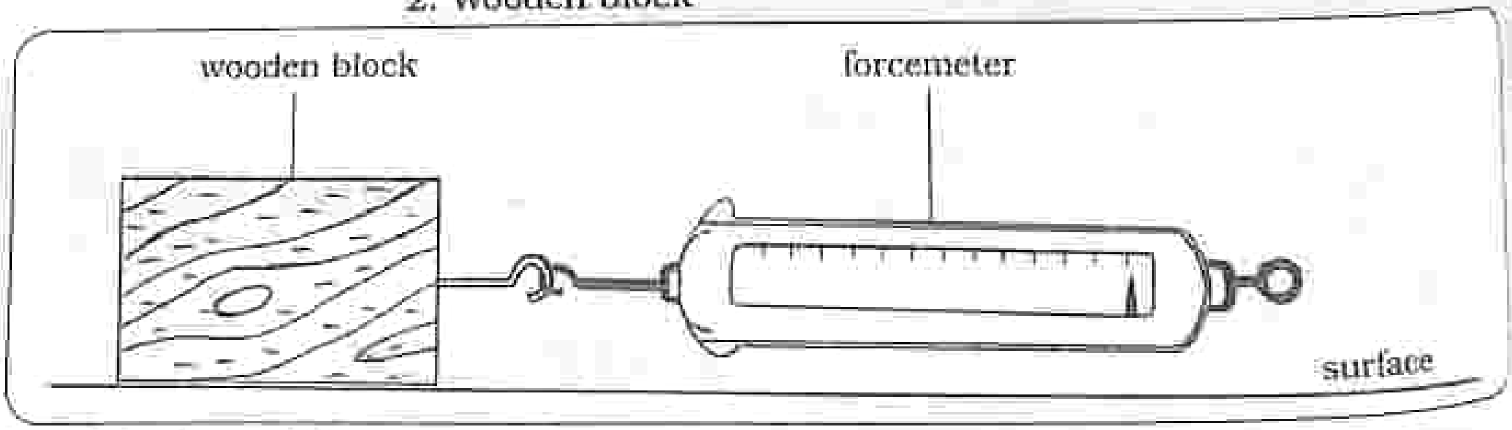


Fig. 23.7: Frictional force

Method

1. Place a wooden block on a smooth surface and attach a forcemeter as shown in Fig 23.7 and pull gently.
2. The forcemeter will show the force required to just move the wooden block.
3. Repeat (1) but placing the wooden block on a rough surface.
   - Note the forcemeter reading

Results

A reading is noted on the forcemeter

Nature of surface	Forcemeter reading
smooth	6N
rough	10N

Conclusion

- The indicated force on the forcemeter is the friction force acting on the wooden block.
- Frictional force on smooth surfaces is less than on rough surfaces.

Practical questions

a) The wooden block is easy to move on which surface?

b) The wooden block is difficult to move on which surface?

Application of frictional force

Road surface

- Friction is needed to move vehicles.
- Road surfaces must be rough such that friction between the road and wheels is increased so that vehicles will not skid.

Soles of shoes

- Friction between our shoes and the ground stops us from slipping.

Tyre treads

- Tyre treads offer a rough surface to the tyre.
- Friction between the tyres and the road stops cars from skidding.
- Tyres with used up treads must be replaced to avoid the car from skidding because the tyre would be smooth without treads.

Car braking system

- Friction between the brakes and wheel helps the vehicles to slow down or to stop.

Hindrance of frictional force

- Friction can be a hindrance or not useful for example friction between moving parts of machines causes energy to be transferred to the surrounding resulting in heating.
- The machine parts will tear causing the machine to become noisy and difficult to operate.

Non- contact forces

- Two objects exert a force on each other without being in contact.

Gravitational force

- It is the force that attracts any two objects with mass.
- It is a universal force therefore every object on the universe experiences this force due to other objects.

Example

- When you throw a tennis ball upwards from earth, it falls down to the earth due to the gravitational force of the earth.

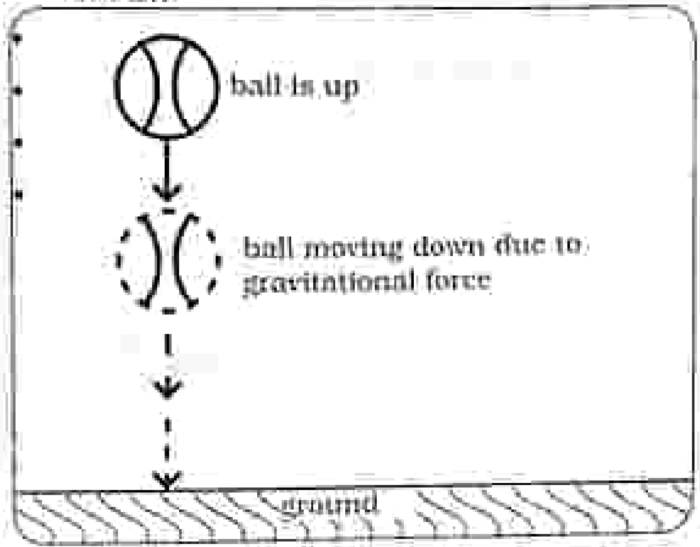


Fig. 23.8: Gravitational force acting on a ball

Electrostatic

- It is the attractive or repulsive force between two electrically charged objects.
- The electric charges can be:
  - positive (+) and positive (+) or negative (-) and negative (-) resulting in repulsive forces.
  - Charges can also be positive (+) and negative (-) resulting in attractive forces.
- The charges are separated by a certain distance making this type of force non contact force.

Magnetic force

- It is a force exerted between two magnetic objects.
- Magnets have two poles that is the north pole (N) and south pole (S)
- Like poles that is the south and south or north and north repel each other when brought close to each other.
- Unlike poles that is the south and north attract each other when brought close to each other.

Weight

- It is the force acting on the body's mass due to the gravitational force of attraction of the earth.

Weight = mass × acceleration due to gravity

Distinguish mass and weight

Weight	Mass
• is the force of gravity acting on the object	• is the amount of matter in an object
• measured in newtons(N)	• measured in kilograms (kg)
• is not constant but depends on acceleration due to gravity at the location	• is constant at any location
• is a vector quantity (have both magnitude and direction)	• is a scalar quantity (have magnitude only)
• measured using a spring balance	• measured using a beam balance

Moments of force

- It is the turning effect of a force.
- Moment of a force depends on the size of the force and the perpendicular distance from the pivot.

Moment of a force

Force × perpendicular distance from the pivot

Example

Calculate the moment of a force of an object which exerts a force of 6N, 2m away from the pivot:

Moment of a force

= force × perpendicular distance from the pivot

= 6N × 2m

= 12Nm

Principles of moments

- It states that at equilibrium sum of clockwise moment is equal to the sum of anticlockwise moment about the same point.

Example

A 60N object is placed 2m away from one end of a pivot. Where must a 40N object be placed to balance the see-saw?

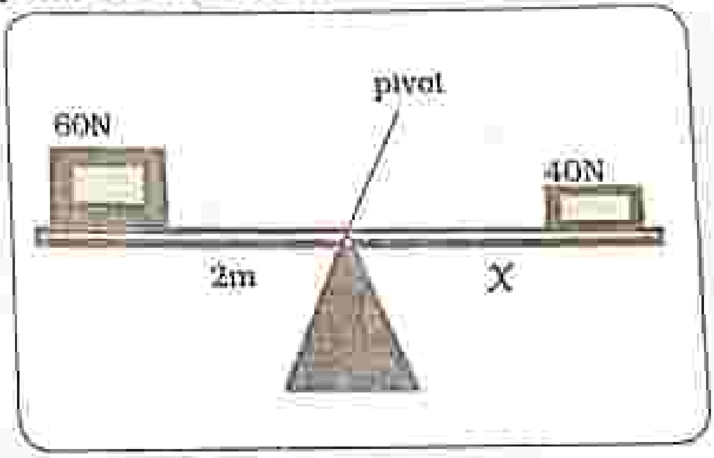


Fig. 23.9 Principle of moments

At equilibrium

Clockwise moments	=	Anticlockwise moments
Force × distance from pivot	=	Force × distance from pivot
60N × 2m	=	40N × X
120Nm	=	40X
$\frac{120Nm}{40N}$	=	X
3m	=	X

The 40N object must be placed 3m away from the other end of the pivot in order to balance the 60N object.

Resultant force

- It is the total force acting on a body along with their directions
- The resultant force can cause a stationary object to move.
- It can cause a moving object to slowdown, move faster, change direction or stop.

Balanced force

- Equal and opposite forces balance (at equilibrium).
- Balanced forces cause an object to stay at rest or to continue moving with uniform velocity.
- Balanced forces have zero resultant.

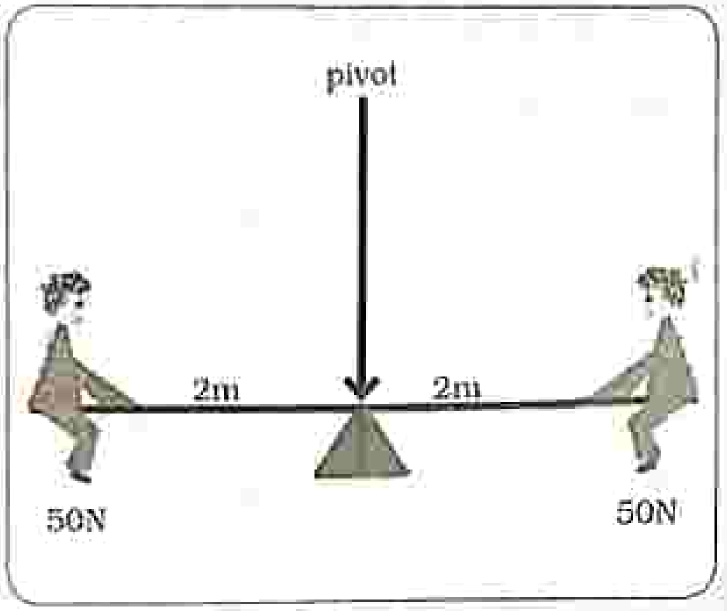


Fig. 23.13: Children on a balanced see-saw

Unbalanced forces

- Unequal forces have a resultant force and the forces are unbalanced.
- There is movement in the direction of the resultant.

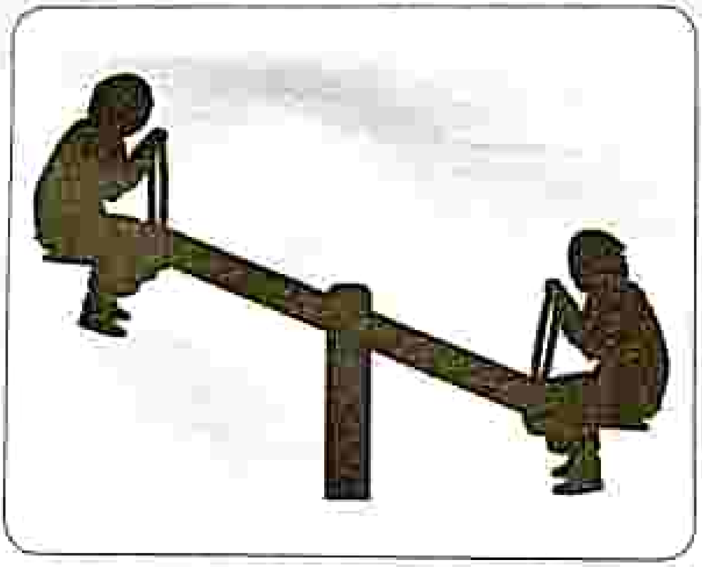


Fig. 23.14: Children on an unbalanced see-saw

In line forces

Same direction

- When forces act in the same line and same direction; the forces are added to give a resultant force.
- The resultant force is in the direction of the forces.

$$\text{Resultant force} = \overset{\longrightarrow}{10N} + \overset{\longrightarrow}{7N} = 17N \longrightarrow$$

Opposite direction

- When forces act in the same line but in opposite direction, the forces are subtracted to give a resultant force.
- The resultant force is in the direction of the larger force.

$$\text{Resultant force} = \overset{\longleftarrow}{12N} - \overset{\longrightarrow}{2N} = \longleftarrow 10N$$

Momentum

It is the mass of a body multiplied by its velocity.

Momentum is useful when bodies are involved in a collision or explosion.

Fig. 23.15: Colliding cars

Momentum = mass x velocity

p = mv

- Momentum is measured in kilogram metre per second (kgm/s).

Inertia

- It is the reluctance of an object to change either its state of rest or if moving in a straight line, its reluctance to stop.

Newton's laws of motion

Newton's first law of motion

- An object at rest will remain at rest, or if moving will continue to move at constant motion in a straight line unless acted upon by an external force.

Application

- A stone resting on the ground will remain stationary forever unless moved by some external force.

Newton's second law

- The resultant force acting upon an object is equal to the product of the mass and the acceleration of the object.
- The force is in the same direction as that of the object's acceleration.

Force = mass × acceleration

F = m × a

Find the force of an object with a mass of 10kg and an acceleration of $0.2m/s^2$

$$F = ma$$
$$= 10kg \times 0.2m/s^2$$
$$= 2N$$

Application of Newton's second law

It is easy to push an empty wheelbarrow than the one full of bricks inside because it has more mass than an empty wheelbarrow.

Newton's third law

For every action there is an equal and opposite reaction.

OR

If object **A** exerts a force on object **B**, then object B will exert an equal but opposite force on A.

Application of Newton's third law

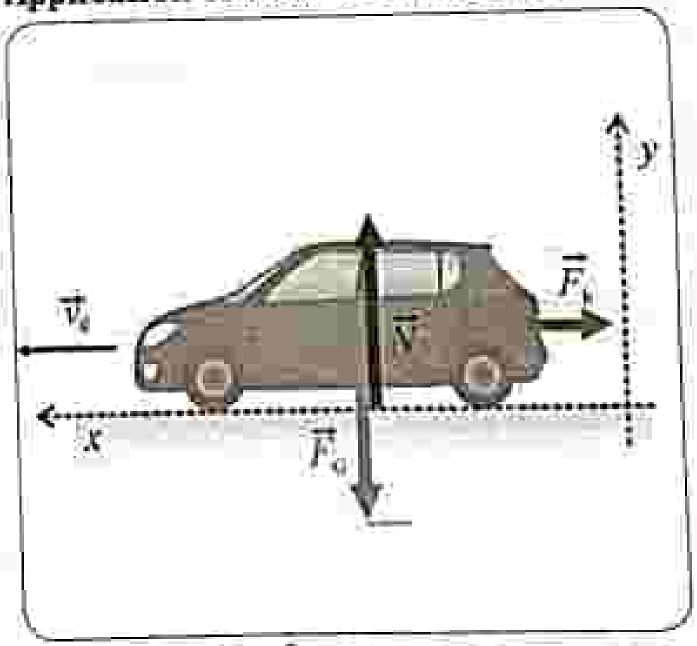


Fig. 23.16: Braking force

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.
The number of marks is given in brackets[ ] at the end of each question.

Multiple choice questions

1. Identify the SI unit for force
   A. watt
   B. pascal
   C. newton
   D. second
2. Which instrument is used to measure force?
   A. voltmeter
   B. forcemeter
   C. ammeter
   D. stop watch
3. Which one is not a type of force?
   A. magnetic
   B. electrostatic
   C. gravitational
   D. heat
4. Moment of force is ___________.
   A. the turning effect of a force
   B. type of force
   C. resultant force
   D. force acting on the body's mass
5. Newton's second law of motion can be summarised as ___________.
   A. Force = mass/acceleration
   B. Force = momentum × acceleration
   C. Force = momentum/acceleration
   D. Force = mass × acceleration
6. What is the resultant force of balanced forces?
   A. 2N
   B. zero
   C. 5N
   D. 3N
7. Calculate the resultant force of 10N force and 15N force moving downward in line.
   A. 25N
   B. 25N
   C. 5N
   D. 5N

8. Calculate the moment of force of an object which exerts a force of 10N, 3m away from the pivot.
   A. 13Nm
   B. 3.3Nm
   C. 7Nm
   D. 30Nm
9. Momentum is ____________.
   A. mass x velocity
   B. force x velocity
   C. mass x acceleration
   D. velocity x acceleration
10. A stone resting on ground will remain in that position forever unless acted upon by an external force, is an application of ____________.
   A. Newton's first law
   B. Newton's second law
   C. Newton's third law
   D. Momentum

Section B: Structured questions

1. Distinguish between mass and weight. [4]
2. State Newton's three laws of motion and their applications. [6]
3. Explain the following terms:
   a) Weight
   b) Momentum
   c) Inertia [3]
4. a) Define moment of a force. [1]
   b) State the principle of moments. [1]
5 a) Define friction. [1]
   b) Describe applications of frictional force. [3]

CHAPTER 24

Machines

Key concepts

- Machines
- Work Done
- Simple machines
  - Levers
  - Pulley system
  - Inclined plane
  - Gears
- Mechanical Advantage (MA)
- Velocity Ratio (VR)
- Efficiency
- Uses and applications of machines
- Energy loses in machines
- Ways of improving efficiency in machines

Summary notes

Machine

- Is a device which is constructed to make work easier by reducing human effort.
- Machines usually change energy from one form to another.

Work done

- Work done is force multiplied by the distance.
- Work done is calculated in newton meters(Nm) or joules (J).

When using a machine, two types of work are identified that is

- Work done by machine
  = Load × Distance moved by load
- Work done by effort
  = Effort × Distance moved by effort

Simple machines

- Levers
- Pulley system
- Inclined plane
- Gears

Levers

- These are machines which can turn about a pivot
- A lever acts as a force multiplier and it consists of :
  - pivot or fulcrum
  - load
  - effort

Effort is the force which is used to overcome a resisting force called the load.

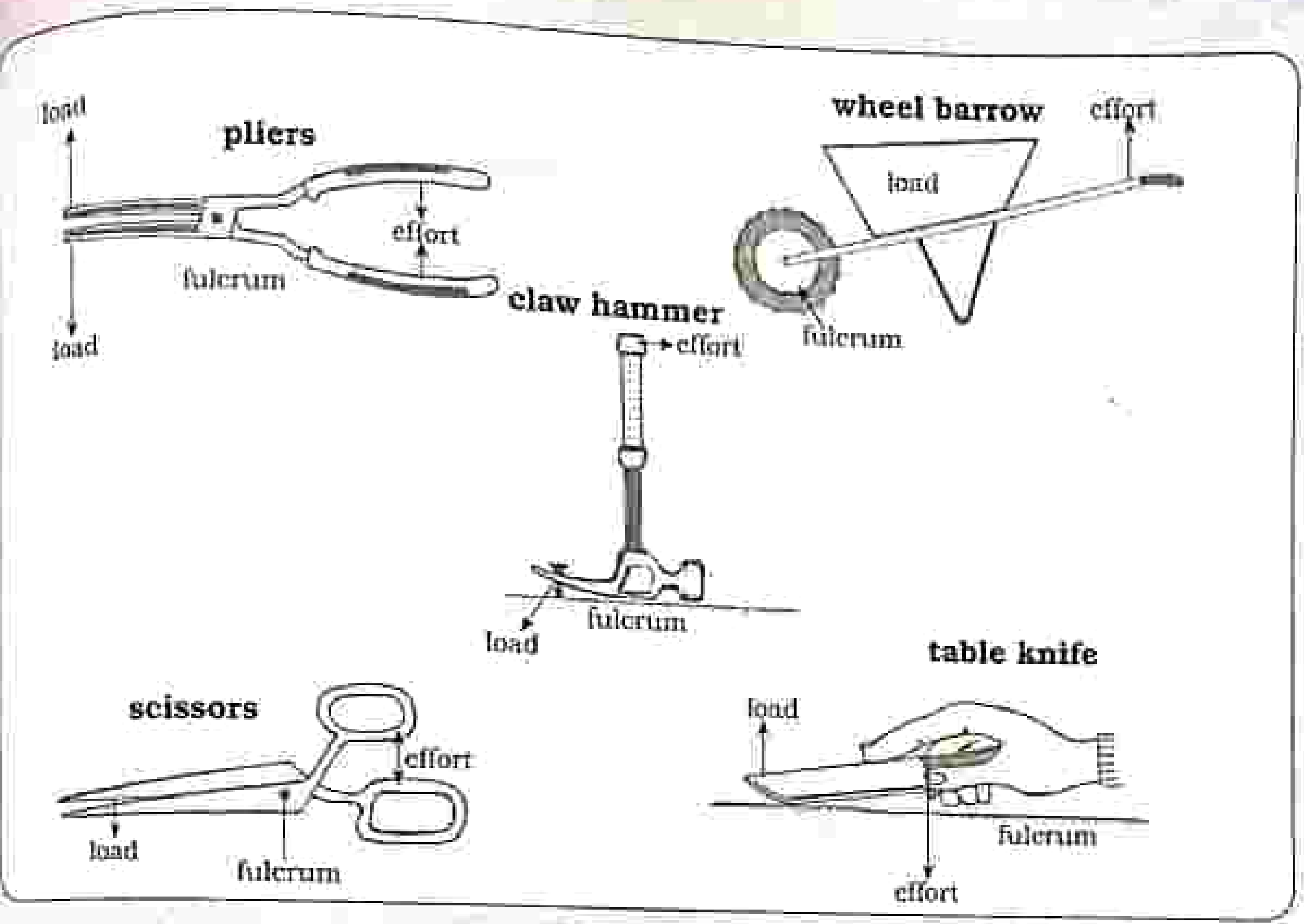


Fig. 24.1 Examples of levers

Mechanical advantage – is the ratio of the load to the effort.

Mechanical Advantage (MA) = $\frac{\text{Load}}{\text{Effort}}$

- a more useful machine has a large MA therefore less effort is required to lift the load
- MA does not have units because it is a ratio

Example

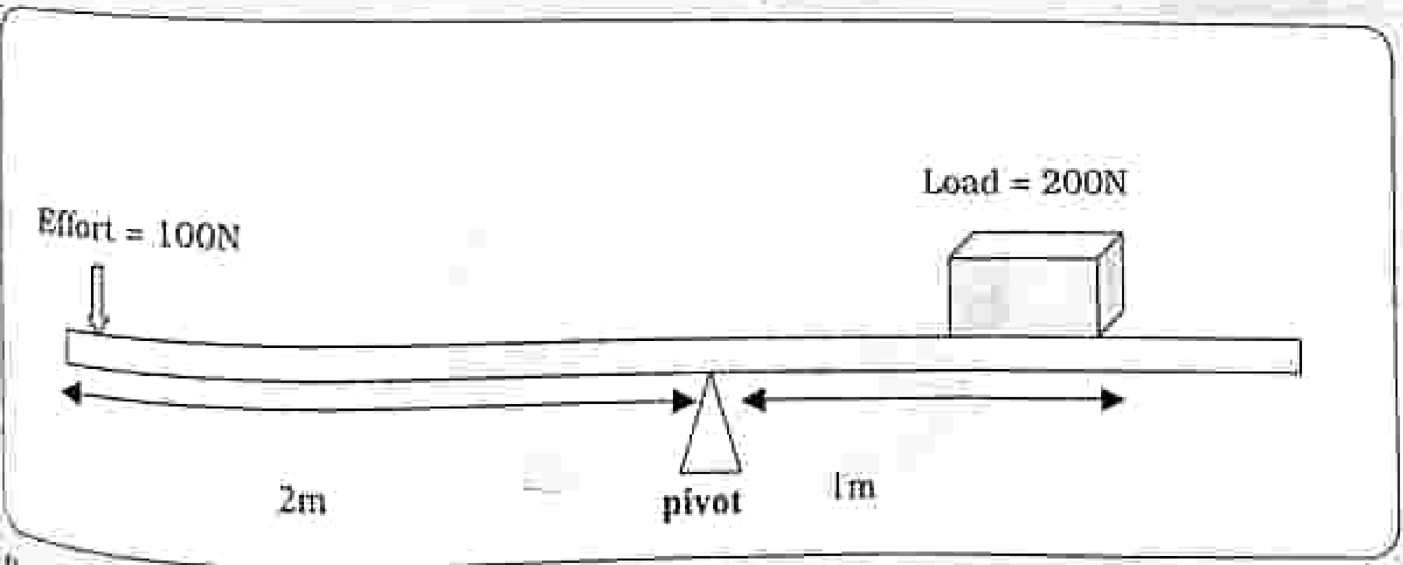


Fig. 24.2

$$MA = \frac{\text{Load}}{\text{Effort}}$$
$$= \frac{200N}{100N}$$
$$= 2$$

If MA = 2, then the load force will be twice as big as the effort force.

Velocity ratio

- Is the ratio of the distance moved by the effort to the distance moved by the load.
- It measures how useful a machine is

$$\text{Velocity ratio} = \frac{\text{distance moved by effort}}{\text{distance moved by load}}$$

Example

Using Fig 24.2 calculate VR if the distance moved by the effort is 2m and the distance moved by the load is 1m.

$$VR = \frac{\text{distance moved by effort}}{\text{distance moved by load}}$$
$$= \frac{2}{1}$$
$$= 2$$

Note: MA and VR do not have units because ratio values do not have units.

Efficiency

- It is a measure of how well a machine works.
- No machine is 100% efficient.
- The efficiency of a machine is calculated using the formula below:

$$\text{Efficiency} = \frac{\text{work output}}{\text{work input}} \times 100$$
$$= \frac{\text{work done on load}}{\text{work done by effort}} \times 100$$
$$= \frac{\text{load} \times \text{distance moved by effort}}{\text{effort} \times \text{distance moved by load}}$$
$$= \frac{MA}{VA} \times 100$$

Note : efficiency is usually expressed as a percentage.

- It does not have units because it is a ratio.

Example

Using Fig 24.2 calculate the efficiency if the distance moved by the effort is 2m and the distance moved by the load is 1m.

$$\text{Efficiency} = \frac{\text{load} \times \text{distance moved by effort}}{\text{effort} \times \text{distance moved by load}} \times 100$$
$$= \frac{200 \times 1}{100 \times 2} \times \frac{100}{1}$$
$$= 100\%$$

Note : Practically a lever can never be 100% efficient because some energy is lost due to friction.

Pulley

- It is used to make work easier by changing the direction of the force.
- Is used for lifting the load for example a pulley can be used by mechanics to lift engines out of cars.
- A pulley is a force multiplier.

Single fixed pulley

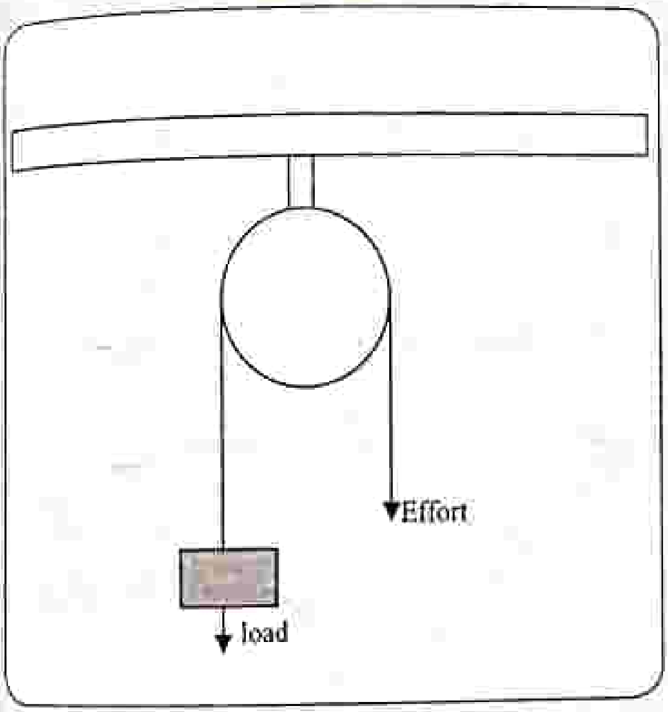


Fig 24.3 Single fixed pulley

- Changes the direction of the effort and the effort is slightly bigger than the load.
- If the pulley is 100% efficient, the values of the load and the effort are equal therefore:

  Mechanical Advantage (MA) = 1
- Distance moved by the load and the effort are the same so:

  Velocity Ratio (VR) = 1

Single movable pulley

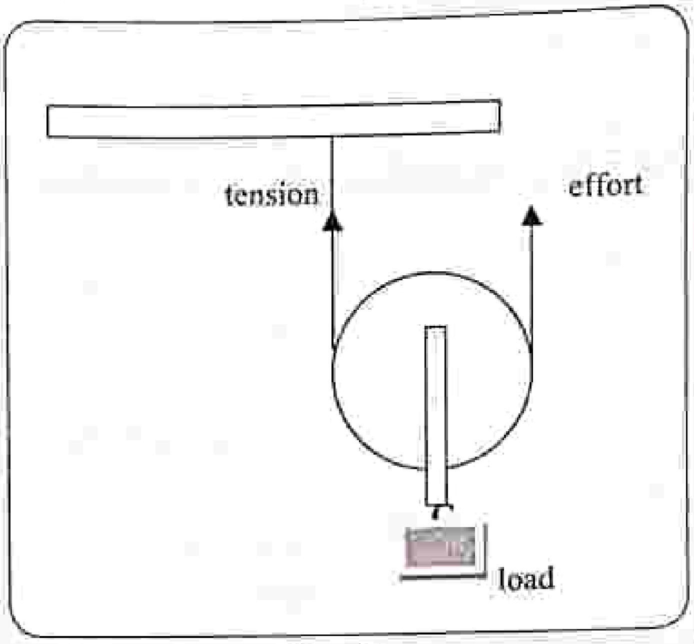


Fig 24.4 Single movable pulley

- The load is attached on the pulley which can be moved.
- Tension in the string and effort are the same and both are upwards forces.
- As the pulley moves, the rope must move twice the distance moved by the load giving a

  Velocity Ratio = 2

  Assuming the pulley is 100% efficient

  Load = 2 × Effort

  $$MA = \frac{\text{Load}}{\text{Effort}} = \frac{2E}{E} = 2$$

Block and Tackle

- Two or more pulleys are used in a block and tackle.
- Each fixed pulley in the block and tackle is a direction changer.

- Increase in the number of pulleys makes it easier to lift a load.

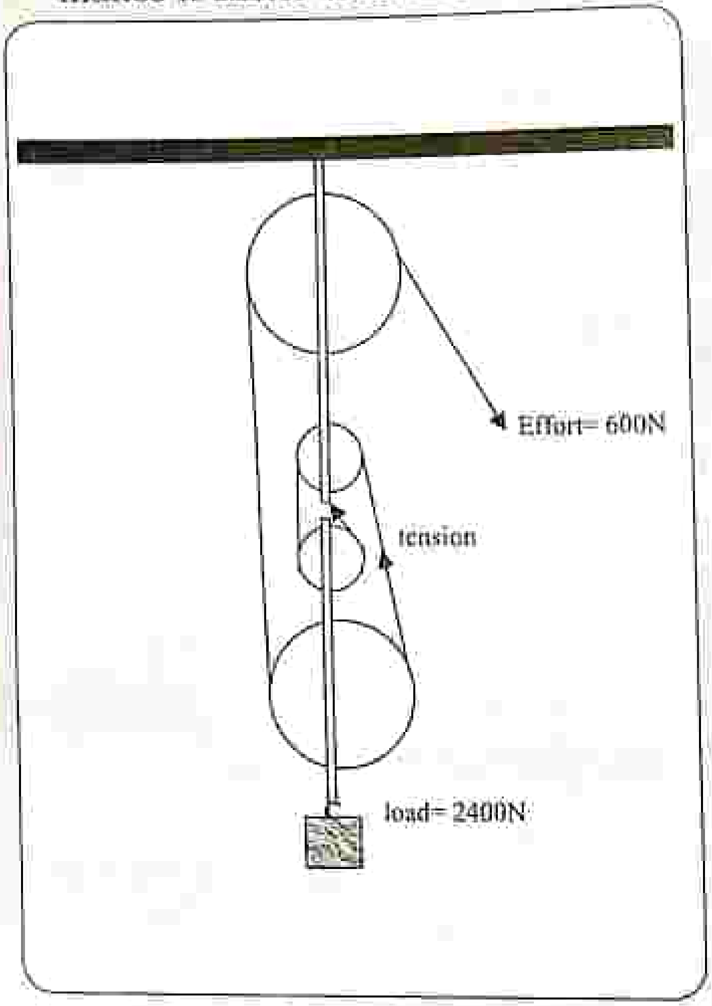


Fig 24.6 Block and tackle pulley

Assuming the pulley is 100% efficient

Load = 4 × effort

Note : 4 is the number of pulleys

Use Fig 24.6 to calculate MA, VR and Efficiency

Mechanical Advantage (MA) $= \frac{\text{Load}}{\text{Effort}}$

$= \frac{2E}{E}$

$= 4$

Velocity Ratio (VR)

- is equal to the number of pulleys in the block and tackle system

  Velocity Ratio = 4

Efficiency

Efficiency $= \frac{\text{load} \times \text{distance moved by effort}}{\text{effort} \times \text{distance moved by load}} \times 100$

$= \frac{MA}{VA} \times 100$

$= \frac{4}{4}\ 100$

$= 100\%$

Inclined Plane

It is a flat plank tilted at an angle with one end higher than the other which is used to reduce the force required to lift a load.

- A smaller angle for the inclined plane results in a small force being used to lift the load but the load will move a long distance.
- It is easier to push a drum of water up a plank onto a lorry than to lift it vertically.

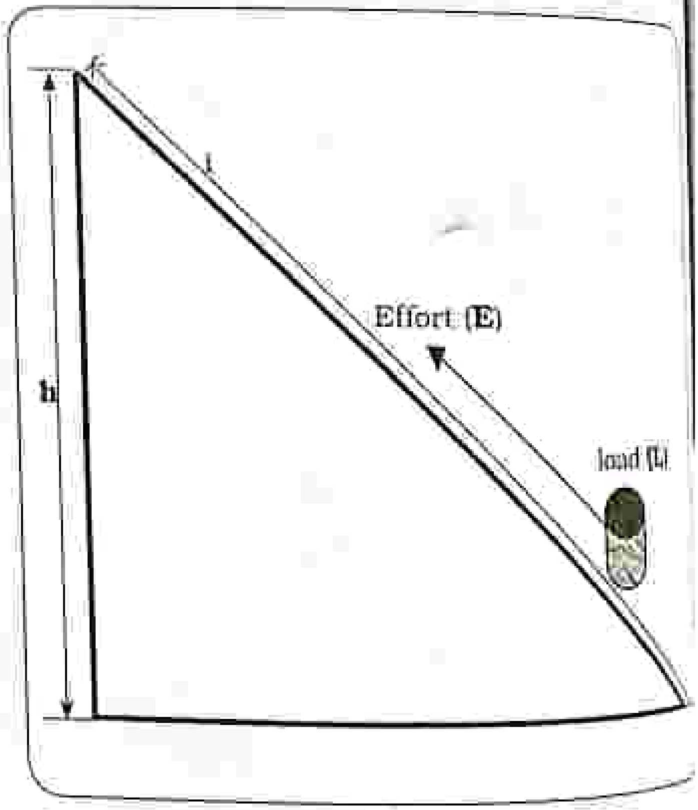


Fig 24.7 Inclined plane

Mechanical Advantage (MA) $= \frac{\text{Load (L)}}{\text{Effort (E)}}$

Velocity Ratio (VR) $= \frac{\text{length of incline (l)}}{\text{height of incline (h)}}$

Efficiency

Work done by load

= Load × length of incline

Work done by effort

= Effort × height of incline

Efficiency = $\frac{L \times l}{E \times h}$ × 100

Where

L = load

l = length of incline

E = effort

h = height of incline

Example

A drum of weight 10N is pulled from the bottom to the top of the inclined plane in Fig 24.7 by a force of 5N. If **h** is 4m and **l** is 2m. Calculate the:

a) MA

b) VR

c) Efficiency

Solution

a) Mechanical Advantage = $\frac{\text{Load}}{\text{Effort}}$

= $\frac{10N}{5N}$

= 2

b) Velocity Ratio (VR) = $\frac{\text{length of incline (L)}}{\text{height of incline (h)}}$

= $\frac{2m}{4m}$

= 0.5

c) Efficiency = $\frac{\text{load} \times \text{length of incline}}{\text{effort} \times \text{height of incline}}$ × 100

= $\frac{10 \times 2}{5 \times 4}$ ×100

= $\frac{20}{20}$ × 100

= 1×100

= 100%

Gears

- Gear wheels consist of cog wheels mounted on shafts.
- A gear is a rotating machine which has cut teeth on the circumference.
- Gear teeth act like small levers.
- Gears of different sizes can transfer force from one gear to another when they interlock.
- The gears consist of the load gear and the effort gear.
- Of most importance is not the size but the number of teeth the gear has.
- When the teeth of the gears interlock properly they prevent slipping and can exhibit efficiencies of up to 98%.
- Always remember that the load gear will turn in the opposite direction of the effort gear.

Low gear

- The load gear is larger than the effort gear.
- Effort gear has to turn more for the load gear to turn once.
- The velocity ratio will be high.
- Low gear is used on very heavy load for example when going uphill.

High gear

- Both the effort gear and the load gear are of the same size.
- One turn of the effort gear will turn the load gear.

- High gear is used when the load is light for example accelerating an already moving car.

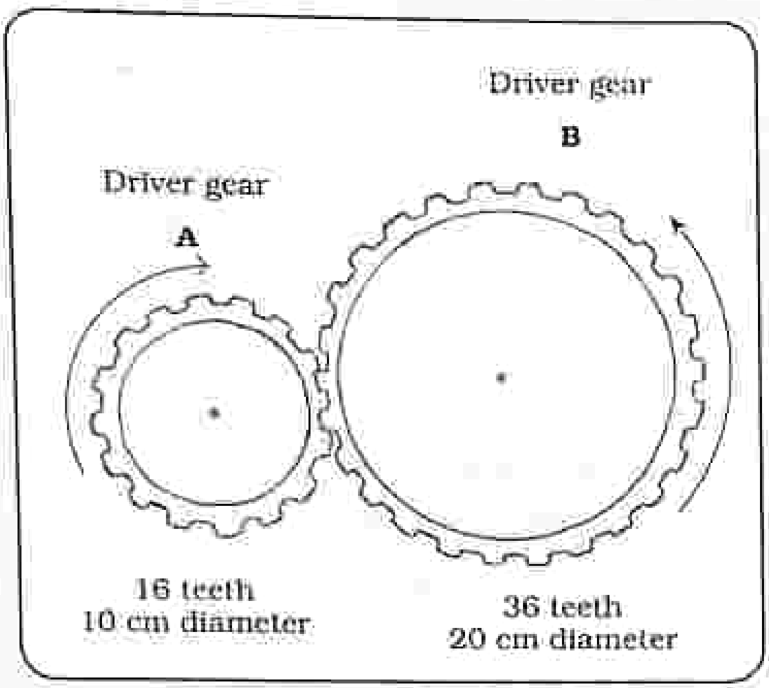


Fig. 24.8 Gear wheels

Example

Using Fig 24.8 find the:

a) Velocity Ratio (VR)

b) Mechanical Advantage (MA) of the gears

Solution

a) Velocity Ratio (VR)

$$= \frac{\text{number of teeth on the load gear}}{\text{number of teeth on the effort gear}}$$

$$= \frac{24}{12}$$

$$= 2$$

b) Mechanical Advantage (MA)

$$= \frac{\text{diameter of the load gear}}{\text{diameter of the load gear}}$$

$$= \frac{20}{10}$$

$$= 2$$

Uses of simple machines

- To reduce the effort (force) required to perform a task.
- Changes the direction of a force.
- Increases the distance or the speed of the force.
- Move forces from one form to another.

Applications of simple machines

Levers

- Used as see saw for children to play on.

Pulley

- Used on flag poles to raise or lower the flag.

Inclined plane

- Ramp for wheelchairs.

Gears

- Used in cars.

Energy loses in machines

No machine is 100% efficient because some energy is lost due to:

- Friction and the energy is lost in the form of heat.
- The mass of moving parts.

Ways of improving efficiency in machines

- Lubrication using grease or oil.
- Use of bearings and roller.
- Reduction of the mass of the moving parts.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

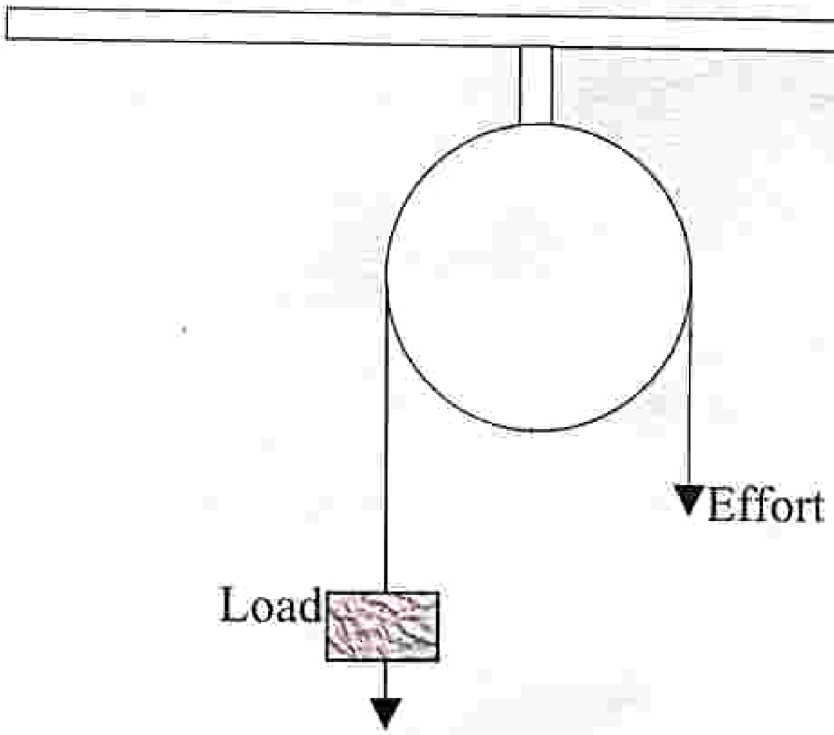
The number of marks is given in brackets[ ] at the end of each question.

Multiple choice questions

1. Which one is not an example of a lever
   A. Scissors B. Wheelbarrow
   C. Forearm D. Ramp

2. How much work is done to lift a drum of water which is 20N through 15m?
   A. 300J B. 3000J
   C. 35J D. 1.33J

Use the diagram below of a single fixed pulley to answer question 3 and question 4



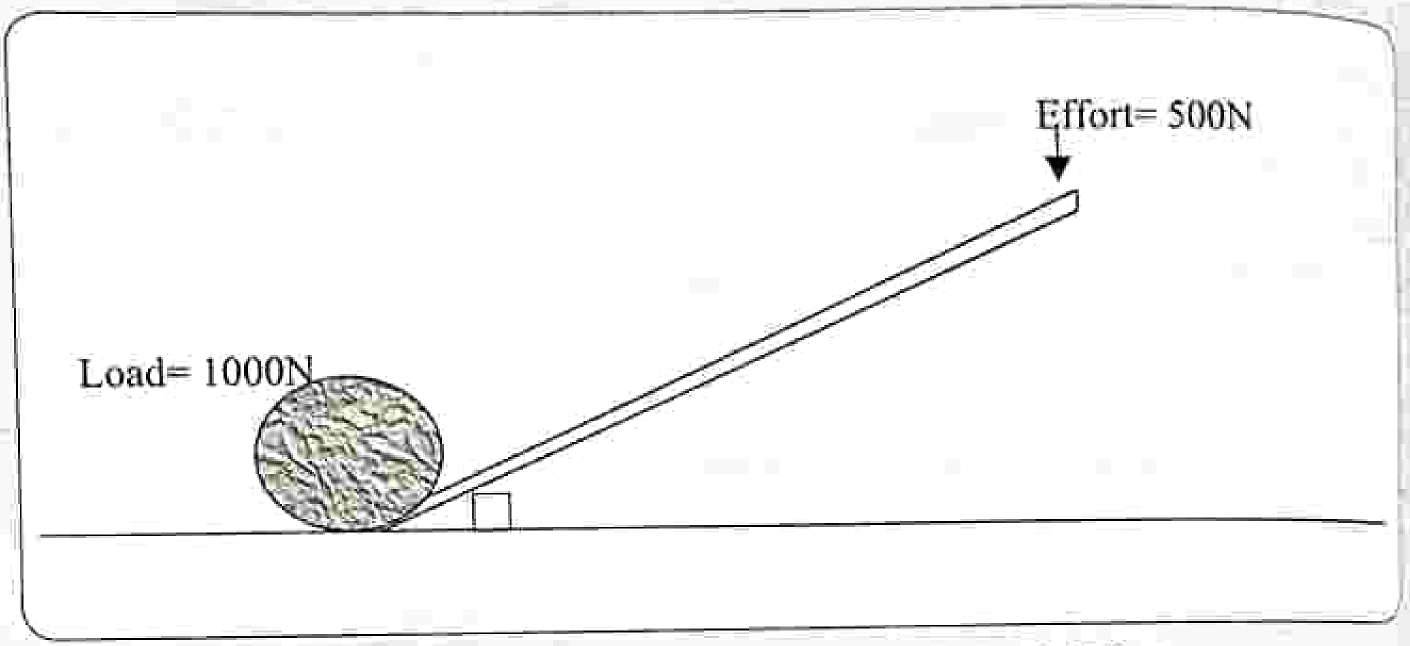
3. What is the Velocity Ratio

A. 1 B. 2 C. 3 D. 4

4. What is the Mechanical Advantage

A. 1 B. 2 C. 3 D. 4

5. The diagram shows a crowbar being used to move a load



What is the Mechanical Advantage of the lever?

A. 4 B. 3 C. 2 D. 1

6. The mechanical advantage of an inclined plane can be increased by
   A. Decreasing the gradient of the plane
   B. Increasing the gradient of the plane
   C. Making the plane long
   D. Making the plane short

7. Which one is not a way of improving efficiency in machines
   A. Lubrication
   B. Use of bearings
   C. Reduction of mass of moving parts
   D. Increasing mass of moving parts

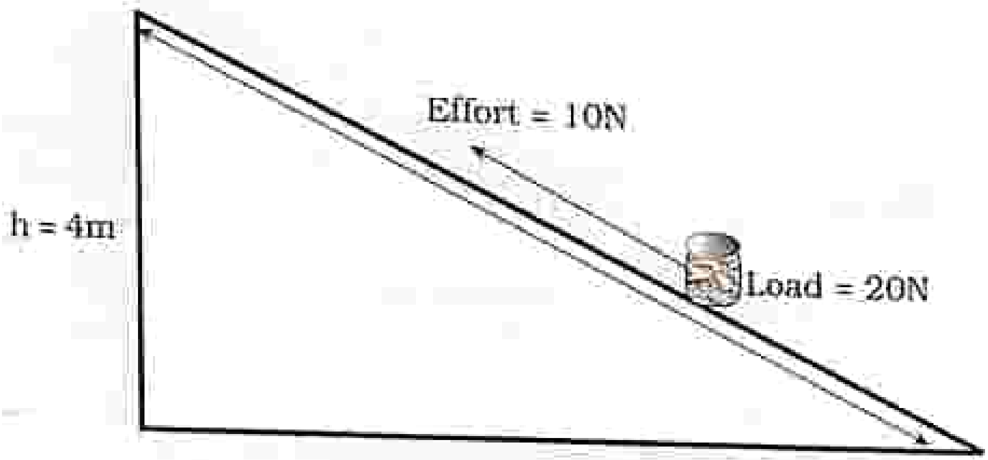
8. Levers consist of
   A. Pivot, Load, Incline
   B. Pivot, Load, Effort
   C. Pivot, Incline, Effort
   D. Pivot, Load, Pulley

9. Energy in machines is lost through
   A. Friction
   B. Distance
   C. Effort
   D. Load

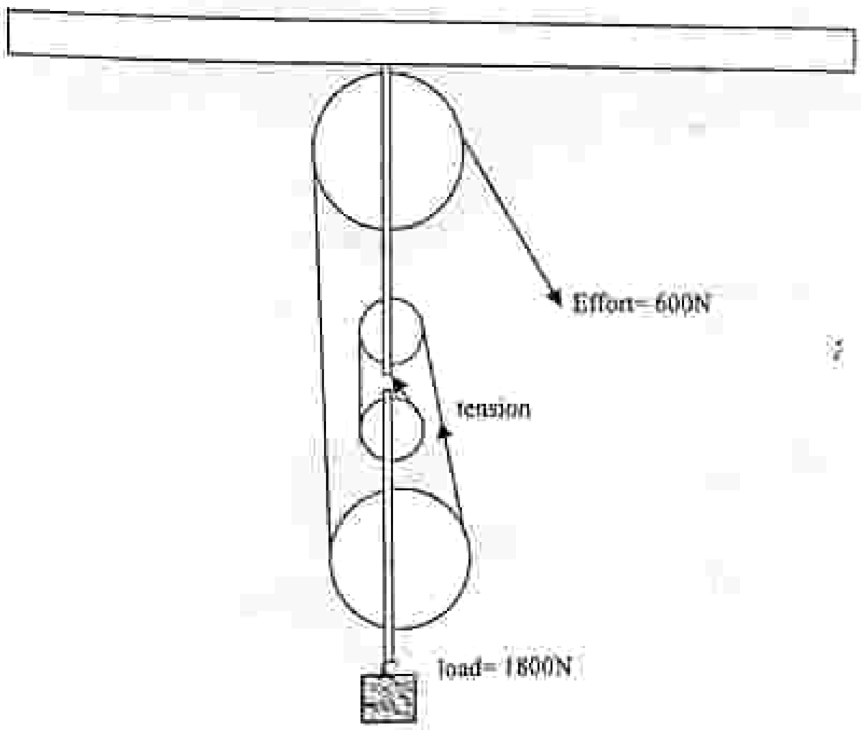
10. Identify where levers are used in everyday life
   A. As see saw for children to play on
   B. On flag poles to raise or lower the flag
   C. Ramp for wheel chairs
   D. In cars to change the motion when moving uphill

Section B: Structured questions

1. a) Define a machine.
   b) Describe uses and applications of simple machines.
2. a) Explain energy losses in machines.
   b) Describe ways of improving efficiency in machines.
3. A drum of weight 20N is pulled from the bottom to the top of the inclined plane by a force of 10N. If **h** is 4m and **l** is 2m,calculate the :
   a) Mechanical Advantage
   b) Velocity Ratio



4. Using the block and tackle pulley calculate the
   a) Mechanical Advantage
   b) Velocity Ratio
   c) Efficiency



CHAPTER 25

Pressure

Key concepts

- Pressure
  - $P = F/A$
  - pressure of solid objects
- Pressure in liquids
  - $P = gh$
  - Effect of depth on pressure
- Atmospheric pressure
- Fluid system
  - Siphon
  - Hydraulic jack
  - Car braking system
- Simple pumps
  - Lift pump : blair pump
  - Force pump : bicycle pump

Summary notes

Pressure

- It is defined as the force acting per unit area
- Units of pressure are Pascals (Pa)

Pressure (P) = $\frac{\text{Force}}{\text{Area}}$ $\frac{F}{A}$

Pressure of solid objects

a)

b)

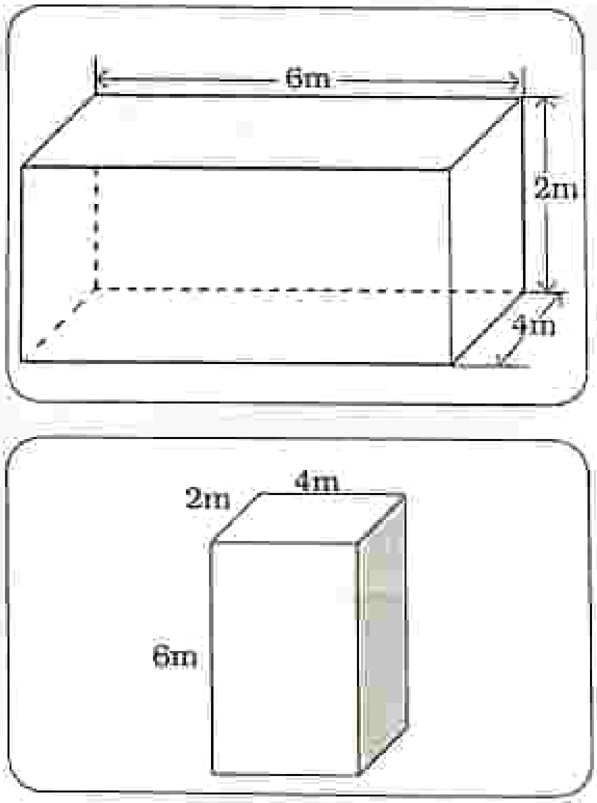


Fig. 25.1: Cuboids

Calculate the pressure exerted on the floor by the box in Fig. 25.1 which weighs 48N.

a) Area = L × W
= 6m × 4m
= $24m^2$

Pressure = Force/Area
= $48N/24m^2$
= 2Pa

b) Area = L × W
= 4m × 2m
= $8m^2$

Pressure = Force/Area
= $48N/8m^2$
= 6Pa

- The greater the area over which a force acts, the less the pressure and the smaller the area over which a force acts, the greater the pressure.
- Pressure can be increased by:
  - Increasing the force
  - Decreasing the area.

Pressure in fluids

- Pressure in fluids is measured in Pascals (Pa).
- Pressure in liquids

  = density × gravitational force × height of fluid column

$P = \rho gh$

- Where

P is pressure in liquid

ρ is density

h is height of liquid column

g is gravitational force

- Calculate the pressure 100m below sea water which has a density of 1150kg/m³. Take g =10m/s²

Pressure = ρgh

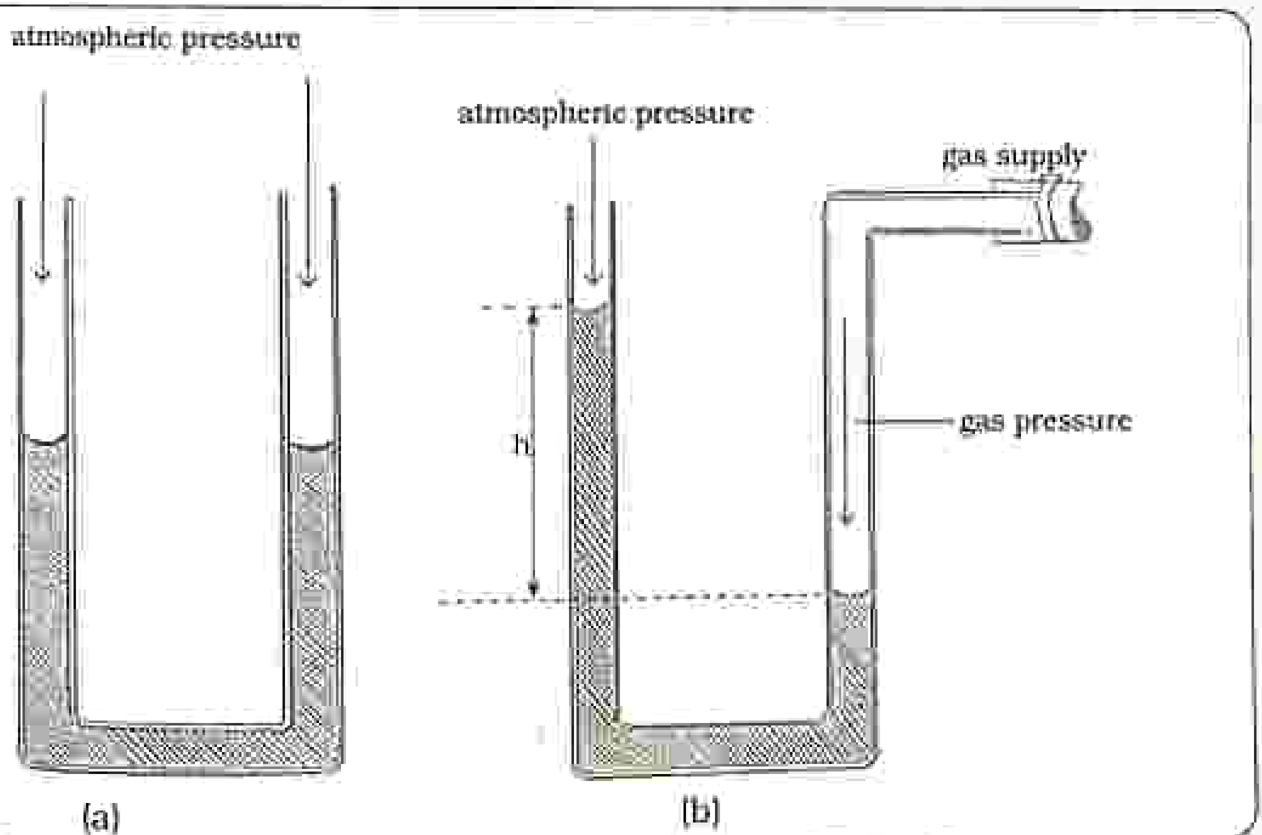
= 1150kg/m³ × 10m/s² × 100m

= 1 150 000Pa

= 1150KPa

Simple manometer

- It is used to measure gas pressure.
- It consists of a U-tube which can be made using a piece of a clear plastic tube.
- A coloured liquid is added into the U tube.
- When both arms of the manometer are exposed to atmosphere the liquid surfaces are at the same level.
- If one arm is connected to a gas supply, the liquid is pushed up in the other arm because the gas exerts a pressure.
- h is the difference in height of the surfaces of the water in both arms.
- The height difference gives the pressure difference between the gas pressure and the atmospheric pressure.
- Gas pressure = Atmospheric pressure + ρgh



25.2: Manometer

Calculate the gas pressure when a U tube manometer containing water of density $1000kg/m^3$ is connected to a gas supply.

The difference in manometer level is 3m.

- The atmospheric pressure is 1300Pa.
- Gas Pressure = atmospheric pressure + ρgh

$= 1300Pa + 1000kg/m^3 \times 10m/s^2 \times 3m$

$= 1300 + 30000$

$= 31300Pa$

$= 31.3KPa$

Effect of depth on Pressure

- Pressure in a liquid increases with depth.
- The further down you go, the greater the weight of the liquid above thus an increase in pressure.

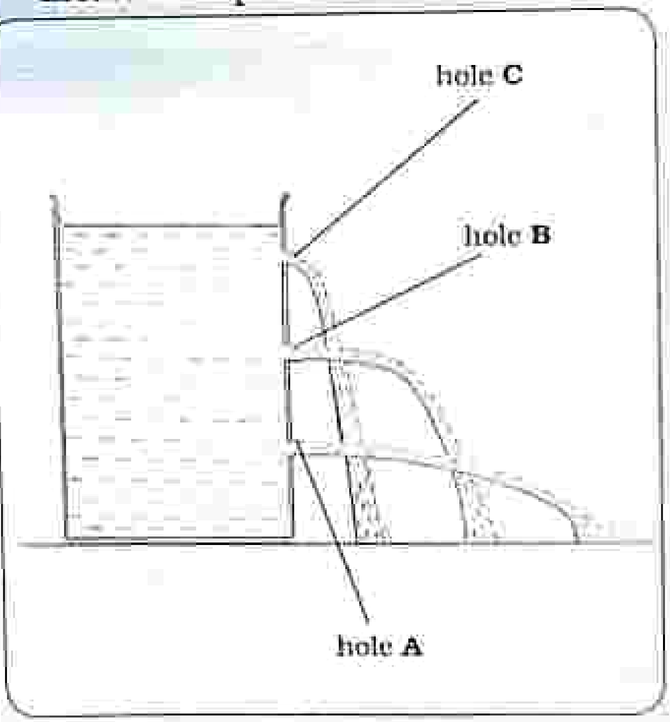


Fig. 25.3: Effect of depth on pressure

Water from hole **A** sprout out fastest and furthest from the container.

Water from hole **C** sprout out slowest and nearest to the container.

Dam wall

The design of a dam wall is in such a way that:

a) The bottom of the dam wall is thick in order to be able to with stand the high pressure at the bottom because pressure increases with depth.

b) The top of the dam wall is thin because the water at the top has less pressure.

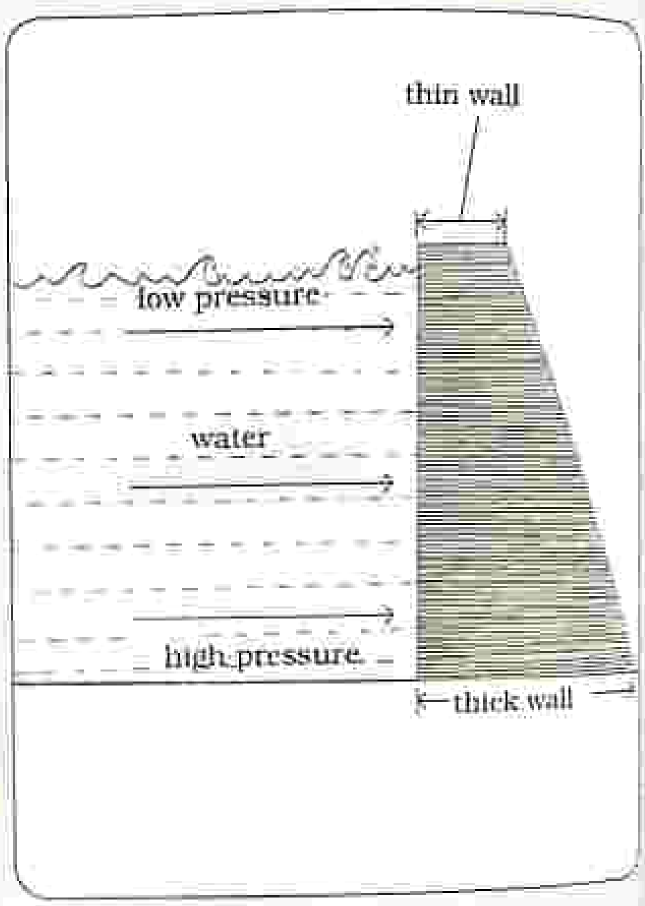


Fig. 25.4: Dam wall

Atmospheric pressure

- It is the pressure exerted by the weight of air on the surface of the earth.
- It can be measured using a barometer.
- A tube is filled with mercury and then inverted into a trough as shown in Fig 25.5.
- XY is the atmospheric pressure in mm of mercury (mmHg).

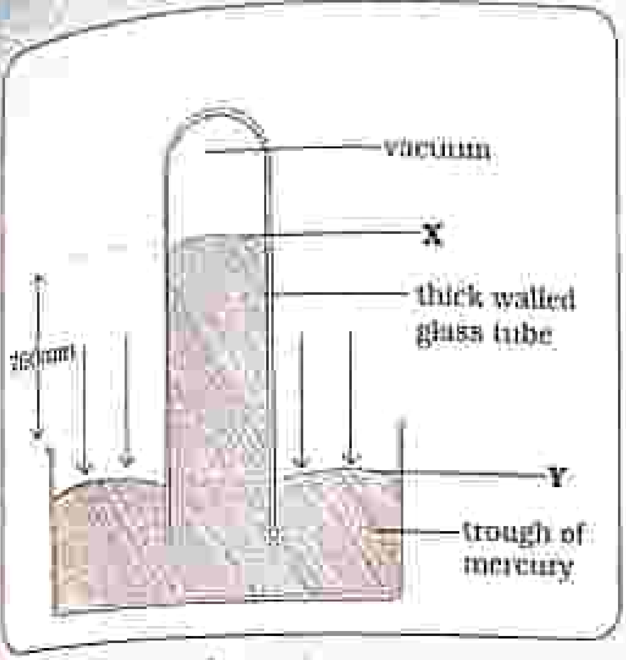


Fig. 25.5: Mercury barometer

Fluid systems

- They work by using fluids under pressure.

Siphon

It is used to move liquids from a higher level to a lower level by using the pressure exerted on the liquid.

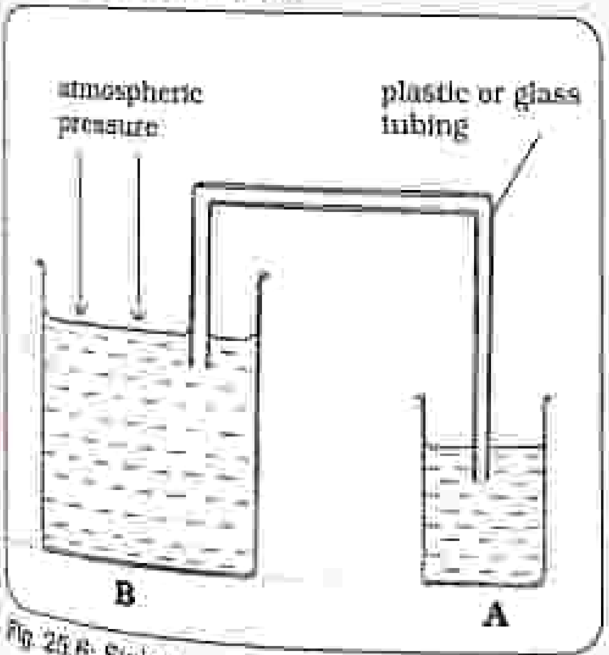


Fig. 25.6: Siphon

- Fill the tube with liquid.
- Gently suck end **A** of the tube until a vacuum is created.
- The liquid will move through the tube from B freely through **A** as long as the end of **B** remains in the liquid, the liquid will continue to move.

Hydraulic jack

- It is used to move large loads using small effort.

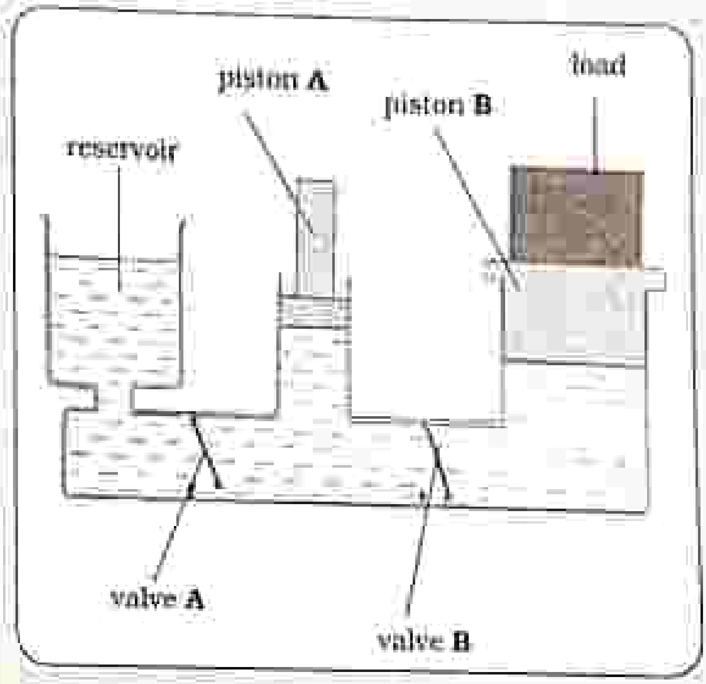


Fig. 25.7: Hydraulic jack

- When piston **A** moves down:
  - It exerts pressure on the liquid and the liquid pressure is transmitted equally through the liquid.
  - Valve **A** closes.
  - Valve **B** opens.
  - Piston **B** is forced to move up thereby lifting the load.

- When valve **A** moves up:
  - Valve **A** opens and liquid from the reservoir moves in through valve **A**.

  Valve **B** closes.
  - Piston **B** moves down.

Car braking system

- When the brake pedal is pressed, the force on the small piston of the master cylinder exerts pressure on the brake fluid.
- The pressure is transmitted equally throughout the fluid.
- The pressure exerts a greater force on the larger pistons of the slave cylinder.
- Which in turn clamps the discs of the brake shoe and slows the car down or stops the car.

NOTE: liquid is incompressible so no air bubble must be present in the liquid because any transmitted force would compress the bubble instead of transmitting the pressure, resulting in the brake pedal becoming spongy and will not stop the car.

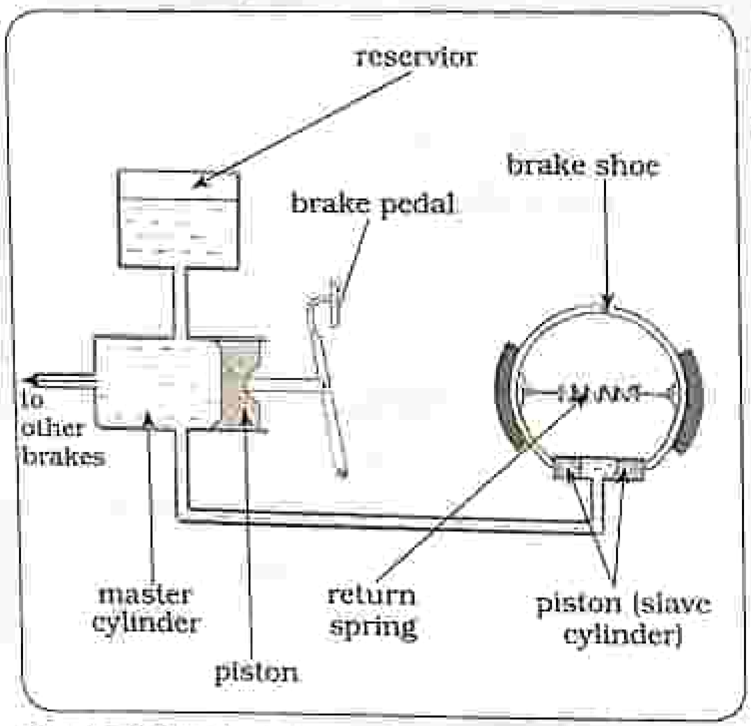


Fig. 25.8: Car braking system

Simple pumps

- Fluids under pressure are used in simple pumps.
- There are two main types of pumps that is:
  - Lift pump
  - Force pump.

Lift pump

- A lift pump has two valves, one in the piston and the other at the bottom of the barrel.
- The pump is primed first, that is adding of water to the pump.

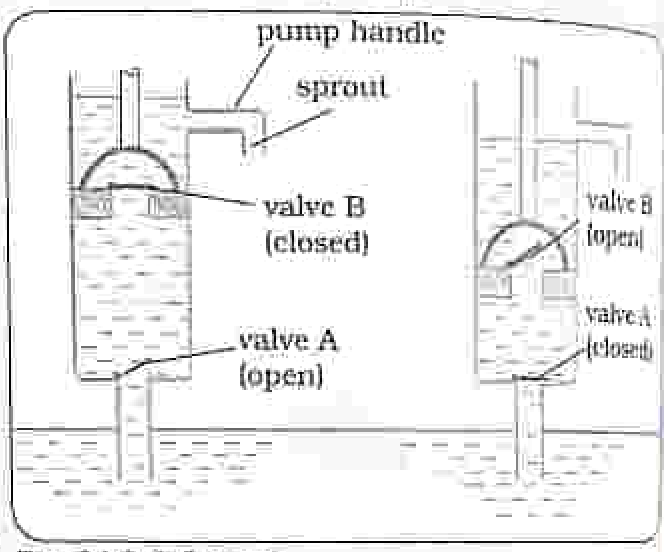


Fig. 25.9: Lift pump

Upstroke

- Valve **A** is closed and valve **B** is opened allowing water into the cylinder which has been pushed by atmospheric pressure acting on the surface of the water.
- On top of the piston, water moves upwards and flows out of the spout.

Downstroke

- Valve **B** closes and valve A opens allowing water to move upwards into the cylinder above the piston.

  Note : a lift pump can only lift water up to 10m because thats far atmospheric pressure can support a column of water.

Blair pump

- It is a lift pump which pulls water out of a well.
- Water is moved from underground to a higher level above the ground.
- The handle is continuously moved up and down to ensure water continues to be pumped out.

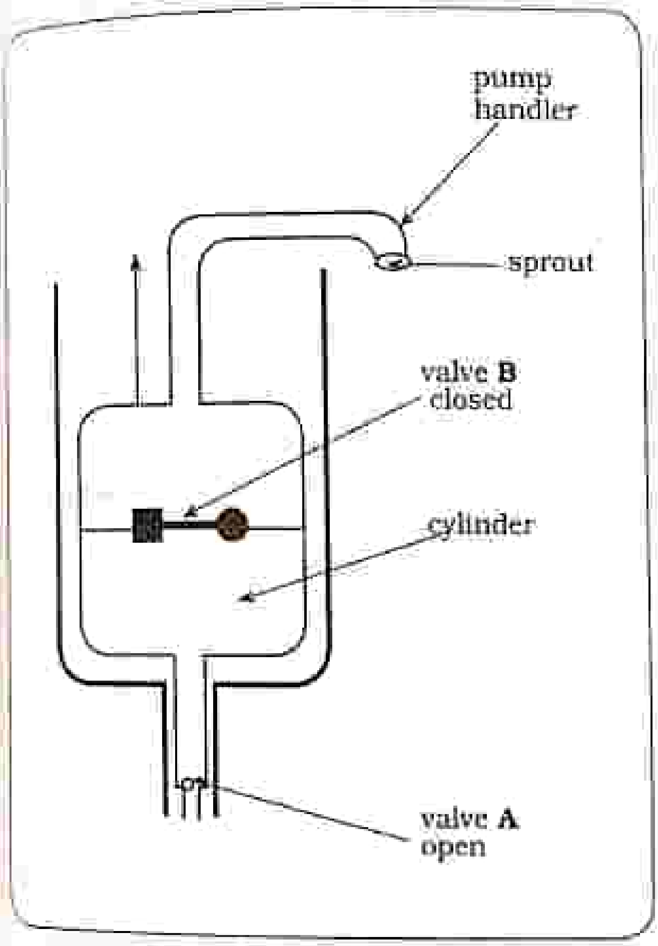


Fig. 25.10: Blair pump

- Handle up:
  Valve **B** closes.
  Pressure causes valve A to open and allows water to enter into the cylinder.
- Handle down:
  valve **A** closes
  water in the cylinder forces valve B to open thereby forcing water out through the spout.

Force pumps

- A force pump can lift water up more than 10metres.
- It does not need to be primed (filled with water) in order to work.

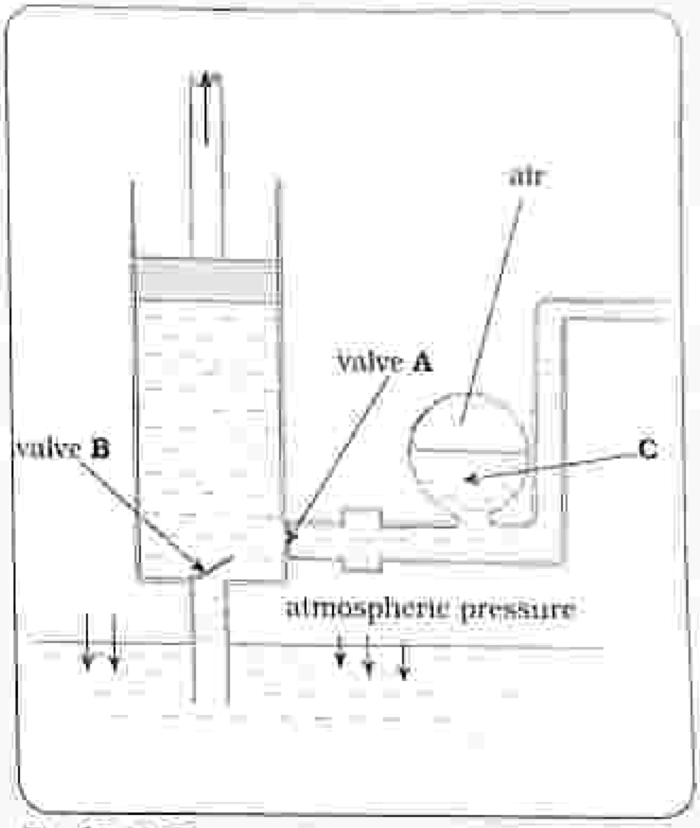


Fig. 25.11: Force pump

Upstroke

- Valve **A** closes and valve **B** opens.
- Atmospheric pressure forces water to move upwards into the cylinder through valve **B**.
- Air in **C** expands to ensure a continuous supply of water to the spout.

Downstroke

- Valve **B** closes and valve **A** opens.
- Water is forced through valve **A** into reservoir **C** and out of the spout.
- Air in **C** is compressed.

Bicycle pump

- A bicycle pump is a force pump which uses fluid under pressure to function.

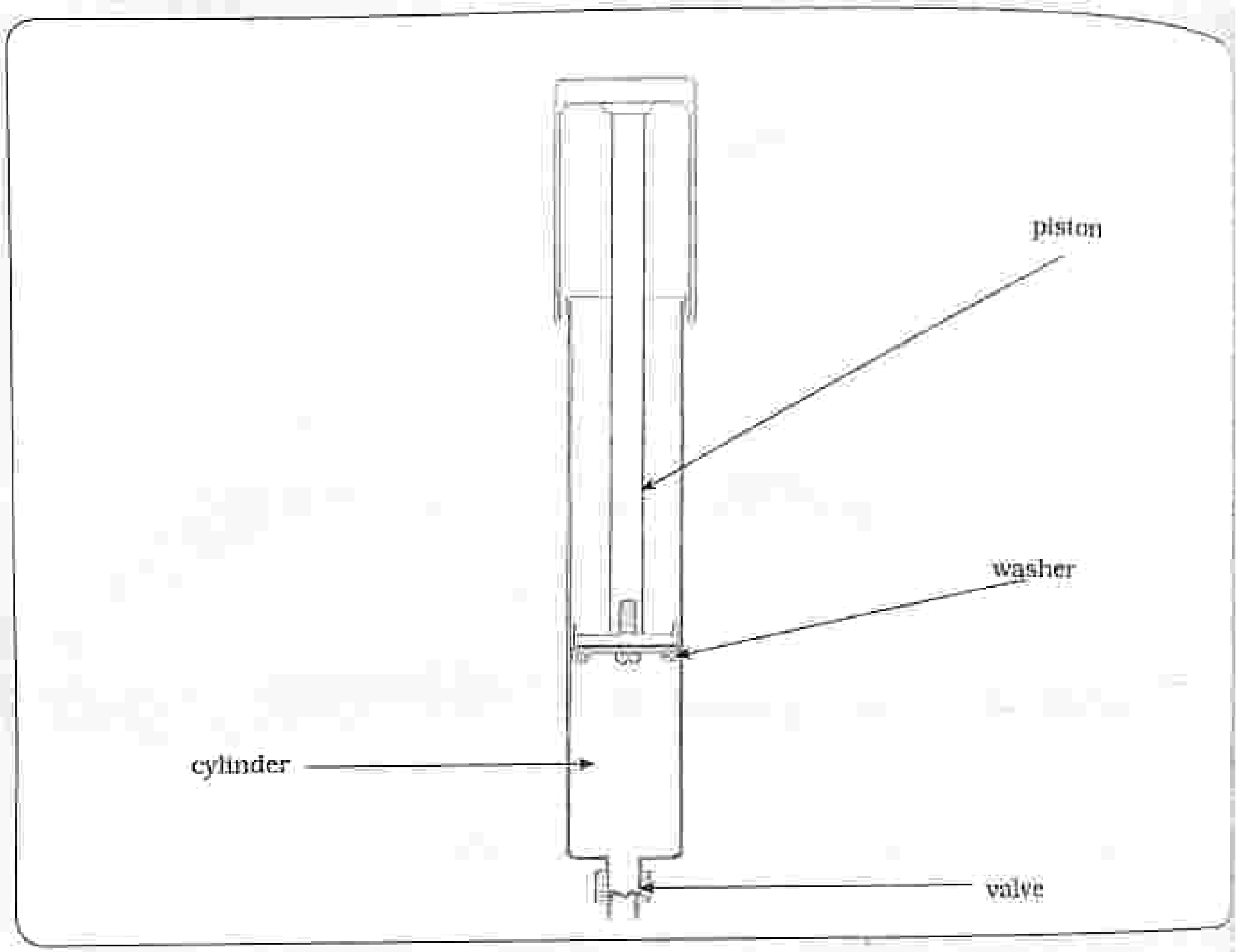


Fig. 25.12: Bicycle pump

Piston pushed in

- When the the piston is pushed down, the plastic washer presses against the walls of the cylinder to form an airtight seal.
- Air in the cylinder is forced out past the tyre valve into the tyre.

Piston pulled out

- The tyre valve is closed by the greater pressure in the tyre.
- Atmospheric pressure causes air to move past the plastic washer this time not pressing against the walls of the cylinder.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

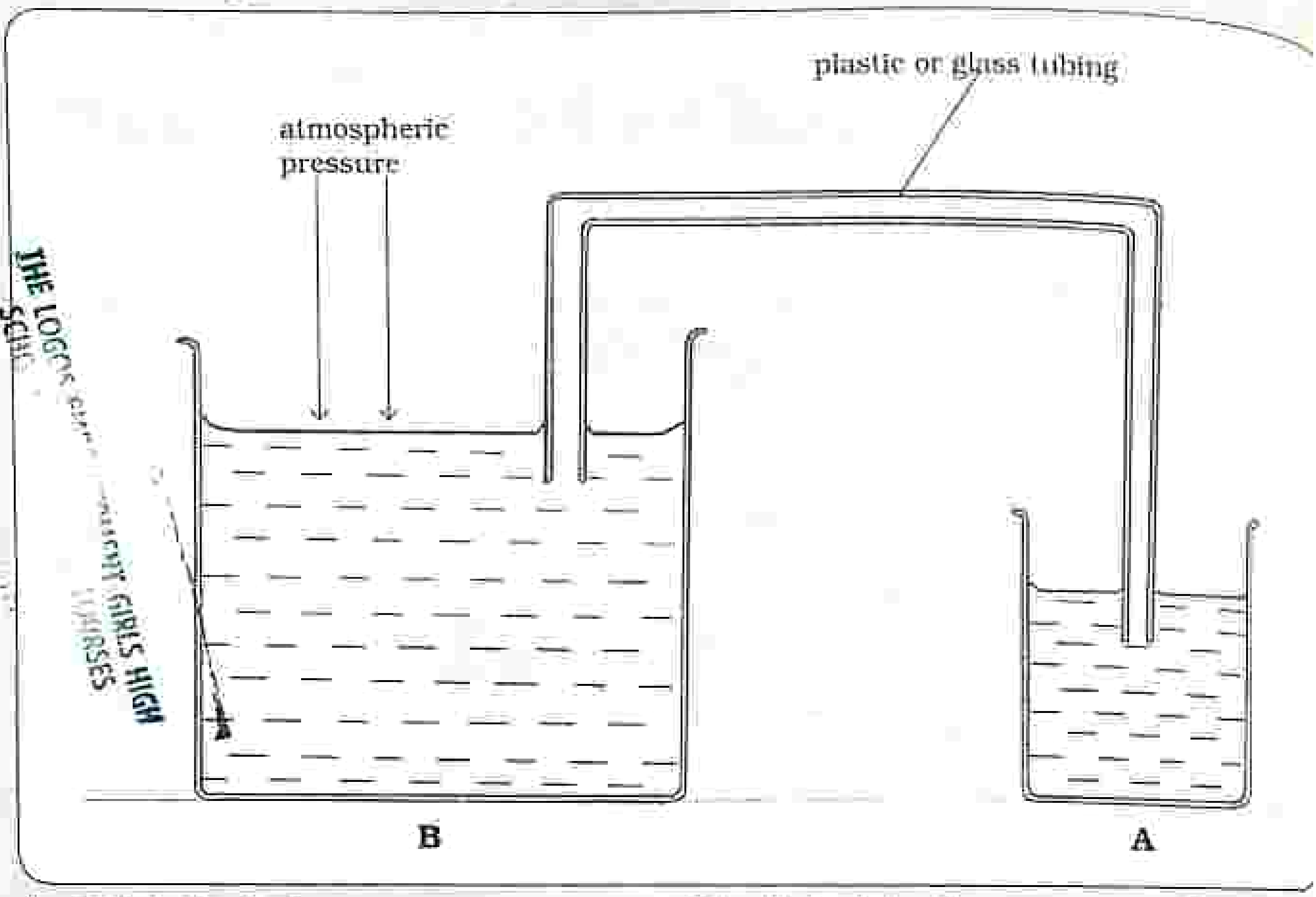
Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Define pressure
   A. force acting per unit mass
   B. force acting per unit area
   C. force acting per unit volume
   D. force multiplied by area
2. Pressure is measured in __________.
   A. watts
   B. pascals
   C. kilogram
   D. metres
3. Pressure in a liquid increases with _____.
   A. depth
   B. area
   C. size
   D. volume
4. Pressure in a liquid is calculated as
   A. pressure = gh.
   B. pressure = Agh.
   C. pressure = gA.
   D. pressure = ρg.
5. A simple manometer is used to measure __________.
   A. pressure of a solid
   B. atmospheric pressure
   C. gas pressure
   D. depth of pressure
6. Hydraulic jack is used to
   A. move large loads using a small effort.
   B. move liquids from higher level to lower level.
   C. measure atmospheric pressure.
   D. measure gas pressure.
7. Identify a lift pump
   A. siphon
   B. bicycle pump
   C. blair pump
   D. hydraulic jack
8. Identify a force pump
   A. siphon
   B. bicycle pump
   C. hydraulic jack
   D. blair pump

9. Name the diagram below.



A. hydraulic jack

B. barometer

C. manometer

D. siphon

10. Calculate the pressure of sea wate 50m below which has a density o $1150kg/m^3$. Take $g = 10m/s^2$

A. 57.5kPa B. 11.5kPa

C. 575kPa D. 5750kPa

Section B: Structured questions

1. a) Define pressure. [1]

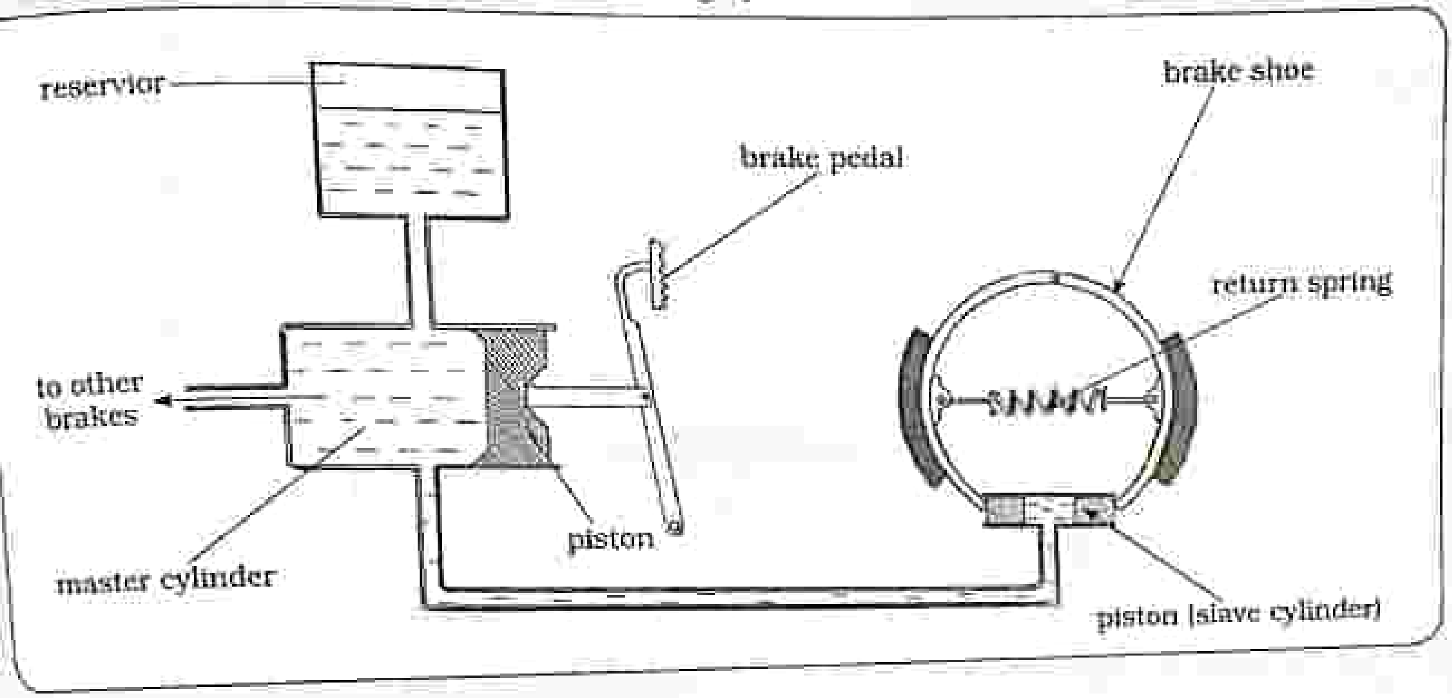
   b) Describe the effect of depth on pressure. [3]

2. a) What is the function of a blair pump. [1]

   b) Describe the structure and operation of a blair pump. [3]

3. Describe the function and operation of a bicycle pump. [3]

The diagram below shows a car braking system.



Explain how it operates. [5]

5. Explain the function and operation of a siphon. [3]

CHAPTER 26

Petrol and diesel engines

Key concepts

- Operation of a four stroke petrol engine
- Role of a carburetor
- Operation of a four stroke diesel engine
- Role of fuel injectors
- Comparison of petrol engine versus diesel engine
- Operation of a modern diesel engine
- Operation of a modern petrol engine
- Advantages of modern petrol engines over old petrol engines

Summary notes

Four stroke engine

- A four stroke engine consists of four piston strokes that include
  - Intake stroke
  - Compression stroke
  - Power stroke
  - Exhaust stroke

Operation of a four stroke petrol engine

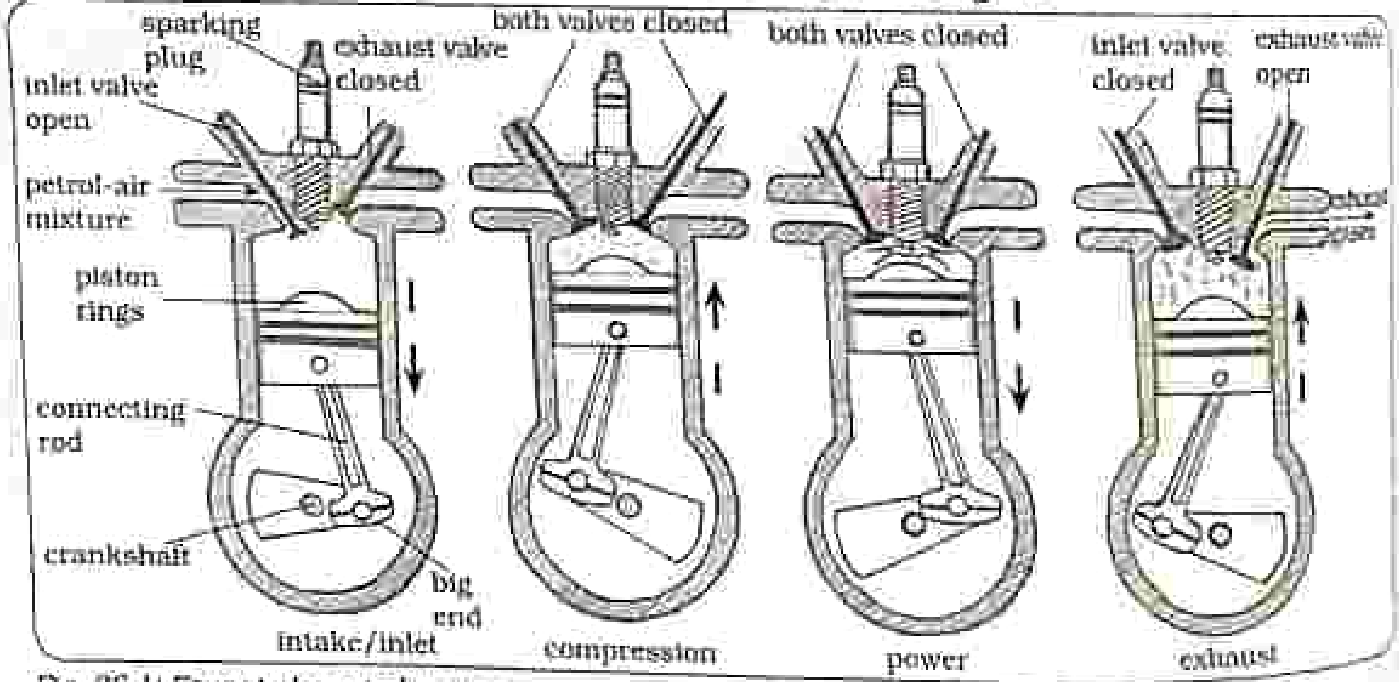


Fig. 26.1: Four stroke petrol engine

Intake stroke

- The inlet valve opens.
- Petrol-air mixture is forced into the cylinder.
- Piston moves down and pressure is reduced inside the cylinder.

Compression stroke

- Both valves are closed.
- Piston moves up and pressure inside the cylinder is increased.
- Petrol- air mixture is compressed.

Power stroke

- Both valves are closed.
- Petrol-air mixture is ignited by sparks from the spark plug. Piston moves down and pressure inside the cylinder is reduced.

Exhaust stroke

- Exhaust valve (outlet valve) opens and inlet valve remains closed.
- Piston moves up and the pressure inside the cylinder is increased.
- Exhaust gases are forced out of the cylinder through the exhaust valve.

Role of a carburetor

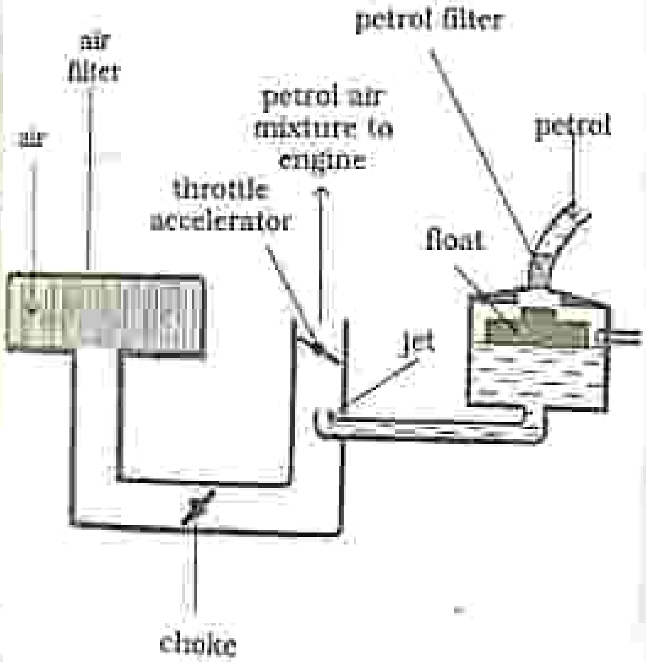


Fig. 26.2: A carburetor

- A carburetor is connected to the petrol engine.
- It allows the engine to run efficiently by allowing the mixture of petrol and air to be produced in correct proportions.

Petrol filter

- Petrol passes through the petrol filter where it is cleaned to prevent the jets from getting blocked by dirt.

Air filter

- Air passes through the air filters where dirt is removed.

Choke

- It controls the amount of air passing through to go and mix with petrol.
- Closing the choke makes the mixture of petrol and air of good proportion which makes starting of the car much easier.

Throttle

- Opening the throttle allows more petrol-air mixture to get to the engine thereby increasing the speed of the engine.

NOTE: Only petrol engines have carburetors.

Operation of a four stroke diesel engine

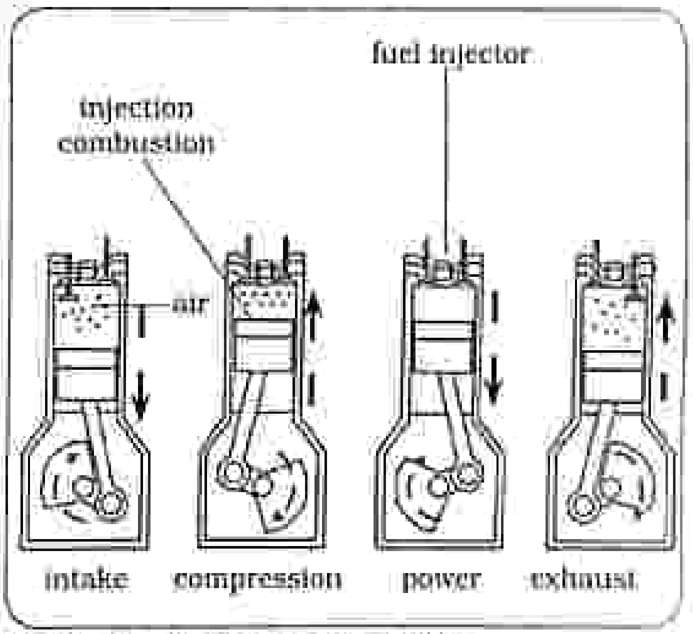


Fig. 26.3: Four stroke diesel engine

Intake/Inlet stroke

- The intake valve opens.
- Air is forced into the cylinder.
- Piston moves down and pressure is reduced inside the cylinder.

Air undergoes very high compression (double the compression in petrol engines) which results in the temperature of the compressed air to rise.

Power stroke

- Both valves are closed.
- Fuel is injected through the fuel injector into the hot air.
- Air-fuel mixture is ignited or will explode due to the high temperature inside the cylinder.
- Piston moves down because of the explosion and pressure inside the cylinder is reduced.
- Chemical energy is converted to kinetic energy causing the car to move.

Exhaust stroke

- Exhaust valve (outlet valve) opens and inlet valve remains closed.
- Piston moves up and the pressure inside the cylinder is increased.
- Exhaust gases are forced out of the cylinder through the exhaust valve.

Note

- Diesel engines are therefore called compression ignition engines because the ignition or explosion is caused by heat generated by the compressed air.
- Diesel engines are heavier than petrol engines.
- Diesel engines are also reliable.

Role of fuel injectors

- Fuel injectors allow the correct amount of fuel to be injected into the cylinder at the correct time.
- Fuel is sprayed into the combustion chamber using the fuel injector.

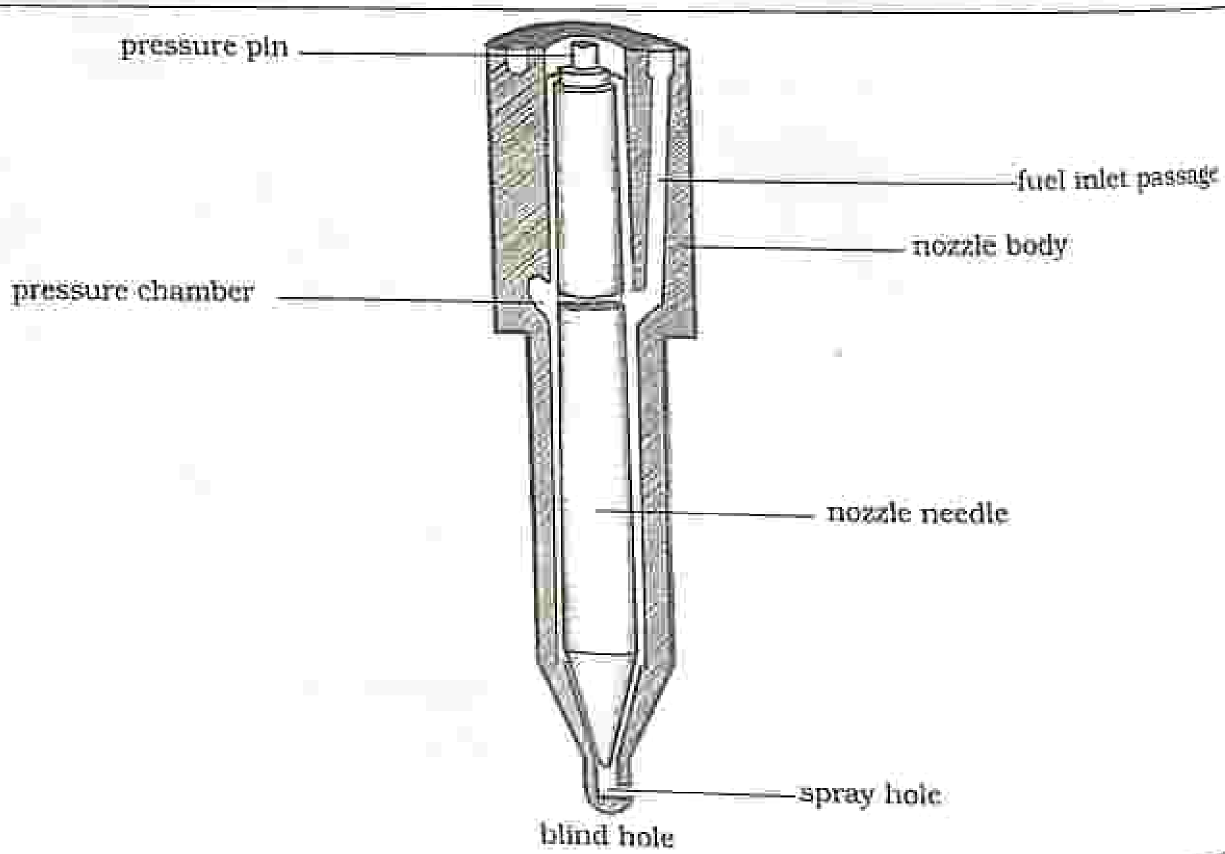


Fig. 26.4: Fuel injector

Comparison of petrol engine versus diesel engine

	Diesel engine	Petrol engine
Ignition methods	• fuel is injected into the cylinder through the fuel injector and mixes with air which is compressed and have very high temperatures causing an explosion or ignition	• petrol-air mixture is ignited by sparks from the spark plugs
Efficiency	• 40% efficient therefore more economical because the car will travel for a longer distance per each liter of fuel	• 25% efficient therefore less economical because the car will travel for a short distance per each liter of fuel
Carbon monoxide production	• because of increased efficiency, less carbon monoxide is released during combustion	• because of less efficiency, more carbon monoxide is released during combustion

Operation of modern diesel engines

- The modern diesel engine has an addition of:
  - two-valve pre-combustion chamber
  - mechanical low pressure fuel injection
  - equipped with particulate filters which reduce the emission of pollutants.
- Stronger material is used in modern engines allowing temperatures inside the cylinder to increase even more during the compression of air resulting in a more powerful explosion therefore increasing the efficiency of the engine.

Operation of modern petrol engines

- The carburetor has been replaced with throttle bodies and electronic fuel injection system which means the petrol-air mixture can be adjusted to give better power output.
- There is a replacement of the mechanical parts with electronic parts which are less prone to wear and tear and makes the running of the engine more accurate.

Advantages of modern petrol engines over old petrol engines

- fast start
- better fuel economy therefore increasing the efficiency
- better power output
- less emission of carbon monoxide
- high performance

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers. A. B. C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

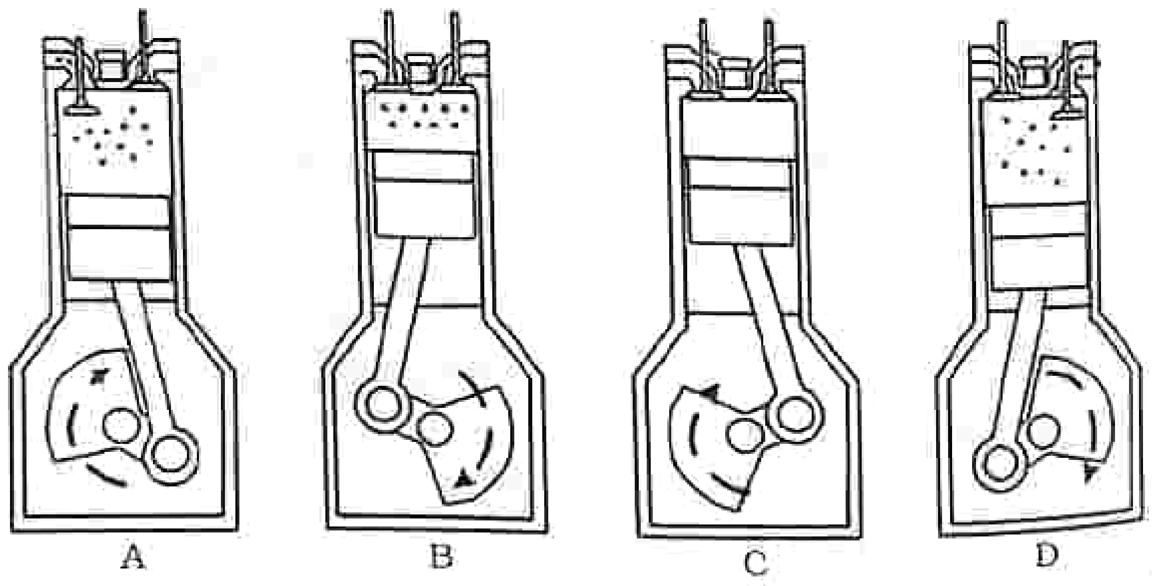
Answer all the questions.
The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. What is the role of fuel injectors?
   A. Allow the mixture of petrol and air to be produced in correct proportions.
   B. Allow the correct amount of fuel to be injected into the cylinder at the correct time.
   C. Filter dirty air
   D. Filter dirty petrol

2. What is the role of the carburetor?
   A. Allow the mixture of petrol and air to be produced in correct proportions.
   B. Allow the correct amount of fuel to be injected into the cylinder at the correct time.
   C. Filter dirty air
   D. Filter dirty petrol

Use the diagram below to answer question 3, 4 and 5



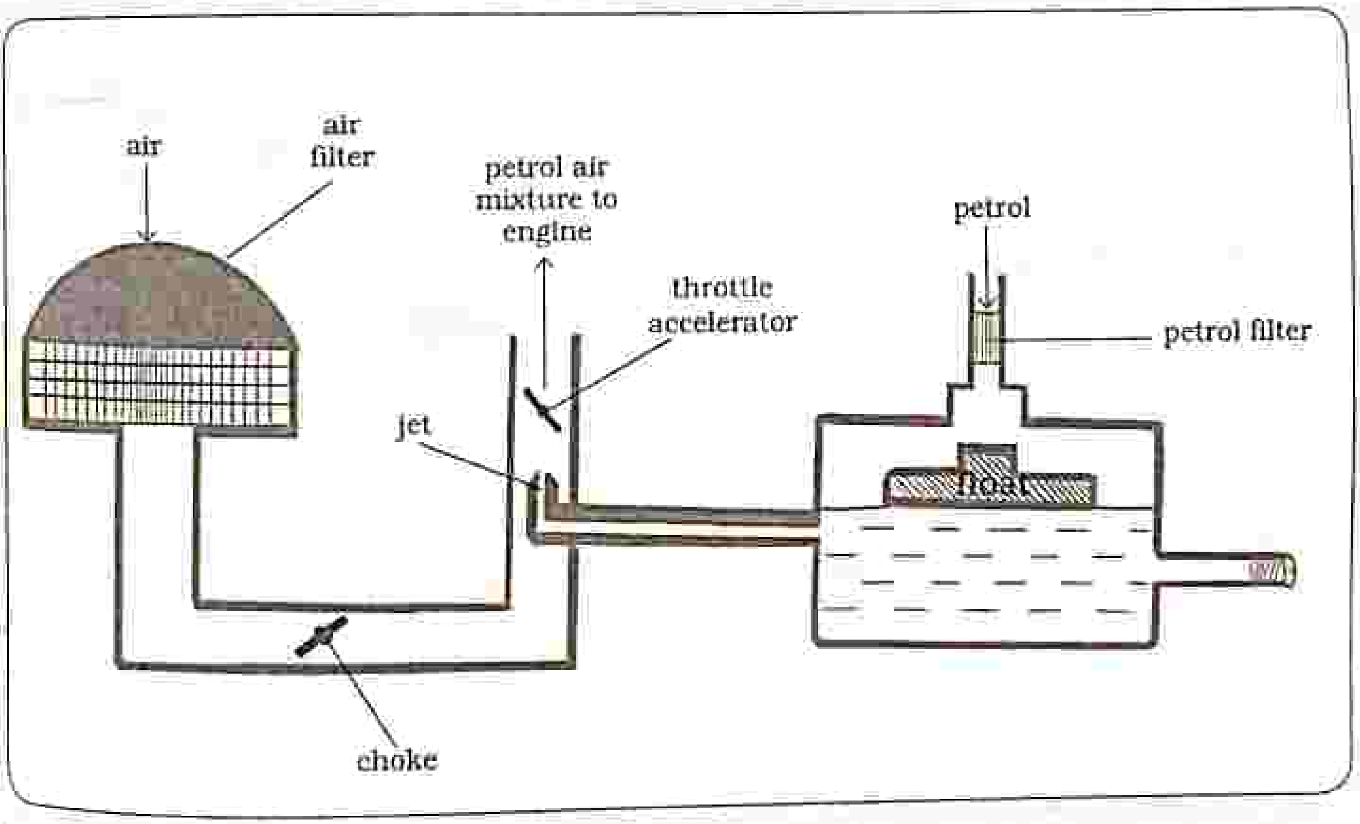
3. Name the strokes in the correct sequence
   A. Power stroke, Exhaust stroke, Inlet stroke, Compression stroke
   B. Exhaust stroke, Inlet stroke, Compression stroke, Power stroke
   C. Power stroke, Exhaust stroke, Compression stroke, Inlet stroke
   D. Inlet stroke, Compression stroke, Exhaust stroke, Power stroke

4. Identify the stroke where ignition takes place
   A. A   B. B
   C. C   D. D

5. Describe the movement of the piston in B.
   A. Piston moves up
   B. Piston moves down
   C. Piston moves to the right
   D. Piston moves to the left

6. Which one is not an advantage of modern petrol engine over old petrol engine?
   A. Fast start
   B. Better fuel economy therefore increasing the efficiency
   C. Better power output
   D. Emission of more carbon monoxide

Use the diagam below of a carburetor to answer question 7 and 8.



7. What is the function of the throttle in a carburetor?
   A. Opening the throttle allows more petrol-air mixture to get to the engine thereby increasing the speed of the engine
   B. Controls the amount of air passing through to go and mix with petrol
   C. Where dirt in the petrol is cleaned
   D. Where dirt in the air is cleaned

8. In a carburetor, dirty petrol will cause
   A. Blocked jets
   B. Blocked choke
   C. Blocked throttle
   D. Blocked air
9. Identify the ignition method in a petrol engine
   A. Sparks from spark plug cause an ignition
   B. Fuel mixes with air which has very high temperatures causing an explosion
   C. Hot petrol
   D. Fuel injectors
10. Which of the following is the ignition method in a diesel engine?
   A. Sparks from spark plug causes an ignition
   B. Fuel mixes with air which has very high temperatures causing an explosion.
   C. Hot petrol
   D. Fuel injectors

Section B: Structured questions

1. Describe the operation of a four stroke petrol engine. [4]
2. Describe the operation of a four stroke diesel engine. [4]
3. Explain the role of the fuel injectors. [2]
4. Explain the role of the carburetor. [2]
5. Outline the advantages of modern petrol engines over old petrol engines. [3]

CHAPTER 27 Energy

Key concepts

- Effects of energy
- Forms of energy and their sources
- Forms of potential energy
- Energy conversion
  - Energy chains
  - Energy converters
- Law of conservation of energy
- Work and energy
- Heat transfer
  - conduction: good and bad conductors
  - convection
  - radiation: good and bad absorbers, emitters and reflectors of different surfaces
- Solar cooker
- Solar water heater

Summary notes

Effects of energy

- Energy cannot be seen but its effects can be seen.
  - Energy makes objects:
  - move
  - make a sound
  - change shape
  - produce light
  - get hot

Forms of energy

Energy exists in many different forms as listed below:

- Kinetic energy
- Electrical energy
- Heat energy
- Potential energy
- Light energy
- Sound energy

Kinetic energy – is energy possessed by moving objects

Sources of kinetic energy

- Wind – moving windmill
- Water – moving turbines in a hydro power station
- Coal – moving turbines in a thermal power station

Electrical energy – is energy derived from kinetic energy (flow of electric charge).

Sources of electrical energy

- solar panel
- generator
- battery
- hydro electric power station
- torch cell
- thermal power station – coal is used to heat water and the steam produced turns generators which produce electricity

Heat energy – is when energy is transferred from one system to another and a temperature change is experienced.

Sources of heat energy

- Burning fuels
- rubbing two objects together

Potential energy – is energy stored in an object but has the ability to make things happen when it is released.

Forms of potential energy

There are three forms of potential energy which include

- Gravitational potential
- Elastic potential
- Chemical potential

Gravitational potential

It is energy stored in an object due to its position or location.

Example: - rock on a cliff
- fruit in a tree

Elastic potential

It is energy stored in a stretched object.

Example: - catapult
- pulled spring

Chemical potential

It is energy stored in fuels.

Example: - petrol
- torch cell
- food

Light energy – is energy produced by burning things and very hot materials.

Sources of light energy

- sun
- fire
- electric bulb
- glow worm
- torch

Practical Activity 27.1

Aim : To show that light travels in a straight line

Apparatus : 1. Light source as shown in Fig. 27.1
2. 3 boxes with a hole on the centre
3. Cotton thread

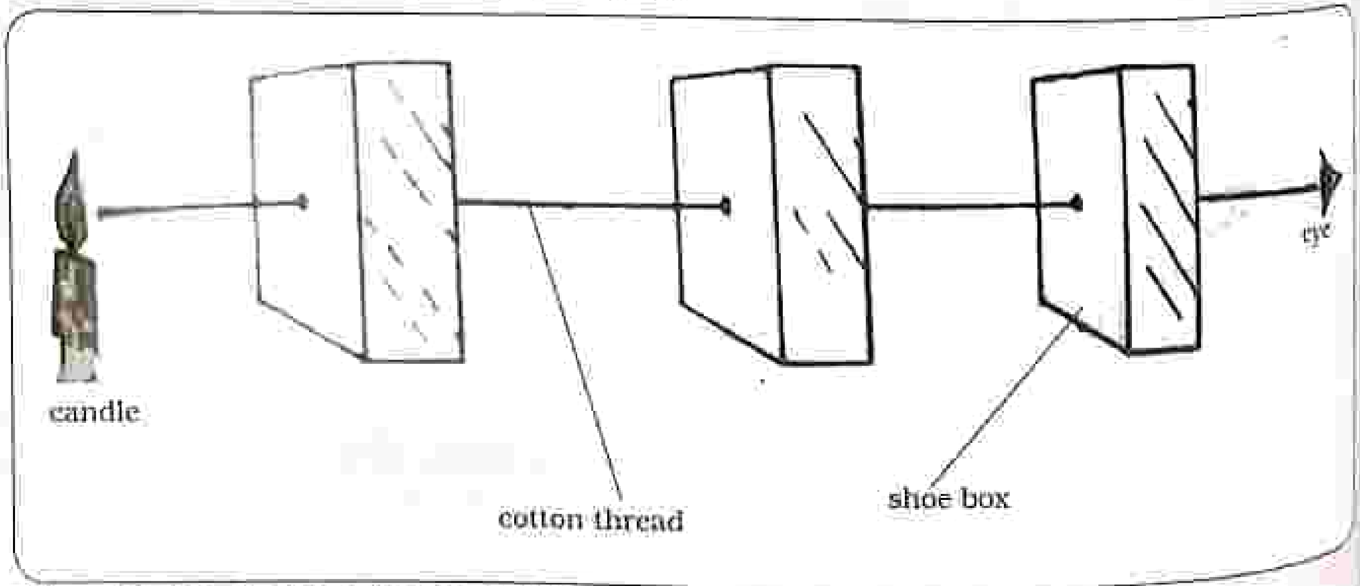


Fig. 27.1: How light travels in a straight line

Method

1. Line up the boxes and put a straight cotton thread through the holes as shown in Fig 27.1.
2. Stand a lit candle behind the hole in the box.
3. View the candle through the three holes lined up by the cotton thread.
4. Observe what happens when you move one hole out of the line.

Results

Three holes lined up

- The burning candle can be seen.

One hole out of line

- The burning candle cannot be seen.

Conclusion

- Light travels in a straight line it cannot bend around an object.

Practical questions

Can you see the burning candle when:

1. the three holes are lined up?
2. one hole is out of line?

Shadows

Practical activity 27.2

Aim	:	Production of a shadow
Apparatus	:	1. Screen (white wall)
		2. Opaque ball
		3. Light source
		4. Black cardboard with a hole

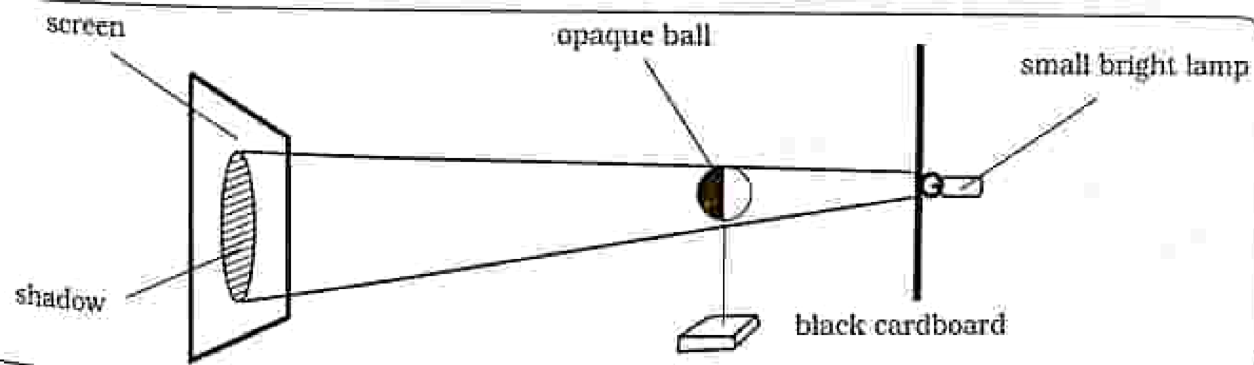


Fig. 27.2: Shadow production

Method

1. Set up apparatus as shown in Fig 27.3 light source, opaque, ball and the screen in line.
2. Adjust the position of the light source and ball until you observe a change on the screen.

Results

A shadow forms on the screen.

Conclusion

- Shadow formation is evidence that light travels in a straight line.
- A shadow is formed when light is blocked because light cannot bend around an object.

Sound energy – energy produced by vibrations.

Source of sound energy

- Vibrating objects produce sound.

Production of sound

- Sound can be produced by different instruments.
- Human voice is produced by the vibration of thin stretched membranes found in the voice box or larynx.
- When air passes through the membranes in the voice box, they vibrate producing sound.

Fig. 27.3: Some instruments which produce sound

Table 27:1

Type of instrument	How sound is produced
drums	beating
guitar	plucking strings
bottles	blowing
mbira	plucking
marimba	hitting

Transmission of sound

- Sound cannot travel in a vacuum (a vacuum is a space which does not have any material or air).
- For sound to be transmitted, it needs a medium.

Medium for the transmission of sound

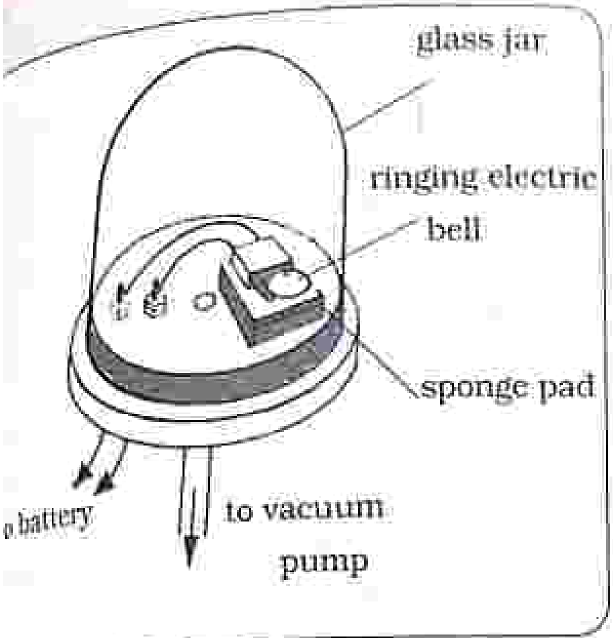
1. Solid-sound travels fastest through solids.
2. Liquid-sound travels fast through liquids.
3. Gas-sound travels less through gases

   Note: sound travels fastest in solids then liquids and slowest in gases

Practical activity 27.3

Aim : To show that sound requires material medium for transmission

Apparatus :
1. Bell jar
2. Electric bell
3. Vacuum pump



g 27.4: *Material medium for carrying sound*

ethod

Set up apparatus as shown in Fig. 27.4.

Observe the sound as the vacuum pump continues to remove air from inside the bell jar.

esults

At first the sound of the ringing bell could be heard clearly.

After some time the sound of the electric bell disappears although the striker can still be seen hitting the gong.

Conclusion

- The vacuum pump is used to remove all the air inside the bell jar leaving a vacuum.
- When all the air is removed, the sound of the electric bell disappears showing that sound cannot be transmitted in a vacuum, it needs a medium.

Energy conversion

- It is a process of changing energy from one form to another.

Energy converter

- It is the machine or device which causes the change of energy from one form to another to happen.

Energy chain

- When several energy conversions take place in one converter, they are linked together by an energy chain.

ble 27:3 *Energy conversions*

nergy convertor	Energy conversion
orch to light bulb	Chemical → light
ynamo (generator)	Kinetic → Electrical → light
atapult	Chemical → kinetic → potential → kinetic
olar panel to light a bulb	Solar → Electrical → Light
lotor	Electrical → kinetic
reen plants	Solar energy → chemical energy

Practical activity 27.5

Aim : To show convection in gases

Apparatus : 1. Smoke chamber
2. Candle
3. Smouldering paper

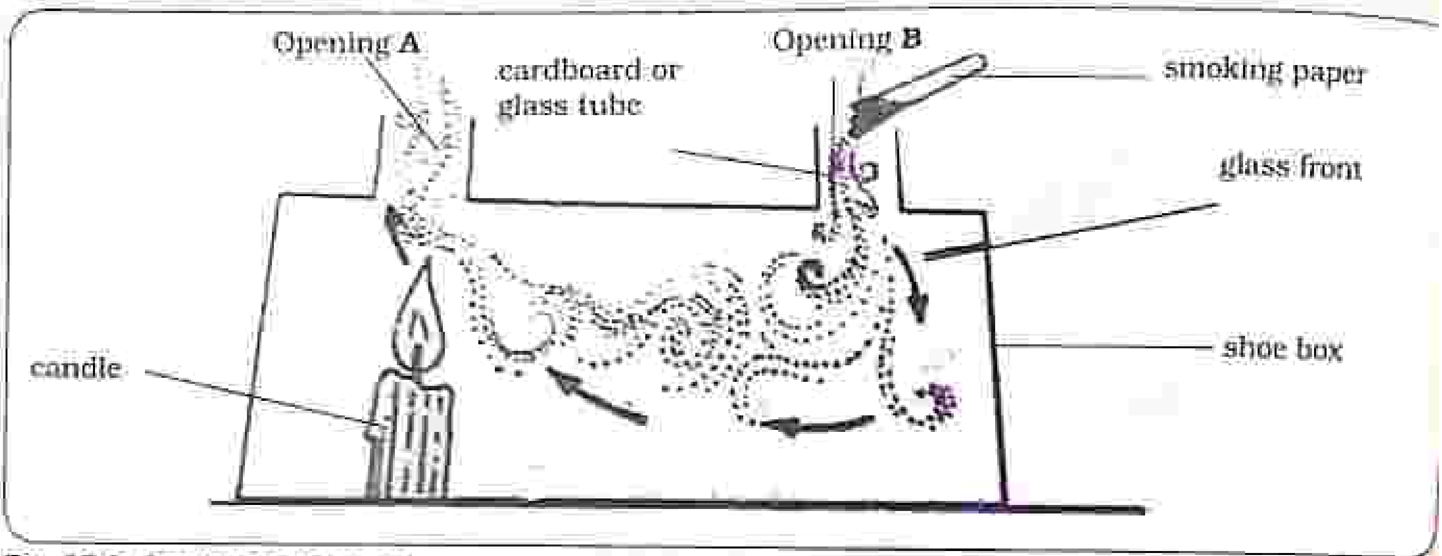


Fig. 27.6: Convection in gases

Method

1. Light the candle inside the smoke chamber on one of the chimneys.
2. On the other chimney of the smoke chamber hold over a smouldering paper as shown in Fig 27.6.
3. Observe any changes in the smoke chamber.

Results

- Smoke moves into the smoke chamber and out after being heated by the burning candle.

Conclusion

- Air above the candle is heated and it expands, becomes less dense, gains more kinetic energy and moves upwards and out through chimney A of the smoke chamber.
- On chimney **B**, cold and denser air moves down into the smoke chamber to take the place left by the heated air and this creates a convection current shown by arrows in the diagram.

NOTE

- The purpose of the smoke is to make the convection currents visible.
- The purpose of the candle is to heat the air thereby creating convection currents

Practical questions

1. Explain what happens at chimney **A**.
2. Explain what happens at chimney **B**.

nduction

It is the movement of heat through matter from an area of high temperature to an area of low temperature.

Practical activity 27.6

im : To show good and bad conductors of heat

pparatus :
1. Iron
2. Candle wax
3. Copper
4. Match stick
5. Brass
6. Wood

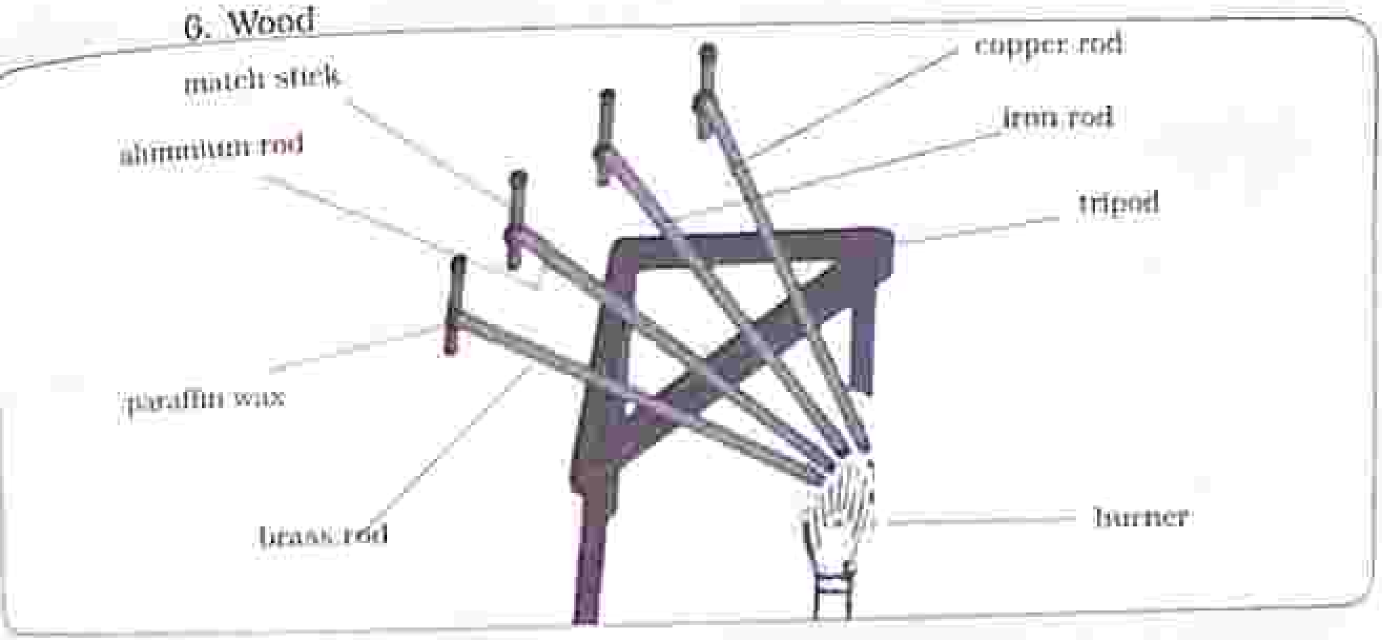


Fig 27.7: Good and bad conductors of heat

Method

- Set up apparatus as shown in Fig 27.7 where different types of materials are heated.

Results

The candle wax melted and the match stick fell in the order below starting with:

- Copper
- Aluminium
- Brass
- Iron
- Wood

Conclusion

- Most metals are good conductors of heat for example copper and most non-metals are bad conductors of heat for example wood, glass, plastics and fabrics.

Practical questions

1. The match stick fell first on which material?
2. The match stick fell last on which material?

Good conductors of heat

Heat travels fastest through the materials below.

Fig. 27.8: Good conductors of heat

Bad conductors

They are also known as insulators.

Heat cannot be easily passed from one point to another.

Fig. 27.9: Bad conductors of heat

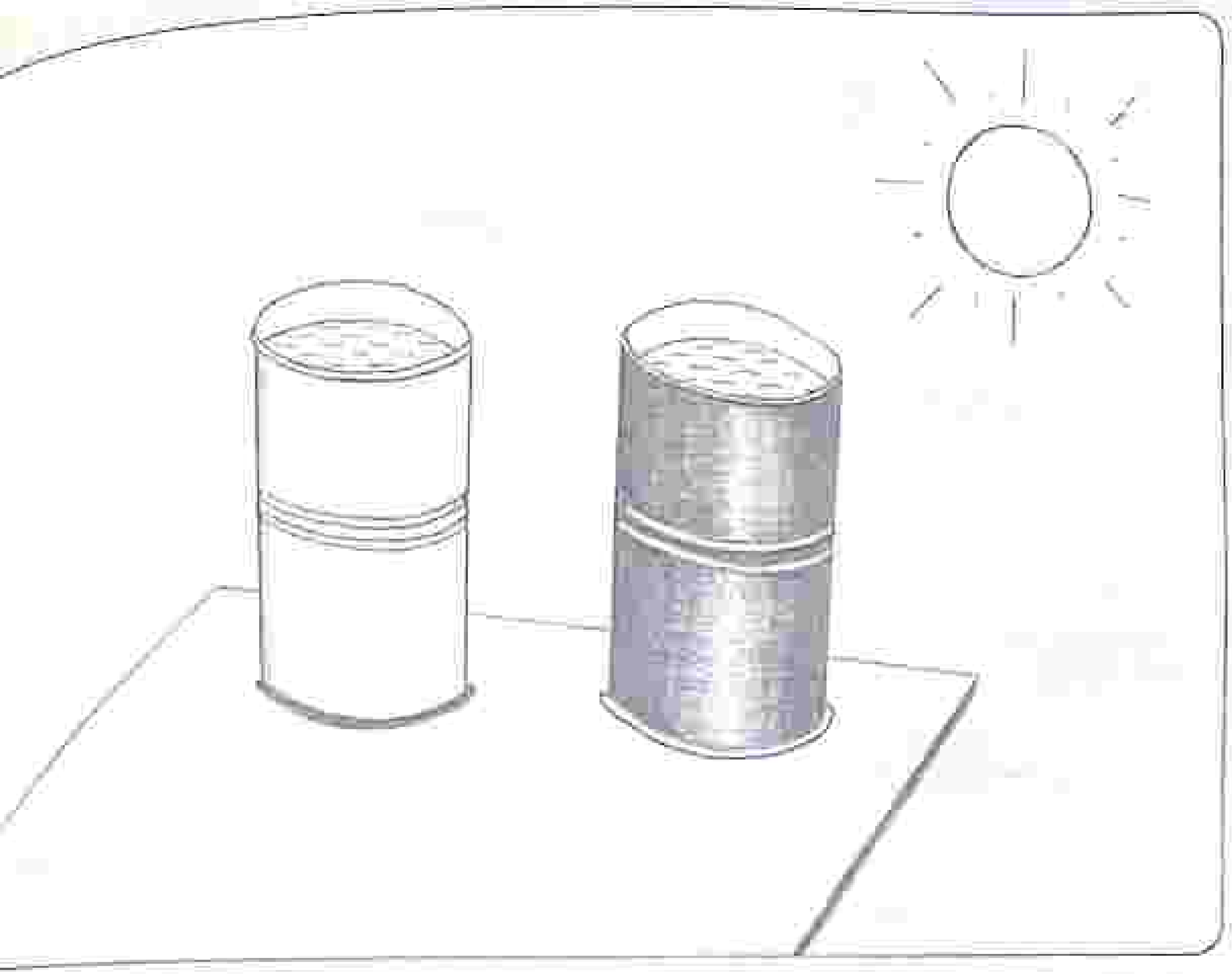
Radiation

- It is the movement of heat from one place to another by means of electromagnetic waves.
- No medium is needed for heat to be transmitted by radiation therefore heat can move in a vacuum.

Practical activity 27.7

Aim : To show good and bad absorbers and good and bad emitters of heat

Apparatus :
1. Black painted tin
2. White painted tin or silver shiny
3. Thermometer
4. Water



...g. 27.10: Black tin and white tin exposed to sunlight

...ethod

- Fill both tins three quarter full with water and using the thermometer, measure the initial temperature.
- Cover both tins with lids and place them where they are exposed to sunlight.
- After 15 minutes, record the temperature of the water again from both tins.
- Remove the two tins from exposure to sunlight and place them in a cool place covered with lids for 5 minutes.
- After 5 minutes, record the temperature of the water for the two tins.

Results

Nature of tin surface	Temperature in sunlight/°C	Temperature in a cool place/°C
Black tin	30	25
White or Silver shiny tin	28	27

...ote: The initial temperature of the water in both tins was 25°C

Conclusion

Absorption of heat energy

- Black tin absorbed more heat energy than the white of silver shiny tin when exposed to sunlight.
- Therefore black surfaces are good absorbers of heat energy whereas white or silver shiny surfaces are bad absorbers of heat energy.

Emission of heat energy

- Black tin emitted heat energy faster than the white or silver shiny tin.
- Therefore black surfaces are good emitters of heat energy whereas white or silver shiny surfaces are bad emitters of heat energy.

Practical questions

1. Which surface is a good absorber of heat energy?
2. Which surface is a good emitter of heat energy?

Practical activity 27.8

Aim : To show good and bad reflectors of heat energy

Apparatus : 1. Black painted tin

2. White painted tin or silver shiny

Fig. 27.11: Good and bad reflectors of heat

Method

1. Three quarter fill both tins with water and using the thermometer, measure the initial temperature.
2. Cover both tins with lids and place them where they are exposed to sunlight.
3. After 15 minutes, record the temperature of the water again from both tins.

Results

Nature of tin surface	Temperature in sunlight (°C)
Black tin	30
White or Silver shiny tin	28

Note: The initial temperature of the water in both tins was 25°C.

Conclusion

- White or silver shiny tin reflects more heat energy when exposed to sunlight than black surfaces.
- Therefore white or silver shiny surfaces are good reflectors of heat energy whereas black surfaces are good absorbers of heat energy.

Practical questions

1. Which surface is a good reflector of heat energy?
2. Which surface is a bad reflector of heat energy?

Solar cooker

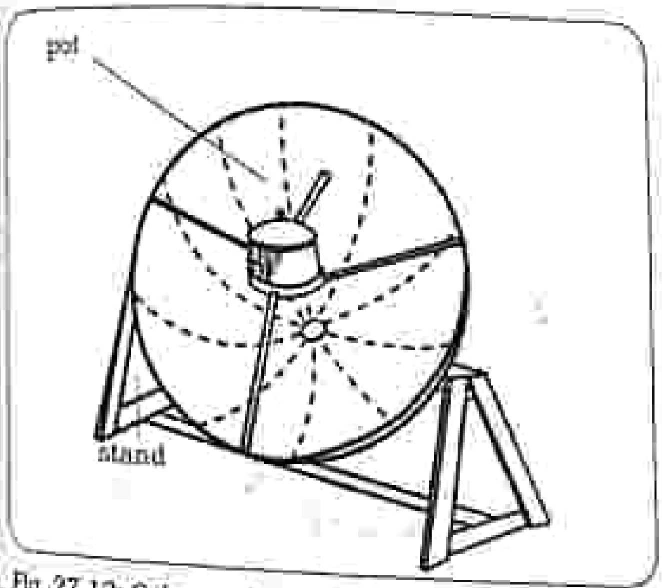


Fig. 27.12: Solar cooker

- A solar cooker converts solar energy to heat energy which is then used for cooking.
- The rays from the sun are focused to one position where maximum heat energy is captured which is then used for heating.

Design of a solar cooker

Shiny surface – to reflect the solar rays

Curved dish – reflects and focuses the solar rays to one position.

Black pot – for maximum absorption of heat energy because black surfaces are good absorbers of heat energy.

Solar water

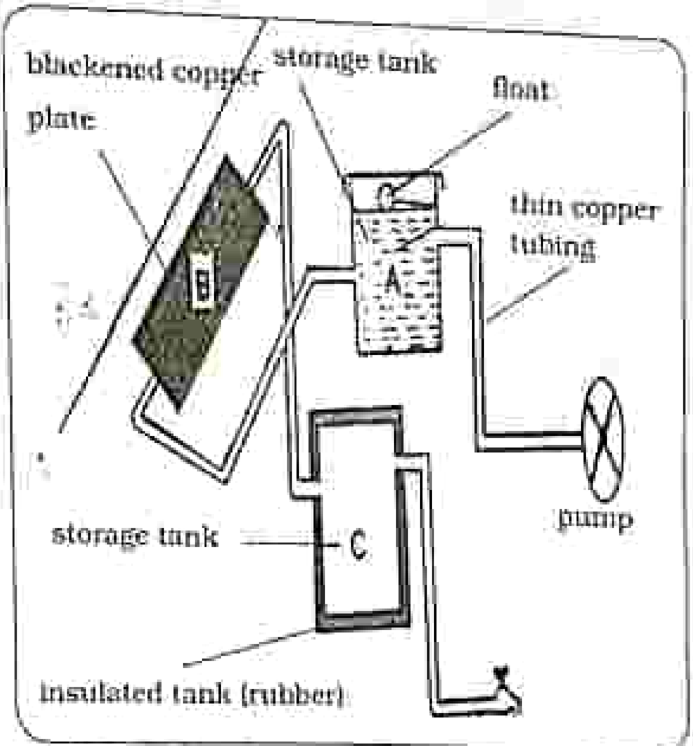


Fig. 27.13: Solar water heater

- Solar water heater converts solar energy to heat energy which is then used for heating water.
- Design of a solar water heater.

Solar panel

- It is positioned on the roof top to trap maximum solar energy.
- The solar panel is painted black to maximize the absorption of solar energy because black surfaces are good absorbers of heat energy.

Section through the solar panel:

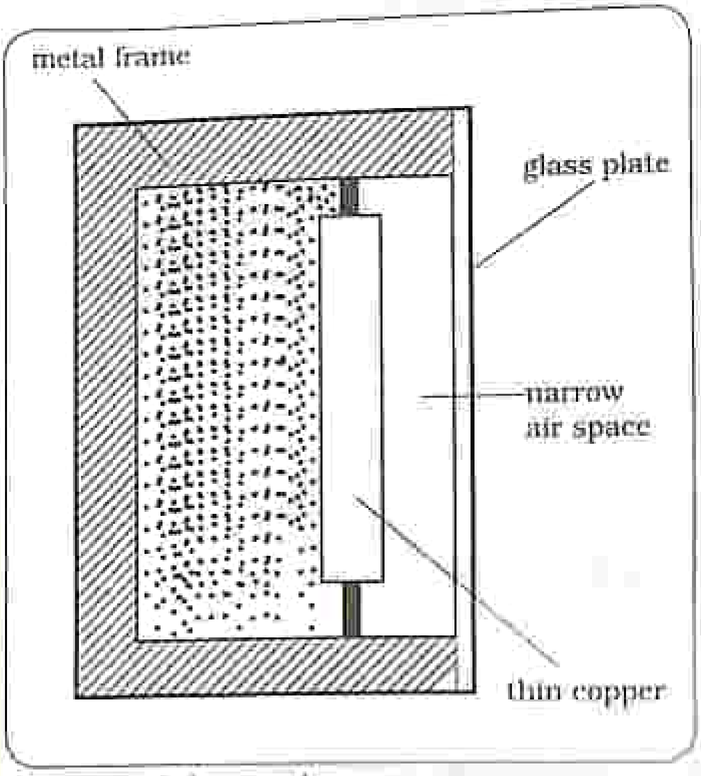


Fig. 27.14: Solar panel

Air space

- Prevents heat loss by convection.

Glass covered panel

- To increase heat absorption into the panel and reducing heat loss by conduction and convection.

Expanded polystyrene base

- It reduces heat loss by convection.

Storage tank A

- Water is pumped into storage tank A before being heated.

Thin copper tubes

- Water flows through thin copper tubing.
- Copper is used because it is a good conductor of heat.
- Thin tubes are used in order to minimize the amount of water passing through the solar panel so that the water warms quickly.

Blacked copper plate

- When water reaches the blackened copper plate, it is strongly heated.
- The copper plate is painted black to increase heat absorption because black surfaces are good heat absorbers.

Insulated storage tank

- When the water has been warmed, it flows into an insulated storage tank where it is stored for use when needed.
- The storage tank is insulated to prevent heat loss to the surroundings.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets[ ] at the end of each question.

Multiple choice questions

1. Which one is not a form of energy?
   A. Kinetic
   B. Light
   C. Heat
   D. Emitter
2. Which one is not a form of potential energy?
   A. Chemical
   B. Sound
   C. Elastic
   D. Gravitational
3. Which one is not a method of heat transfer?
   A. Convection
   B. Conduction
   C. Radiation
   D. Sound
4. Sound is produced by ___________.
   A. Light
   B. Heat
   C. Vibrations
   D. Kinetic
5. When light is blocked a _____ is formed.
   A. Fire B. Shadow
   C. Sound D. Fuel
6. Work done is calculated by
   A. Work = Force × Distance
   B. Work = Force + Distance
   C. Work = Force – Distance
   D. Work = Force ÷ Distance
7. A solar cooker converts solar energy to ______________.
   A. Kinetic energy
   B. Heat energy
   C. Electrical energy
   D. Elastic potential energy
8. How much energy is used to lift a bucket weighing 300N up a well that is 10m deep?
   A. 3000J B. 310J
   C. 30J D. 290J

9. Identify a good absorber of heat energy.

A. Black pot

B. Silver pot

C. Shiny pot

D. White pot

10. Name a good emitter of heat energy.

A. Shiny pot

B. White pot

C. Black pot

D. Silver pot

Section B: Structured questions

1. a) List the 6 forms of energy. [6]
   b) List the 3 forms of potential energy. [3]
   c) List any three sources of light energy: [3]
2. a) State the law of conservation of energy. [2]
   b) Identify the three mediums for sound transmission from the fastest to the slowest. [3]
3. a) Identify three methods of heat transmission. [3]
   b) Explain convection in terms of the kinetic theory of matter. [3]
4. Describe the function and design of a solar cooker. [4]
5. Describe the function and design of a water heater. [5]

CHAPTER 28

Magnetism

Key concepts

- Magnetic and non-magnetic materials
- Poles of a magnet
- Properties of magnets
- The law of magnetism
- Magnetic fields
- Magnetic fields on current carrying conductor

Summary notes

Magnetic and non-magnetic material

- Materials can be grouped based on their response towards a magnet.
- All materials can be grouped as magnetic or non-magnetic.

Magnetic materials

- These are materials that are attracted by a magnet.
- These materials can be magnetized or can be converted into magnets. Example of magnetic materials are iron, nickel, cobalt.
- Some materials can keep their magnetism for a long time and are called permanent magnets.

Non-magnetic materials

- These are materials that are not attracted by a magnet. These materials cannot be magnetized. Examples of non-magnetic materials are plastic, rubber, copper, glass, silver, gold and wood.

Different forms of magnets

Magnets are found in many forms which include:

- Bar magnet
- Horse shoe magnet
- U shaped magnets
- Electromagnets
- Cylindrical magnet

Poles of a magnet

- A magnetic pole is a region at the end of a magnet where magnetism is very strong.
- All magnets have two poles namely the North Pole and the South Pole and the two poles are different.

Properties of magnets

Magnets have certain important properties which are:

- They attract magnetic materials.
- Magnetism is stronger at the poles than in the middle.
- Like poles repel each other and unlike poles attract each other.
- A freely suspended magnet always points in the north south direction of the earth, with the South Pole pointing towards the south pole of the earth and the North Pole pointing towards the north pole of the earth.

The law of magnetism

- The law of magnetism is that like poles repel and unlike poles attract.
- The north pole of a magnet is repelled by the north pole of another magnet.
- The north pole of a magnet is attracted by the south pole of another magnet.

Magnetic field

- A magnetic field is an area around a magnet where a magnetic material experiences magnetic force.
- Magnetic field is characterised by magnetic lines of force which:
  - start from the north pole and end at the south pole
  - cannot intersect each other
  - Are close if the magnetic field is strong.

To find the direction of a magnetic field iron fillings and a plotting compass may be used.

Using iron fillings to find the direction of a magnetic field

- A bar of magnet is placed underneath a piece of paper.
- Iron fillings are then sprinkled over the paper, the paper is then gently tapped.
- The iron fillings will form a pattern shown below.

Fig. 28.1: Magnetic field lines on a bar of magnet

Using a plotting compass to find the direction of a magnetic field

A plotting compass has a needle that is a magnet, with the head being the North Pole and the tail being the South Pole.

- A bar of magnet is placed on a piece of paper with its north pointing north and its south pointing south.
- The compass is placed near one end of the magnet, the positions of the tail (south) and head (North Pole) of the compass are marked by pencil dots.
- The compass is moved such that its tail rests on a dot that was made for position of the head previously and a dot is placed in front of the head arrow.
- The process is repeated along the whole length of the magnet.
- The magnetic field is determined by joining the dots.

Magnetic field lines around a bar of magnet

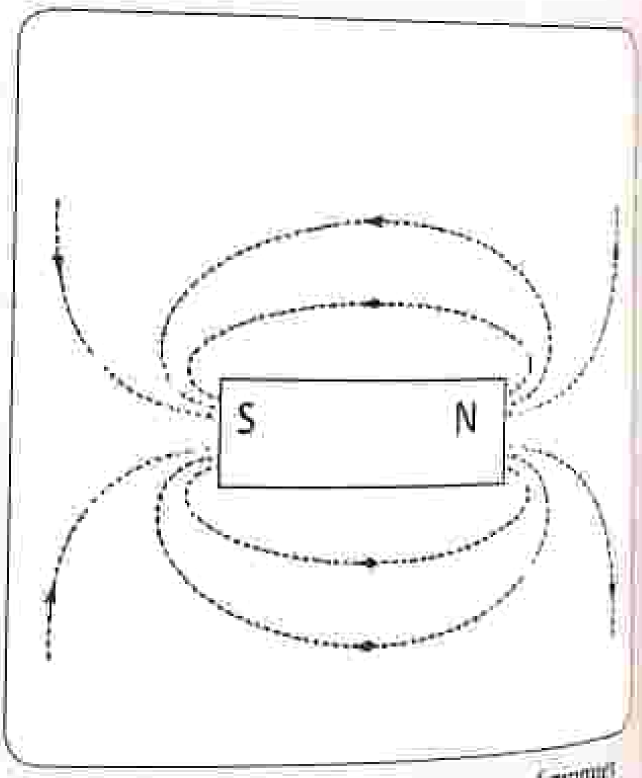


Fig. 28. 2: Magnetic field around a bar of magnet

Magnetic field lines of a pair of like poles together

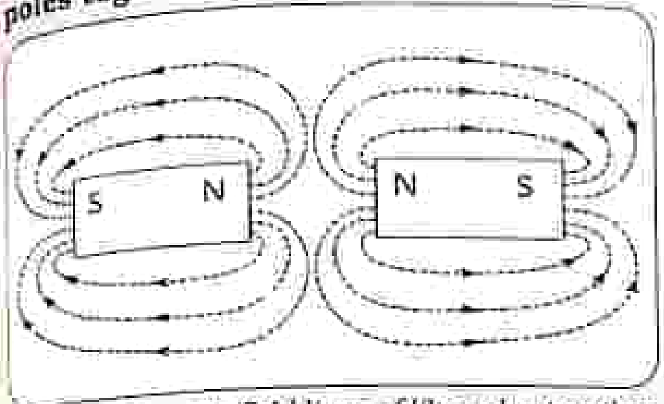


Fig. 28.3: Magnetic field lines of like poles together

Magnetic field lines of a pair of unlike poles together

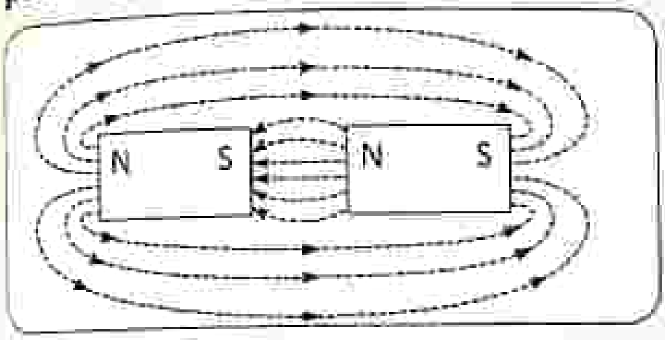


Fig. 28.4: Magnetic field lines of a pair of unlike poles together

Magnetic field on a current carrying conductor

- When current flows inside a conductor a magnetic field is produced around the wire.
- The magnetic field pattern depends on the shape of the conductor and on the direction of current flowing in the conductor.

Magnetic field of a long straight conductor

- The direction of magnetic field around a long straight conducting conductor is determined by using the right hand grip rule.
- The conductor is gripped with the right hand, with the thumb pointing in the direction of the current and the fingers curling in the direction of the magnetic field.
- The field form concentric lines around the conductor.

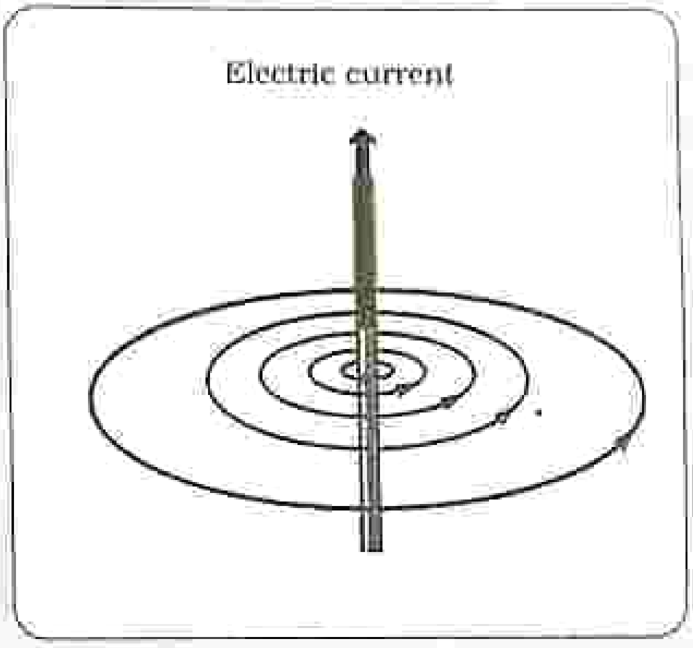


Fig. 28.5: Magnetic field lines on a straight conducting wire

- Changing the direction of current changes the direction of the magnetic field.
- The strength of the field depends on the magnitude of current flowing. Increasing current increases the field strength.

Magnetic field of a solenoid

- Magnetic field around a solenoid resembles that of a bar magnet.
- The solenoid just like a bar magnet has poles.
- The solenoid is sometimes called an electromagnet and is a temporary magnet.

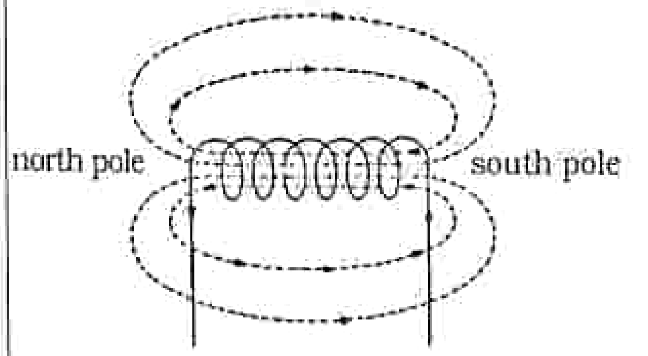


Fig. 28.6: Magnetic field lines on a solenoid

- There are two ways of determining the direction of field of a solenoid.

The right hand grip rule

- The solenoid is gripped by the right hand in such a way that the fingers curl in the direction of current and the end of the solenoid that the thumb points to is the North Pole.

The direction of current at the ends of the solenoid

- When looking at the ends of a solenoid, current flows in a clockwise direction on one end and in an anticlockwise on the other end.
- At the North Pole current flows in an anticlockwise and at the South Pole current flows in a clockwise direction.

Factors affecting the of field of a solenoid

- Magnitude of current – increasing current increases the strength of the field.
- Number of turns – increasing the number of turns increases the strength of the field.
- Use of a soft iron core – the presence of a soft iron core inside the solenoid increases the strength of the field.

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

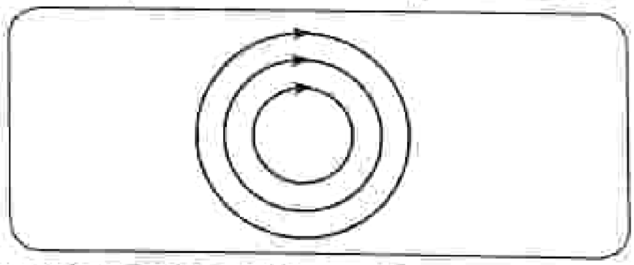
Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. A substance that retains its magnetism for a long period is called
   A. temporary magnet.
   B. permanent magnet.
   C. magnetic substance.
   D. non-magnetic substance.
2. Which is true about magnetic field lines of force?
   A. magnetic lines of force move from South Pole to North Pole.
   B. magnetic lines of force can intersect one another.
   C. closer lines indicate a stronger magnetic field.
   D. there are more field lines at the North Pole than at the South Pole.
3. A freely suspended magnet always aligns itself in which direction?
   A. north east
   B. north south
   C. south east
   D. south west
4. The law of magnetism states that _________________.
   A. like poles attract and unlike poles repel
   B. like poles repel and unlike poles attract
   C. like charges repel and unlike charges attract
   D. like charges attract and unlike charges repel
5. The diagram shows magnetic field on a straight current carrying conductor.



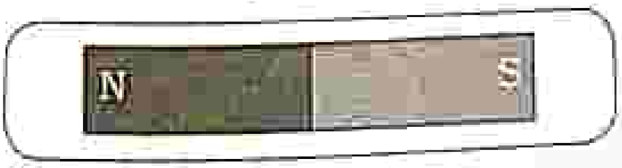
   The direction of current is:
   A. into page.
   B. out of page.
   C. up the page.
   D. down the page.

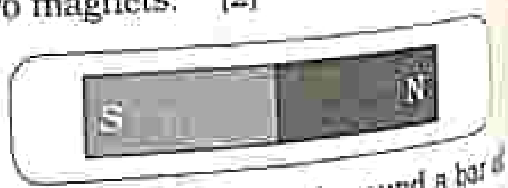
6. The magnitude of the magnetic field produced by a strait current carrying conductor depends on ________.
   A. direction of current flow
   B. magnitude of current flowing
   C. the type of conducting material used for the conductor
   D. the type of the voltage supply
7. The magnetic field of a solenoid resembles that of ________.
   A. a straight conductor
   B. flat circular coil
   C. a bar magnet
   D. horse shoe magnet
8. The diagram below shows a current carrying conductor.

The direction of magnetic field at point **X** above a current carrying conductor is
   A. up the page.
   B. down the page.
   C. into the page.
   D. out of the page.
9. The direction of the magnetic field around a current carrying straight wire can be determined using ____
   A. Fleming's left hand rule
   B. Fleming's right hand rule
   C. right hand grip rule
   D. left hand grip rule
10. A current flows in the direction out of page. What is the direction of the magnetic field?
   A. into page
   B. out of page
   C. clockwise
   D. anticlockwise

Section B: Structured questions

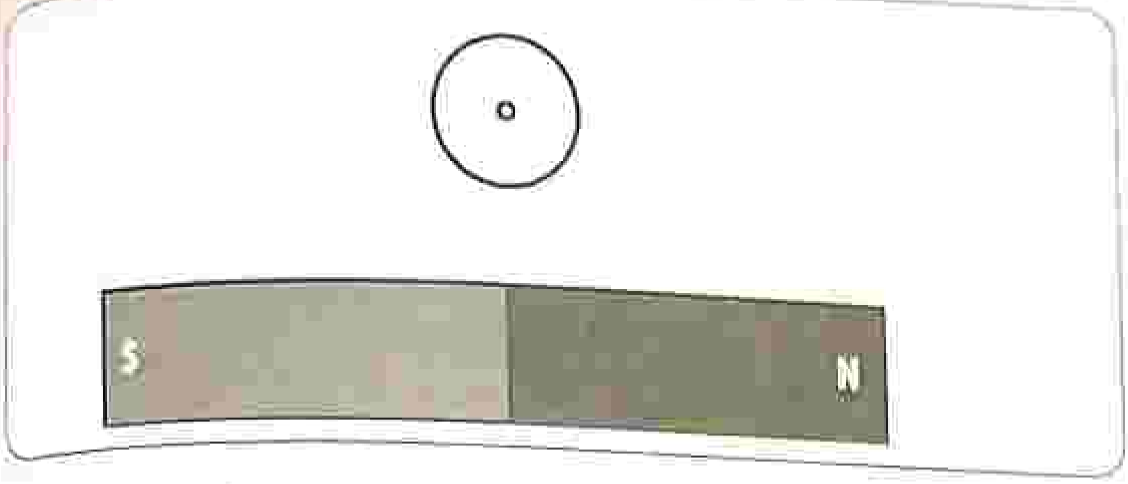
1. a) What is a magnetic material? [1]
   b) Name any three magnetic materials. [3]
2. Draw the magnetic field lines around a bar magnet. [2]
3. Draw the magnetic field pattern around the two magnets. [2]





4. Describe an experiment you would carryout to determine the filed around a bar magnet. [3]

5. The diagram shows a plotting compass placed above a bar magnet.



a) Which part of a compass is a magnet? [1]

b) Draw on the diagram the direction of the arrow of the plotting compass. [1]

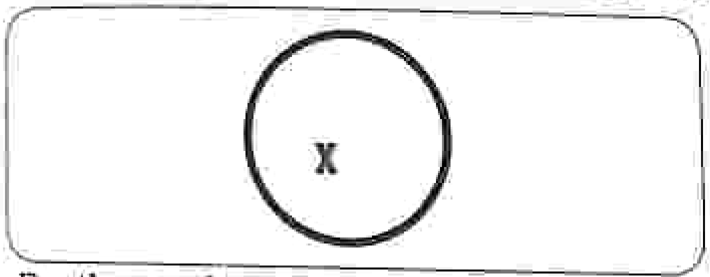
6. a) What is the other name for a solenoid? [1]

b) Describe how you would determine the magnetic field of a solenoid. [2]

7. State and explain any two factors that affect the strength of a field round a solenoid. [4]

8. The diagram shows current flowing through a conductor into paper,

a) draw the magnetic field around the conductor [2]



b) For the conductor in (4 a) state the factor that affects:

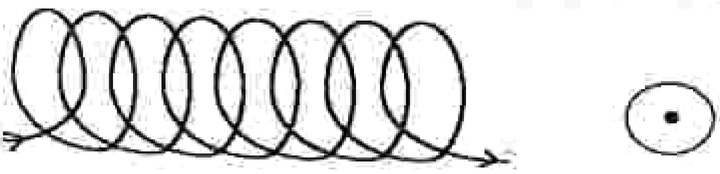
(i) The direction of field lines of force. [1]

(ii) The strength of the magnetic field. [1]

9. a) State the law of magnetism. [1]

b) Illustrate using diagrams the law of magnetism. [2]

10. The diagram below shows a compass placed near one end of a solenoid.



a) Indicate on the diagram the South Pole and the North Pole. [2]

b) Show the direction of the compass needle. [1]

CHAPTER 29 Electromagnetism

Key concepts

- Motor effect: Force on current carrying conductor in magnetic field
- dc motor generator
- Factors that affect the rotation of the coil
- Generator effect: electromagnetic induction
- Factors that affect the induced emf
- Operation of an ac and a dc generator
- Hydro and thermal power generation

Summary notes

The motor effect

- If a current carrying conductor is placed in a magnetic field will experience a force (motor effect).
- The force is as a result of the interaction between the field of the conductor and the field in which the conductor is placed.
- The direction of the force is determined by means of Fleming's left hand rule, the thumb, forefinger and second finger are held at right angles to each other. The forefinger points in the direction of the field, the second finger points in the direction of current and then thumb points in the direction of the force/motion.
- The force is always at right angles to both current direction and field direction.
- The force changes direction when either current or field changes direction.

The d.c motor

See Fig. 29.1

- When current flows through the loop **ABCD**, side AB will experience a downward force and side **CD** will experience an upward force.
- The loop will rotate clockwise until it is vertical and no current flows in the loop.
- Momentum of the loop makes it return to the horizontal position.
- The direction of current in the sides is reversed by the commutator.
- Side AB now experiences an upward force whilst the side **CD** experiences a downward force.
- The loop continues to rotate in a clockwise direction.

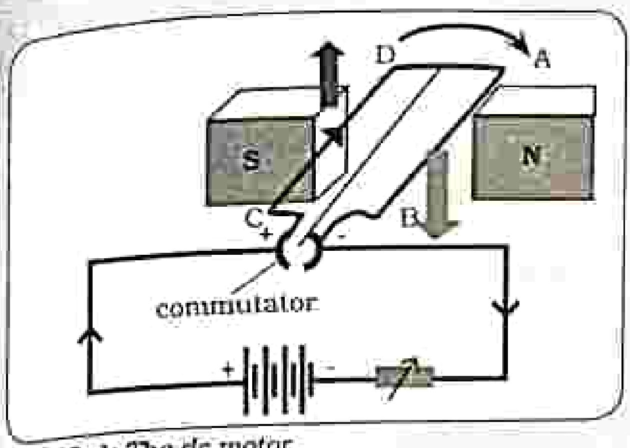


Fig. 29. 1: The dc motor

Factors that affect the rotation of a coil

- Number of turns – increasing the number of turns increases the rotation.
- Magnitude of current – increasing current increases the rotation.
- Use of a soft iron core – winding the conductor on a soft iron core increases the rotation.

Generator effect

- When a conductor placed in a magnetic field is moved or the field around it changes, a voltage or electromotive force (e.m.f) is setup along the conductor and current flows in the conductor.
- This is known as the generator effect or electromagnetic induction.

Factors affecting the magnitude of the induced e.m.f

- Length/number of turns of the conductor – the greater the number of turns the greater the magnitude of induced e.m.f.
- The strength of the magnet – the greater the strength of magnetic field the greater the magnitude of induced e.m.f.
- The speed at which the conductor moves – the faster the speed the greater the magnitude of induced e.m.f.
- Number of turns in the coil – the greater the number of turns, the greater the magnitude of induced e.m.f.

Generators

- A generator converts mechanical energy into electrical energy.
- Two types of generators are: the direct current (dc) generator and the alternating current (ac) generator.

dc generator

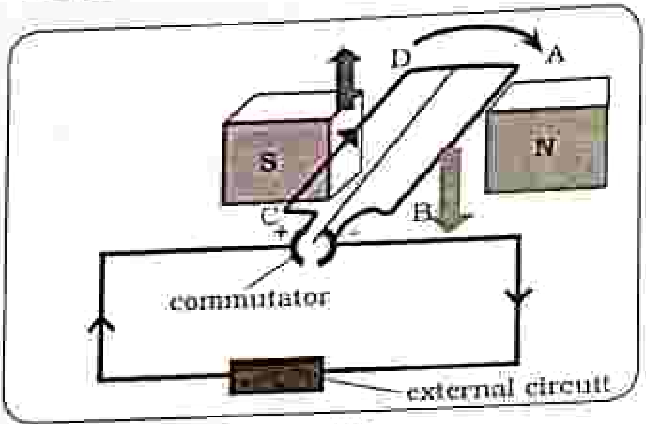


Fig. 29. 2: The dc generator

- The dc generator is a reverse of a dc motor.
- It converts mechanical energy to electrical energy.
- The coil is made to rotate in a magnetic field; as a result an e.m.f is induced in the coil. Fleming's left hand rule is used to determine the direction of the induced current.
- The coil is attached to a split ring / commutator and it allows current to flow in one direction in an external circuit.

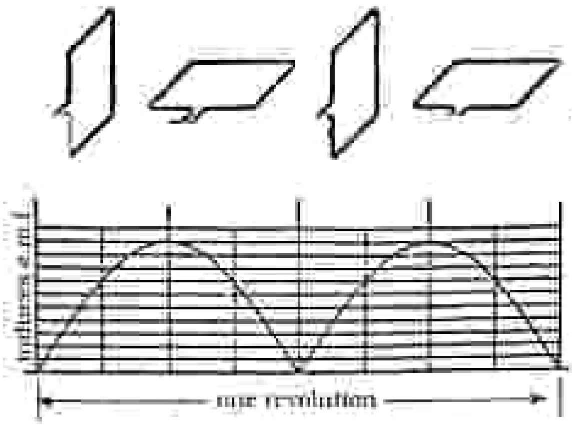


Fig. 29. 3: Output voltage graphical presentation for a dc generator

- The magnitude of the induced e.m.f varies with variation in the position of the coil.
- The induced e.m.f is maximum when the plane of the coil is parallel to the field as the sides of the coil will be cutting field lines at a maximum rate.
- The induced e.m.f is zero when the coil is perpendicular to the field as the side of the coil will be moving parallel to the filed.
- The e.m.f is always positive and current flows in one direction only.

Ac generator

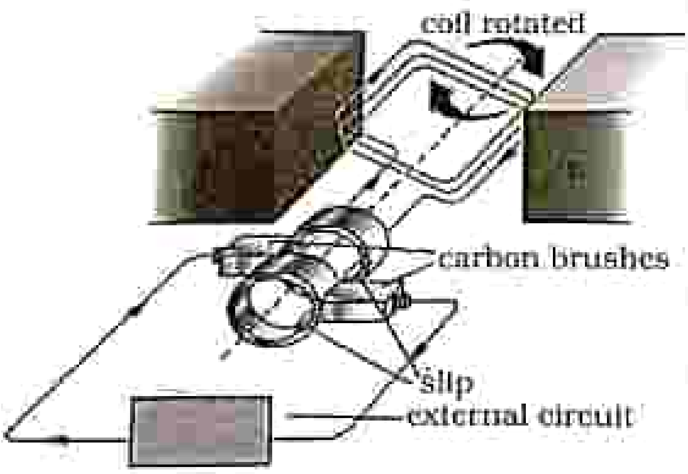


Fig. 29. 4: ac generator

- The coil is made to rotate in a magnetic field, the sides of the coil move in opposite directions and cut the magnetic field lines of force.
- As a result an e.m.f is induced in the coil side of the coil.
- Fleming's left hand rule is used to determine the direction of the induced current in the coil.
- As the coil rotates its side changes direction and as the direction changes the direction of the induced current also changes.
- Slip rings allow the continuously changing current to flow in the external circuit.

Output voltage graphical presentation for an ac generator

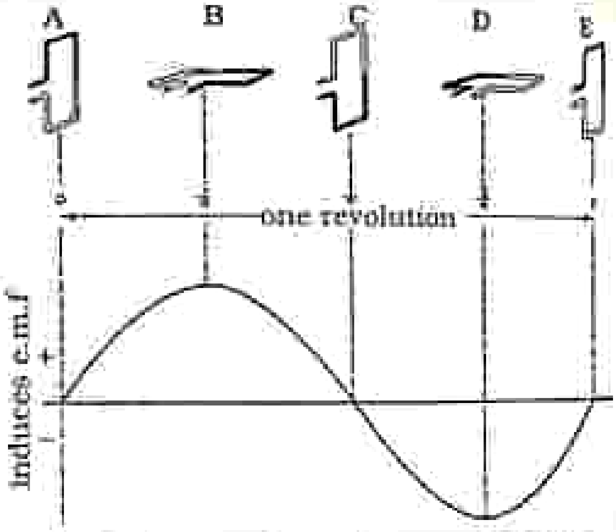


Fig. 29. 5: Output voltage for an ac generator

- The magnitude of the induced e.m.f varies with variation in the position of the coil.
- The induced e.m.f is maximum when the plane of the coil is parallel to the field as the sides of the coil will be cutting field lines at a maximum rate.

- The induced e.m.f is zero when the coil is perpendicular to the field as the side of the coil will be moving parallel to the filed.
- After every half cycle the induced e.m.f changes direction.

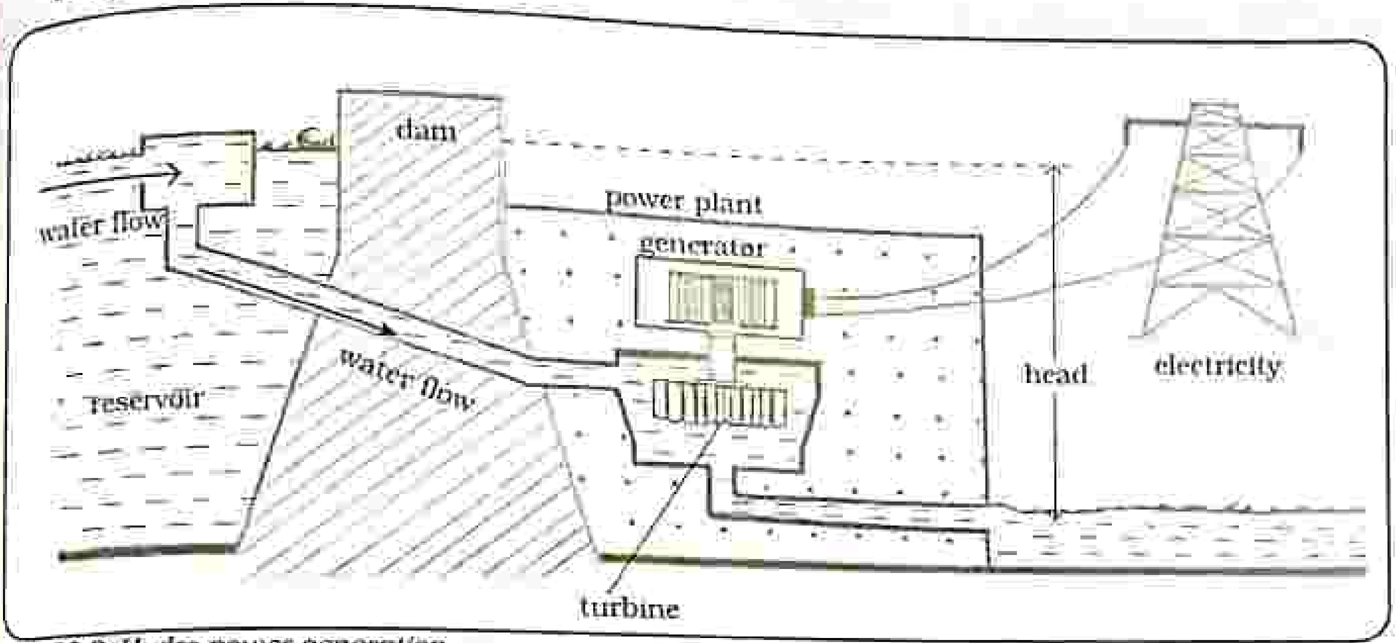


Fig. 29.6: Hydro power generation

- Falling water is used to produce electricity.
- Water is made to hit the turbines causing them to rotate.
- The turbines will rotate the magnet inside the generator as a result e.m.f is induced.
- The amount of electricity produced depends on:
- How far the water falls – the further the water falls the more power produced.
- Amount of water falling – more water falling through the turbine will produce more power.

Energy conversions in hydro power stations

potential energy → kinetic energy → mechanical energy → electrical energy

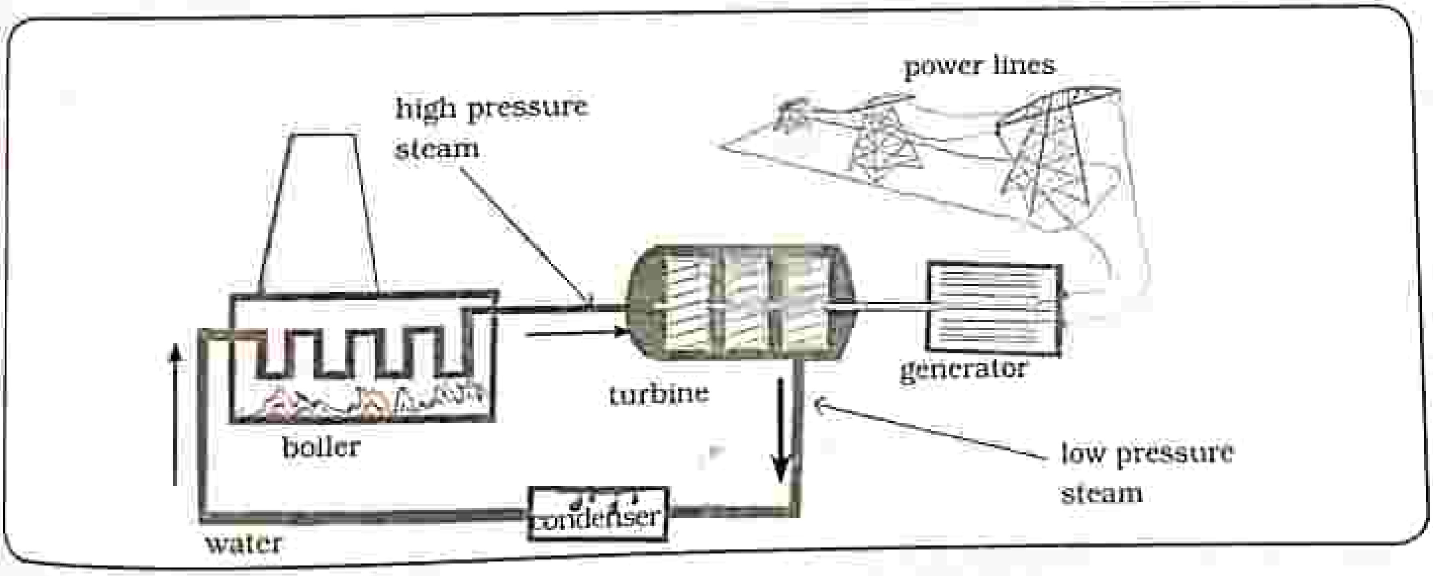


Fig. 29.7: Thermal power generation

- At a thermal power station heat energy is converted to electrical energy.
- Fossil fuels such as coal or coke are burnt to produce heat which boils water into steam.
- The pressure of the steam then turns the turbines.
- The turbines are attached to the big magnet inside the generator and current is created.
- The steam then condenses into liquid water and is recycled.

Energy conversions in thermal power stations

chemical potential energy ⟶ kinetic energy ⟶ mechanical energy ⟶ electrical energy

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

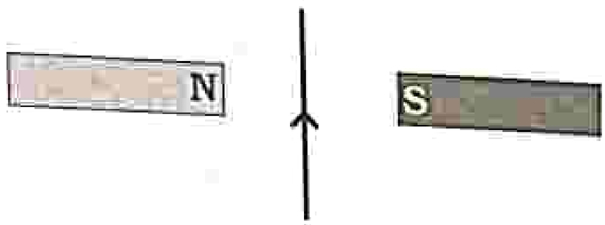
Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Force acts on a current carrying conductor when it is:
   A. placed perpendicular to the field.
   B. placed parallel to the field.
   C. moving in a direction perpendicular to the field.
   D. moving in a direction parallel to the field.
2. In the Flemings left hand rule, the fore finger represents
   A. current.
   B. force.
   C. magnetic field.
   D. voltage.
3. The direction of force on a current carrying conductor is given by:
   A. Fleming's left hand rule.
   B. Flemings right hand rule.
   C. right hand grip rule.
   D. left hand grip rule.
4. Direction of force on a current carrying conductor depends on
   A. the length of conductor in the magnetic field.
   B. the strength of the magnetic field.
   C. the magnitude of current flowing.
   D. the direction of current flow.
5. The diagram shows a current carrying conductor in a magnetic field



In which direction will the conductor move?
   A. upwards
   B. downwards
   C. into page
   D. out of page

CHAPTER 30

Static electricity

Key concepts

- Electrostatic charging
- The electroscope
- Force between charges
- The lightning concept
- Precautions against lightning

Summary notes

Static electricity

- It is an imbalance of stationary electrical charges within or on the surface of a material.
- This charge may be able to move away by means of an electrical discharge or electrical current.
- Static electricity is the major cause of fires and explosions.

Electrostatic charging

- There are two types of electric charge; positive and negative charge.
- The unit of charge is the coulomb(C).
- All matter is made up of atoms. Atoms have protons that are positively charged and electrons that are negatively charged.
- Usually the number of electrons is equal to the number of protons and the object is said to be neutral.
- When a body loses electrons it becomes positively charged and when a body gains electrons it becomes negatively charged.

The electroscope

- An electroscope is used to detect electric charge, it works on the principle of like charges repelling.
- When a negatively charged object is brought nearer a metal cap, electrons from the cap and stem repel and move down to the gold leaf and the metal plate.
- The gold leaf and the metal plate become negatively charged.
- The gold leaf will diverge because like charges repel. This shows that the object is charged.

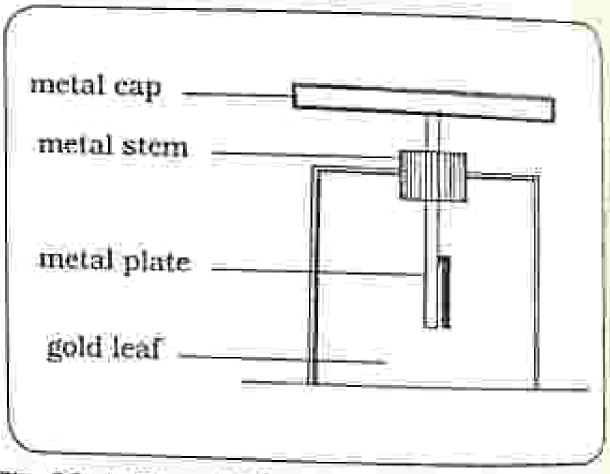


Fig. 30. 1: The gold leaf electroscope

Charging by contact

- When there is contact between the charged object and the metal cap there will be transfer of electrons and the electroscope becomes permanently charged.

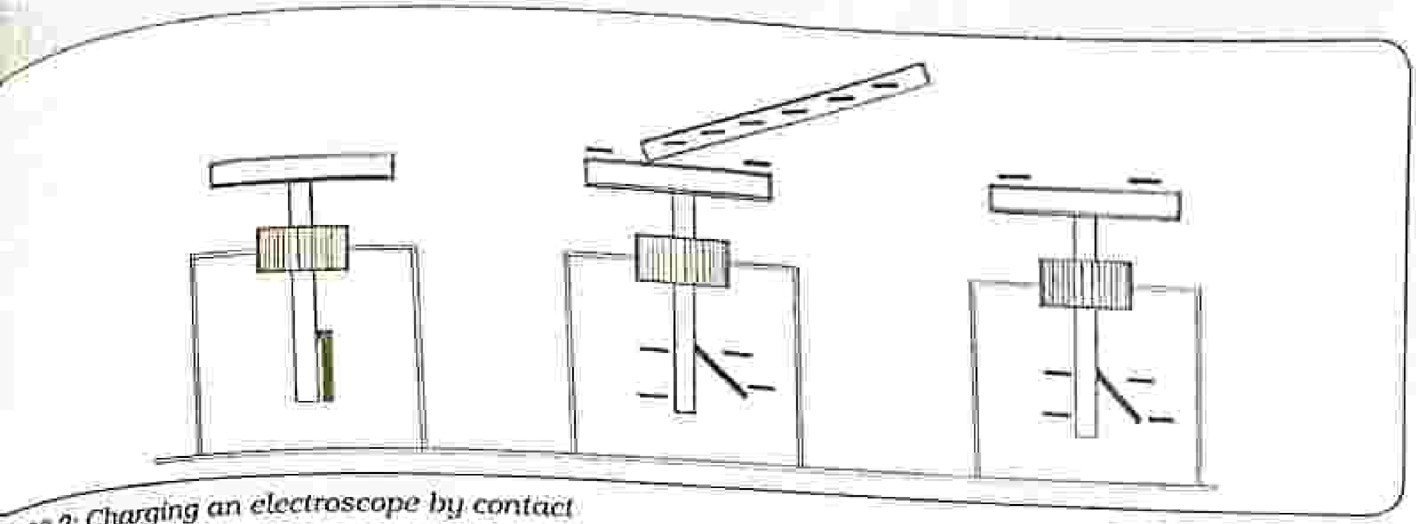


Fig. 30.2: Charging an electroscope by contact

Charging by induction

An electroscope can also be charged without there being contact between it and the charged object, this is called induction.

- A charged object is brought near the electroscope which is connected to the earth such that the repelled negative charge flows to the earth leaving the electroscope positively charged.

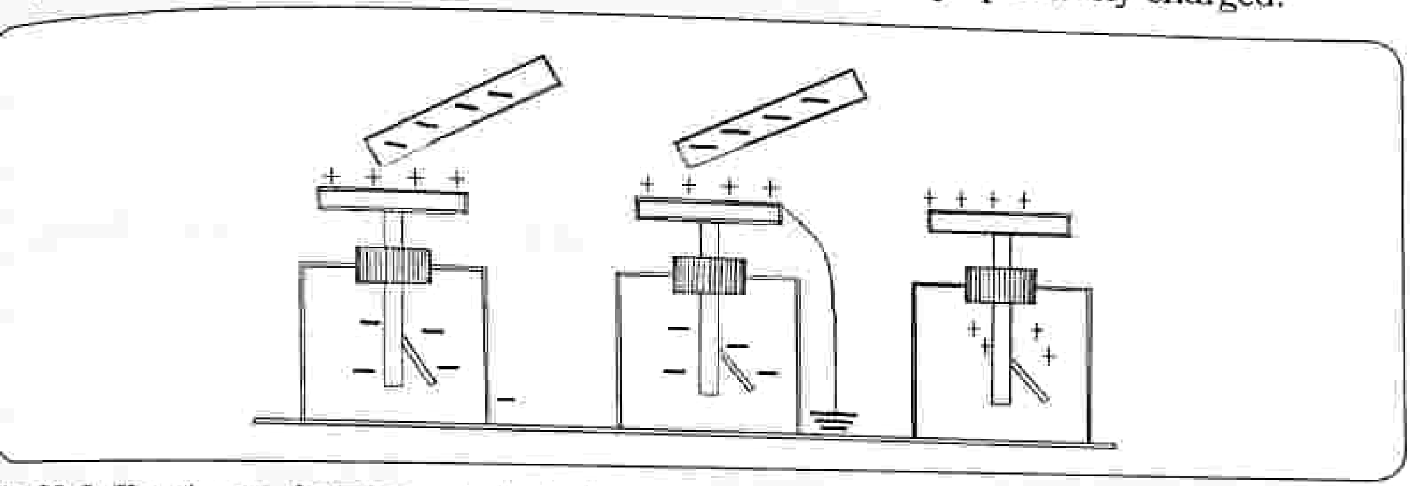


Fig. 30. 3: Charging an electroscope by induction

Charging by friction

- It involves rubbing two surfaces together, as a result electrons will move from one surface to the other.
- One surface becomes negatively charged and the other becomes positively charged.
- This method is useful for charging insulators.

Before rubbing

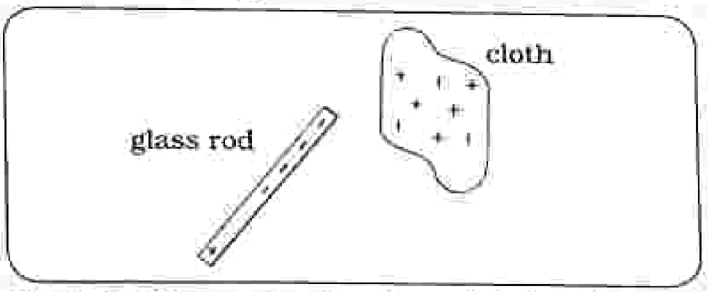


Fig. 30.4: Glass rod and a piece of cloth before rubbing

After rubbing

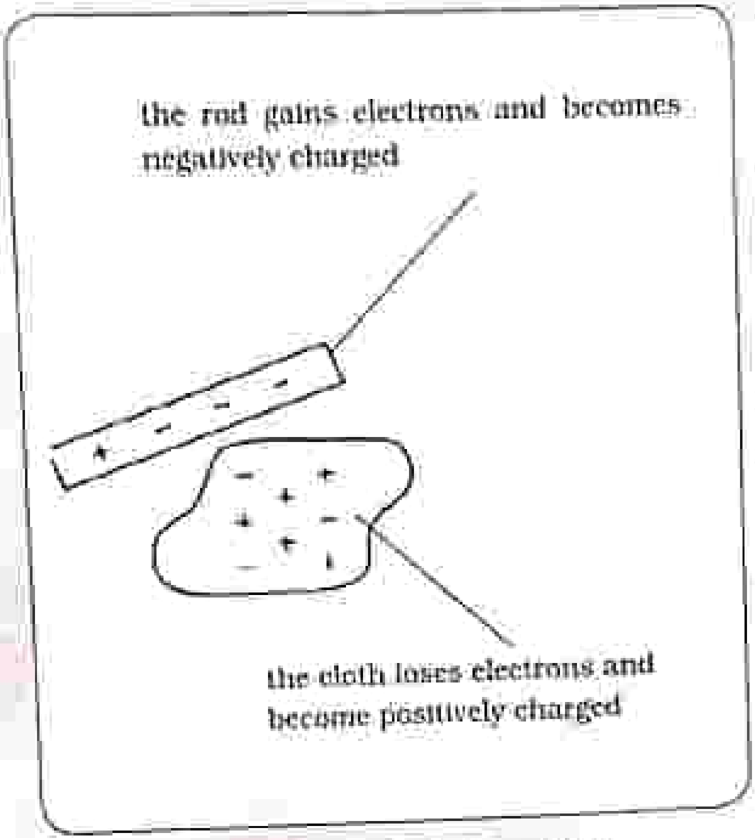


Fig. 30.5: Glass rod and cloth after rubbing

Forces between charges

- A charged object brought near another charged object will experience a force called electrostatic force.
- This force is a non-contact force and it can be repulsive or attractive depending on the type of charge on the objects.

Law of charges

- Like charges repel and unlike charges attract.
- When a positively charged object is brought near a negatively charged object, the objects will attract each other.
- When a positively charged object is brought near a positively charged object, the objects will repel each other.

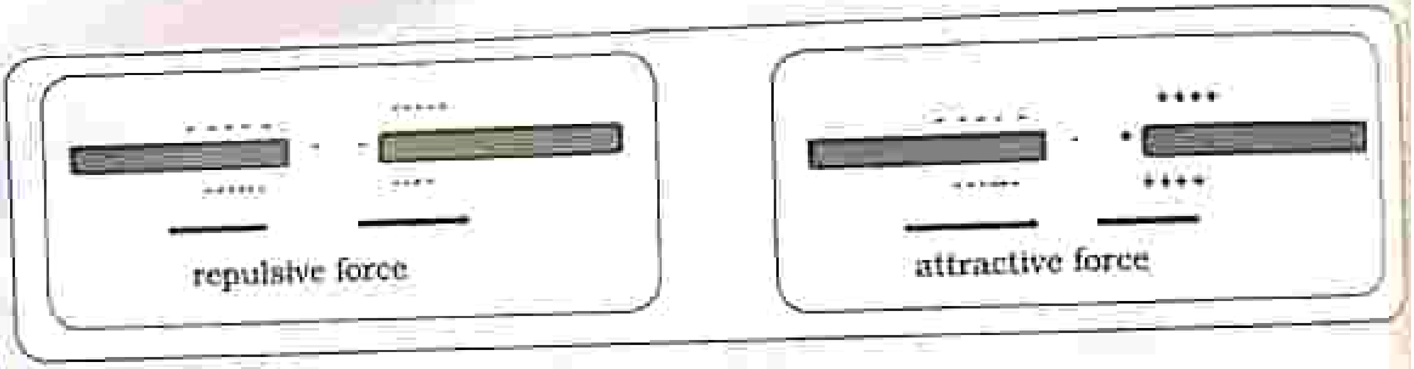


Fig. 30.6: Law of charges

Production of lightning

- Lightning is a bright flash of electricity produced by thunderstorm.
- It can occur between cloud and cloud and between cloud and the earth.

Cloud to earth lightning

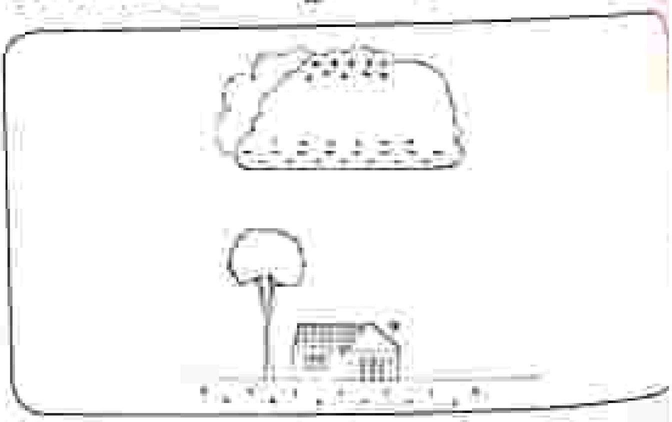


Fig. 30.7: Production of cloud to earth lightning

- Friction between wind and thunder clouds results in the build up of electrostatic charge in the thunder clouds.
- The top of the cloud becomes positively charged and the bottom of the cloud becomes negatively charged.
- By induction the ground becomes positively charged due to repulsion of electrons by the negative charge on the thunder clouds.
- There will be attraction between the positive charge on the surface of the earth and the negative charge on the bottom of the cloud, as a result electrons are discharged from the cloud to the ground.

Lightning conductor

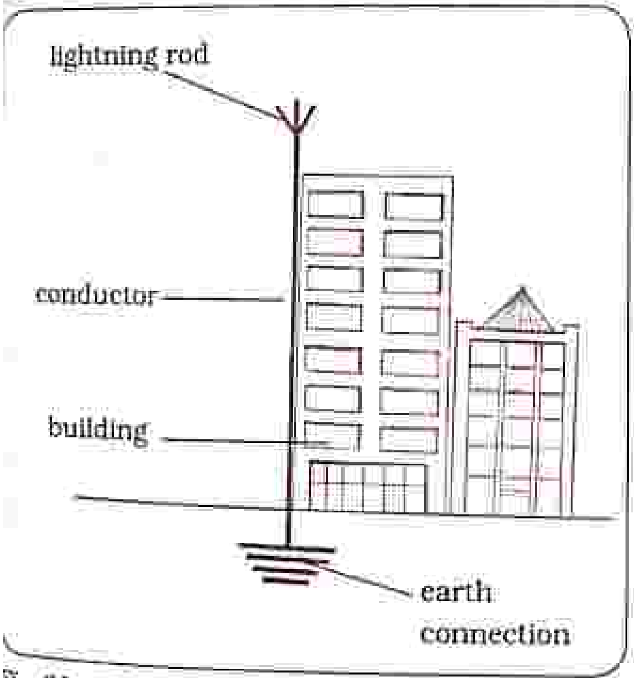


Fig. 30. 8: Lightning conductor

- A lightning conductor is a metal rod that is mounted on top of a building and connected to the ground to protect the building from lightning strikes.
- It provides a safe path of charge to travel to the earth.
- It should be the tallest object around because lightning strikes the tallest object.

Dangers of lightning

Lightning can have devastating effects such as:

- Destruction of property and infrastructure.
- Loss of life and injuries.

Safety precautions against lightning

During a thunder storm:

- Do not take shelter under a tree because lighting strikes the tallest objects.
- Do not take shelter near electrical conductors: avoid fences, poles and machinery.
- Do not swim in water.
- Take shelter inside a house or a car and stay there, at least 30 minutes after the storm passes without returning outside.
- Stay away from windows and keep them closed.
- Do not touch metal or electrical objects.
- If outside, get down on your knees and keep your head tucked to your chest covering your ears, but do not lay flat because it gives lightning a better chance of striking you.
- If you are caught in a lightning storm as a group, maintain a distance of 15 – 30m between each person to reduce the risk of lightning travelling from one person to another.

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

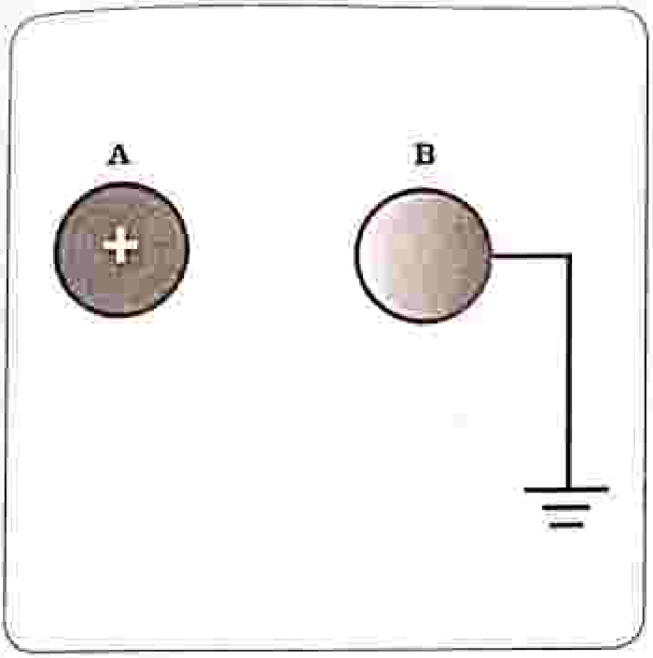
Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. If an object has a higher number of protons than electrons, what type of charge does it have?
   A. negative charge
   B. positive charge
   C. neutral charge
   D. no charge
2. The unit of electric charge is ________.
   A. coulomb
   B. ohm's
   C. volts
   D. amperes
3. The electroscope detects ________.
   A. negative charges only
   B. positive charges only
   C. static electricity
   D. current electricity
4. The major cause of fire and explosions at many places is:
   A. thunderstorm.
   B. heat.
   C. static electricity.
   D. rainfall.
5. Which law states that like charges repel and unlike charges attract?
   A. Conservation of charge law
   B. Ohm's law
   C. Law of electrical charge
   D. Newton's second law.
6. A plastic rod and a piece of cloth are rubbed together and the plastic rod becomes negatively charged. The charge on the cloth is________.
   A. positive of greater magnitude
   B. positive of equal magnitude
   C. neutral
   D. no charge

7. A positively charged sphere A is brought close to a neutral but earthed sphere B.



What is the charge on sphere B when the earth connection is removed?

A. Positive
B. Negative
C. Neutral
D. Depends on contact time

8. A negatively charged sphere is brought close to a neutral sphere. Which of the following statements is true?
   A. there is no electrostatic force between the two spheres.
   B. there is a repulsive electrostatic force between the two spheres.
   C. there is an attractive electrostatic force between the two spheres.
   D. there is a repulsive force on the charged sphere only.

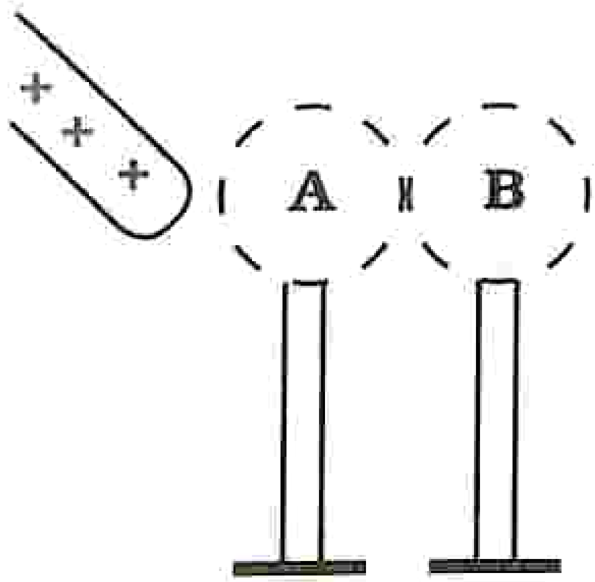
9. There is repulsive force between two charged objects when they have:
   A. like charges.
   B. unlike charges.
   C. same number of electrons.
   D. the same number of protons.

10. What is the safest place to be during a thunderstorm?
   A. Under a tree
   B. Inside a car
   C. In a swimming pool
   D. On top of a mountain

Section B: Structured questions

1. What charge is carried by
   a) Electrons
   b) Protons [2]
2. Explain how an object becomes positively charged [1]
3. Describe the function of an electroscope. [2]
4. State and explain the method used to charge insulators. [2]

5. A positively charged rod is brought close to conducting spheres **A** and **B** which in contact as shown. Sphere **B** is then moved away from sphere **A** and the rod also moved away.



a) What is the charge on sphere **B**? [1]

b) Explain the process that is responsible for the charging of sphere **B**. [2]

6. a) State the law of charges. [1]

b) Explain the condition in which two charged objects experience a repulsive force. [1]

7. With the aid of a well labelled diagram describe the production of lightning. [4]

8. What are the dangers of lightning? [2]

9. State and explain any three safety precautions against lightning. [6]

10. Describe the principle operation of a lightning conductor. [3]

CHAPTER 31 Current electricity

Key concepts

- Ohm's law
- Resistance
- Circuit symbols for components
- Resistors in series
- Resistors in parallel

Summary notes

Current electricity

- Current electricity deals with charge that flows through a conductor.
- Electric current is the flow of charge through an electric circuit.
- In an electrical circuit electrons flow from the negative terminal of the battery to the positive terminal however, the conventional flow of current is from the positive terminal to the negative terminal.
- Electrons and current flow in opposite directions in a circuit.
- When charge moves in a circuit, work is done.
- The work done in moving a unit charge from one point to another in a circuit is called voltage or potential difference (p.d).

Ohm's law

- Relates the resistance of a conducting material to its voltage and current.
- Ohm's law states that current is proportional to voltage and inversely proportional to resistance.
- Increasing voltage will increase current and increasing resistance will decrease current. A Conductor that obeys Ohm's law is called an ohmic conductor.

Ohm's law calculations

The quantities of voltage, current and resistance have the relationship

Voltage = current x resistance

$$V = IR$$

This relationship is called Ohm's law

- Voltage or p.d is measured in volts (V) using a voltmeter. A voltmeter is always connected parallel to an electrical component.
  - V is a symbol for voltage.
  - R is a symbol for resistance and it is measured in Ohm's (Ω.)
- Current is measured in amperes (A) using an ammeter. An ammeter is connected in series with electrical components. I is the symbol for current

Example 1

A bulb takes up 0.2A and is connected to a 2V battery. Calculate its resistance:

$R = \frac{V}{I}$ $R = \frac{2}{0.2}$ $R = 10\Omega$

Example 2

A resistor of 4Ω is connected to a 3V battery. Calculate the current flowing through the resistor.

$$I = \frac{V}{R}$$

$$R = \frac{3}{4}$$

$$I = 0.75A$$

Limitations of Ohm's law

- The major limitation of Ohm's law is that temperature of a conductor has to remain constant for the relationship to be true.
- This is because the resistance of any material varies with temperature.
- Resistance of a conductor increases with an increase in temperature.

Resistance

- It is that which opposes the flow of electrical current in materials.
- There are several factors that affect the resistance of a conductor.

Factors that affect resistance

- Length of conducting wire – the longer the wire the greater the resistance.
- Thickness of conducting wire – thinner wires have greater resistance.
- Material of the conductor – different types of conductors have different resistance e.g. copper has lower resistance than steel.
- Temperature of the conductor – increasing temperature of a conductor will increase resistance.

Circuit symbols for components

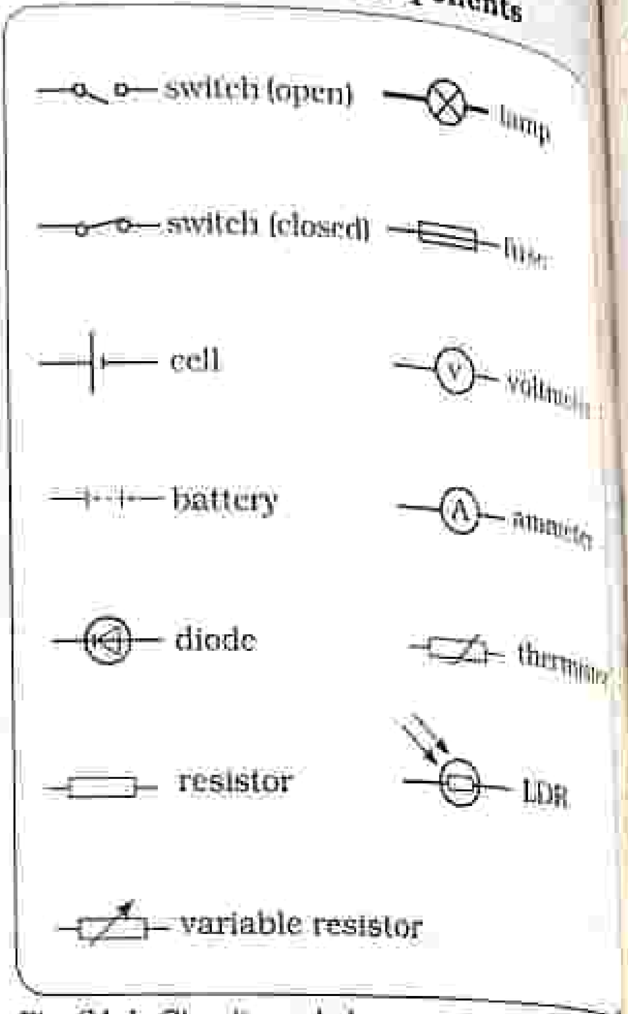


Fig. 31.1: Circuit symbols

Resistance calculation

- Resistors can be connected in series or in parallel.

Series circuit

- Current only flows along one path.
- There are no branches in the circuit.

Resistors in series

When resistors are connected in series:

- Same current passes through all the resistors.
- Sum of the voltages across each resistor is equal to the voltage of the battery.
- The total resistance is equal to the sum of individual resistors.

Example

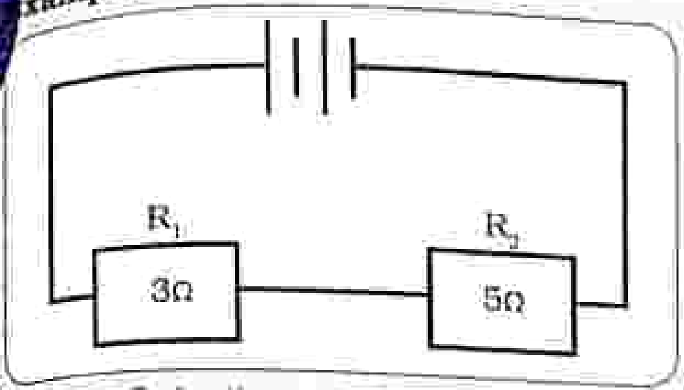


Fig. 31.2: Series circuit

Total resistance of the circuit is

$$R_T = R_1 + R_2$$

$$R_T = 3+5$$

$$R_T = 8\Omega$$

Parallel circuit

- There is more than one path for the flow of current.
- There are branches in the circuit and current split into the different branches.

Resistors in parallel

When resistors are connected in parallel:

- Current flowing through each resistor may be different.
- The voltage across each resistor is equal to the voltage of the battery.
- Total resistance is calculated using the formula $\frac{1}{RT} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3}$

Example

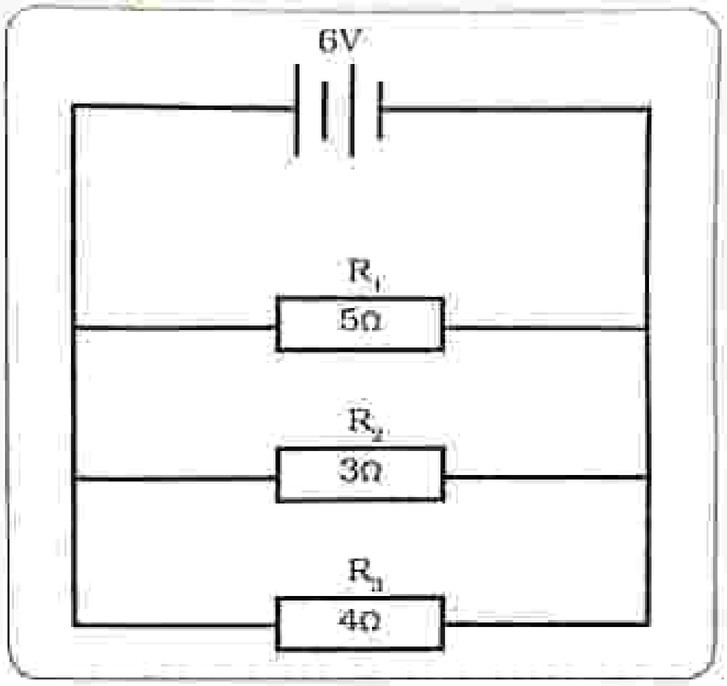


Fig. 31.3: Parallel circuit

The total resistance of the circuit is

$$\frac{1}{RT} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3}$$

$$\frac{1}{RT} = \frac{1}{4} + \frac{1}{3} + \frac{1}{5}$$

$$R_T = \frac{1}{0.78}$$

$$R_T = 1.28\Omega$$

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Another name for potential difference is
   A. current.
   B. voltage.
   C. charge.
   D. resistance.
2. When using Ohm's law which physical quantity should remain constant?
   A. temperature.
   B. voltage.
   C. current.
   D. pressure.
3. The resistance of a conductor does not depend on:
   A. the type of material that makes up the conductor.
   B. length of the conductor.
   C. cross sectional area of the conductor.
   D. the amount of current flowing through the conductor.
4. Which of the electrical symbols represents a fuse?
   A. 
   B. 
   C. 
   D. 
5. What happens to the resistance of a wire when its length is doubled?
   A. doubled
   B. halved
   C. stays the same
   D. increase by half the original value
6. What is the p.d across a 4 Ohm's resistor in which a 5amp current flows?
   A. 0.8V
   B. 1V
   C. 1.25V
   D. 20V
7. Two resistors of 10kΩ and 15kΩ are connected in parallel across a 15V battery. What is the potential difference

across the 10kΩ resistor?

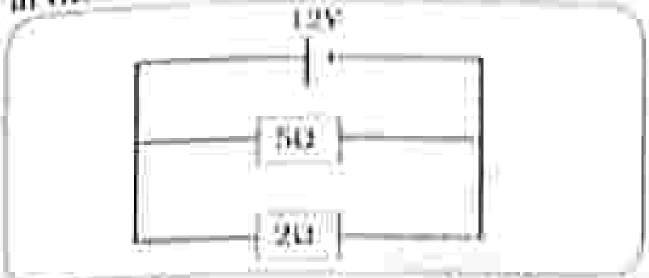
A. 15V

B. 9V

C. 7.5V

D. 6V

8. Two resistors are connected as shown in the circuit.



What is the current passing through the 2Ω resistor?

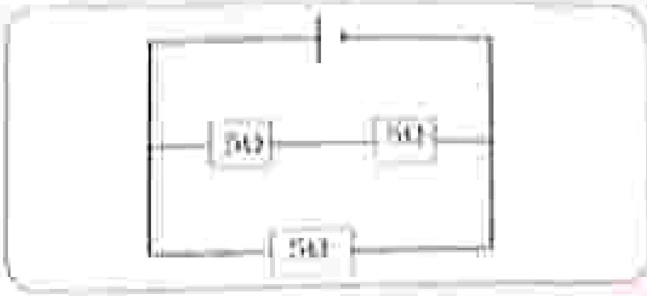
A. 1.71Ω

B. 2.4Ω

C. 6.0Ω

D. 8.4Ω

9. The effective resistance of the circuit below is:



A. 1.67Ω

B. 3.33Ω

C. 5Ω

D. 15Ω

10. An electric current has a single path through a circuit in:

A. parallel combination.

B. series combination.

C. bridge combination.

D. all combinations.

Section B: Structured questions

1. a) State Ohm's law. [1]

   b) State the limitations of Ohm's law. [1]

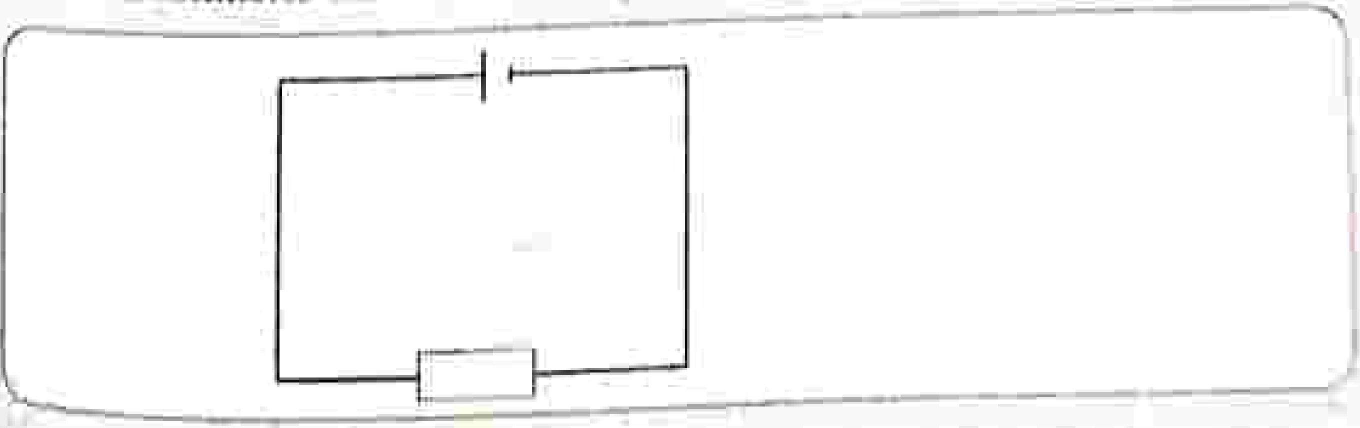
2. Describe the characteristics of an ohmic conductor. [2]

3. A 2A current flows through a resistor connected across a 6V battery. What is the resistance of the resistor? [3]

4. a) Draw circuit symbols for the following: open switch, cell, an ammeter and a variable resistor. [4]

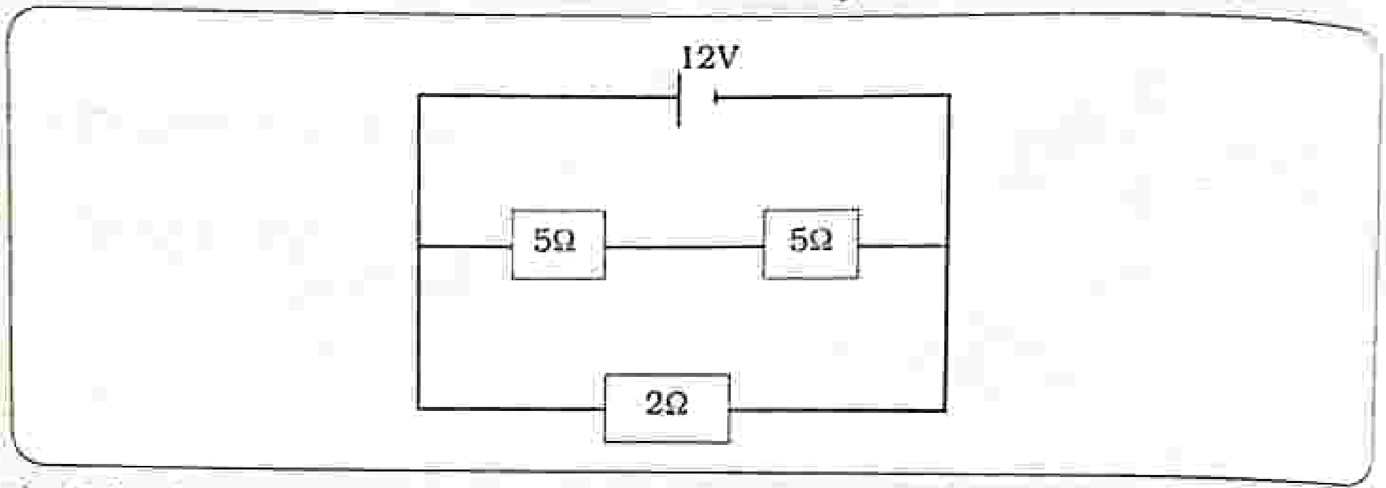
   b) Connect the components in a) to form a complete circuit. [2]

5. Connect a voltmeter on the circuit below. [1]



6. a) State any two factors that affect the resistance of a conducting wire. [2]
   b) Explain how each factor stated in a) affects the wire's resistance. [2]

7. Three resistors are connected across a 12V battery as shown.



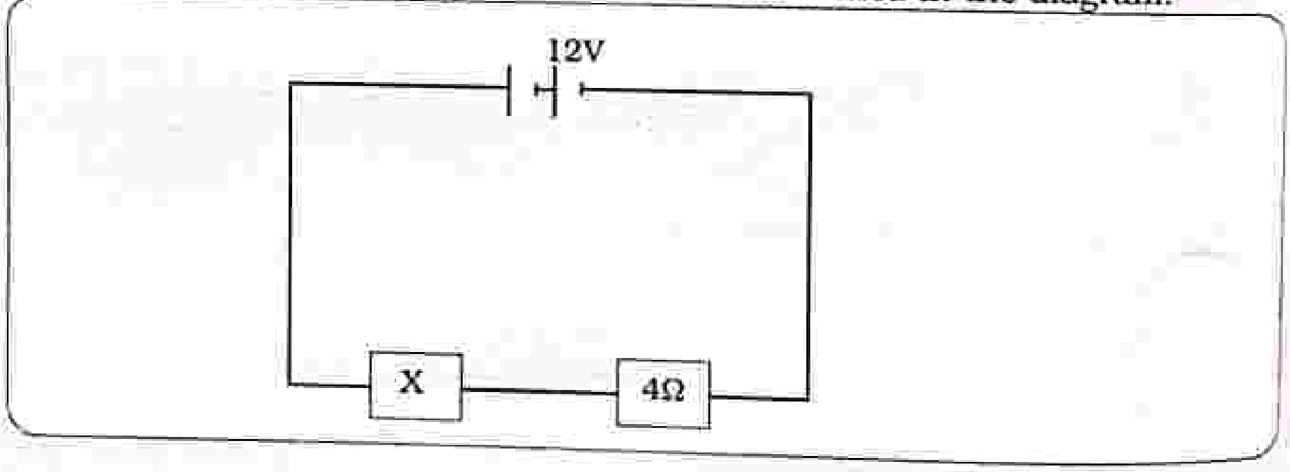
a) Calculate the total resistance of the circuit. [3]
b) Find the total current flowing from the battery. [2]
c) Calculate the p.d across one of the 5Ω resistors. [3]
d) Calculate the current flowing through the 2Ω resistor. [3]

8. You are given two 5Ω resistors. find their equivalent resistance when they are connected:

a) In series [1]
b) In parallel [2]

9. Distinguish between a series and a parallel circuit. [2]

10. A current of 2A flows through the circuit as illustrated in the diagram.



a) What is the voltage across the 4Ω resistor? [2]
b) Find the resistance of resistor X. [3]

CHAPTER 32

Electrical power and energy

Key concepts

- Power and energy used by appliances
- Calculating electrical power and energy

Summary notes

Power and energy

- Electrical energy is measured in joules however the other units for electrical energy are kilowatt hours (kWh).
- Electrical power is the rate at which electrical energy is converted to other forms of energy in an electrical circuit.
- The amount of power dissipated by an appliance is always shown on the appliance and is known as its power rating.
- Appliances that converts a lot of electrical energy will have greater power ratings.
- Appliances that use the heating effect of current have a very high power rating. Electrical power is measured in Watts (W)
  1W = 1J/s and 1 Kw = 1000J

Electrical energy calculations

Electrical energy = power × time

Electrical energy = voltage × time × current

$$E = V \times I \times t$$

Example

A lamp is connected to 12V battery and a current of 1.5A flows through it for 5 minutes. Calculate the energy that is converted to light and heat.

In joules

Convert time to seconds

Electrical energy = voltage × current × time

$$E = V \times I \times t$$

$$E = 12 \times 1.5 \times (60 \times 5)$$

$$E = 318\ J$$

In kiloWatt-hours

Convert time to hours and power to kilowatts

$E = V \times I \times t$

$E = 12 \times 1.5 \times (5/60)$

$E = 1.5Wh$

$E = 0.0015kWh$

Electrical power calculations

Electrical power = voltage × current

$$P = V \times I$$

Example

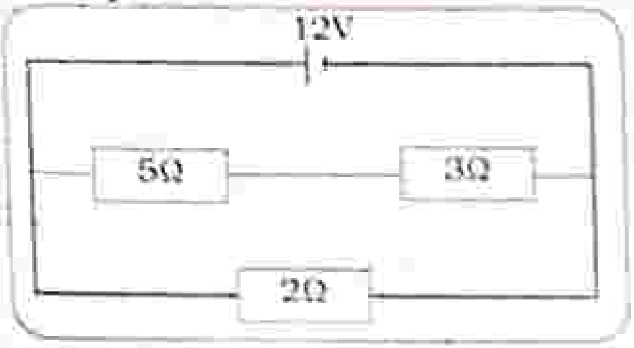


Fig. 32.1: The resistor

The resistor in Fig. 32.1 draws a current of 6 A.

Calculate the electrical power of the resistor.

P= voltage × current

$P = 12 \times 6$

$P = 72W$

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

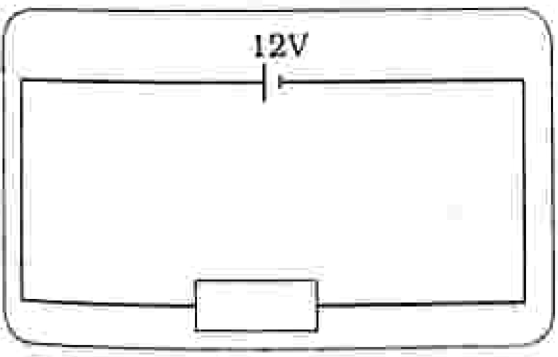
1. Which of the following are not units of electrical energy?
   - A. kWh
   - B. Joules
   - C. Watts
   - D. Watts seconds
2. Which of the following appliances use the heating effect of current?
   - A. Electric stove
   - B. Washing machine
   - C. Television
   - D. Fan
3. Which is the correct formula to calculate electrical power?
   - A. V × I
   - B. V/I
   - C. E×I
   - D. E/t
4. Which of the following quantities is not needed in calculating electrical energy?
   - A. Potential difference
   - B. Current
   - C. Resistance
   - D. Time taken

5. What is the power dissipated by a 25Ω resistor through which a 3A current passes?
   A. 0.12W
   B. 8.3W
   C. 75W
   D. 225W
6. An electrical appliance is rated 40W, 10V, is operated for 20 seconds. What is the amount of electrical energy converted?
   A. 800J
   B. 400J
   C. 200J
   D. 4J
7. An electrical appliance rated 20W operates for 20 minutes how much energy does the appliance convert?
   A. 0.0067kWh
   B. 0.4kWh
   C. 0.67kWh
   D. 400kWh
8. An appliance with a large power rating
   A. draws a very large current.
   B. draws very little current.
   C. has a large resistance.
   D. has a small resistance.
9. 10W is dissipated by a resistor when a current of 0.5A flows through it. What is the resistance of the resistor?
   A. 5Ω
   B. 10Ω
   C. 20Ω
   D. 40Ω

Section B: Structured questions

1. Define electrical power. [1]
2. State any two appliance that use the heating effect of current. [2]
3. What is the relationship between electrical power and electrical energy? [1]

The circuit below shows three resistors connected to a 12 V supply.



4. What is the current flowing through the 2Ω resistor? Current = [3]
5. What is the power dissipated in the 2Ω resistor? Power = [2]
6. An electrical appliance rated 240V, 1200W is operated for 5 hours.
   a) Calculate the current flowing through the appliance. Current = [2]
   b) Calculate the energy converted in kWh. Energy = [3]

CHAPTER 33

Hazards and precautions

Key concepts

- Hazards and safety precautions
- Wiring the Three pin plug
- Use of the two pin plug

Summary notes

Hazards of electricity

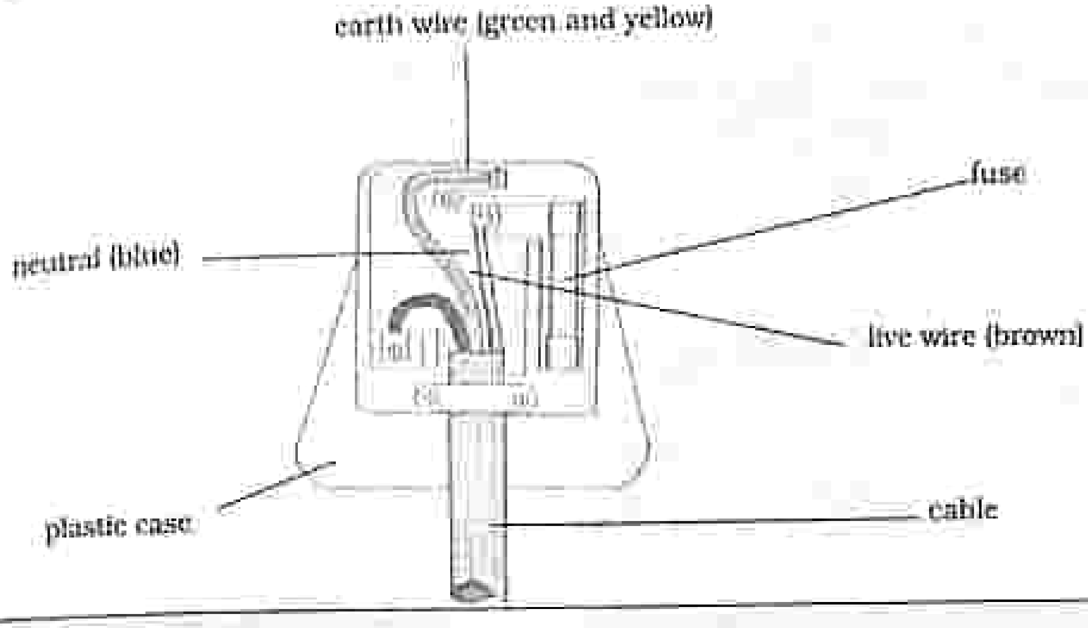
- When using electricity we need to be very cautious because if not it may result in electric shocks and fire.
- When the electric shock is severe it can result in death.
- Electrical hazards include:
  - Damaged insulation: exposed wiring can result in electrical shocks or fire.
  - Overloading circuits: this causes a large amount of current to flow in the wires, as a result the wires may overheat, melt and burn.
  - Damp conditions: water conducts electricity such that handling an appliance with wet hands increase the chances of electric shocks.

Safety precautions

- When using electricity safety precautions should be followed, these include:
  - Earthing: the earth wire is connected to the metal casing of the appliance. If the live wire is loose and touches the metal casing current will flow through the metal casing, the earth wire directs the current to the ground. It protects the user from electrical shocks.
  - Avoid overloading circuits by connecting several appliances with high power ratings to one power outlet.
  - Use insulated cables.
  - Repair damaged insulation.
  - Do not handle appliances with wet hands.
  - Use fuse with proper ratings to avoid overheating of cables.
  - Put on protective clothing such as rubber foot wear when using appliances.

Three pin plug

- Three pin plug is found on electrical appliances.
- It has three pins that are fitted into a wall socket.
- The main cable contains three wires of different colour codes namely: the live wire, earth wire and neutral wire.
- The wires and a fuse are fitted on the pins.



g 33.1: The three pin plug

omponents of the three pin plug

use

- It is a safety device that protects appliances from being damaged by large currents and is installed along the live wire.
- It is designed to break when excessive current flows through it.
- When the fuse breaks current stops flowing and hence the appliance is protected from damage.
- The amount of current needed to blow the fuse is called fuse rating. Different appliances require different amounts of current so fuses have different fuse ratings.
- Fuse rating must be slightly greater than the amount of current flowing in the electrical appliance

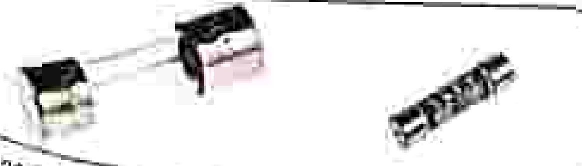


Fig. 33.2 Fuse

Live wire

- The live wire is at a high voltage.
- All electrical components and the fuse are connected along the live wire. Its colour code is brown or red.

Neutral wire

- It is at a very low voltage which is almost zero.
- It completes the current by carrying current away from the appliance.
- No electrical components are connected along the neutral wire. Its colour code is blue.

Earth wire

- It is a low resistance wire connected to the outer metal casing of the appliance and connects it to the earth.
- It protects the user from electrical shocks in the event that the live wire gets in contact with the metal casing. Its colour code is green and yellow.

Wiring the three pin plug

- Using a cutter or a stripper remove some insulation from each wire ensuring that the wires do not break, and gently twist the wires together.
- Secure each wire to the correct terminal according to the colour codes and tighten the cord grip.
- Put the fuse in its position and cover the plug by screwing the cover on.

Two pin plug

- Some appliances use the two pin plug.
- It has two pins and does not have the earth wire.
- Appliances that use the two pin plug have double insulation.

Fig. 33.3: Two pin plug

Double insulation

- Double insulation is a safety feature that is a substitute for earth wire and protects the user from electric shocks.
- Appliances that have double insulation are covered by two layers of insulation which prevent the live wire from coming in contact with any part of the appliance.
- Firstly an electric cable is insulated from the internal components of the appliances. Secondly the metal parts of the appliance are insulated from the outer casing.

Symbol for double insulation

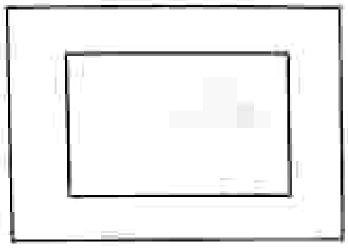


Fig. 33. 4: Double insulation symbol

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.
The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Overloading circuits may result in
   A. some appliances not working properly
   B. less current flowing in the circuits
   C. cables melting and burning
   D. voltage decrease
2. ________ wire is connected to the fuse in a three pin plug?
   A. Live
   B. Earth
   C. Neutral
   D. Both earth and live
3. What colour is the neutral wire?
   A. Blue
   B. Yellow
   C. Brown
   D. Green
4. An appliance uses a current of 12A for its operation, the most suitable fuse rating is:
   A. 13A.
   B. 11A.
   C. 15A.
   D. 16A.
5. Which wire is at very high voltage?
   A. Live
   B. Earth
   C. Neutral
   D. Both the live and neutral
6. When the live wire touches the metal casing of an appliance, which wire conducts the current in the casing?
   A. Earth
   B. Live
   C. Neutral
   D. All the three wires
7. A fuse ______________.
   A. prevents the flow of current from the live wire to the metal casing
   B. protects the user from electric shocks
   C. protects the appliance from excessive current flow

D. is always connected to the earth wire

8. A switch is connected to the
   A. earth wire.
   B. live wire.
   C. neutral wire.
   D. both the live and the neutral.

9. Two pin plugs do not have
   A. earth wire.
   B. live wire.
   C. neutral wire.
   D. put the fourth option

10. The symbol for double insulation is
   A. 
   B. 
   C. 
   D. 

Section B: Structured questions

1. State and explain any two electrical hazards. [4]
2. Explain any two safety measures that need to be considered when using electricity. [4]
3. How does the earth wire protect the user from electric shock? [2]
4. Explain why electrical components such as fuses and switches are installed along the live wire. [1]
5. An electric kettle is connected to a 240V supply and draws a current of 12A:
   a) What is the resistance of the kettle? [2]
   b) What is the most suitable fuse rating for the kettle? [2]
6. a) What is the function of a fuse? [1]
   b) Explain how the fuse is connected in an electrical circuit. [2]
7. a) Identify the three wires that are mounted on the three pin plug. [3]
   b) State the colour code for each of the wires in a). [3]
8. Distinguish between the two pin plug and the three pin plug. [2]
9. Describe the correct way of wiring the three pin plug. [3]
10. Some appliances have double insulation.
   a) Draw the symbol for double insulation. [1]
   b) What is the purpose of double insulation? [2]

CHAPTER 34 Uses and costs of electricity

Key concepts

- Uses of electricity at home
- Calculating cost of electricity
- Methods of saving electricity
- Solar photo voltaic system

Summary notes

Uses of electricity at home

- Electrical energy is used for various purposes which include heating, lighting and powering of various electrical appliances.
- Most appliances we use in everyday life are electrical and require electricity for their operation such as: television, refrigerator, washing machines, iron, heater, computer and microwave.

Cost of electricity

- To calculate the cost of electricity, the amount of electrical energy consumed by a household needs to be measured.
- Every household that uses electricity has an electricity meter which measures the amount of electrical energy used in kilowatt-hours.
- Electrical meter can be analogue or digital. The more electrical energy used the more the money paid.

Fig. 34.1 An electric meter

Meter reading

- The meter reading is taken every month.
- The meter will continuously measure the energy used from the day of its installation, hence there is need to calculate the electrical energy used in a month.
- The previous reading is subtracted from the current reading.

Calculating

Cost of electricity = energy used x the cost of one unit (1kWh)

Example

Electricity costs 10c per kWh, calculate the total cost of electricity for the month of June given that:

- Meter reading for the month of May was 74327kWh
- Meter reading for the month of June was 74752kWh
- The energy used in June is: 74752kWh -74327 kWh = 425kWh

Therefore the total cost = 425 kWh x 10c

Total cost = 4250c = \$42.50

Methods of saving electricity

- Electricity can be expensive if not used sparingly. there is need to take measures to save electricity in order to reduce the amount of money going towards paying for electricity.
- Ways that can help save electricity include:
  - Replacing all light bulbs with energy saving bulbs: incandescent light bulbs use more energy as compared to fluorescent and LED light bulbs.
  - Appliances not in use should be switched off: saves electricity as the energy does no go to waste.
  - Using alternative sources of energy such as solar where possible: solar energy is cheaper and readily available
  - Using gas such as biogas for cooking: gas is much cheaper than electricity and biogas is a renewable source of energy.
  - Using appliances with low power rating: low power rating electricity consumes less electrical energy.

Solar photovoltaic system

- Photovoltaic (PV) system is a system that converts light energy from the sun to electrical energy. It consists of individual photovoltaic / solar cells.
- Each cell generates a small amount of electricity.
- Photovoltaic cells are connected to form a solar panel, which generates a greater amount of electricity.
- When solar panels are connected together an array is formed.
- PV systems vary in size depending on application, energy produced depends on how big the system is.
- Solar energy has many uses which include lighting, heating and powering electrical appliances, hence solar energy can be used as an alternative energy source to electrical energy.

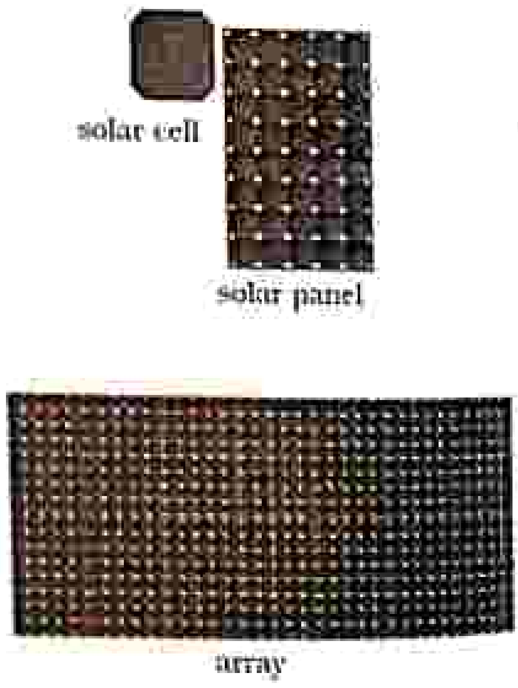


Fig. 34. 2: Photovoltaic system

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.
The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. ________ will help save electricity?
   A. Using appliances with three pin plugs
   B. Using appliances with double insulation
   C. Using appliances with low power rating
   D. Using appliances that draw little current
2. Which of the following is not a renewable energy source?
   A. Solar
   B. Biogas
   C. Firewood
   D. Petroleum
3. Electrical energy can be used for cooking. Which is the best alternative energy source?
   A. Biogas
   B. Firewood
   C. Solar
   D. Paraffin
4. Which is the formula for electrical cost?
   A. Energy x time
   B. Energy x cost of 1kWh
   C. Power x cost of 1kWh
   D. Power x time
5. Electrical energy costs 90c/kWh. What is the cost of using an electric heater rated 1200W, 240V for 4 hours?
   A. $432
   B. $108
   C. $4.32
   D. $1.08

The table below shows electrical meter readings for four months. The meter reading is taken on the first day of each month. Use the table to answer question 6, 7 and 8.

Month	Energy (kWh)
August	784085
September	784251
October	784406
November	784585

6. The least amount of energy was used in:
   A. August
   B. September
   C. October
   D. November
7. In which month was the greatest amount of energy used?
   A. November
   B. October
   C. September
   D. August
8. Calculate the costs of electricity used in October, if electricity costs 10c/kWh.
   A. $1.79 B. $17.90 C. $179 D. $1790
9. What is the advantage of solar energy over electrical energy?
   A. Solar is cheaper.
   B. Solar is more efficient.
   C. All appliances can be powered directly by solar.
   D. Electrical energy is unsafe.
10. An array is a:
   A. Photovoltaic cell.
   B. Group of photovoltaic cells.
   C. Group of photovoltaic systems combined.
   D. Group of solar panels combined.

Section B: Structured questions

1. State and explain any two methods of saving electricity. [4]
2. Explain the importance of saving electricity at home. [2]
3. Identify some alternative sources of energy to electrical energy. [2]
4. An electric iron rated 1000W, 240V is operated for 2 hours. What is the total cost of electrical energy used if electrical energy costs 98c/Kw? [3]

The table below shows electricity meter readings for January, February and March

Month	Energy (kWh)
January	434670
February	434892
March	435021

a) Calculate the electrical energy consumed in February and March. [2]
b) Calculate the electricity bill for March if electricity costs 10c per unit. [3]
c) In which month is the cost of electricity low? [1]

6. What is a photovoltaic system? [1]
7. Distinguish between a photovoltaic system and an array. [2]
8. Identify some of the uses of solar energy. [2]
9. Biogas is an alternative source of energy which can be used instead of electricity
   a) State any one use of biogas. [1]
   b) What is used to produce biogas? [1]
10. Explain whether or not biogas is a renewable energy source. [2]

CHAPTER 35 Telecommunications

Key concepts

- Communication over a distance
  - Cell phones operations and related signal transmitters and receivers
  - E-mail
- Signal transmission in different media transmitters
  - Unguided transmission media

Energy conversions
- Types of media for signal transmission
- Guided transmission media

Energy conversions
- Types of media for signal transmission

Summary notes

Communication over distance

- Communication – is the means of sending or receiving information.
- Sender – is the person who sends a message.
- Receiver – is the person who receives the message.
- Telecommunication
  - is when information is passed from sender to receiver over a distance using technology.
  - can either be through wireless media transmitters or wired media transmitters.
- Cell phones and emails can be used for communication over a distance.

Cellphones

- Sound, text messages, pictures and videos can be sent using a cell phone.
- Cell phones send information as electromagnetic waves through radio waves.
- The radio waves travel in all directions until they reach the cell tower.
- The cell tower then sends the radio waves to the receiver.

Advantages of cell phones

- Very portable hence you can carry it around.
- Information is shared quickly.
- Retrieve information quickly.
- Reliable.
- Connects to the world.

E-mail

- It is a message which is sent by electronic means from one computer user to another or more receivers over a network.
- A smart phone (a cell phone that has many functions similar to that of a computer) can also be used to send an e-mail.

Advantages of E-mail

- Quick
- Easy
- Reliable

Transmission of signal in different media transmitters

The process for signal or information transmission is shown below.

Sender ⟶ Encoding ⟶ Medium of transmission ⟶ Decoding ⟶ Receiver

Encoding – is done by the sender

The sender translates information into a message that can be understood by others.

Decoding – is done by the receiver

The receiver interprets the message by reading, listening or seeing.

Categories of Transmission media

Transmission media is classified into two categories such as unguided transmission media and guided transmission media.

Unguided transmission media

- It is also known as wireless media because the transmitter and the receiver are not connected by a wire or a cable but information is transmitted through free space in form of electromagnetic signal.
- In unguided transmission media, information is converted into radio waves through the use of an antenna by the transmitter.

Energy conversions in unguided transmission – flows as shown below:

Light or sound energy **(sender)** ⟶ Electrical energy ⟶ Electromagnetic waves ⟶ Electrical energy ⟶ Light or sound energy **(receiver)**

Types of media for signal transmission

WI-FI – enables devices to be connected to the internet without the use of a wire or a cable.

- A wireless transmitter receives information from the internet and converts it into radio waves.
- The radio waves are emitted by the wireless transmitter to a certain distance around it.
- Within the area, devices can now connect to the internet.

Examples of devices that use unguided transmission media

- Cell phone
- Computers
- Radio
- Television

Guided transmission media

- It is also known as wired media.
- Information is transmitted through wires or cables which have a fixed path.
- In guided transmission media.

information from the transmitter is converted into an electric current which is transmitted through the wire or cable to the receiver.

- The electric current is converted back to information by the receiver.

Energy conversion in guided transmission media flows as shown below:

Sound energy **(sender)** ⟶ Electrical energy ⟶ Sound energy **(receiver)**

Types of media for signal transmission

1. Optic fibre

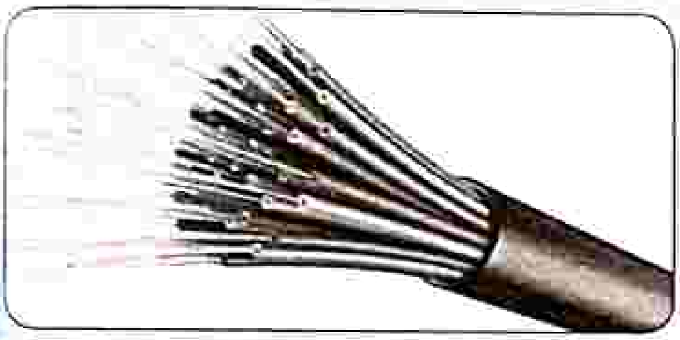


Fig. 35.1: Optic Fibre

- It carries information using pulses of light from one place to another.
- Optic fibre allows high quality transmission of information for long distances with negligible signal loss.
- They have a great carrying capacity hence, large volumes of telephone calls and information can be transmitted in a short time.

2. Coaxial cables

Fig. 35.2: Coaxial cable

- It is an insulated copper wire.
- Information is transmitted through the copper wire as an electric current.
- The insulation also known as a di-electric prevents information from being disrupted or changed before reaching the receiver.

3. Sheathed pair cables

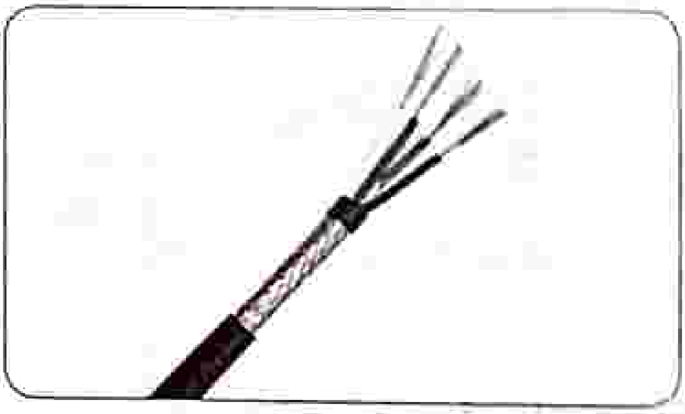


Fig. 35.3: Sheathed pair cables

- It is made up of two insulated copper wires which are covered in a sheath of foil.
- The sheath of foil minimises the amount of interference from other signals.
- Sheathed pair cables are used in telephone systems.

Examples of devices that use guided transmission media

- Telegraph
- Telephone

Revision

INSTRUCTIONS

Section A

Multiple choice

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct. Each correct answer will score one mark.

Section B

Structured questions

Answer all the questions.

The number of marks is given in brackets [ ] at the end of each question.

Multiple choice questions

1. Encoding of information is done by ______.
   A. receiver
   B. sender
   C. transmission media
   D. cable
2. Decoding of information is done by ______.
   A. receiver
   B. sender
   C. transmission media
   D. cable
3. Which one is an example of an unguided transmission media?
   A. optic fibre
   B. Wi-Fi
   C. coaxial cables
   D. sheathed pair cables
4. Which one is not an example of a guided transmission media?
   A. coaxial cable
   B. Wi-Fi
   C. sheathed pair cable
   D. optic fibre
5. Identify the energy conversion which takes place in an unguided transmission media.
   A. sound → electrical → electromagnetic → electrical → sound
   B. sound → electrical → sound
   C. light → sound → electrical
   D. sound → electromagnetic → sound
6. Identify the energy conversion which takes place in a guided transmission media.
   A. sound → electrical → electromagnetic electrical → sound
   B. sound → electrical → sound
   C. light → sound → electrical
   D. sound → electromagnetic → sound
7. ______ is a device which uses wireless transmission media?
   A. telephone
   B. cell phone
   C. telegraph
   D. cable

8. Which device uses wired transmission media?

   A. cell phone
   B. telephone
   C. radio
   D. television

9. Messages sent through cellphones are in the form of ____________.

   A. longitudinal waves
   B. electromagnetic waves
   C. electrical waves
   D. heat waves

10. Which one is an advantage of E-mail over ordinary mail?

   A. not reliable
   B. difficult to use
   C. reliable
   D. very slow

Section B: Structured questions

1. Describe the operations of a cell phone. [2]
2. Describe communication over a distance using e-mail. [2]
3. List down types of media for signal transmission. [4]
4. Outline the energy conversions in an unguided transmission media. [2]
5. Outline the energy conversions in guided transmission media. [2]

PAPER 1 Multiple choice questions

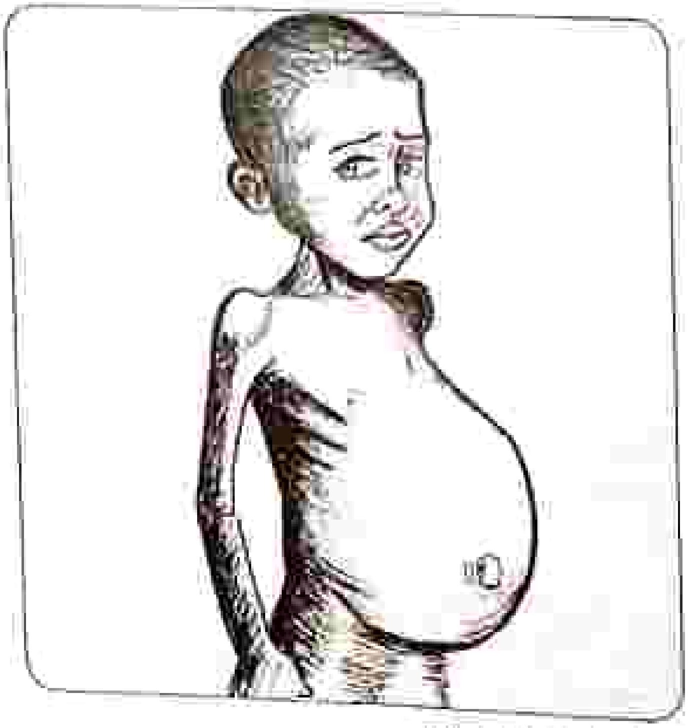
TIME: 1 HOUR

INSTRUCTIONS

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct.

Each correct answer will score one mark.

1. Which nutrient is responsible for growth and repair of worn out tissues?
   A. Protein B. Carbohydrates C. Water D. Fats
2. What is the function of the liver?
   A. Produces bile which helps to break down fat
   B. Stores undigested food
   C. Store bile
   D. Absorption of soluble food
3. Name the condition shown.



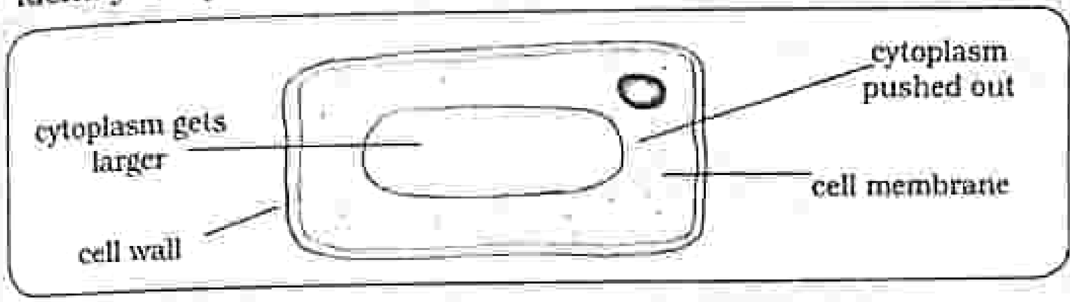
A. Obesity
B. Kwashiorkor
C. Anorexia nervosa
D. Diabetes

4. When iodine solution was added to a food sample, iodine changed colour to blue-black. Which food component does the food sample belong?
A. Starch B. Fat C. Protein D. Glucose

5. During anaerobic respiration_______________.
A. alcohol is used
B. oxygen is used
C. a large amount of energy is released
D. lactic acid is produced in plant cells

6. What is the function of the alveoli in gaseous exchange?
A. Diffusion of carbon dioxide B. Digestion
C. Assimilation D. Photosynthesis

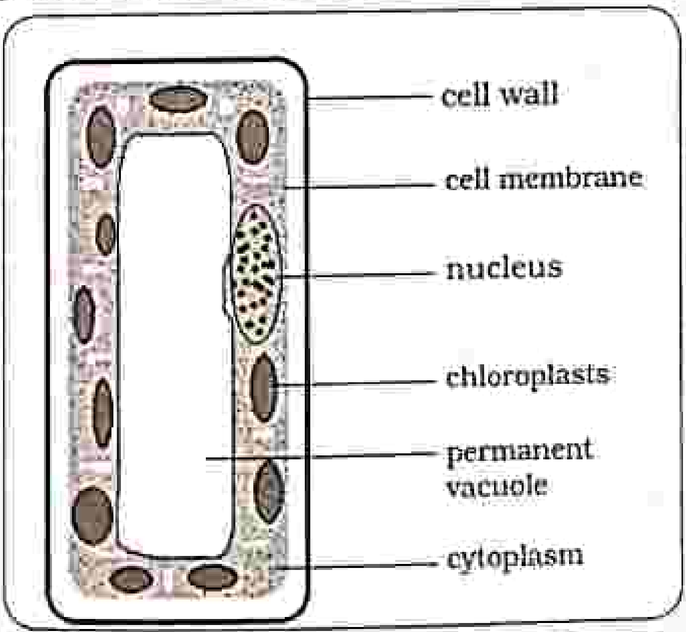
7. Identify the process shown in the diagram.



A.Turgidity B. Plasmolysis C. Transpiration D. Respiration

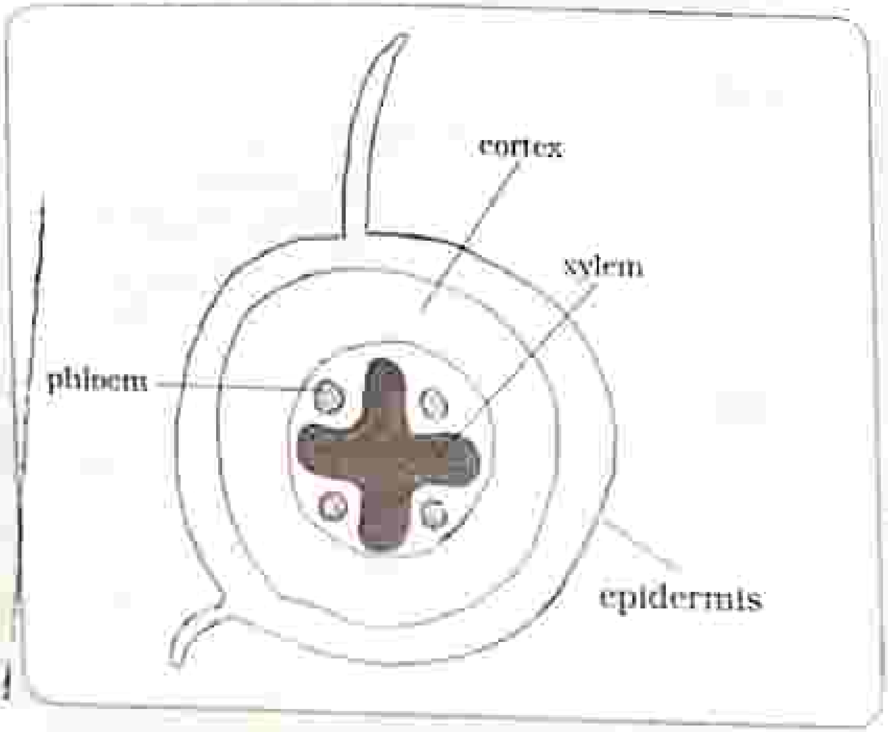
8. In what way is the alveolus adapted for gaseous exchange
A. Has few blood capillaries. B. Has a thick alveolus wall.
C. Has a large surface area. D. Has a dry surface

9. The diagram shows a palisade cell.



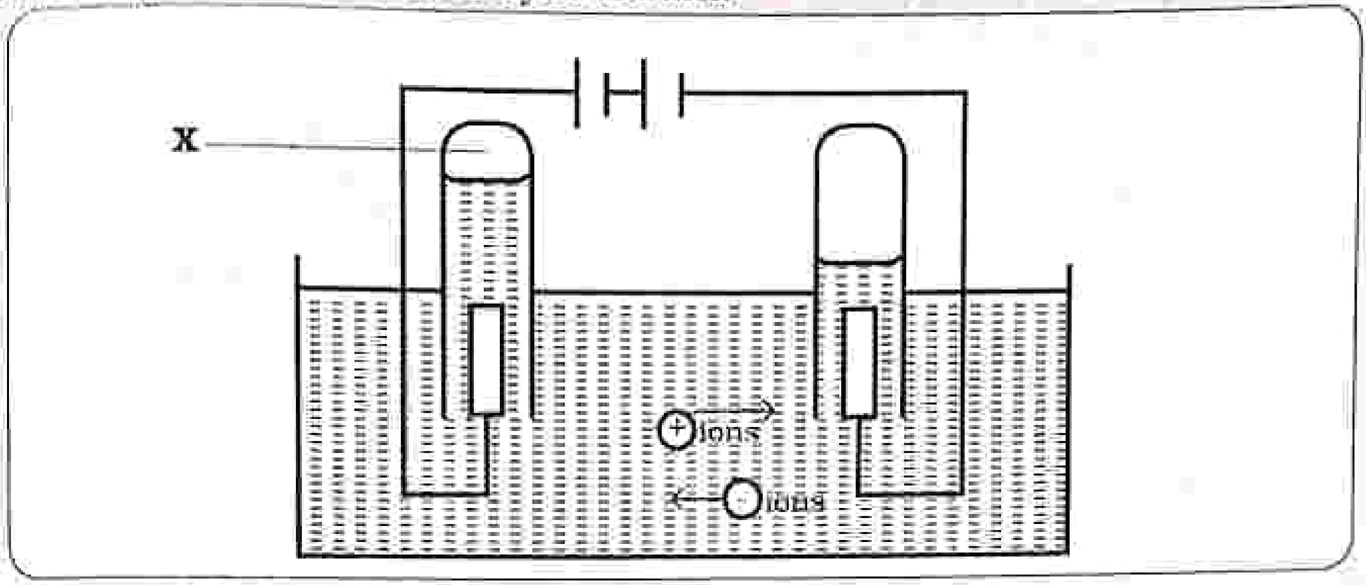
What is the function of the cell membrane?
A. Controls the activities of the cell B. Controls what gets in and out of the cell
C. For photosynthesis D. Stores sugars

10. Which one is an example of a natural ecosystem
    A. pond   B. garden   C. plantation   D. river
11. Identify the end products of photosynthesis
    A. Carbon dioxide and water   B. Carbohydrates and oxygen
    C. Oxygen and water   D. Carbohydrates and Carbon dioxide
12. The diagram shows the root structure of a dicotyledonous plant. Identify the xylem.



    A. A   B. B   C. C   D. D
13. Diffusion is ______________________.
    A. when water moves from an area of high concentration to an area of low concentration through a semi permeable membrane
    B. a process by which plants lose water to the atmosphere in form of water vapour
    C. the movement of particles from a region of low concentration down a diffusion gradient
    D. when mineral salts are absorbed into the cell against a concentration gradient and the processes need energy which comes from the food.
14. Transpiration is ______________________.
    A. when water moves from an area of high concentration to an area of low concentration through a semi permeable membrane
    B. a process by which plants lose water to the atmosphere in form of water vapour
    C. the movement of particles from a region of low concentration down a diffusion gradient
    D. when mineral salts are absorbed into the cell against a concentration gradient and the processes need energy which comes from the food.

15. The diagram shows the electrolysis of water.



Name gas X

A. Chlorine B. Oxygen C. Nitrogen D. Hydrogen

16. Identify the gas produced during fractional distillation of liquid air.

A. Oxygen B. Hydrogen

C. Carbon monoxide D. Chlorine

17. Carbon dioxide is used________

A. as a refrigerant

B. in fire extinguishers

C. in fizzy drinks

D. to manufacture nitric acid and ammonia

18. Electroplating is done ____________

A. to prevent corrosion. B. to allow corrosion.

C. to produce gas. D. to lose weight.

19. During the Contact process, a pressure of __________ is used.

A. 200atm B. 1atm C. 300atm D. 450atm

20. Identify the raw materials used in the manufacture of ammonia.

A. Hydrogen and Nitrogen B. Oxygen and Carbon dioxide

C. Sulphur dioxide and Oxygen D. Chlorine and Oxygen

21. Which process is prevented by painting?

A. Rusting B. Reduction C. Osmosis D. Decomposition

22. Oxidation is ________________

A. loss of electron. B. loss of oxygen.

C. gain of hydrogen. D. gain of electrons.

23. Which statement about ethane is true?

A. Has a double bond B. Forms polythene

C. It is a fuel D. It is an alcohol

24. What is the role of yeast in the fermentation process?

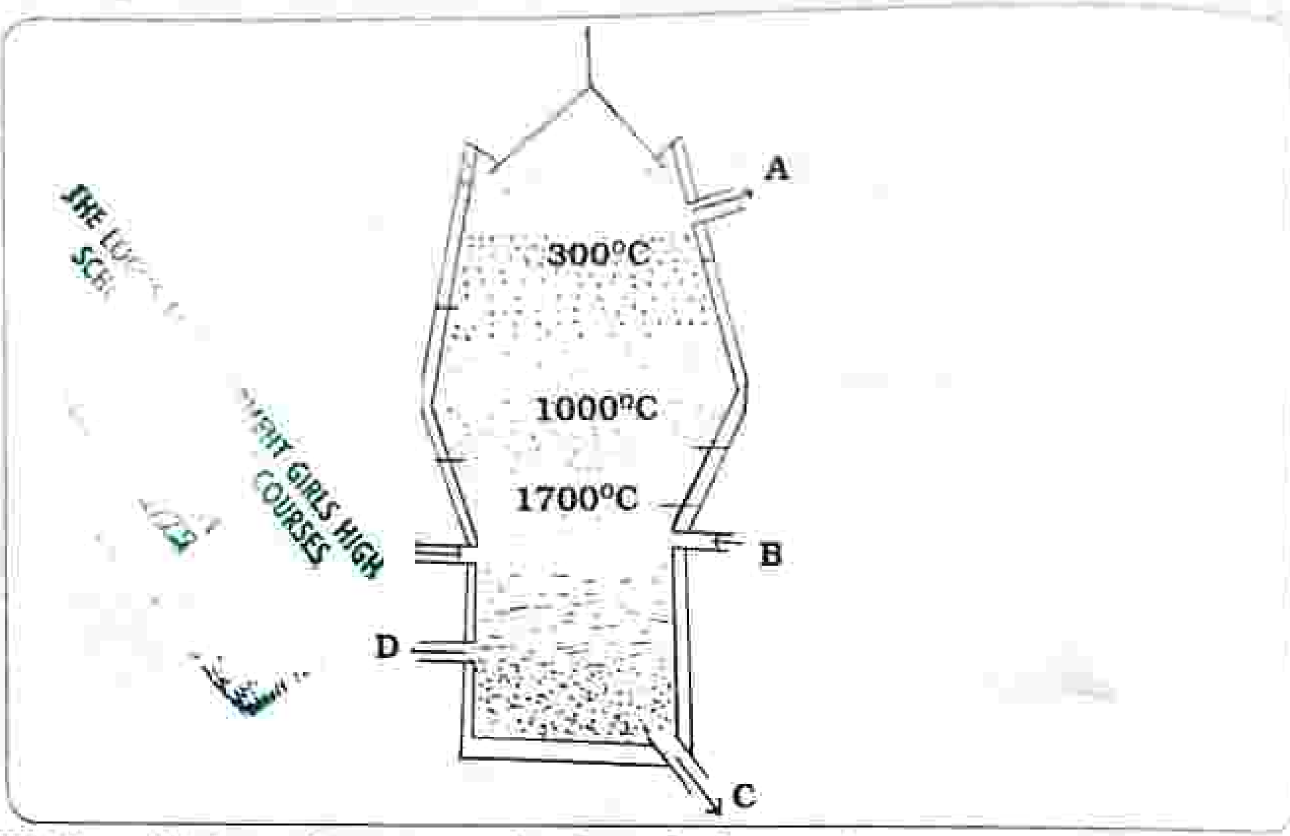
A. Act as a catalyst in the breakdown of glucose

B. Add heat

C. Add colour

D. Add flavour

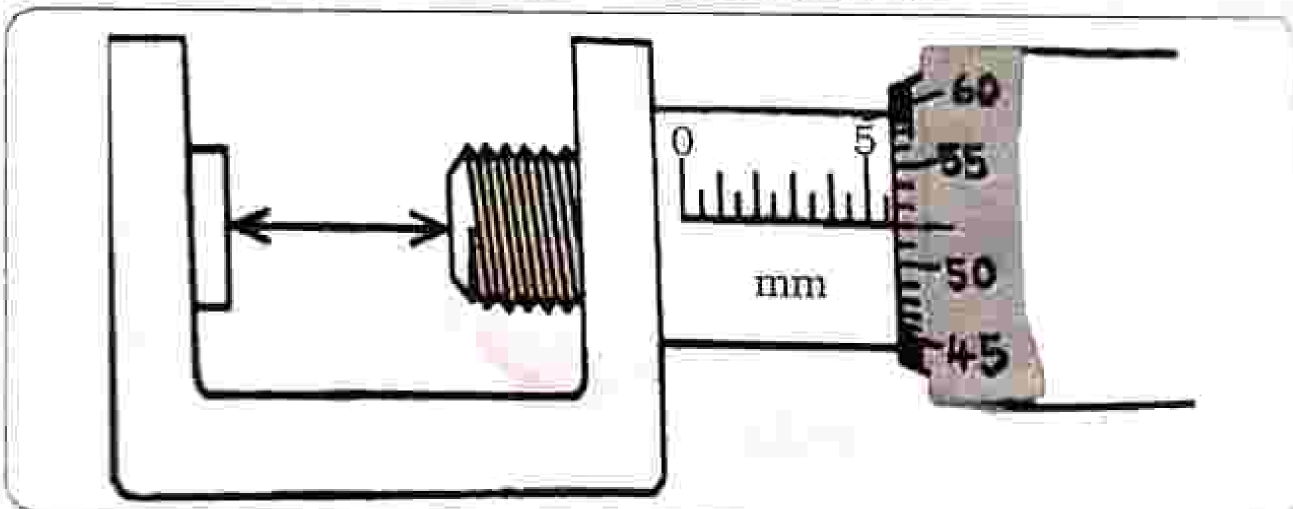
25. The diagram shows the blast furnace



Which outlet is used to remove slag from the furnace?

A. A B. B C. C D. D

26. What is the reading shown on the micrometer screw gauge?

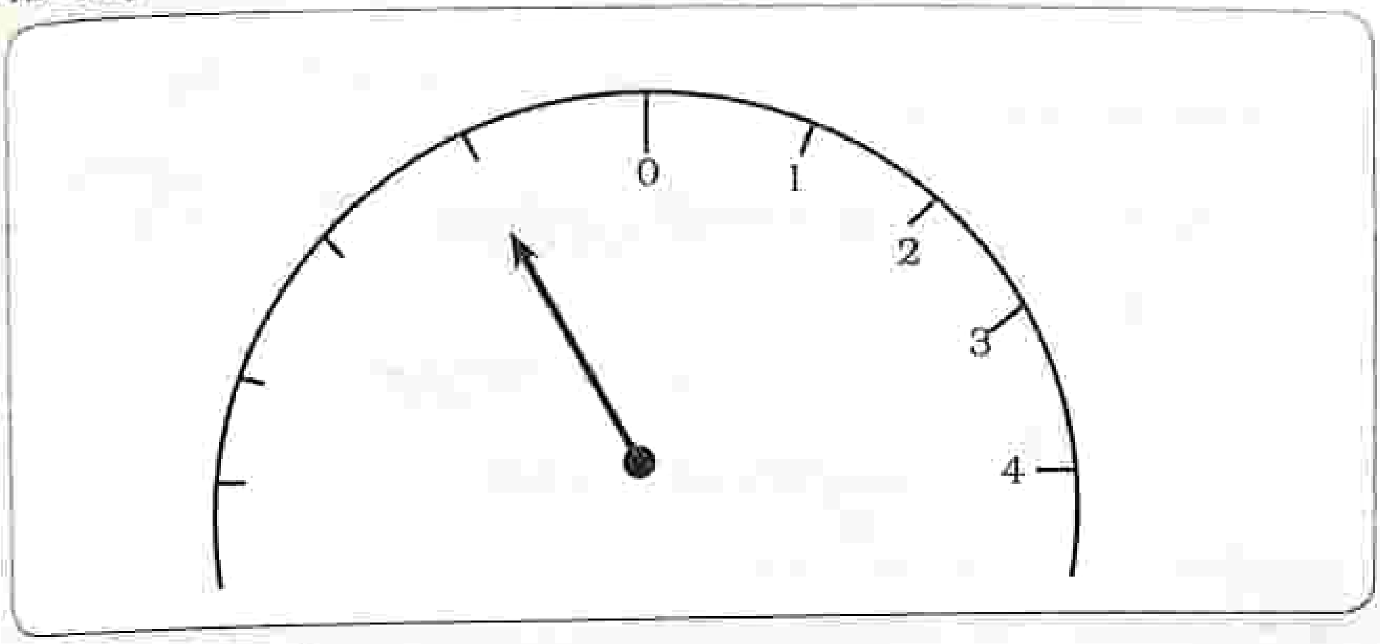


A. 6.02mm B. 5.52mm C. 5.50mm D. 6.55mm

27. What is the SI unit of temperature?

A. Kilogram B. Second C. Kelvin D. Ampere

28. The diagram shows an ammeter.



Identify the type of error.

A. Parallax error B. Zero error

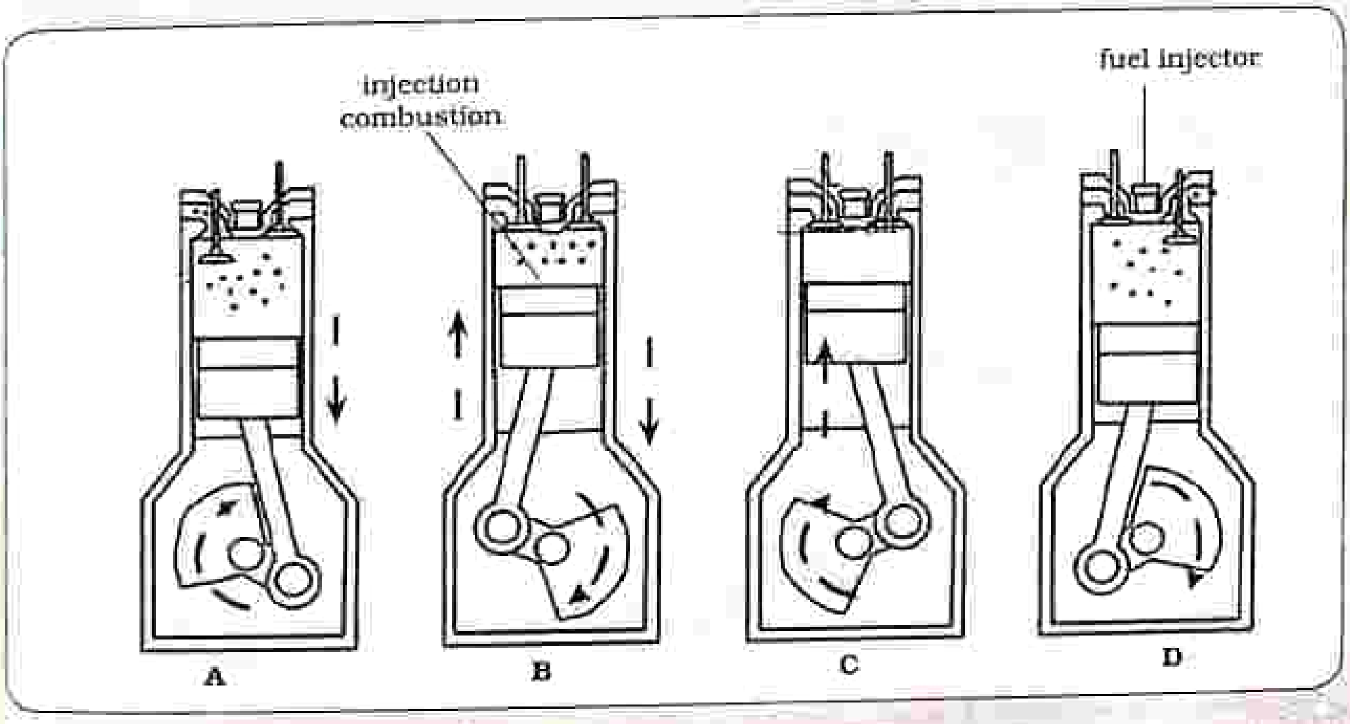
C. Negative D. Positive

29. Define density.

A. Mass per unit volume B. Mass multiplied by volume

C. Volume per unit mass D. Volume subtract mass

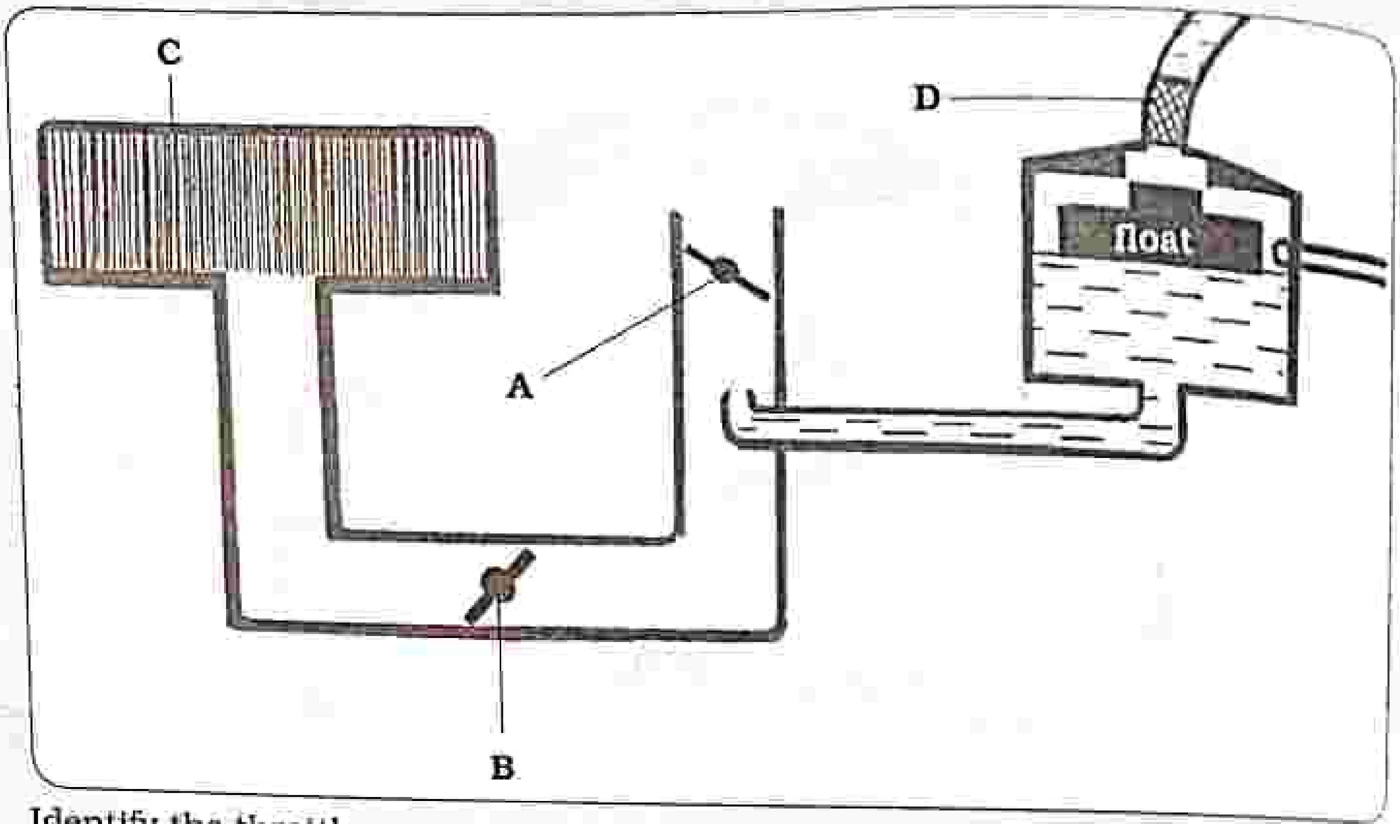
30. The diagram shows a four stroke petrol engine.



Identify the exhaust stroke.

A. **B** B. **C** C. **D** D. **A**

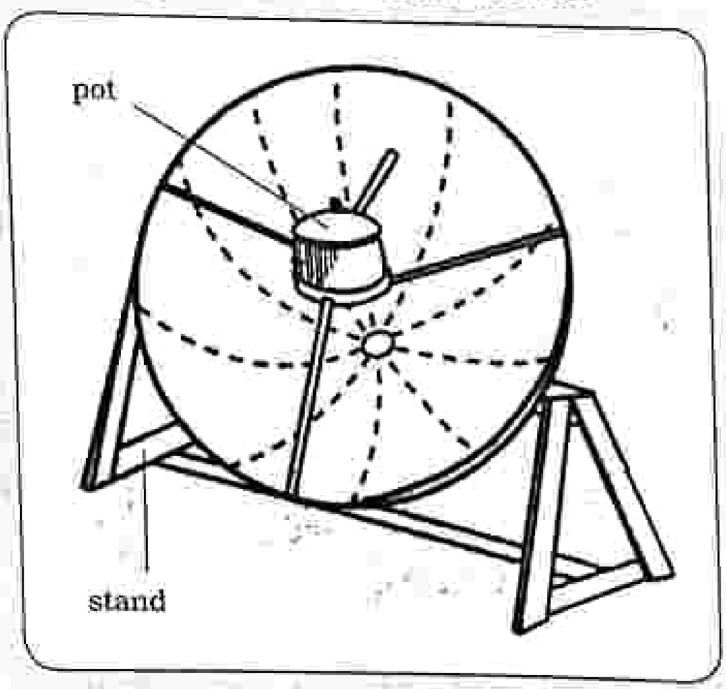
31. The diagram shows a carburetor.



Identify the throttle.

A. B B. C C. D D. A

32. The diagram shows a solar cooker.



Why is a black pot used?

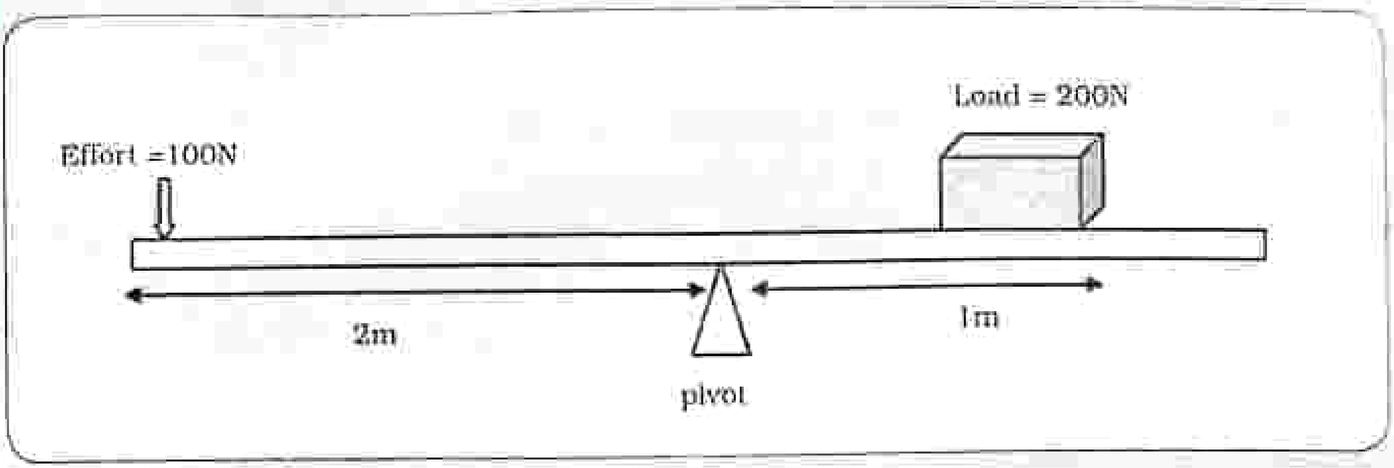
A. For maximum absorption of heat energy

B. Focuses the solar rays to one position

C. Reflects the solar rays

D. For minimum absorption of heat energy

33. The diagram shows a lever.



Calculate the Mechanical Advantage of the lever?

A. 1 B. 2 C. 3 D. 4

34. Calculate the gas pressure when a U tube manometer containing water of density $1000kg/m^3$ is connected to a gas supply. The difference in manometer level is 3m. The atmospheric pressure is 1300Pa.

A. 33kPa B. 32.1kPa C. 31300Pa D. 313.3kPa

35. Which one is an example of a lever?

A. Gears B. Scissors C. Inclined plane D. Pulley

36. Which one is a way of improving efficiency in machines?

A. Reduction of the mass of the moving parts.

B. Not using bearings and roller.

C. No lubrication.

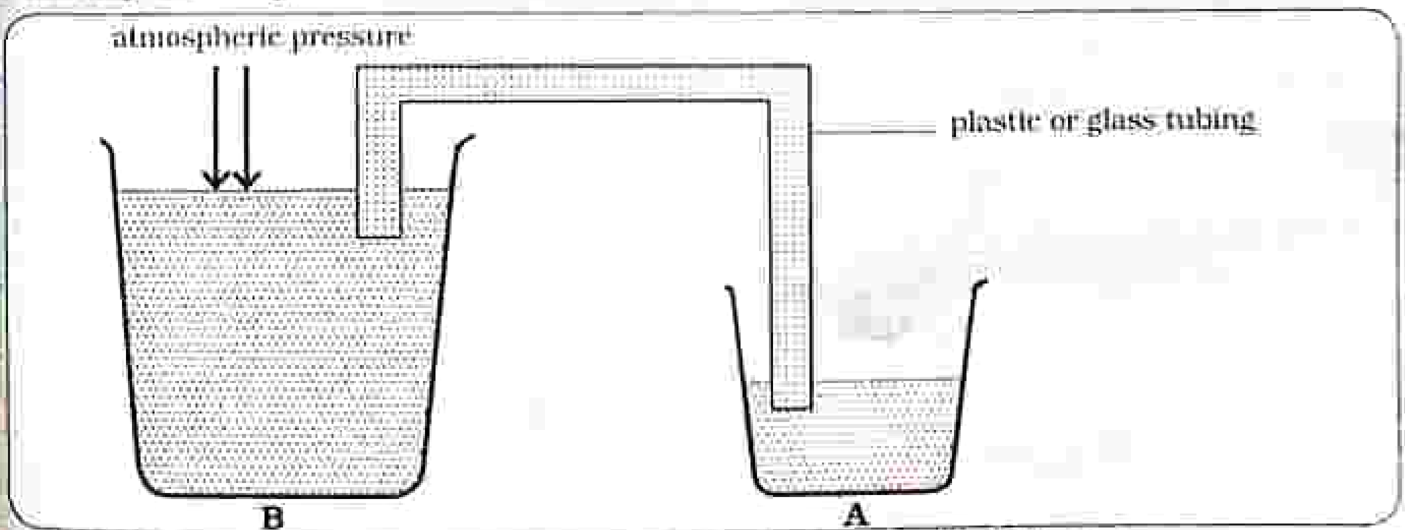
D. Increasing the mass of moving parts.

37. Pressure is defined as ____________________.

A. force acting per unit area B. force multiplied by area

C. area acting per unit force D. force minus area

38. Identify the diagram shown.



A. Hydraulic jack B. Siphon C. Bucket D. Lift pump

39. State the principle of moments.

A. At equilibrium sum of clockwise moment is equal to the sum of anticlockwise moment about the same point.

B. Force acting on the body's mass due to the gravitational force.

C. An object at rest will remain at rest.

D. Resultant force acting upon an object is equal to the product of the mass and the acceleration of the object.

40. Identify an alloy with the following properties; malleable, ductile and can be welded.

A. Stainless steel

B. Mild steel

C. Cast iron

D. Titanium steel

PAPER 1 Multiple choice questions

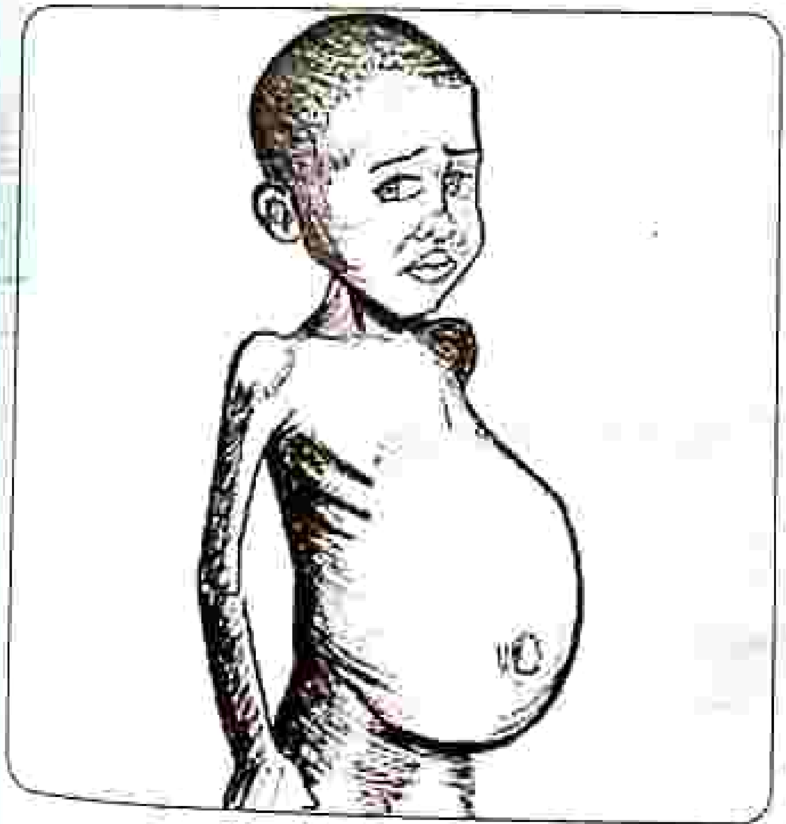
TIME: 1 HOUR

INSTRUCTIONS

For each question, there are four possible answers, A, B, C and D. Choose the one you consider correct.

Each correct answer will score one mark.

1. Name the nutrient which acts as an energy store?
   A. Fats
   B. Water
   C. Carbohydrates
   D. Protein
2. What is the function of the large intestine?
   A. Store bile
   B. Absorption of water
   C. Stores undigested food
   D. Absorption of soluble food
3. Name the condition shown.



   A. Obesity
   B. Kwashiorkor
   C. Anorexia nervosa
   D. Diabetes

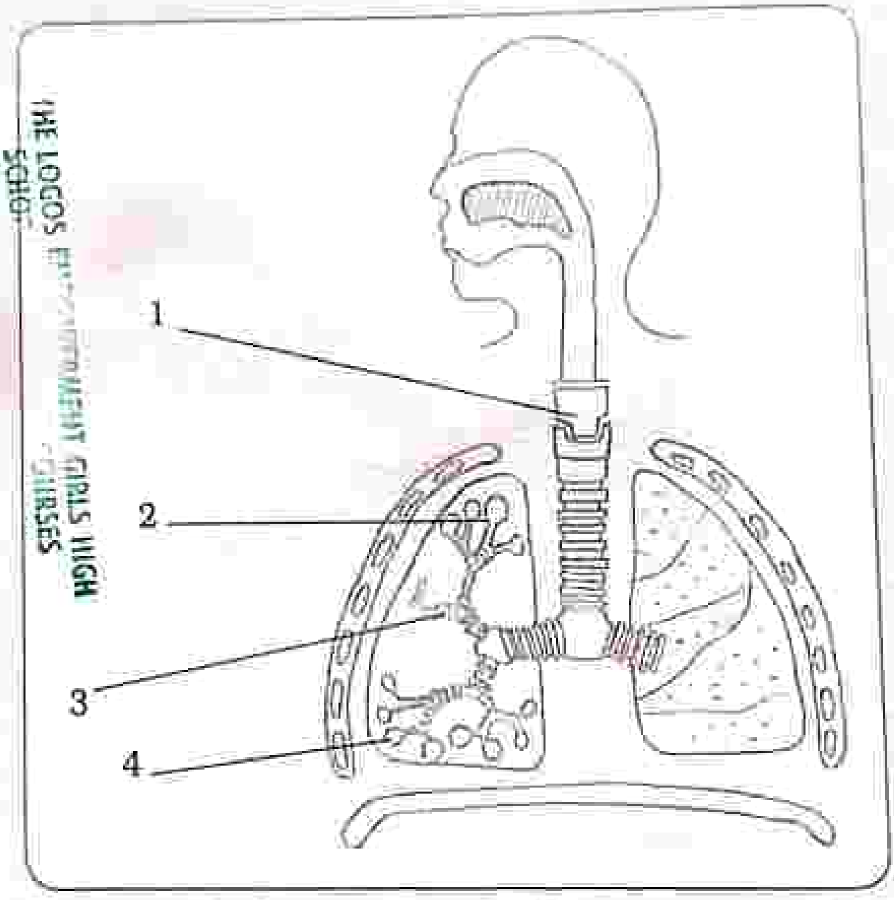
4. A clinistix was dipped into a food solution, the clinistix turned purple. Identify the food component present in the solution.

A. Glucose
B. Protein
C. Starch
D. Fats

5. During aerobic respiration ________.

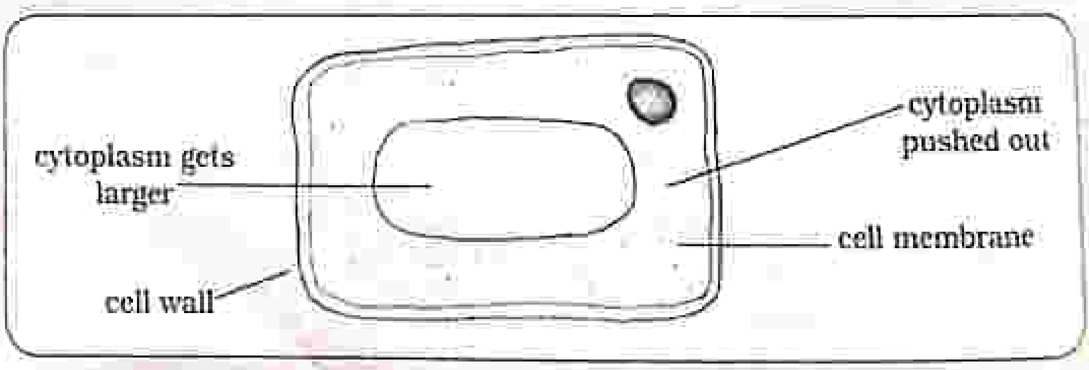
A. carbon dioxide and water are produced
B. oxygen is absent
C. less energy is released
D. lactic acid is produced

6. Identify the trachea.



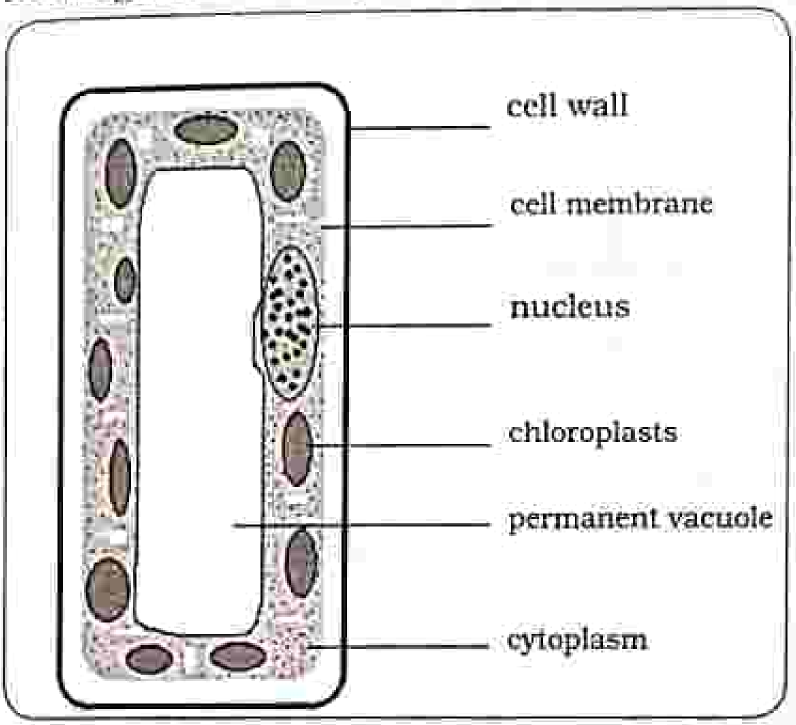
A. 1
B. 2
C. 3
D. 4

7. Name the process the plant cell has undergone.



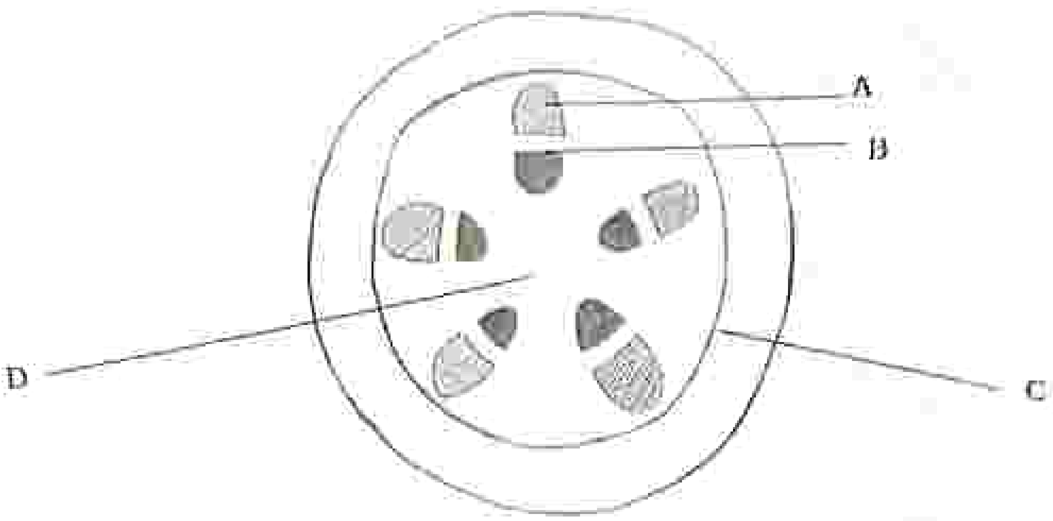
A. Turgidity
B. Plasmolysis
C. Digestion
D. Respiration

8. How is a leaf adapted to reduce transpiration?
   A. Large number of stomata on the leaf.
   B. Thin cuticle layer on the leaf.
   C. Small surface area of the leaf.
   D. Thick cuticle layer on the leaf.
9. The diagram shows a palisade cell.



What is the function of the chloroplasts?
   A. Controls the activities of the cell
   B. Controls what gets in and out of the cell
   C. For photosynthesis
   D. Stores sugars
10. Which one is an example of a natural ecosystem?
   A. pond   B. garden
   C. desert   D. plantation
11. Identify the word equation for photosynthesis.
   A. Carbon dioxide + water $\longrightarrow$ Carbohydrates + Oxygen
   B. Carbon dioxide + Oxygen $\longrightarrow$ Carbohydrates + Water
   C. Oxygen + water $\longrightarrow$ Carbohydrates + Carbon dioxide
   D. Carbohydrates + Oxygen $\longrightarrow$ Carbon dioxide + Water

12. The diagram shows the stem structure of a dicotyledonous plant.
Identify the phloem.



A. A B. B C. C D. D

13. Osmosis is

A. when water moves from an area of high concentration to an area of low concentration through a semi permeable membrane.

B. a process by which plants lose water to the atmosphere in form of water vapour.

C. the movement of particles from a region of low concentration down a diffusion gradient.

D. when mineral salts are absorbed into the cell against a concentration gradient and the processes need energy which comes from the food.

14. Active transport is

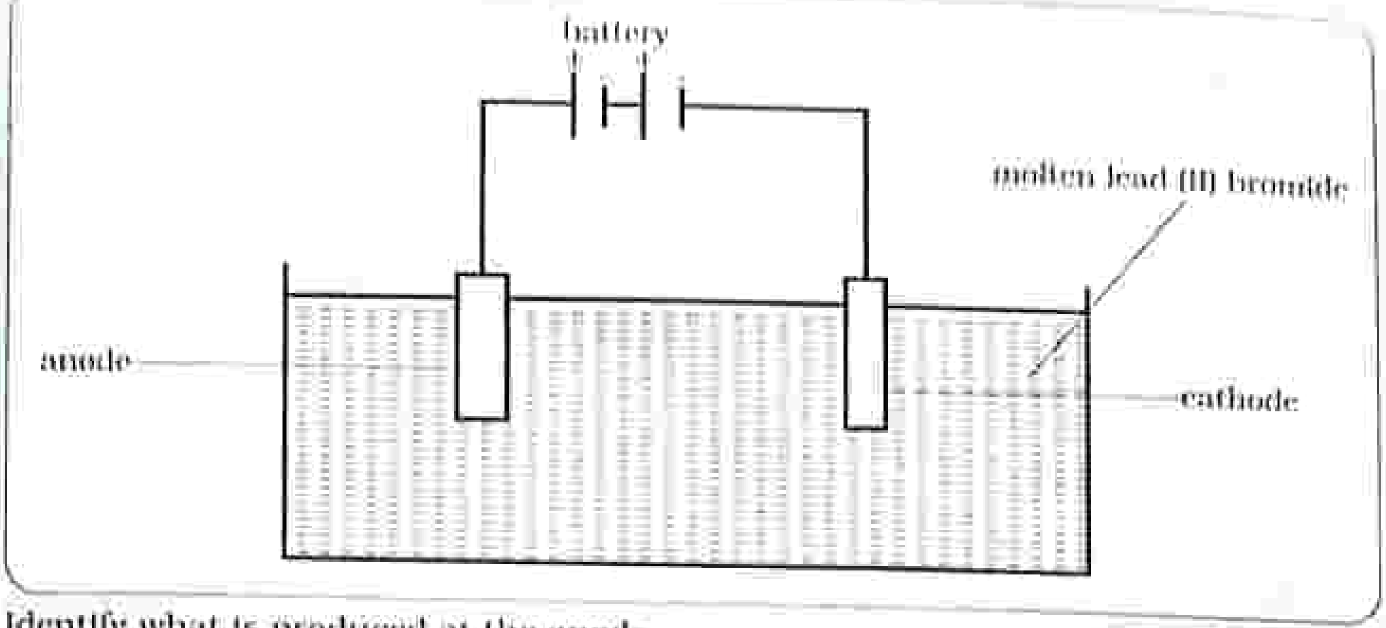
A. when water moves from an area of high concentration to an area of low concentration through a semi permeable membrane.

B. a process by which plants lose water to the atmosphere in form of water vapour.

C. the movement of particles from a region of low concentration down a diffusion gradient.

D. when mineral salts are absorbed into the cell against a concentration gradient and the processes need energy which comes from the food.

15. The diagram shows the electrolysis of molten lead bromide



Identify what is produced at the anode

A. Solid lead
B. Bromine fumes
C. Oxygen
D. Carbon dioxide

16. Identify a gas produced during fractional distillation of liquid air.

A. Hydrogen
B. Nitrogen
C. Chlorine
D. Carbon monoxide

17. Nitrogen is used for ________.

A. Freezing vegetables
B. Medical purposes
C. Manufacture of nitric acid and ammonia
D. Used in fizzy drinks

18. Electroplating is done ____________.

A. for decoration
B. to allow corrosion
C. to produce gas
D. to produce ions

19. During the Haber process, a pressure of --------------is used

A. 200atm B. 1atm C. 450atm D. 300atm

20. Identify raw materials used in the manufacture of sulphuric acid.

A. Hydrogen and Nitrogen
B. Oxygen and Carbon dioxide
C. Sulphur dioxide and Oxygen
D. Chlorine and Oxygen

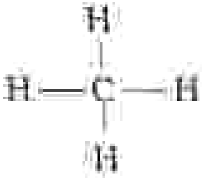
21. Which process is prevented by plating a metal?

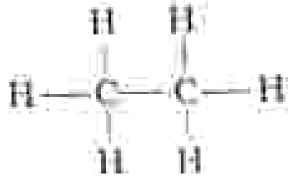
A. Decomposition
B. Diffusion
C. Rusting
D. Reduction

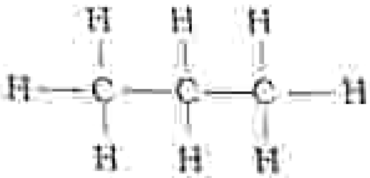
22. Reduction is ________________.

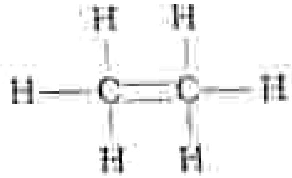
A. gain of oxygen
B. gain of electrons
C. loss of hydrogen
D. loss of electrons

23. Which one is the correct structural formula of methane?

A. 

B. 

C. 

D. 

24. Alcohol belongs to which group?

A. Alkenes
B. Alkanes
C. Hydroxyl
D. Propene

25. When glucose solution is acted upon by yeast, it is converted to ethanol
________.

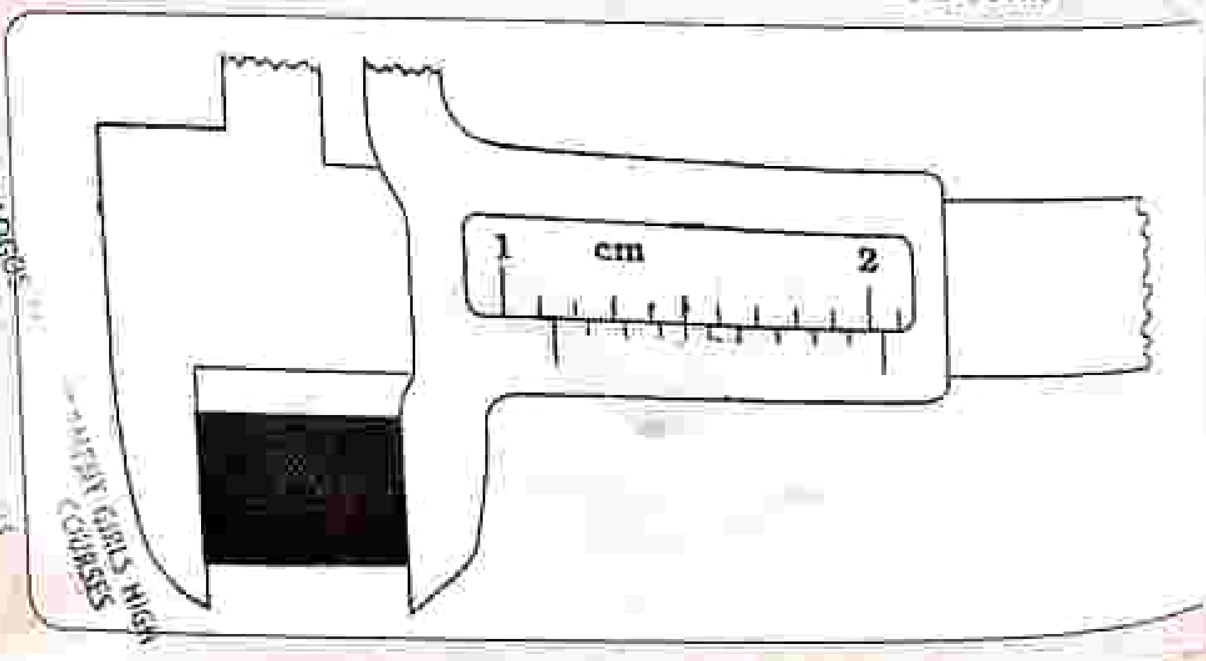
A. Oxygen
B. Carbon dioxide
C. Water
D. Nitrogen

26. In the extraction of iron, what is the function of limestone?

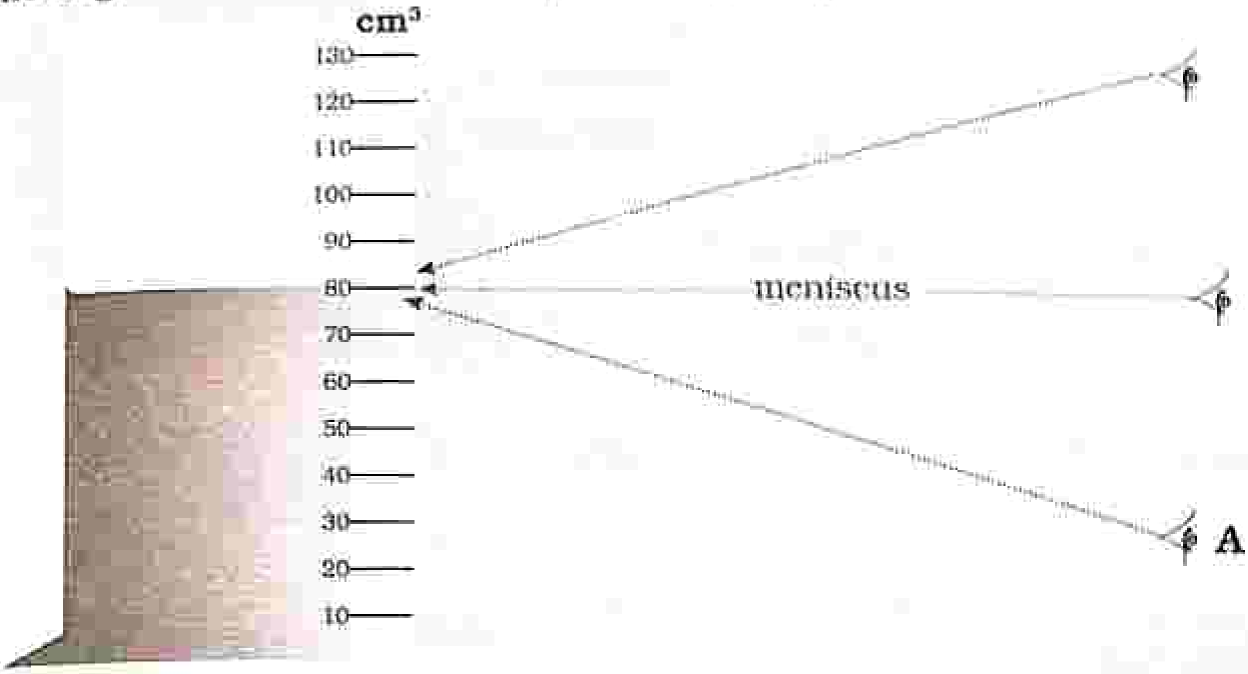
A. Source of iron
B. To remove impurities
C. For producing carbon monoxide
D. Contains oxygen

27. What is the reading shown on the vernier calipers?

A. 1.50cm
B. 2.10cm
C. 1.43cm
D. 2.70cm



28. The diagram shows a learner taking a reading from position A.



Identify the type of error.

A. Zero error B. Parallax error

C. Side D. Negative

29. Calculate the density of an object with a mass of 2kg and a volume of $10m^3$.

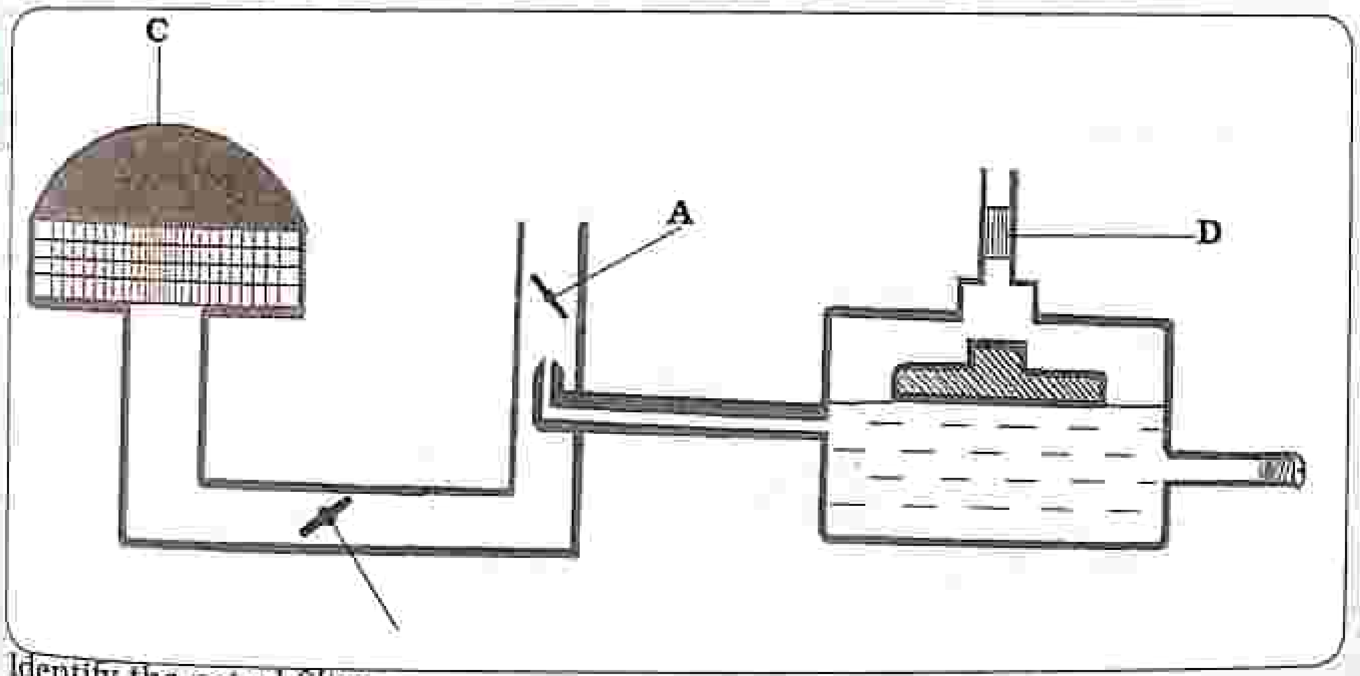
A. $0.2kg/m^3$ B. $20kg/m^3$ C. $5kg/m^3$ D. $50kg/m^3$

30. In a four stroke diesel engine, an explosion takes place in which stroke?

A. Compression stroke. B. Inlet stroke.

C. Power stroke. D. Exhaust stroke.

31. The diagram shows a carburetor.



Identify the petrol filter.

A. A B. B C. C D. D

32. Why are the thin tubes made of copper in a solar water heater?

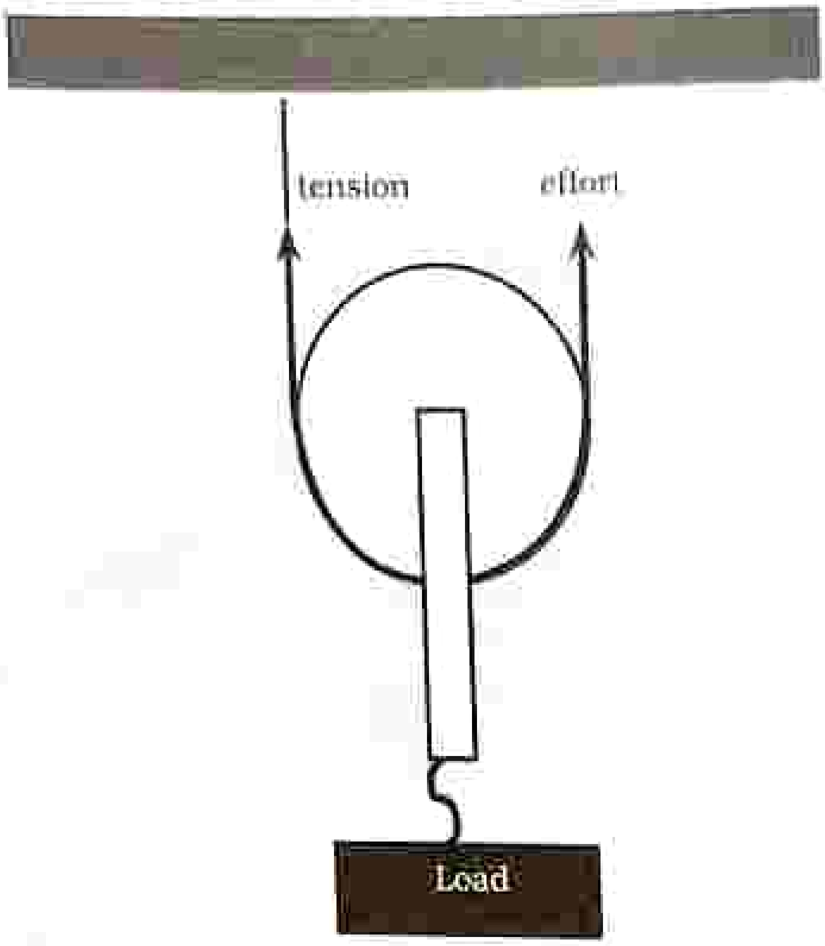
A. Copper is heavy.

B. Copper is a bad conductor of heat energy.

C. Copper is a good conductor of heat energy.

D. Copper corrodes easily.

33. The diagram shows a single movable pulley.



What is the Velocity Ratio for the pulley?

A. 1 B. 2 C. 3 D. 4

34. Which one is not a simple machine?

A. Lever B. Inclined plane

C. Pulley D. Rock

35. Which one is not a use of simple machines?

A. To reduce the effort (force) required to perform a task.

B. Move forces from one form to another.

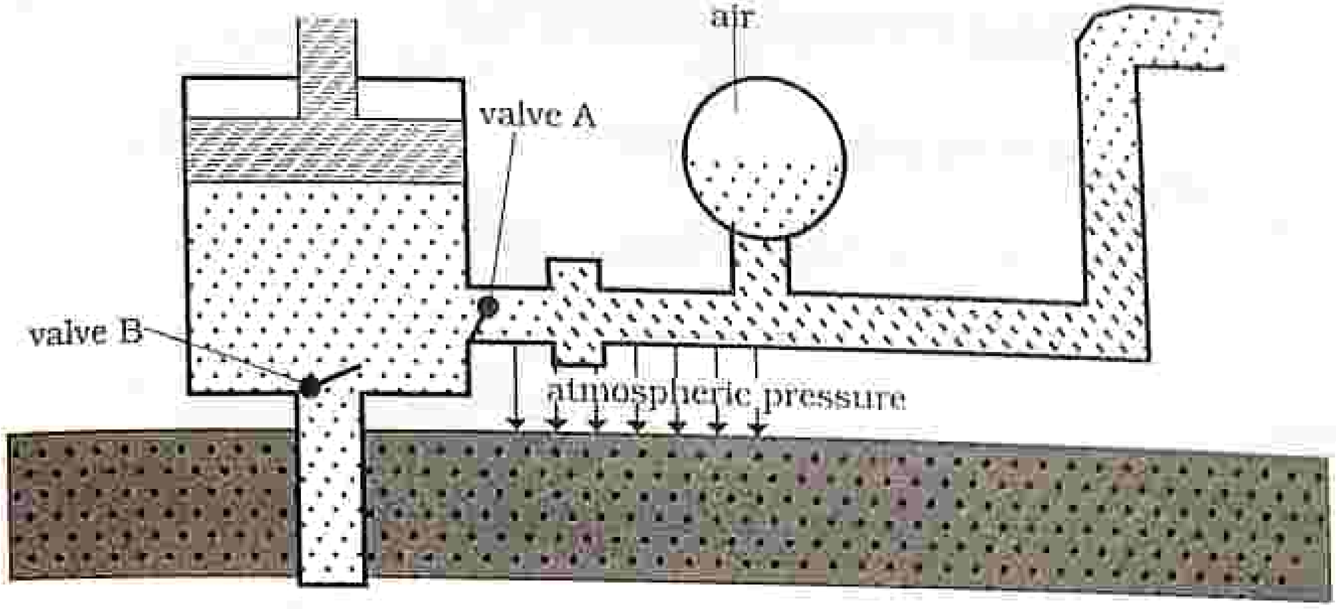
C. Changes the direction of a force.

D. Inclined plane

36. Units for pressure are ____________.

A. Newton B. Square meter C. Pascal D. Watt

37. The diagram shows a force pump.



What happens to the valves during up stroke?

A. Valve A opens valve B closes.

B. Valve B opens valve A closes.

C. Both valves opens.

D. Both valves closes.

38. Calculate the gas pressure when a U tube manometer containing water of density $1000kg/m^3$ is connected to a gas supply. The difference in manometer level is 2m. The atmospheric pressure is 1100Pa

A. 2200kPa B. 2.2kPa C. 200000Pa D. 2000kPa

39. State Newton's third law of motion.

A. For every action there is an equal and opposite reaction.

B. The resultant force acting upon an object is equal to the product of the mass and the acceleration of the object.

C. An object at rest will remain at rest, or if moving will continue to move at constant motion in a straight line unless acted upon by an external force.

D. Is the force acting on the body's mass due to the gravitational force of attraction of the earth.

40. What is the SI unit of mass.

A. meter B. kilogram C. newton D. kelvin

PAPER 2

TIME: 2 HOURS

INSTRUCTIONS TO CANDIDATES

SECTION A

Answer **all** questions

SECTION B

Answer any **two** questions

SECTION C

Answer any **two** questions

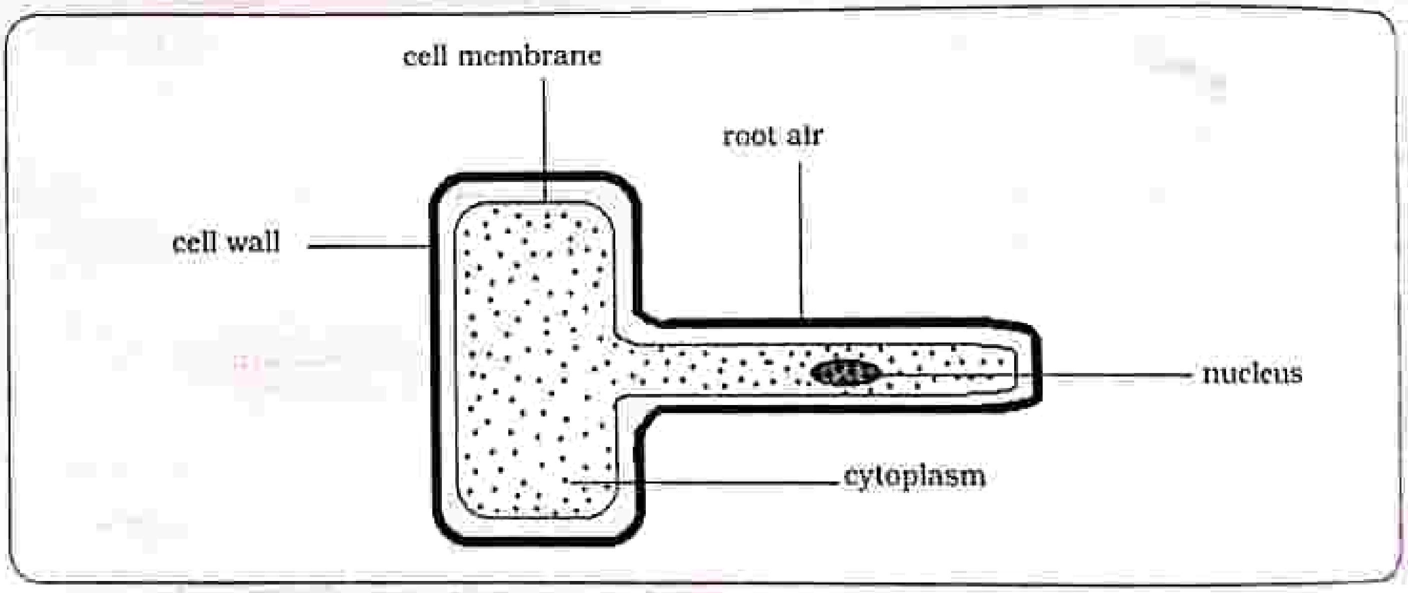
SECTION D

Answer any **two** questions

Section A

Answer all questions in this section

1. Fig 1.1 shows a specialised cell.



a) i) Name the specialised cell shown in Fig 1.1 [1]

ii) Explain how the cell in Fig 1.1 is adapted for its function? [3]

b) i) Define an ecosystem. [1]

ii) List any two physical components of the ecosystem. [2]

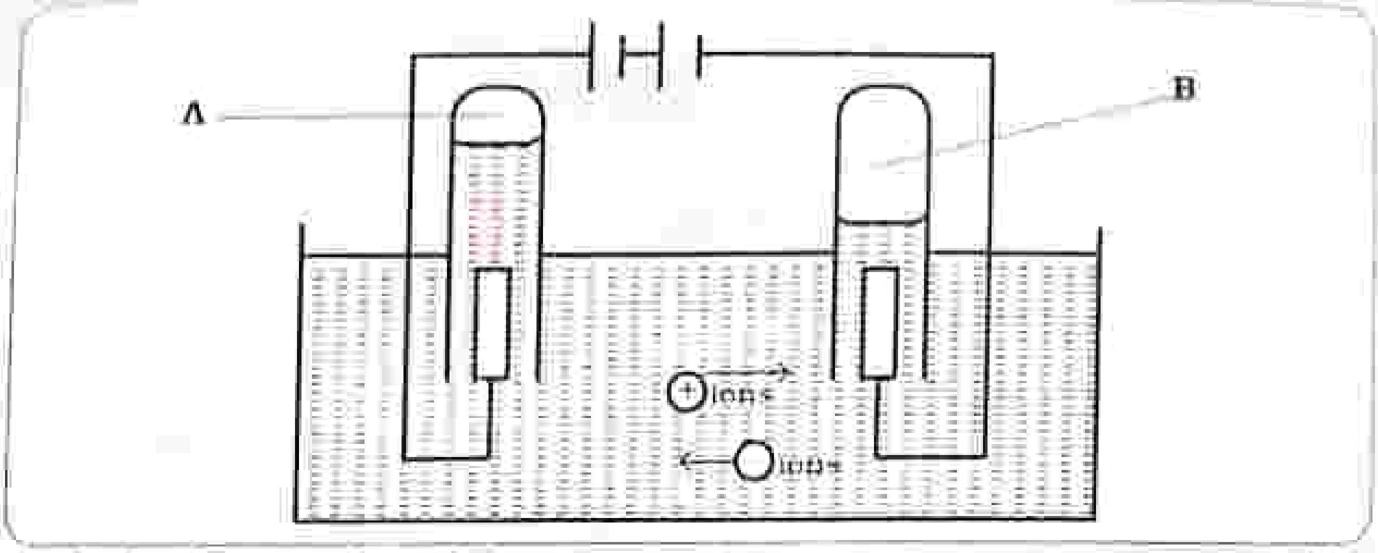
2. i) Identify the factors which affect photosynthesis. [3]

ii) Explain the fate of the end products of photosynthesis

Carbohydrates

Oxygen [4]

3. Fig below shows the diagram for the electrolysis of water.



a) Define the term electrolysis. [1]

b) Name the products formed at

i) A [1]

ii) B [1]

c) Briefly explain how to test for the products formed at

A. [2]

B. [2]

4. a) Define;

i) Oxidation

ii) Reduction [2]

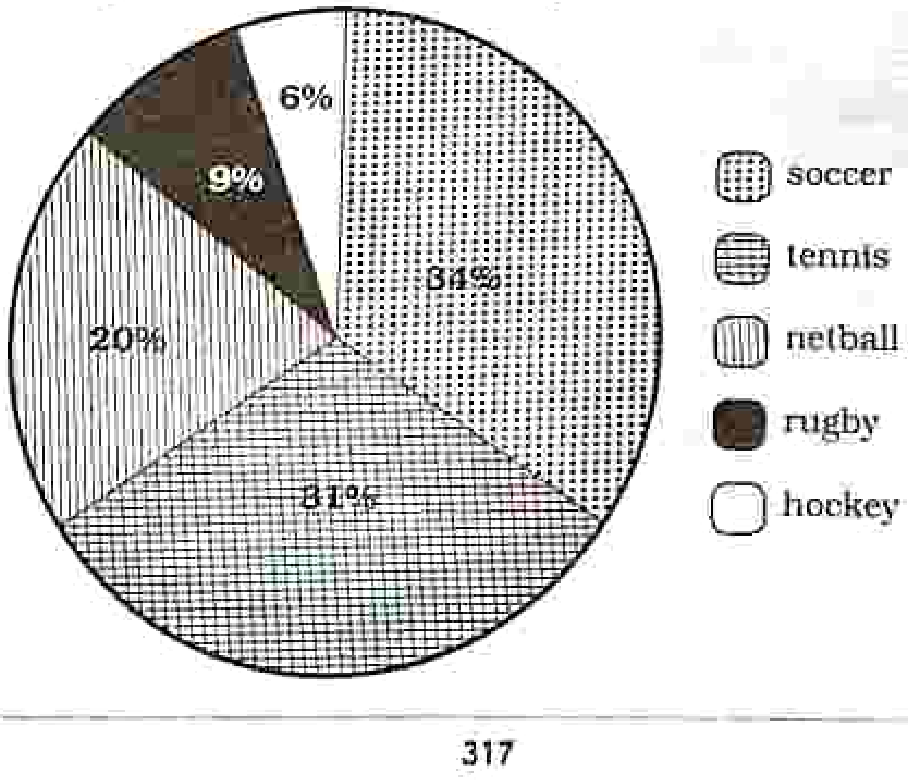
b) State the conditions necessary for rusting to occur [2]

c) What is the function of the following raw materials used in the extraction of iron?

i) coke

ii) iron or [2]

5. The favourite sport for learners in 3Y is represented below:



i) Name the type of data presentation shown [1]
ii) Identify the least favourite sport [1]
iii) Identify the most favourite sport [1]
iv) What percentage represents netball [1]
v) What percentage represents tennis [1]
vi) How many types of sports are represented [1]

6. i) Complete the table below

Physical quantity	SI Unit
Length	....
Mass	....
Time	....
Temperature	....
Electric current	....

[5]

ii) Identify the derived SI units for
Force [1]
Power [1]

Section B

Answer any two questions

7. a) Define the term photosynthesis. [1]
b) Write the word equation for photosynthesis. [2]
c) Explain how a leaf is adapted for photosynthesis. [4]
d) Outline the function of the following components of a balanced diet.
Fats
Proteins
Carbohydrates [3]

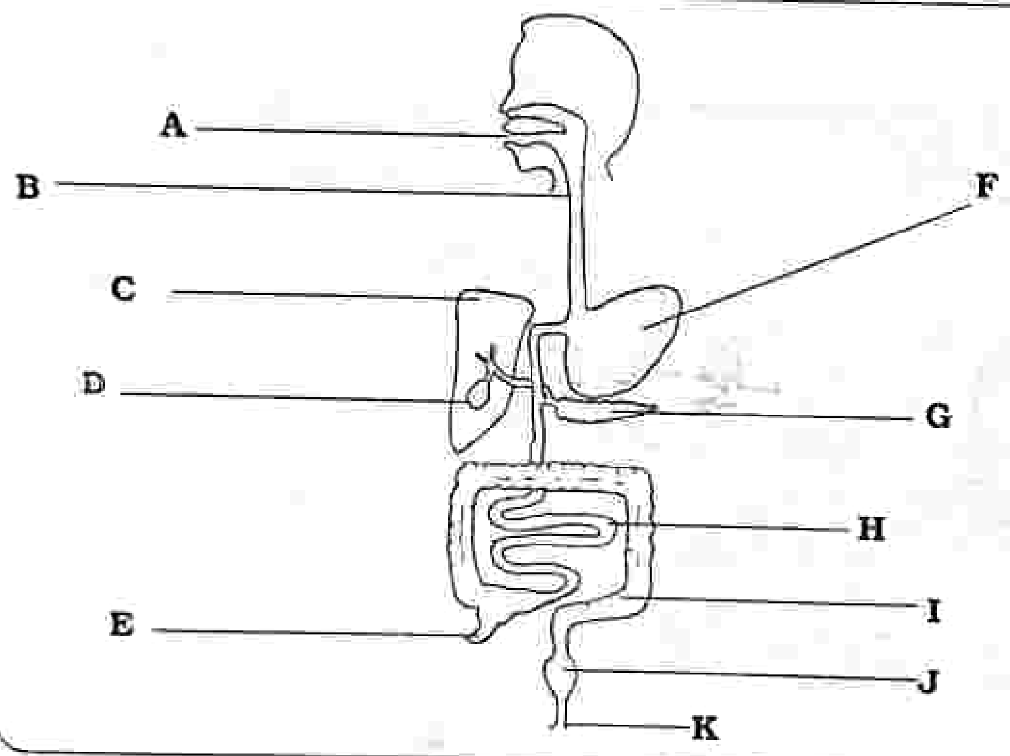
8. a) Define the term digestion. [1]

b) Complete the table of the end products of digestion.

Food Group	End products
Carbohydrates	....
Proteins	....
Fats	....

[3

c) Label the diagram of the human alimentary canal.



[6]

9. a) Define the following terms.

i) Diffusion

ii) Osmosis

iii) Active uptake

iv) Transpiration [4]

b) Describe any four factors which affect the rate of transpiration. [4]

c) Identify the importance of transpiration. [2]

SECTION C

Answer any two questions

10. a) Identify the three forms of fuel. [3]
    b) Methane, Ethane and Propane are alkanes. Write the displayed structure for each.
    c) Write the general formula for alkenes. [1]

11. a) Complete the table on the alloys of iron.

Alloy of iron	Composition	Properties	Uses
Mild steel	0.2 % carbon	Malleable Ductile	--------- ---------
Stainless steel	18% chromium 6% nickel	-----------------	-----------
Cast iron	Up to 4% carbon	------------------	------------

[6]

    b) Explain how iron and slag separates [4]

12. The Haber process is an industrial process.
    a) What are the raw materials used in the manufacture of ammonia. [2]
    b) What are the conditions needed for the manufacture of ammonia. [3]
    c) Outline the stages in the manufacture of ammonia. [4]
    d) What are the uses of ammonia? [1]

SECTION D

Answer any two questions

13. a) State Newton's three laws of motion
    First law
    Second law
    Third law [6]
    b) Distinguish between mass and weight [4]

14. A diesel engine has four strokes.

a) Name the four strokes [4]

A, B, C and D

b) Briefly explain what happens in each stroke. [6]

15. a) List down types of media for signal transmission. [4]

b) Describe two advantages of using a cell phone over a telephone. [2]

c) Describe the operation of a cell phone. [2]

d) Complete the process followed in the transmission of a signal.

Sender ⟶ X ⟶ Medium of transmission ⟶ Decoding ⟶ Y

X [2]

Y